

Namespace FireBlade.WinInteropUtils

Classes

[EnumExtensions](#)

Provides extension methods for [Enum](#) .

[IconExtensions](#)

Adds extension methods to the [Icon](#)  class.

[Macros](#)

Defines Win32 macros.

[MessageLoop](#)

Provides methods for managing message loops.

[Shell32](#)

Declares P/Invoke functions from [Shell32.dll](#).

[StockIconHelper](#)

Allows you to get a shell stock icon. Unlike the built-in .NET [GetStockIcon\(StockIconId, int\)](#)  or [GetStockIcon\(StockIconId, StockIconOptions\)](#)  functions, this wrapper allows you to enter any icon ID you want.

[TitleBarInfo](#)

Stores data about a window's title bar.

[User32](#)

Declares P/Invoke functions from [User32.dll](#).

[VisualStyle](#)

Provides methods for managing and drawing with visual styles.

[WinConstants](#)

Defines common Win32 constants.

[Window](#)

Provides methods for managing Win32 windows.

[WindowClass](#)

Represents a window class.

[WindowLongPtr](#)

Defines default index values for [SetWindowLongPtr\(int, nint\)](#) and [GetWindowLongPtr\(int\)](#).

[WindowStyles](#)

Represents the various standard styles that can be applied to a window (after the window has been created, these styles cannot be modified, except as documented).

[WindowsFile](#)

Stores data about a Windows file.

[ZOrderWindows](#)

Represents windows used for Z position ordering.

Structs

[CreateStruct](#)

The structure passed to the `lParam` parameter of the `WM_CREATE` message of a window.

[DrawTextAdvancedOptions](#)

Represents a structure that contains extended formatting options

[HResult](#)

Represents a COM error code (`HResult`).

[Shell32.SHFILEINFO](#)

The internal Windows API struct for [SHGetFileInfoW\(string, uint, ref SHFILEINFO, uint, uint\)](#)[↗]. Use with [GetFileInfoEx\(string, WindowsFileAttributes, SHGetFileInfoFlags, ref SHFILEINFO\)](#).

[WinMessage](#)

Provides information about a Windows message.

Interfaces

[IHandle](#)

Represents an interface for classes wrapping unmanaged handles.

Enums

[ClassStyle](#)

Represents class styles. Class styles define additional elements of the window class.

[ExWindowStyles](#)

Specifies extended window styles.

[Facility](#)

Represents the facility code of a [HResult](#).

[MessageProcessingLimit](#)

Specifies what message types should be processed by [PeekMessage\(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit\)](#).

[MessageRemoval](#)

Specifies message removal options for [PeekMessage\(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit\)](#).

[Shell32.SHGetFileInfoFlags](#)

Provides flags for the Windows API function [SHGetFileInfoW\(string, uint, ref SHFILEINFO, uint, uint\)](#) [↗].

To use this function, call the [GetFileInfoEx\(string, WindowsFileAttributes, SHGetFileInfoFlags, ref SHFILEINFO\)](#) wrapper instead.

[StockIcon](#)

Icon identifiers for the [GetIcon\(StockIcon, StockIconOptions\)](#) function.

[SystemColorId](#)

Defines the identifiers for system colors.

[SystemFontId](#)

Defines the identifiers for system fonts.

[ThemeTextOptions](#)

Represents the options for [DrawThemeText\(Graphics, int, int, string, int, ThemeTextOptions, Rectangle\)](#).

[TitleBarElement](#)

Defines elements on a title bar.

[TitleBarElementState](#)

Defines the states of a title bar element.

[Win32ErrorCode](#)

Defines Win32 error codes.

[WindowSetPosOptions](#)

Window sizing and positioning flags for [SetPos\(Point, Size, WindowSetPosOptions, Window?\)](#).

[WindowShowCommand](#)

Represents the show command for [Show\(WindowShowCommand\)](#) and [ShowAsync\(WindowShowCommand\)](#).

[WindowState](#)

Defines states a window can be in.

[WindowsFileAttributes](#)

Specifies attributes for Windows files.

Delegates

[WindowClass.WndProc](#)

A callback function, which you define in your application, that processes messages sent to a window. The `WndProc` name is a placeholder for the name of the function that you define in your application.

Enum ClassStyle

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Represents class styles. Class styles define additional elements of the window class.

```
[Flags]  
public enum ClassStyle
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

ByteAlignClient = 4096

Aligns the window's client area on a byte boundary (in the x direction). This style affects the width of the window and its horizontal placement on the display.

ByteAlignWindow = 8192

Aligns the window on a byte boundary (in the x direction). This style affects the width of the window and its horizontal placement on the display.

ClassDeviceContext = 64

Allocates one device context to be shared by all windows in the class. Because window classes are process specific, it is possible for multiple threads of an application to create a window of the same class. It is also possible for the threads to attempt to use the device context simultaneously. When this happens, the system allows only one thread to successfully finish its drawing operation.

DoubleClicks = 8

Sends a double-click message to the window procedure when the user double-clicks the mouse while the cursor is within a window belonging to the class.

DropShadow = 131072

Enables the drop shadow effect on a window. Typically, this is enabled for small, short-lived windows such as menus to emphasize their Z-order relationship to other windows. Windows created from a class with this style must be top-level windows; they may not be child windows.

GlobalClass = 16384

Indicates that the window class is an application global class. For more information, see [Application Global Classes](#)↗.

HRedraw = 2

Redraws the entire window if a movement or size adjustment changes the width of the client area.

NoClose = 512

Disables Close on the window menu.

OwnDeviceContext = 32

Allocates a unique device context for each window in the class.

ParentDC = 128

Sets the clipping rectangle of the child window to that of the parent window so that the child can draw on the parent. A window from a class containing the [ParentDC](#) style receives a regular device context from the system's cache of device contexts. It does not give the child the parent's device context or device context settings. Specifying [ParentDC](#) enhances an application's performance.

SaveBits = 2048

Saves, as a bitmap, the portion of the screen image obscured by a window of this class. When the window is removed, the system uses the saved bitmap to restore the screen image, including other windows that were obscured. Therefore, the system does not send **WM_PAINT** messages to windows that were obscured if the memory used by the bitmap has not been discarded and if other screen actions have not invalidated the stored image.

This style is useful for small windows (for example, menus or dialog boxes) that are displayed briefly and then removed before other screen activity takes place. This style increases the time required to display the window, because the system must first allocate memory to store the bitmap.

VRedraw = 1

Redraws the entire window if a movement or size adjustment changes the height of the client area.

Struct CreateStruct

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

The structure passed to the `lParam` parameter of the `WM_CREATE` message of a window.

```
public struct CreateStruct
```

Inherited Members

[ValueType.Equals\(object\)](#)[↗] , [ValueType.GetHashCode\(\)](#)[↗] , [ValueType.ToString\(\)](#)[↗] ,
[object.Equals\(object, object\)](#)[↗] , [object.GetType\(\)](#)[↗] , [object.ReferenceEquals\(object, object\)](#)[↗]

Fields

Class

The class of the new window.

```
public WindowClass Class
```

Field Value

[WindowClass](#)

CreateParam

The parameter passed to the `param` parameter of [Window](#).

```
public object? CreateParam
```

Field Value

[object](#)[↗]

ExStyle

The extended window style for the new window.

```
public ExWindowStyles ExStyle
```

Field Value

[ExWindowStyles](#)

HInstance

A handle to the module that owns the new window.

```
public nint HInstance
```

Field Value

[nint](#)

HMenu

A handle to the menu to be used by the new window.

```
public nint HMenu
```

Field Value

[nint](#)

Location

The location of the new window. If the new window is a child window, coordinates are relative to the parent window. Otherwise, the coordinates are relative to the screen origin.

```
public Point Location
```

Field Value

[Point](#)

Parent

The parent window, if the window is a child window. If the window is owned, this member identifies the owner window. If the window is not a child or owned window, this member is [null](#).

```
public Window? Parent
```

Field Value

[Window](#)

Size

The size of the new window, in pixels.

```
public Size Size
```

Field Value

[Size](#)

Style

The style for the new window. For a list of the common shared values, see [WindowStyles](#).

```
public long Style
```

Field Value

[long](#)

Text

The text of the new window.

```
public string Text
```

Field Value

[string](#) 

Struct DrawTextAdvancedOptions

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Represents a structure that contains extended formatting options

```
public struct DrawTextAdvancedOptions
```

Inherited Members

[ValueType.Equals\(object\)](#)[☞] , [ValueType.GetHashCode\(\)](#)[☞] , [ValueType.ToString\(\)](#)[☞] , [object.Equals\(object, object\)](#)[☞] , [object.GetType\(\)](#)[☞] , [object.ReferenceEquals\(object, object\)](#)[☞]

Properties

LeftMargin

The left margin, in units equal to the average character width.

```
public int LeftMargin { readonly get; set; }
```

Property Value

[int](#)[☞]

LengthDrawn

Receives the number of characters processed by [DrawText\(Graphics, string, ref Rectangle, ThemeTextOptions, ref DrawTextAdvancedOptions\)](#), including white-space characters. The number can be the length of the string or the index of the first line that falls below the drawing area. Note that [DrawText\(Graphics, string, ref Rectangle, ThemeTextOptions, ref DrawTextAdvancedOptions\)](#) always processes the entire string if the [NoClip](#) formatting flag is specified.

```
public readonly uint LengthDrawn { get; }
```

Property Value

[uint](#)

RightMargin

The right margin, in units equal to the average character width.

```
public int RightMargin { readonly get; set; }
```

Property Value

[int](#)

TabLength

The size of each tab stop, in units equal to the average character width.

```
public int TabLength { readonly get; set; }
```

Property Value

[int](#)

Class EnumExtensions

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Provides extension methods for [Enum](#).

```
public static class EnumExtensions
```

Inheritance

[object](#) ← EnumExtensions

Inherited Members

[object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Methods

GetValues(Enum)

Gets a list of values that were combined for an enum with the [\[Flags\] attribute](#).

```
public static IEnumerable<Enum> GetValues(this Enum e)
```

Parameters

e [Enum](#)

The enum.

Returns

[IEnumerable](#) <[Enum](#)>

The list of combined values that were making up the Enum.

Enum ExWindowStyles

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Specifies extended window styles.

[Flags]

```
public enum ExWindowStyles : long
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

AcceptFiles = 16

The window accepts drag-drop files.

AppWindow = 262144

Forces a top-level window onto the taskbar when the window is visible.

ClientEdge = 512

The window has a border with a sunken edge.

Composited = 33554432

Paints all descendants of a window in bottom-to-top painting order using double-buffering. Bottom-to-top painting order allows a descendent window to have translucency (alpha) and transparency (color-key) effects, but only if the descendent window also has the [Transparent](#) extended style set. Double-buffering allows the window and its descendants to be painted without flicker. This cannot be used if the window has a class style of [OwnDeviceContext](#), [ClassDeviceContext](#), or [ParentDC](#).

ContextHelp = 1024

The title bar of the window includes a question mark. When the user clicks the question mark, the cursor changes to a question mark with a pointer. If the user then clicks a child window, the child receives a WM_HELP message. The child window should pass the message to the parent window procedure, which should call the `WinHelp` function using the `HELP_WM_HELP` command. The Help application displays a pop-up window that typically contains help for the child window.

[ContextHelp](#) cannot be used with the [MaximizeBox](#) or [MinimizeBox](#) styles.

ControlParent = 65536

The window itself contains child windows that should take part in dialog box navigation. If this style is specified, the dialog manager recurses into children of this window when performing navigation operations such as handling the **TAB** key, an arrow key, or a keyboard mnemonic.

DlgModalFrame = 1

The window has a double border; the window can, optionally, be created with a title bar by specifying the [Caption](#) style in the **style** parameter.

LTRReading = 0

The window text is displayed using left-to-right reading-order properties. This is the default.

Layered = 524288

The window is a layered window. This style cannot be used if the window has a class style of either [OwnDeviceContext](#) or [ClassDeviceContext](#).

** NOTE**

Windows 8: The [Layered](#) style is supported for top-level windows and child windows. Previous Windows versions support [Layered](#) only for top-level windows.

LayoutRTL = 4194304

If the shell language is Hebrew, Arabic, or another language that supports reading order alignment, the horizontal origin of the window is on the right edge. Increasing horizontal values advance to the left.

Left = 0

The window has generic left-aligned properties. This is the default.

LeftScrollBar = 16384

If the shell language is Hebrew, Arabic, or another language that supports reading order alignment, the vertical scroll bar (if present) is to the left of the client area. For other languages, the style is ignored.

MdiChild = 64

The window is a MDI (Multi-Document Interface) child window.

NoActivate = 134217728

A top-level window created with this style does not become the foreground window when the user clicks it. The system does not bring this window to the foreground when the user minimizes or closes the foreground window. The window should not be activated through programmatic access or via keyboard navigation by accessible technology, such as Narrator.

To activate the window, use the **SetActiveWindow** or **SetForegroundWindow** function. The window does not appear on the taskbar by default. To force the window to appear on the taskbar, use the [App Window](#) extended style.

NoInheritLayout = 1048576

The window does not pass its window layout to its child windows.

NoParentNotify = 4

The child window created with this style does not send the **WM_PARENTNOTIFY** message to its parent window when it is created or destroyed.

NoRedirectionBitmap = 2097152

The window does not render to a redirection surface. This is for windows that do not have visible content or that use mechanisms other than surfaces to provide their visual.

OverlappedWindow = WindowEdge | ClientEdge

The window is an overlapped window.

PaletteWindow = TopMost | ToolWindow | WindowEdge

The window is palette window, which is a modeless dialog box that presents an array of commands.

RTLReading = 8192

If the shell language is Hebrew, Arabic, or another language that supports reading-order alignment, the window text is displayed using right-to-left reading-order properties. For other languages, the style is ignored.

Right = 4096

The window has generic "right-aligned" properties. This depends on the window class. This style has an effect only if the shell language is Hebrew, Arabic, or another language that supports reading-order alignment; otherwise, the style is ignored.

Using the [Right](#) style for static or edit controls has the same effect as using the `SS_RIGHT` or `ES_RIGHT` style, respectively. Using this style with button controls has the same effect as using `BS_RIGHT` and `BS_RIGHTBUTTON` styles.

`RightScrollBar = 0`

The vertical scroll bar (if present) is to the right of the client area. This is the default.

`StaticEdge = 131072`

The window has a three-dimensional border style intended to be used for items that do not accept user input.

`ToolWindow = 128`

The window is intended to be used as a floating toolbar. A tool window has a title bar that is shorter than a normal title bar, and the window title is drawn using a smaller font. A tool window does not appear in the taskbar or in the dialog that appears when the user presses `ALT+TAB`. If a tool window has a system menu, its icon is not displayed on the title bar. However, you can display the system menu by right-clicking or by typing `ALT+SPACE`.

`TopMost = 8`

The window should be placed above all non-topmost windows and should stay above them, even when the window is deactivated. To add or remove this style, use the [SetPos\(Point, Size, WindowSet PosOptions, Window?\)](#) function.

`Transparent = 32`

The window should not be painted until siblings beneath the window (that were created by the same thread) have been painted. The window appears transparent because the bits of underlying sibling windows have already been painted.

`WindowEdge = 256`

The window has a border with a raised edge.

Enum Facility

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Represents the facility code of a [HResult](#).

```
public enum Facility
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

AAF = 18

Advanced Authoring Format (AAF).

ACS = 20

Access Control Services.

Accelerator = 1536

Hardware acceleration.

ActionQueue = 44

Action queue infrastructure.

Audio = 102

Audio subsystem.

AudioClient = 2185

Windows Audio Client (WASAPI).

AudioStreaming = 1094

Audio streaming.

BCD = 57

Boot Configuration Data (BCD).

BLBCommandLine = 121

Blob command-line interface (alias of BlbCLI).

BackgroundCopy = 32

Background Intelligent Transfer Service (BITS).

Blb = 120

Blob storage infrastructure.

BlbCLI = 121

Blob storage command-line interface.

BlbUI = 128

Blob storage user interface.

BluetoothATT = 101

Bluetooth Attribute Protocol (ATT).

CMI = 54

Configuration Management Infrastructure.

COMPlus = 17

COM+ services.

Cert = 11

Certificate services.

Configuration = 33

System configuration services.

Control = 10

Control mechanism and device control.

DLS = 153

Dynamic Link Services.

DMServer = 256

Device Management Server.

DPlay = 21

DirectPlay (network gaming).

DXGI = 2170

DirectX Graphics Infrastructure (DXGI).

DXGIDDI = 2171

DXGI Device Driver Interface.

Debuggers = 176

Debugger infrastructure.

Defragmenter = 2304

Disk defragmentation service.

DeploymentServicesBinlService = 261

Deployment Services BINL service.

DeploymentServicesContentProvider = 293

Deployment Services content provider.

DeploymentServicesDriverProvisioning = 278

Deployment Services driver provisioning.

DeploymentServicesImaging = 258

Deployment Services imaging.

DeploymentServicesManagement = 259

Deployment Services management.

DeploymentServicesMulticastClient = 290

Deployment Services multicast client.

DeploymentServicesMulticastServer = 289

Deployment Services multicast server.

DeploymentServicesPXE = 263

Deployment Services PXE provider.

DeploymentServicesServer = 257

Windows Deployment Services (server).

DeploymentServicesTFTP = 264

Deployment Services TFTP service.

DeploymentServicesTransportManagement = 272

Deployment Services transport management.

DeploymentServicesUtil = 260

Deployment Services utilities.

DirService = 37

Directory Services (e.g., Active Directory).

Direct2D = 2201

Direct2D graphics API.

Direct3D10 = 2169

Direct3D 10.

Direct3D11 = 2172

Direct3D 11.

DirectMusic = 2168

DirectMusic API.

Dispatch = 2

COM IDispatch (late binding) subsystem.

EAP = 66

Extensible Authentication Protocol (EAP).

EAS = 85

Exchange ActiveSync.

FAG = 100

File Access Gateway.

FVE = 49

Full Volume Encryption (BitLocker).

FWP = 50

Windows Filtering Platform (firewall).

Graphics = 38

Graphics subsystem.

HTTP = 25

HTTP services.

ITF = 4

Interface-specific (custom) error codes.

Input = 64

Input subsystem.

Internet = 12

Internet (WinINet / networking).

JScript = 2306

JScript engine.

Leap = 2184

Project Leap / internal graphics component.

LinguisticServices = 305

Linguistic services.

MSMQ = 14

Microsoft Message Queuing (MSMQ).

MediaServer = 13

Media server services.

MetaDirectory = 35

Meta-directory services.

Mobile = 1793

Mobile platform.

NDIS = 52

Network Driver Interface Specification (NDIS).

OPC = 81

Open Packaging Conventions (OPC).

OnlineID = 134

Online identity services.

P2P = 99

Peer-to-peer networking.

P2PInt = 98

Peer-to-peer infrastructure (internal).

PIDGeneratorX = 2561

Product ID generation and licensing.

Parse = 113

Parsing subsystem.

PowerShell = 84

PowerShell engine.

RAS = 83

Remote Access Service (RAS).

RPC = 1

Remote Procedure Call (RPC) subsystem.

Restore = 256

System restore (alias of DMServer).

SCard = 16

Smart card subsystem.

SDiagnostics = 60

System diagnostics.

SOS = 160

Son of Strike (SOS) debugging extension.

SPP = 256

Software Protection Platform (alias of DMServer).

SSPI = 9

Security Support Provider Interface (alias of Security).

Script = 112

Scripting engine.

Security = 9

Security and authentication subsystem.

SetupAPI = 15

Setup API (device installation).

Shell = 39

Windows Shell.

StateManagement = 34

State management infrastructure.

Storage = 3

Structured storage subsystem.

SxS = 23

Side-by-side assembly (SxS) infrastructure.

SyncEngine = 2050

Synchronization engine.

TPMServices = 40

Trusted Platform Module (TPM) services.

TPMSoftware = 41

TPM software stack.

Tiering = 131

Storage tiering.

UI = 42

User interface framework.

UMI = 22

Unified Messaging infrastructure.

URT = 19

.NET Common Language Runtime (CLR).

USN = 129

Update Sequence Number (USN) journal.

`Unspecified = 0`

The facility code is not specified.

`UserModeCommonLog = 26`

User-mode Common Log File System (CLFS).

`UserModeFilterManager = 31`

User-mode Filter Manager.

`UserModeHypervisor = 53`

User-mode Hyper-V (virtualization).

`UserModeSDBus = 2305`

User-mode SD bus driver.

`UserModeVHD = 58`

User-mode Virtual Hard Disk (VHD).

`UserModeVirtualization = 55`

User-mode virtualization.

`UserModeVolumeManager = 56`

User-mode volume manager.

`UserModeVolumeSnap = 130`

User-mode Volume Snapshot Service.

`VisualCPP = 109`

Microsoft Visual C++ runtime.

`WEP = 2049`

Windows Encryption Platform.

`WMAAECMA = 1996`

Windows Media Audio (AECMA).

WPN = 62

Windows Push Notifications.

WSBApp = 122

Windows Sandbox application.

WSBOnline = 133

Windows Sandbox online services.

Web = 885

Web platform.

WebServices = 61

Web Services (WS-* APIs).

WebSocket = 886

WebSocket protocol.

Win32 = 7

Win32 error codes mapped to HRESULT.

WinCE = 24

Windows CE subsystem.

WinCodecDWriteDwm = 2200

Alias of WinCodecDirectWriteDesktopWindowManager.

WinCodecDirectWriteDesktopWindowManager = 2200

Windows Imaging / DirectWrite / DWM integration.

WinDefender = 80

Windows Defender.

WinPE = 61

Windows Preinstallation Environment (alias of WebServices).

WinRM = 51

Windows Remote Management (WinRM).

WinStore = 63

Windows Store / app platform.

WinUpdate = 36

Windows Update subsystem.

Windows = 8

General Windows subsystem.

WindowsErrorReport = 27

Windows Error Reporting (WER).

WindowsSetup = 48

Windows Setup and installation.

XAML = 43

XAML UI framework.

XPS = 82

XML Paper Specification (XPS).

XmlPrinterSpec = 82

XML Paper Specification (alias of XPS).

Struct HRESULT

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Represents a COM error code ([HRESULT](#)).

```
public struct HRESULT
```

Inherited Members

[object.Equals\(object, object\)](#) , [object.GetType\(\)](#) , [object.ReferenceEquals\(object, object\)](#)

Constructors

HRESULT(int)

Creates a new [HRESULT](#) from an error code.

```
public HRESULT(int code)
```

Parameters

code [int](#)

The error code of the [HRESULT](#).

HRESULT(int, Facility, int)

Creates a new [HRESULT](#) from the specified facility, severity, and base codes.

```
public HRESULT(int baseCode, Facility facility = Facility.Unspecified, int severity = 0)
```

Parameters

baseCode [int](#)

The base error code.

facility [Facility](#)

The facility code of the error.

severity [int](#)

The severity of the error.

Properties

BaseCode

Gets the base error code, which specifies the type of error.

```
public int BaseCode { get; }
```

Property Value

[int](#)

Exception

Gets the .NET [Exception](#) represented by this [HResult](#).

```
public Exception? Exception { get; }
```

Property Value

[Exception](#)

Facility

Gets the facility code, which specifies the location where the error came from.

```
public Facility Facility { get; }
```

Property Value

[Facility](#)

See Also

[FACILITY_WIN32](#)

FullCode

Gets the full [HRESULT](#) code.

```
public int FullCode { get; }
```

Property Value

[int](#)

Message

Gets the message of the [HRESULT](#).

```
public string Message { get; }
```

Property Value

[string](#)

Severity

Gets the severity of the error.

```
public int Severity { get; }
```

Property Value

[int](#)

Methods

Equals(object?)

Indicates whether this [HResult](#) and the specified [HResult](#) represent the same code.

```
public override bool Equals(object? obj)
```

Parameters

obj [object](#)[↗]

The [HResult](#) to compare to.

Returns

[bool](#)[↗]

[true](#)[↗] if this [HResult](#) and the specified [HResult](#) represent the same code; otherwise, [false](#)[↗].

GetHashCode()

Returns the hash code for this instance.

```
public override int GetHashCode()
```

Returns

[int](#)[↗]

A 32-bit signed integer that is the hash code for this instance.

ToString()

Returns the fully qualified type name of this instance.

```
public override string ToString()
```

Returns

[string](#)[↗]

The fully qualified type name.

Operators

operator ==(HResult?, HResult?)

Indicates whether two [HResult](#) values represent the same code.

```
public static bool operator ==(HResult? left, HResult? right)
```

Parameters

left [HResult?](#)

The first [HResult](#) to compare.

right [HResult?](#)

The second [HResult](#) to compare to.

Returns

[bool](#)[↗]

[true](#)[↗] if both [HResult](#) values represent the same code; otherwise, [false](#)[↗].

implicit operator Exception(HResult)

Converts this [HResult](#) to its .NET [Exception](#)[↗] representation.

```
public static implicit operator Exception(HResult hr)
```

Parameters

hr [HResult](#)

The [HResult](#) to convert.

Returns

[Exception](#) 

implicit operator int(HResult)

Converts this [HResult](#) to its full code.

```
public static implicit operator int(HResult hr)
```

Parameters

hr [HResult](#)

The [HResult](#) to convert.

Returns

[int](#) 

implicit operator HResult(int)

Converts this code to the equivalent [HResult](#).

```
public static implicit operator HResult(int c)
```

Parameters

c [int](#) 

The code to convert.

Returns

[HResult](#)

operator !=(HResult?, HResult?)

Indicates whether two [HResult](#) values don't represent the same code.

```
public static bool operator !=(HResult? left, HResult? right)
```

Parameters

left [HResult?](#)

The first [HResult](#) to compare.

right [HResult?](#)

The second [HResult](#) to compare to.

Returns

[bool](#)[↗]

[true](#)[↗] if both [HResult](#) values don't represent the same code; otherwise, [false](#)[↗].

Interface IHandle

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Represents an interface for classes wrapping unmanaged handles.

```
public interface IHandle
```

Properties

Handle

The handle of the object.

```
nint Handle { get; }
```

Property Value

[nint](#)

Class IconExtensions

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Adds extension methods to the [Icon](#) class.

```
public static class IconExtensions
```

Inheritance

[object](#) ← IconExtensions

Inherited Members

[object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Methods

Destroy(Icon)

Destroys the icon's GDI handle.

```
public static bool Destroy(this Icon icon)
```

Parameters

icon [Icon](#)

The icon whose handle is to be destroyed.

Returns

[bool](#)

[true](#) if successful; otherwise, [false](#). To get extended error information, call [GetLastPInvokeError\(\)](#).

Class Macros

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Defines Win32 macros.

```
public static class Macros
```

Inheritance

[object](#)  ← Macros

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Methods

Failed(HResult)

Provides a generic test for failure on any [HResult](#) status value.

```
public static bool Failed(HResult hr)
```

Parameters

hr [HResult](#)

The status code. This value can be a [HResult](#). A non-negative number indicates success.

Returns

[bool](#) 

[true](#)  if the status code is successful; otherwise, [false](#) .

Failed(Win32ErrorCode)

Provides a generic test for failure on any [Win32ErrorCode](#) status value.

```
public static bool Failed(Win32ErrorCode hr)
```

Parameters

hr [Win32ErrorCode](#)

The status code. This value can be a [Win32ErrorCode](#). A negative number indicates failure.

Returns

[bool](#)

[true](#) if the status code indicates failure; otherwise, [false](#).

HResultFromWin32(Win32ErrorCode)

Maps a system error code to an [HResult](#) value.

```
public static HResult HResultFromWin32(this Win32ErrorCode error)
```

Parameters

error [Win32ErrorCode](#)

The system error code.

Returns

[HResult](#)

The created [HResult](#).

HighByte(decimal)

Retrieves the high-order byte from the given value.

```
public static byte HighByte(decimal w)
```

Parameters

w [decimal](#)

The value to be converted.

Returns

[byte](#)

The high-order byte of the specified value.

HighByte(double)

Retrieves the high-order byte from the given value.

```
public static byte HighByte(double w)
```

Parameters

w [double](#)

The value to be converted.

Returns

[byte](#)

The high-order byte of the specified value.

HighByte(int)

Retrieves the high-order byte from the given value.

```
public static byte HighByte(int w)
```

Parameters

w [int](#)

The value to be converted.

Returns

[byte](#)

The high-order byte of the specified value.

HighByte(float)

Retrieves the high-order byte from the given value.

```
public static byte HighByte(float w)
```

Parameters

w [float](#)

The value to be converted.

Returns

[byte](#)

The high-order byte of the specified value.

HighByte<TNum>(TNum)

Retrieves the high-order byte from the given value.

```
public static byte HighByte<TNum>(TNum w) where TNum : INumber<TNum>
```

Parameters

w TNum

The value to be converted.

Returns

[byte](#)

The high-order byte of the specified value.

Type Parameters

TNum

The number type.

HighWord(decimal)

Retrieves the high-order word from the specified 32-bit value.

```
public static ushort HighWord(decimal 1)
```

Parameters

1 [decimal](#)

The value to be converted.

Returns

[ushort](#)

The high-order word of the value specified by **1**.

HighWord(double)

Retrieves the high-order word from the specified 32-bit value.

```
public static ushort HighWord(double 1)
```

Parameters

1 [double](#)

The value to be converted.

Returns

[ushort](#)

The high-order word of the value specified by 1.

HighWord(int)

Retrieves the high-order word from the specified 32-bit value.

```
public static ushort HighWord(int 1)
```

Parameters

1 [int](#)

The value to be converted.

Returns

[ushort](#)

The high-order word of the value specified by 1.

HighWord(float)

Retrieves the high-order word from the specified 32-bit value.

```
public static ushort HighWord(float 1)
```

Parameters

1 [float](#)

The value to be converted.

Returns

[ushort](#)

The high-order word of the value specified by **1**.

HighWord<TNum>(TNum)

Retrieves the high-order word from the specified 32-bit value.

```
public static ushort HighWord<TNum>(TNum 1) where TNum : INumber<TNum>
```

Parameters

1 TNum

The value to be converted.

Returns

[ushort](#)

The high-order word of the value specified by **1**.

Type Parameters

TNum

The number type.

LowByte(decimal)

Retrieves the low-order byte from the specified value.

```
public static byte LowByte(decimal w)
```

Parameters

w [decimal](#)

The value to be converted.

Returns

[byte](#)

The low-order byte of the value specified by `w`.

LowByte(double)

Retrieves the low-order byte from the specified value.

```
public static byte LowByte(double w)
```

Parameters

`w` [double](#)

The value to be converted.

Returns

[byte](#)

The low-order byte of the value specified by `w`.

LowByte(int)

Retrieves the low-order byte from the specified value.

```
public static byte LowByte(int w)
```

Parameters

`w` [int](#)

The value to be converted.

Returns

[byte](#)

The low-order byte of the value specified by `w`.

LowByte(float)

Retrieves the low-order byte from the specified value.

```
public static byte LowByte(float w)
```

Parameters

`w` [float](#)

The value to be converted.

Returns

[byte](#)

The low-order byte of the value specified by `w`.

LowByte<TNum>(TNum)

Retrieves the low-order byte from the specified value.

```
public static byte LowByte<TNum>(TNum w) where TNum : INumber<TNum>
```

Parameters

`w` TNum

The value to be converted.

Returns

[byte](#)

The low-order byte of the value specified by `w`.

Type Parameters

TNum

The number type.

LowWord(decimal)

Retrieves the low-order word from the specified value.

```
public static ushort LowWord(decimal 1)
```

Parameters

1 [decimal](#)

The value to be converted.

Returns

[ushort](#)

The low-order word of the value specified by 1.

LowWord(double)

Retrieves the low-order word from the specified value.

```
public static ushort LowWord(double 1)
```

Parameters

1 [double](#)

The value to be converted.

Returns

[ushort](#)

The low-order word of the value specified by [1](#).

LowWord(int)

Retrieves the low-order word from the specified value.

```
public static ushort LowWord(int 1)
```

Parameters

[1](#) [int](#)

The value to be converted.

Returns

[ushort](#)

The low-order word of the value specified by [1](#).

LowWord(float)

Retrieves the low-order word from the specified value.

```
public static ushort LowWord(float 1)
```

Parameters

[1](#) [float](#)

The value to be converted.

Returns

[ushort](#)

The low-order word of the value specified by [1](#).

LowWord<TNum>(TNum)

Retrieves the low-order word from the specified value.

```
public static ushort LowWord<TNum>(TNum 1) where TNum : INumber<TNum>
```

Parameters

1 TNum

The value to be converted.

Returns

[ushort](#)

The low-order word of the value specified by **1**.

Type Parameters

TNum

The number type.

MakeIntAtom<TNum>(TNum)

Converts the specified atom into a string, so it can be passed to functions which accept either atoms or strings.

```
public static string MakeIntAtom<TNum>(TNum num) where TNum : INumber<TNum>
```

Parameters

num TNum

The numeric value to be made into an integer atom. This parameter can be either an integer atom or a string atom.

Returns

[string](#)

The atom, converted to a value that can be passed to functions that accept atoms or strings.

Type Parameters

TNum

The number type.

Remarks

Although the return value of the MAKEINTATOM macro is cast as a string value, it cannot be used as a string pointer except when it is passed to atom-management functions that require a string argument.

MakeIntResource(int)

Converts an integer value to a resource type compatible with the resource-management functions. This macro is used in place of a string containing the name of the resource.

```
public static string MakeIntResource(int i)
```

Parameters

i [int](#)

The integer value to be converted.

Returns

[string](#)

The created resource.

Remarks

The return value should be passed only to functions which explicitly indicate that they accept [MakeIntResource\(int\)](#) as a parameter. For example, the resource management functions allow the return value of [MakeIntResource\(int\)](#) to be passed as the **lpType** or **lpName** parameters.

Succeeded(HResult)

Provides a generic test for success on any [HResult](#) status value.

```
public static bool Succeeded(HResult hr)
```

Parameters

hr [HResult](#)

The status code. This value can be a [HResult](#). A non-negative number indicates success.

Returns

[bool](#)

[true](#) if the status code is successful; otherwise, [false](#).

Succeeded(Win32ErrorCode)

Provides a generic test for success on any [Win32ErrorCode](#) status value.

```
public static bool Succeeded(Win32ErrorCode hr)
```

Parameters

hr [Win32ErrorCode](#)

The status code. This value can be a [Win32ErrorCode](#). A non-negative number indicates success.

Returns

[bool](#)

[true](#) if the status code is successful; otherwise, [false](#).

Class MessageLoop

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Provides methods for managing message loops.

```
public static class MessageLoop
```

Inheritance

[object](#)  ← MessageLoop

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Examples

To create a message loop, use the following pattern:

```
WinMessage msg = new WinMessage();
while (MessageLoop.GetMessage(ref msg, null, 0, 0))
{
    MessageLoop.TranslateMessage(ref msg);
    MessageLoop.DispatchMessage(ref msg);
}
```

Methods

DispatchMessage(ref WinMessage)

Dispatches a message to a window procedure. It is typically used to dispatch a message retrieved by the [GetMessage\(ref WinMessage, Window?, uint, uint\)](#) function.

```
public static nint DispatchMessage(ref WinMessage msg)
```

Parameters

msg [WinMessage](#)

A structure that contains the message.

Returns

uint [↗](#)

The return value specifies the value returned by the window procedure. Although its meaning depends on the message being dispatched, the return value generally is ignored.

Remarks

The [WinMessage](#) structure must contain valid message values. If the **msg** parameter points to a **WM_TIMER** message and the **lParam** parameter of the **WM_TIMER** message is not [null](#) [↗](#), **lParam** points to a function that is called instead of the window procedure.

Note that the application is responsible for retrieving and dispatching input messages to the dialog box. Most applications use the main message loop for this. However, to permit the user to move to and to select controls by using the keyboard, the application must call **IsDialogMessage**. For more information, see [Dialog Box Keyboard Interface](#) [↗](#).

GetMessage(ref WinMessage, Window?, uint, uint)

Retrieves a message from the calling thread's message queue. The function dispatches incoming sent messages until a posted message is available for retrieval.

```
public static bool GetMessage(ref WinMessage msg, Window? wnd, uint minMessageFilter,
uint maxMessageFilter)
```

Parameters

msg [WinMessage](#)

A [WinMessage](#) structure that receives message information from the thread's message queue.

wnd [Window](#)

The window whose messages are to be retrieved. The window must belong to the current thread.

minMessageFilter [uint](#) [↗](#)

The integer value of the lowest message value to be retrieved. Use `WM_KEYFIRST` (0x0100) to specify the first keyboard message or `WM_MOUSEFIRST` (0x0200) to specify the first mouse message.

`maxMessageFilter` [uint](#)

The integer value of the highest message value to be retrieved. Use `WM_KEYLAST` to specify the last keyboard message or `WM_MOUSELAST` to specify the last mouse message.

Returns

[bool](#)

[true](#) if the function retrieves a message other than `WM_QUIT`; otherwise, [false](#).

Remarks

If `wnd` is [null](#), [GetMessage\(ref WinMessage, Window?, uint, uint\)](#) retrieves messages for any window that belongs to the current thread, and any messages on the current thread's message queue whose `hwnd` value is [Zero](#). Therefore if `wnd` is [null](#), both window messages and thread messages are processed.

Use `WM_INPUT` in `minMessageFilter` and `maxMessageFilter` to specify only the `WM_INPUT` messages.

If `minMessageFilter` and `maxMessageFilter` are both zero, [GetMessage\(ref WinMessage, Window?, uint, uint\)](#) returns all available messages (that is, no range filtering is performed).

An application typically uses the return value to determine whether to end the main message loop and exit the program.

The [GetMessage\(ref WinMessage, Window?, uint, uint\)](#) function retrieves messages associated with the window in `wnd` parameter or any of its children, as specified by the `IsChild` function, and within the range of message values given by the `minMessageFilter` and `maxMessageFilter` parameters. Note that an application can only use the low word in the `maxMessageFilter` and `maxMessageFilter` parameters; the high word is reserved for the system.

Note that [GetMessage\(ref WinMessage, Window?, uint, uint\)](#) always retrieves `WM_QUIT` messages, no matter which values you specify for `minMessageFilter` and `maxMessageFilter`.

During this call, the system delivers pending, nonqueued messages, that is, messages sent to windows owned by the calling thread using the [SendMessage\(uint, uint, uint\)](#), [SendMessageCallback](#), [SendMessageTimeout](#), or [SendNotifyMessage](#) function. Then the first queued message that matches the specified filter is retrieved. The system may also process internal events. If no filter is specified, messages are processed in the following order:

1. Sent messages

2. Posted messages
3. Input (hardware) messages and system internal events
4. Sent messages (again)
5. `WM_PAINT` messages
6. `WM_TIMER` messages

To retrieve input messages before posted messages, use the `minMessageFilter` and `maxMessageFilter` parameters.

[GetMessage\(ref WinMessage, Window?, uint, uint\)](#) does not remove `WM_PAINT` messages from the queue. The messages remain in the queue until processed.

If a top-level window stops responding to messages for more than several seconds, the system considers the window to be not responding and replaces it with a ghost window that has the same z-order, location, size, and visual attributes. This allows the user to move it, resize it, or even close the application. However, these are the only actions available because the application is actually not responding. When in the debugger mode, the system does not generate a ghost window.

DPI Virtualization

This API does not participate in DPI virtualization. The output is in the mode of the window that the message is targeting. The calling thread is not taken into consideration.

Exceptions

[Win32Exception](#) 

The function has failed. For example, the function fails if `wnd` is an invalid window.

PeekMessage(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit)

Dispatches incoming nonqueued messages, checks the thread message queue for a posted message, and retrieves the message (if any exist).

```
public static bool PeekMessage(ref WinMessage msg, Window? wnd, uint minMessageFilter, uint maxMessageFilter, MessageRemoval removeMsg, MessageProcessingLimit limit)
```

Parameters

`msg` [WinMessage](#)

A [WinMessage](#) structure that receives message information.

wnd [Window](#)

The window whose messages are to be retrieved. The window must belong to the current thread.

minMessageFilter [uint](#)

The value of the first message in the range of messages to be examined. Use **WM_KEYFIRST** (0x0100) to specify the first keyboard message or **WM_MOUSEFIRST** (0x0200) to specify the first mouse message.

maxMessageFilter [uint](#)

The value of the last message in the range of messages to be examined. Use **WM_KEYLAST** to specify the last keyboard message or **WM_MOUSELAST** to specify the last mouse message.

removeMsg [MessageRemoval](#)

Specifies how messages are to be handled.

limit [MessageProcessingLimit](#)

Specifies what messages should be processed.

Returns

[bool](#)

[true](#) if a message is available; otherwise, [false](#).

Remarks

[PeekMessage\(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit\)](#) retrieves messages associated with the window in the **wnd** parameter or any of its children as specified by the **IsChild** function, and within the range of message values given by the **minMessageFilter** and **maxMessageFilter** parameters. Note that an application can only use the low word in the **minMessageFilter** and **maxMessageFilter** parameters; the high word is reserved for the system.

Note that [PeekMessage\(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit\)](#) always retrieves **WM_QUIT** messages, no matter which values you specify for **minMessageFilter** and **maxMessageFilter**.

During this call, the system dispatches (DispatchMessage) pending, nonqueued messages, that is, messages sent to windows owned by the calling thread using the [SendMessage\(uint, nuint, nint\)](#), [SendMessageCallback](#), [SendMessageTimeout](#), or [SendNotifyMessage](#) function. Then the first queued message

that matches the specified filter is retrieved. The system may also process internal events. If no filter is specified, messages are processed in the following order:

1. Sent messages
2. Posted messages
3. Input (hardware) messages and system internal events
4. Sent messages (again)
5. `WM_PAINT` messages
6. `WM_TIMER` messages

To retrieve input messages before posted messages, use the `minMessageFilter` and `maxMessageFilter` parameters.

The [PeekMessage\(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit\)](#) function normally does not remove `WM_PAINT` messages from the queue. `WM_PAINT` messages remain in the queue until they are processed. However, if a `WM_PAINT` message has a [null](#) update region, [PeekMessage\(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit\)](#) does remove it from the queue.

If a top-level window stops responding to messages for more than several seconds, the system considers the window to be not responding and replaces it with a ghost window that has the same z-order, location, size, and visual attributes. This allows the user to move it, resize it, or even close the application. However, these are the only actions available because the application is actually not responding. When an application is being debugged, the system does not generate a ghost window.

DPI Virtualization

This API does not participate in DPI virtualization. The output is in the mode of the window that the message is targeting. The calling thread is not taken into consideration.

PostQuitMessage(int)

Indicates to the system that a thread has made a request to terminate (quit). It is typically used in response to a `WM_DESTROY` message.

```
public static void PostQuitMessage(int exitCode = 0)
```

Parameters

`exitCode` [int](#)

The application exit code. This value is used as the `wParam` parameter of the `WM_QUIT` message.

Remarks

The [PostQuitMessage\(int\)](#) function posts a `WM_QUIT` message to the thread's message queue and returns immediately; the function simply indicates to the system that the thread is requesting to quit at some time in the future.

When the thread retrieves the `WM_QUIT` message from its message queue, it should exit its message loop and return control to the system. The exit value returned to the system must be the `wParam` parameter of the `WM_QUIT` message.

TranslateMessage(ref WinMessage)

Translates virtual-key messages into character messages. The character messages are posted to the calling thread's message queue, to be read the next time the thread calls the [GetMessage\(ref WinMessage, Window?, uint, uint\)](#) or [PeekMessage\(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit\)](#) function.

```
public static bool TranslateMessage(ref WinMessage msg)
```

Parameters

`msg` [WinMessage](#)

A [WinMessage](#) structure that contains message information retrieved from the calling thread's message queue by using the [GetMessage\(ref WinMessage, Window?, uint, uint\)](#) or [PeekMessage\(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit\)](#) function.

Returns

[bool](#) 

[true](#)  if the message is translated (that is, a character message is posted to the thread's message queue) or the message is `WM_KEYDOWN`, `WM_KEYUP`, `WM_SYSKEYDOWN`, or `WM_SYSKEYUP`; otherwise, [false](#) .

Remarks

The [TranslateMessage\(ref WinMessage\)](#) function does not modify the message pointed to by the `msg` parameter.

`WM_KEYDOWN` and `WM_KEYUP` combinations produce a `WM_CHAR` or `WM_DEADCHAR` message. `WM_SYSKEYDOWN` and `WM_SYSKEYUP` combinations produce a `WM_SYSCHAR` or `WM_SYSDEADCHAR` message.

[TranslateMessage\(ref WinMessage\)](#) produces `WM_CHAR` messages only for keys that are mapped to ASCII characters by the keyboard driver.

If applications process virtual-key messages for some other purpose, they should not call [TranslateMessage\(ref WinMessage\)](#). For instance, an application should not call [TranslateMessage\(ref WinMessage\)](#) if the `TranslateAccelerator` function returns a nonzero value. Note that the application is responsible for retrieving and dispatching input messages to the dialog box. Most applications use the main message loop for this. However, to permit the user to move to and to select controls by using the keyboard, the application must call `IsDialogMessage`. For more information, see [Dialog Box Keyboard Interface](#).[↗](#)

Enum MessageProcessingLimit

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Specifies what message types should be processed by [PeekMessage\(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit\)](#).

[Flags]

```
public enum MessageProcessingLimit
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Input = 470220800

Process mouse and keyboard messages.

None = 0

Process all messages,

Paint = 2097152

Process paint messages.

PostMessage = 9961472

Process all posted messages, including timers and hotkeys.

SendMessage = 4194304

Process all sent messages.

Enum MessageRemoval

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Specifies message removal options for [PeekMessage\(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit\)](#).

[Flags]

```
public enum MessageRemoval : uint
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

NoRemove = 0

Messages are not removed from the queue after processing by [PeekMessage\(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit\)](#).

NoYield = 2

Prevents the system from releasing any thread that is waiting for the caller to go idle. Combine this value with either [NoRemove](#) or [Remove](#).

Remove = 1

Messages are removed from the queue after processing by [PeekMessage\(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit\)](#).

Class Shell32

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Declares P/Invoke functions from [Shell32.dll](#).

```
public static class Shell32
```

Inheritance

[object](#)  ← Shell32

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Methods

DestroyIcon(nint)

Destroys an icon and frees any memory the icon occupied.

```
public static bool DestroyIcon(nint hIcon)
```

Parameters

hIcon [nint](#) 

A handle to the icon to be destroyed. The icon must not be in use.

Returns

[bool](#) 

[true](#)  if successful; otherwise, [false](#) . To get extended error information, call [GetLastPInvokeError\(\)](#) .

Remarks

It is only necessary to call [DestroyIcon\(nint\)](#) for icons and cursors created with the following functions: [CreateIconFromResourceEx](#) (if called without the LR_SHARED flag), [CreateIconIndirect](#), and [CopyIcon](#). Do not use this function to destroy a shared icon. A shared icon is valid as long as the module from which it was loaded remains in memory. The following functions obtain a shared icon:

- [LoadIcon](#)
- [LoadImage](#) (if you use the LR_SHARED flag)
- [CopyImage](#) (if you use the LR_COPYRETURNORG flag and the hImage parameter is a shared icon)
- [CreateIconFromResource](#)
- [CreateIconFromResourceEx](#) (if you use the LR_SHARED flag)

GetFileInfo(string)

Retrieves information about an object in the file system, such as a file, folder, directory, or drive root.

```
[Obsolete("Use the new WinFile APIs instead. This function will be removed in the next  
version of WinInteropUtils.")]  
public static WindowsFile? GetFileInfo(string path)
```

Parameters

path [string](#)

A string of maximum length [MAX_PATH](#) that contains the path and file name. Both absolute and relative paths are valid.

This [string](#) can use either [short](#) (the 8.3 form) or [long file](#) names.

Returns

[WindowsFile](#)

The [WindowsFile](#) instance containing the data about the file if successful; otherwise, [null](#).

GetFileInfoEx(string, WindowsFileAttributes, SHGetFileInfoFlags, ref SHFILEINFO)

Retrieves information about an object in the file system, such as a file, folder, directory, or drive root. This function provides all the customization of the Windows API function [SHGetFileInfoW\(string, uint, ref](#)

[SHFILEINFO, uint, uint\)](#), unlike [GetFileInfo\(string\)](#).

```
[Obsolete("Use the new WinFile APIs instead. This function will be removed in the next  
version of WinInteropUtils.")]  
public static nuint GetFileInfoEx(string pszPath, WindowsFileAttributes dwFileAttributes,  
Shell32.SHGetFileInfoFlags uFlags, ref Shell32.SHFILEINFO psfi)
```

Parameters

pszPath [string](#)

A string of maximum length [MAX_PATH](#) that contains the path and file name. Both absolute and relative paths are valid.

If the `uFlags` parameter includes the [FireBlade.WinInteropUtils.Shell32.SHGetFileInfoFlags.SHGFI_PIDL](#) flag, this parameter must be the address of an ITEMIDLIST(PIDL) structure that contains the list of item identifiers that uniquely identifies the file within the Shell's namespace. The PIDL must be a fully qualified PIDL. Relative PIDLs are not allowed.

If the `uFlags` parameter includes the [FireBlade.WinInteropUtils.Shell32.SHGetFileInfoFlags.SHGFI_USEFILEATTRIBUTES](#) flag, this parameter does not have to be a valid file name.

The function will proceed as if the file exists with the specified name and with the file attributes passed in the `dwFileAttributes` parameter. This allows you to obtain information about a file type by passing just the extension for `pszPath` and passing `FILE_ATTRIBUTE_NORMAL` in `dwFileAttributes`.

This string can use either short (the 8.3 form) or long file names.

dwFileAttributes [WindowsFileAttributes](#)

A combination of one or more file attribute flags ([WindowsFileAttributes](#) values). If `uFlags` does not include the [SHGFI_USEFILEATTRIBUTES](#) flag, this parameter is ignored, so in that case you should set it to [None](#) instead.

uFlags [Shell32.SHGetFileInfoFlags](#)

The flags that specify the file information to retrieve. This should be a combination of [Shell32.SHGetFileInfoFlags](#) values.

`psfi` [Shell32.SHFILEINFO](#)

Pointer to a [Shell32.SHFILEINFO](#) structure to receive the file information.

Returns

[nuint](#)[↗]

A value whose meaning depends on the `uFlags` parameter.

If `<code class="paramref">uFlags</code>` does not contain `<xref href="FireBlade.WinInteropUtils.Shell32.SHGetFileInfoFlags.SHGFI_EXETYPE" data-throw-if-not-resolved="false"></xref>` or `<xref href="FireBlade.WinInteropUtils.Shell32.SHGetFileInfoFlags.SHGFI_SYSICONINDEX" data-throw-if-not-resolved="false"></xref>`, the return value is non-zero if successful, or zero otherwise.

If `<code class="paramref">uFlags</code>` contains the `<xref href="FireBlade.WinInteropUtils.Shell32.SHGetFileInfoFlags.SHGFI_EXETYPE" data-throw-if-not-resolved="false"></xref>` flag, the return value specifies the type of the executable file.

It will be one of the following values:

<code><code class="term">0</code></code>	<code><code class="description">Nonexecutable file or an error condition.</code></code>
<code><code class="term">LOWORD = NE or PE and HIWORD = Windows version</code></code>	<code><code class="description">Windows application.</code></code>
<code><code class="term">LOWORD = PE and HIWORD = 0</code></code>	<code><code class="description">Console application or .bat file</code></code>

Remarks

You should call this function from a background thread. Failure to do so could cause the UI to stop responding.

If [GetFileInfoEx\(string, WindowsFileAttributes, SHGetFileInfoFlags, ref SHFILEINFO\)](#) returns an icon handle in the [hIcon](#) member of the [Shell32.SHFILEINFO](#) structure pointed to by `psfi`, you are responsible for freeing it with [DestroyIcon\(nint\)](#) when you no longer need it. The [WindowsFile](#) wrapper automatically cleans up these resources on dispose, so you should ideally use [WindowsFile](#) and convert `psfi` to a [WindowsFile](#) using the [WindowsFile\(SHFILEINFO, string\)](#) constructor. Additionally, [WindowsFile](#) provides managed types for almost all the [Shell32.SHFILEINFO](#) properties, so using it is preferred.

Note: Once you have a handle to a system image list, you can use the [Image List API](#)[↗] to manipulate it like any other image list. Because system image lists are created on a per-process basis, you should treat them as read-only objects. Writing to a system image list may overwrite or delete one of the system images, making it unavailable or incorrect for the remainder of the process.

You must initialize Component Object Model (COM) with [Initialize\(COMInitOptions\)](#) prior to calling [SHGetFileInfoW\(string, uint, ref SHFILEINFO, uint, uint\)](#). [GetFileInfoEx\(string, WindowsFileAttributes, SHGetFileInfoFlags, ref SHFILEINFO\)](#) wraps this automatically.

When you use the [SHGFI_EXETYPE](#) flag with a Windows application, the Windows version of the executable is given in the [HIWORD](#) of the return value. This version is returned as a hexadecimal value. For details on equating this value with a specific Windows version, see [Using the Windows Headers](#).

ShellAbout(nint, string, string?, Icon?)

Displays a [ShellAbout](#) dialog box. For an example of this dialog box, run the [winver](#) command.

```
public static bool ShellAbout(nint hWnd, string app, string? moreText, Icon? icon = null)
```

Parameters

hWnd [nint](#)

A window handle to a parent window.

app [string](#)

A string that contains text to be displayed in the title bar of the [ShellAbout](#) dialog box and on the first line of the dialog box after the text "Microsoft". If the text contains a separator (<#>) that divides it into two parts, the function displays the first part in the title bar and the second part on the first line after the text "Microsoft".

<i>Windows Vista, Windows Server 2008</i>: This [string](#) cannot exceed 200 characters in length. The contents of `<code class="paramref">app</code>` will no longer show after "Microsoft", unless there is a `<code>#</code>` separator, in which case the part after the `<code>#</code>` will completely replace the first line.

moreText [string](#)

A string that contains text to be displayed in the dialog box after the version and copyright information. This parameter can be [null](#).

icon [Icon](#)

An icon that is displayed in the dialog box. This parameter can be [null](#), in which case the dialog displays the Windows icon.

Returns

[bool](#)

[true](#) if successful; otherwise, [false](#).

Examples

For an example of this dialog box, run the `winver` command.

ShellAbout(string, string, Icon?)

Displays a `ShellAbout` dialog box. For an example of this dialog box, run the `winver` command.

```
public static bool ShellAbout(string app, string moreText, Icon? icon = null)
```

Parameters

`app` [string](#)

A string that contains text to be displayed in the title bar of the `ShellAbout` dialog box and on the first line of the dialog box after the text "Microsoft". If the text contains a separator (`#`) that divides it into two parts, the function displays the first part in the title bar and the second part on the first line after the text "Microsoft".

Windows Vista, Windows Server 2008: This `string` cannot exceed 200 characters in length. The contents of `<code class="paramref">app</code>` will no longer show after "Microsoft", unless there is a `<code>#</code>` separator, in which case the part after the `<code>#</code>` will completely replace the first line.

`moreText` [string](#)

A string that contains text to be displayed in the dialog box after the version and copyright information. This parameter can be [null](#).

`icon` [Icon](#)

An icon that is displayed in the dialog box. This parameter can be [null](#), in which case the dialog displays the Windows icon.

Returns

[bool](#)

[true](#) if successful; otherwise, [false](#).

Examples

For an example of this dialog box, run the `winver` command.

ShowFileProperties(params string[])

Displays a merged property sheet for a set of files. Property values common to all the files are shown, while those that differ display the string (multiple values).

```
public static HRESULT ShowFileProperties(params string[] filePaths)
```

Parameters

`filePaths` [string](#)[]

The file paths of the files to show.

Returns

[HRESULT](#)

If this function succeeds, it returns `FireBlade.WinInteropUtils.HRESULT.S_OK`. Otherwise, it returns a [HRESULT](#) error code.

ShowPickIconDialog(nint, string, int, out Icon)

Displays a dialog box that allows the user to choose an icon from the selection available embedded in a resource such as an executable or DLL file.

```
public static bool ShowPickIconDialog(nint hWnd, string defIconPath, int defIconIndex, out  
Icon icon)
```

Parameters

`hWnd` [nint](#)

The handle of the parent window.

defIconPath [string](#)

A string that contains the fully qualified path of the default resource that contains the icons. You should verify that the path is valid before using it.

defIconIndex [int](#)

An integer that specifies the index of the initial selection.

icon [Icon](#)

When this function returns, contains the [Icon](#) that was selected. If loading the icon fails, contains [Error](#).

Returns

[bool](#)

[true](#) if successful; otherwise, [false](#).

ShowPickIconDialog(nint, string, int, out string, out int)

Displays a dialog box that allows the user to choose an icon from the selection available embedded in a resource such as an executable or DLL file.

```
public static bool ShowPickIconDialog(nint hWnd, string defIconPath, int defIconIndex, out string iconPath, out int iconIndex)
```

Parameters

hWnd [nint](#)

The handle of the parent window.

defIconPath [string](#)

A string that contains the fully qualified path of the default resource that contains the icons. You should verify that the path is valid before using it.

defIconIndex [int](#)

An integer that specifies the index of the initial selection.

iconPath [string](#)

If the user chooses a different resource in the dialog, contains the path of that file when the function returns; otherwise, it contains the path stored in **defIconPath**.

iconIndex [int](#)

When this function returns successfully, receives the index of the icon that was selected.

Returns

[bool](#)

[true](#) if successful; otherwise, [false](#).

ShowPickIconDialog(string, int, out Icon)

Displays a dialog box that allows the user to choose an icon from the selection available embedded in a resource such as an executable or DLL file.

```
public static bool ShowPickIconDialog(string defIconPath, int defIconIndex, out Icon icon)
```

Parameters

defIconPath [string](#)

A string that contains the fully qualified path of the default resource that contains the icons. You should verify that the path is valid before using it.

defIconIndex [int](#)

An integer that specifies the index of the initial selection.

icon [Icon](#)

When this function returns, contains the [Icon](#) that was selected. If loading the icon fails, contains [Error](#).

Returns

[bool](#)

[true](#) if successful; otherwise, [false](#).

ShowPickIconDialog(string, int, out string, out int)

Displays a dialog box that allows the user to choose an icon from the selection available embedded in a resource such as an executable or DLL file.

```
public static bool ShowPickIconDialog(string defIconPath, int defIconIndex, out string iconPath, out int iconIndex)
```

Parameters

defIconPath [string](#)

A string that contains the fully qualified path of the default resource that contains the icons. You should verify that the path is valid before using it.

defIconIndex [int](#)

An integer that specifies the index of the initial selection.

iconPath [string](#)

If the user chooses a different resource in the dialog, contains the path of that file when the function returns; otherwise, it contains the path stored in **defIconPath**.

iconIndex [int](#)

When this function returns successfully, receives the index of the icon that was selected.

Returns

[bool](#)

[true](#) if successful; otherwise, [false](#).

Struct Shell32.SHFILEINFO

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

The internal Windows API struct for [SHGetFileInfoW\(string, uint, ref SHFILEINFO, uint, uint\)](#). Use with [GetFileInfoEx\(string, WindowsFileAttributes, SHGetFileInfoFlags, ref SHFILEINFO\)](#).

```
public struct Shell32.SHFILEINFO
```

Inherited Members

[ValueType.Equals\(object\)](#), [ValueType.GetHashCode\(\)](#), [ValueType.ToString\(\)](#), [object.Equals\(object, object\)](#), [object.GetType\(\)](#), [object.ReferenceEquals\(object, object\)](#)

Fields

dwAttributes

```
public uint dwAttributes
```

Field Value

[uint](#)

hIcon

```
public nint hIcon
```

Field Value

[nint](#)

ilcon

```
public int iIcon
```

Field Value

[int](#)↗

szDisplayName

```
public string szDisplayName
```

Field Value

[string](#)↗

szTypeName

```
public string szTypeName
```

Field Value

[string](#)↗

Enum Shell32.SHGetFileInfoFlags

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Provides flags for the Windows API function [SHGetFileInfoW\(string, uint, ref SHFILEINFO, uint, uint\)](#)[↗]. To use this function, call the [GetFileInfoEx\(string, WindowsFileAttributes, SHGetFileInfoFlags, ref SHFILEINFO\)](#) wrapper instead.

```
[Flags]
[Obsolete("Use the new WinFile APIs instead. This enum will be removed in the next version of WinInteropUtils.")]
public enum Shell32.SHGetFileInfoFlags
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

SHGFI_ADDOVERLAYS = 32

Version 5.0. Apply the appropriate overlays to the file's icon. The SHGFI_ICON flag must also be set.

SHGFI_ATTRIBUTES = 2048

Retrieve the item attributes. The attributes are copied to the dwAttributes member of the structure specified in the psfi parameter. These are the same attributes that are obtained from `IShellFolder::GetAttributesOf`.

SHGFI_ATTR_SPECIFIED = 131072

Modify SHGFI_ATTRIBUTES to indicate that the dwAttributes member of the SHFILEINFO structure at psfi contains the specific attributes that are desired. These attributes are passed to `IShellFolder::GetAttributesOf`. If this flag is not specified, 0xFFFFFFFF is passed to `IShellFolder::GetAttributesOf`, requesting all attributes. This flag cannot be specified with the SHGFI_ICON flag.

SHGFI_DISPLAYNAME = 512

Retrieve the display name for the file, which is the name as it appears in Windows Explorer. The name is copied to the szDisplayName member of the structure specified in psfi. The returned display name

uses the long file name, if there is one, rather than the 8.3 form of the file name. Note that the display name can be affected by settings such as whether extensions are shown.

SHGFI_EXETYPE = 8192

Retrieve the type of the executable file if pszPath identifies an executable file. The information is packed into the return value. This flag cannot be specified with any other flags.

SHGFI_ICON = 256

Retrieve the handle to the icon that represents the file and the index of the icon within the system image list. The handle is copied to the hIcon member of the structure specified by psfi, and the index is copied to the iIcon member.

SHGFI_ICONLOCATION = 4096

Retrieve the name of the file that contains the icon representing the file specified by pszPath, as returned by the IExtractIcon::GetIconLocation method of the file's icon handler. Also retrieve the icon index within that file. The name of the file containing the icon is copied to the szDisplayName member of the structure specified by psfi. The icon's index is copied to that structure's iIcon member.

SHGFI_LARGEICON = 0

Modify SHGFI_ICON, causing the function to retrieve the file's large icon. The SHGFI_ICON flag must also be set.

SHGFI_LINKOVERLAY = 32768

Modify SHGFI_ICON, causing the function to add the link overlay to the file's icon. The SHGFI_ICON flag must also be set.

SHGFI_OPENICON = 2

Modify SHGFI_ICON, causing the function to retrieve the file's open icon. Also used to modify SHGFI_SYSICONINDEX, causing the function to return the handle to the system image list that contains the file's small open icon. A container object displays an open icon to indicate that the container is open. The SHGFI_ICON and/or SHGFI_SYSICONINDEX flag must also be set.

SHGFI_OVERLAYINDEX = 64

Version 5.0. Return the index of the overlay icon. The value of the overlay index is returned in the upper eight bits of the iIcon member of the structure specified by psfi. This flag requires that the SHGFI_ICON be set as well.

SHGFI_PIDL = 8

Indicate that pszPath is the address of an ITEMIDLIST structure rather than a path name.

SHGFI_SELECTED = 65536

Modify SHGFI_ICON, causing the function to blend the file's icon with the system highlight color. The SHGFI_ICON flag must also be set.

SHGFI_SHELLICONSIZE = 4

Modify SHGFI_ICON, causing the function to retrieve a Shell-sized icon. If this flag is not specified the function sizes the icon according to the system metric values. The SHGFI_ICON flag must also be set.

SHGFI_SMALLICON = 1

Modify SHGFI_ICON, causing the function to retrieve the file's small icon. Also used to modify SHGFI_SYSICONINDEX, causing the function to return the handle to the system image list that contains small icon images. The SHGFI_ICON and/or SHGFI_SYSICONINDEX flag must also be set.

SHGFI_SYSICONINDEX = 16384

Retrieve the index of a system image list icon. If successful, the index is copied to the ilcon member of psfi. The return value is a handle to the system image list. Only those images whose indices are successfully copied to ilcon are valid. Attempting to access other images in the system image list will result in undefined behavior.

SHGFI_TYPENAME = 1024

Retrieve the string that describes the file's type. The string is copied to the szTypeName member of the structure specified in psfi.

SHGFI_USEFILEATTRIBUTES = 16

Indicates that the function should not attempt to access the file specified by pszPath. Rather, it should act as if the file specified by pszPath exists with the file attributes passed in dwFileAttributes. This flag cannot be combined with the SHGFI_ATTRIBUTES, SHGFI_EXETYPE, or SHGFI_PIDL flags.

Enum StockIcon

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Icon identifiers for the [GetIcon\(StockIcon, StockIconOptions\)](#) function.

```
public enum StockIcon
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Application = 2

Generic application with no custom icon.



AudioFiles = 71

An audio file.



AutoList = 49

A virtual folder that contains the results of a search.



ClusteredDrive = 140

A cluster disk array.



Critical = 80

Error.



Delete = 84

A UI item, such as a button, that issues a delete command.



DesktopPC = 94

A desktop computer.



DeviceAudioPlayer = 102

An audio player.



DeviceCamera = 100

A digital camera.



DeviceCellPhone = 99

A cell phone.



DeviceVideoCamera = 101

A digital video camera.



DocumentNoAssociation = 0

Document of a type with no associated application.



`DocumentWithAssociaton = 1`

Document of a type with an associated application.



`Drive35 = 6`

3.5-inch disk drive.



`Drive525 = 5`

5.25-inch disk drive.



`DriveBD = 133`

High definition DVD drive (any type - BD-ROM, BD-R, BD-RE) that uses the Blu-ray Disc format.



`DriveCD = 11`

CD drive.



`DriveDVD = 59`

DVD drive.



`DriveFixed = 8`

Fixed drive (hard disk).



`DriveHDDVD = 132`

High definition DVD drive (any type - HD DVD-ROM, HD DVD-R, HD-DVD-RAM) that uses the HD DVD format.



`DriveNetwork = 9`

Network drive (connected).



`DriveNetworkDisabled = 10`

Network drive (disconnected).



`DriveRAM = 12`

RAM disk drive.



`DriveRemove = 7`

Removable drive.



`DriveUnknown = 58`

Unknown drive type.



`Error = 80`

Error.



Find = 22

The **Search** feature.



Folder = 3

Folder (generic, unspecified state).



FolderBack = 75

Folder back.



FolderFront = 76

Folder front.



FolderOpen = 4

Folder (open).



Help = 23

The **Help and Support** feature.



ImageFiles = 72

An image file.



Info = 79

Informational.



Information = 79

Informational.



Internet = 104

The **Network and Internet** Control Panel item.



Key = 81

Key.



Link = 29

Overlay for a shortcut.



Lock = 47

Security lock.



MediaAudioDVD = 85

Audio DVD media.



MediaBDR = 138

High definition write-once media in the Blu-ray Disc BD-R format.



MediaBDRE = 139

High definition read/write media in the Blu-ray Disc BD-RE format.



MediaBDROM = 137

High definition DVD-ROM media in the Blu-ray Disc BD-ROM format.



MediaBlankCD = 69

Blank writable CD media.



MediaBluRay = 90

High definition DVD media in the Blu-ray Disc™ format.



MediaCDAudio = 40

Audio CD media.



MediaCDAudioPlus = 65

CD+ (enhanced audio CD) media.



MediaCDBurn = 68

A writable CD in the process of being burned.



MediaCDR = 67

CD-R media.



MediaCDROM = 70

CD-ROM media.



MediaCDRW = 66

CD-RW media.



MediaCompactFlash = 98

CompactFlash media.



MediaDVD = 60

DVD media.



MediaDVDPlusR = 92

DVD+R media.



MediaDVDPlusRW = 93

DVD+RW media.



MediaDVDR = 63

DVD-R media.



MediaDVDRAM = 61

DVD-RAM media.



MediaDVDROM = 64

DVD-ROM media.



MediaDVDRW = 62

DVD-RW media.



MediaEnhancedCD = 87

Enhanced CD media.



MediaHDDVD = 89

High definition DVD media in the HD DVD format.



MediaHDDVDR = 135

High definition DVD-R media in the HD DVD-R format.



MediaHDDVDRAM = 136

High definition DVD-RAM media in the HD DVD-RAM format.



MediaHDDVDROM = 134

High definition DVD-ROM media in the HD DVD-ROM format.



MediaMovieDVD = 86

Movie DVD media.



MediaSVCD = 56

Super Video CD (SVCD) media.



MediaVCD = 91

Video CD (VCD) media.



MixedFiles = 74

A mixed file.



MobilePC = 95

A mobile computer (laptop).

MyNetwork = 17

The **Network** virtual folder.



Network = 17

The **Network** virtual folder.



NetworkConnect = 103

Connect to network.



PrintToFile = 54

A file that receives the output of a **Print to file** operation.



Printer = 16

A local printer or print destination.



PrinterFax = 52

A local fax printer.



PrinterFaxNet = 53

A network fax printer.



PrinterFile = 54

A file that receives the output of a **Print to file** operation.



PrinterNet = 50

A network printer.



RecycleBinEmpty = 31

The Recycle Bin (empty).



RecycleBinFull = 32

The Recycle Bin (not empty).



Recycler = 31

The Recycle Bin (empty).



RecyclerFull = 32

The Recycle Bin (not empty).



Remove = 84

A UI item, such as a button, that issues a delete command.



Rename = 83

A UI item, such as a button, that issues a rename command.



SecurityLock = 47

Security lock.



Server = 15

A computer on the network.



ServerShare = 51

A server shared on a network.



Settings = 106

The **Additional Options** Control Panel item.

Share = 28

Overlay for a shared item.



Shield = 77

Security shield. Use for UAC prompts only.



Shortcut = 29

Overlay for a shortcut.



SlowFile = 30

Overlay for items that are expected to be slow to access.



SmartMedia = 97

Smart media.



Software = 82

Software.



Stack = 55

A category that results from a **Stack by** command to organize the contents of a folder.



StuffedFolder = 57

A folder that contains only subfolders as child items.



UAC = 77

Security shield. Use for UAC prompts only.



UACShield = 77

Security shield. Use for UAC prompts only.



UserAccountControl = 77

Security shield. Use for UAC prompts only.



Users = 96

The **User Accounts** Control Panel item.



VideoFiles = 73

A video file.



Warning = 78

Warning.



World = 13

The entire network.



ZipFile = 105

A compressed file with a **.zip** file name extension.



Class StockIconHelper

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Allows you to get a shell stock icon. Unlike the built-in .NET [GetStockIcon\(StockIconId, int\)](#) or [GetStockIcon\(StockIconId, StockIconOptions\)](#) functions, this wrapper allows you to enter any icon ID you want.

```
public class StockIconHelper
```

Inheritance

[object](#) ← StockIconHelper

Inherited Members

[object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Methods

GetIcon(StockIcon, StockIconOptions)

Gets the icon for the specified value from the [StockIcon](#) enumeration.

```
public static Icon GetIcon(StockIcon icon, StockIconOptions options  
= StockIconOptions.Default)
```

Parameters

icon [StockIcon](#)

The icon to get.

options [StockIconOptions](#)

Additional options for the icon.

Returns

[Icon](#)

The created icon.

Exceptions

[Exception](#)

An error occurred in the `SHGetStockIconInfo` WinAPI function.

GetIcon(int, StockIconOptions)

Gets the icon from the specified custom `stockIconId`.

```
public static Icon GetIcon(int stockIconId, StockIconOptions options  
= StockIconOptions.Default)
```

Parameters

`stockIconId` [int](#)

The ID of the stock icon.

`options` [StockIconOptions](#)

Additional options for the icon.

Returns

[Icon](#)

The created icon.

Exceptions

[Exception](#)

An error occurred in the `SHGetStockIconInfo` WinAPI function.

Enum SystemColorId

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Defines the identifiers for system colors.

```
public enum SystemColorId
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

ActiveBorder = 10

Active window border color.

NOTE

Windows 10 or greater: This value is not supported.

ActiveCaption = 2

Active window title bar color. The associated foreground color is [CaptionText](#).

Specifies the left-side color in the color gradient of an active window's title bar if the gradient effect is enabled.

> [!NOTE] > **Windows 10 or greater:** This value is not supported.

AppWorkspace = 12

Background color of Multi Document Interface (MDI) applications.

NOTE

Windows 10 or greater: This value is not supported.

Background = 1

Desktop background color.

(i) NOTE

Windows 10 or greater: This value is not supported.

ButtonFace = 15

Face color for 3D display elements and for dialog box backgrounds. The associated foreground color is [ButtonText](#).

(i) NOTE

Windows 10 or greater: This value is not supported.

ButtonHiLight = 20

Highlight color for 3D display elements (for edges facing the light source.)

(i) NOTE

Windows 10 or greater: This value is not supported.

ButtonHighlight = 20

Highlight color for 3D display elements (for edges facing the light source.)

(i) NOTE

Windows 10 or greater: This value is not supported.

ButtonText = 18

Text on push buttons. The associated background color is [ButtonFace](#).

CaptionText = 9

Text in caption, size box, and scroll bar arrow box. The associated background color is [ActiveCaption](#).

(i) NOTE

Windows 10 or greater: This value is not supported.

Dark3DShadow = 21

Dark shadow for 3D display elements.

(i) NOTE

Windows 10 or greater: This value is not supported.

Desktop = 1

Desktop background color.

(i) NOTE

Windows 10 or greater: This value is not supported.

Face3D = 15

Face color for 3D display elements and dialog box backgrounds.

GradientActiveCaption = 27

Right side color in the color gradient of an active window's title bar. [ActiveCaption](#) specifies the left side color.

(i) NOTE

Windows 10 or greater: This value is not supported.

GradientInactiveCaption = 28

Right side color in the color gradient of an inactive window's title bar. [InactiveCaption](#) specifies the left side color.

i NOTE

Windows 10 or greater: This value is not supported.

GrayText = 17

Grayed (disabled) text. This color is set to 0 if the current display driver does not support a solid gray color.

HiLight3D = 20

Highlight color for 3D display elements (for edges facing the light source.)

i NOTE

Windows 10 or greater: This value is not supported.

Highlight = 13

Item(s) selected in a control. The associated foreground color is [HighlightText](#).

Highlight3D = 20

Highlight color for 3D display elements (for edges facing the light source.)

i NOTE

Windows 10 or greater: This value is not supported.

HighlightText = 14

Text of item(s) selected in a control. The associated background color is [Highlight](#).

HotLight = 26

Color for a hyperlink or hot-tracked item. The associated background color is [Window](#).

InactiveBorder = 11

Inactive window border.

(i) NOTE

Windows 10 or greater: This value is not supported.

InactiveCaption = 3

Inactive window caption. The associated foreground color is [InactiveCaptionText](#). Specifies the left-side color in the color gradient of an inactive window's title bar if the gradient effect is enabled.

(i) NOTE

Windows 10 or greater: This value is not supported.

InactiveCaptionText = 19

Color of text in an inactive caption. The associated background color is [InactiveCaption](#).

(i) NOTE

Windows 10 or greater: This value is not supported.

InfoBackground = 24

Background color for tooltip controls. The associated foreground color is [InfoText](#).

(i) NOTE

Windows 10 or greater: This value is not supported.

InfoText = 23

Text color for tooltip controls. The associated background color is [InfoBackground](#).

(i) NOTE

Windows 10 or greater: This value is not supported.

Light3D = 22

Light color for 3D display elements (for edges facing the light source.)

(i) NOTE

Windows 10 or greater: This value is not supported.

Menu = 4

Menu background. The associated foreground color is [MenuText](#).

(i) NOTE

Windows 10 or greater: This value is not supported.

MenuBar = 30

The background color for the menu bar when menus appear as flat menus. However, [Menu](#) continues to specify the background color of the menu popup.

(i) NOTE

Windows 10 or greater: This value is not supported.

MenuHiLight = 29

The color used to highlight menu items when the menu appears as a flat menu. The highlighted menu item is outlined with [Highlight](#).

(i) NOTE

Windows 10 or greater: This value is not supported.

MenuText = 7

Text in menus. The associated background color is [Menu](#).

i NOTE

Windows 10 or greater: This value is not supported.

ScrollBar = 0

Scroll bar gray area.

i NOTE

Windows 10 or greater: This value is not supported.

Shadow3D = 16

Shadow color for 3D display elements (for edges facing away from light source.)

i NOTE

Windows 10 or greater: This value is not supported.

Window = 5

Window background. The associated foreground colors are [WindowText](#) and [HotLight](#).

WindowFrame = 6

Window frame.

i NOTE

Windows 10 or greater: This value is not supported.

WindowText = 8

Text in windows. The associated background color is [Window](#).

Enum SystemFontId

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Defines the identifiers for system fonts.

```
public enum SystemFontId
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

CaptionFont = 801

The font used by window captions.

IconTitleFont = 806

The font used for icons.

MenuFont = 803

The font used by menus.

MessageBoxFont = 805

The font used to display messages in a message box.

SmallCaptionFont = 802

The font used by window small captions.

StatusFont = 804

The font used in status messages.

Enum ThemeTextOptions

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Represents the options for [DrawThemeText\(Graphics, int, int, string, int, ThemeTextOptions, Rectangle\)](#).

[Flags]

```
public enum ThemeTextOptions
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

AlignBottom = 8

Renders the text string at the bottom of the display rectangle. This value is used only with the [SingleLine](#) value.

AlignCenter = 1

Centers text horizontally in the display rectangle.

AlignLeft = 0

Aligns text to the left.

AlignMiddle = 4

Centers text vertically. This value is used only with the [SingleLine](#) value.

AlignRight = 2

Aligns text to the right.

AlignTop = 0

Renders the text at the top of the display rectangle.

EditControl = 8192

Duplicates the text-displaying characteristics of a multi-line edit control. Specifically, the average character width is calculated in the same manner as for an edit control, and the function does not display a partially visible last line.

EndEllipsis = 32768

Truncates a text string that is wider than the display rectangle and adds an ellipsis (...) to indicate the truncation. The string is not modified unless the [ModifyString](#) value is also specified.

ExpandTabs = 64

Expands tab characters. The default number of characters per tab is 8. Cannot be used with the [EndEllipsis](#), [WordEllipsis](#), and [PathEllipsis](#) values.

ExternalLeading = 512

Include the external leading of a font in the line height. Normally, external leading is not included in the height of a line of text.

HidePrefix = 1048576

Ignore the prefix character & in the text. The letter that follows is not underlined, but other prefix characters are still processed. For example:

Input string: A&bc&&d

Normal: Abc&d

[HidePrefix](#): Abc&d

ModifyString = 65536

Modifies a string to match the displayed text. This value has no effect unless [EndEllipsis](#) or [PathEllipsis](#) is specified.

NoClip = 256

Draws the text string without clipping the display rectangle.

NoFullWidthCharBreak = 524288

Prevents a line break at a double-byte character set (DBCS), so that the line-breaking rule is equivalent to single-byte character set (SBCS). This can be used, for example, to make icon labels written in Korean text more readable. This value has no effect unless [WordBreak](#) is specified.

NoPrefix = 2048

Turns off processing of prefix characters. Normally, the function interprets the prefix character & as a directive to underscore the character that follows, and the prefix characters && as a directive to print a single &. By specifying [NoPrefix](#), this processing is turned off. For example:

Input string: A&bc&&d

Normal: Abc&d

[NoPrefix](#): A&bc&&d

PathEllipsis = 16384

Replaces characters in the middle of text with an ellipsis so that the result fits in the display rectangle. If the string contains backslash (\) characters, [PathEllipsis](#) preserves as much as possible of the text after the last backslash. The string is not modified unless the [ModifyString](#) is specified.

PrefixOnly = 2097152

Draws only an underline at the position of the character following the prefix character &. Normally the function interprets the & as a directive to underline the character that follows and the prefix characters && as a directive to print a single &. By specifying [PrefixOnly](#), no characters are drawn, only an underline. White spaces are placed in the positions where characters would normally appear. For example:

Input string: A&bc&&d

Normal: Abc&d

[PrefixOnly](#): " "

RtlReading = 131072

Lays out text in right-to-left (RTL) order for bidirectional text, for example, text in a Hebrew or Arabic font. The default direction for text is left-to-right.

SingleLine = 32

Displays text on a single line. Carriage returns (\r) and line feeds (\n) do not break the line.

TabStop = 128

Sets tab stops.

WordBreak = 16

Breaks lines between words if a word would extend past the edge of the display rectangle. A carriage return/line feed (CR/LF, `\r\n`) sequence also breaks the line.

`WordEllipsis = 262144`

Truncates any word that does not fit in the display rectangle and adds an ellipsis.

Enum TitleBarElement

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Defines elements on a title bar.

```
public enum TitleBarElement
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

CloseButton = 5

The close (X) button.

HelpButton = 4

The help (?) button.

MaximizeButton = 3

The maximize/restore button.

MinimizeButton = 2

The minimize (-) button.

[Obsolete("TitleBarElement.Reserved is a reserved value. Do not use.")] Reserved = 1

A reserved value. Do not use.

TitleBar = 0

The title bar itself.

Enum TitleBarElementState

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Defines the states of a title bar element.

[Flags]

```
public enum TitleBarElementState
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Focusable = 1048576

The element can accept the focus.

Invisible = 32768

The element is invisible.

Offscreen = 65536

The element has no visible representation.

Pressed = 8

The element is in the pressed state.

Unavailable = 1

The element is unavailable.

Class TitleBarInfo

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Stores data about a window's title bar.

```
public class TitleBarInfo
```

Inheritance

[object](#)  ← TitleBarInfo

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Remarks

This data does not update live; it stays at the values it was at when [GetTitleBarInfo\(nint\)](#) was called.

Properties

ElementStates

Gets the states of title bar elements (at the time [GetTitleBarInfo\(nint\)](#) was called).

```
public Dictionary<TitleBarElement, TitleBarElementState> ElementStates { get; }
```

Property Value

[Dictionary](#)  <[TitleBarElement](#), [TitleBarElementState](#)>

Position

Gets the position of the title bar (at the time [GetTitleBarInfo\(nint\)](#) was called).

```
public Rectangle Position { get; }
```

Property Value

[Rectangle](#) 

Class User32

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Declares P/Invoke functions from [User32.dll](#).

```
public static class User32
```

Inheritance

[object](#)  ← User32

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Methods

GetTitleBarInfo(nint)

Retrieves information about the specified title bar.

```
public static TitleBarInfo? GetTitleBarInfo(nint hWnd)
```

Parameters

hWnd [nint](#) 

A handle to the title bar whose information is to be retrieved.

Returns

[TitleBarInfo](#)

If the function succeeds, the return value is non-[null](#) .

If the function fails, the return value is null. To get extended error information, call <xref

href="System.Runtime.InteropServices.Marshal.GetLastPInvokeError" data-throw-if-not-resolved="false"></xref>.

IsGuiThread(bool)

Determines whether the calling thread is already a GUI thread. It can also optionally convert the thread to a GUI thread (`bConvert`).

```
public static bool IsGuiThread(bool bConvert = false)
```

Parameters

`bConvert` [bool](#)

If [true](#) and the thread is not a GUI thread, convert the thread to a GUI thread.

Returns

[bool](#)

The function returns a nonzero value in the following situations:

`<p>- If the calling thread is already a GUI thread.</p>`
`<p>- If <code class="paramref">bConvert</code> is <a href="https://learn.microsoft.com/dotnet/csharp/language-reference/builtin-types/bool">>true</a> and the function successfully converts the thread to a GUI thread.</p>`
Otherwise, the function returns zero.
If `<code class="paramref">bConvert</code>` is `<a href="https://learn.microsoft.com/dotnet/csharp/language-reference/builtin-types/bool">>true</a>` and the function cannot successfully convert the thread to a GUI thread, `IsGuiThread` returns `ERROR_NOT_ENOUGH_MEMORY`.

RegisterWindowMessage(string)

Defines a new window message that is guaranteed to be unique throughout the system. The message value can be used when sending or posting messages.

```
public static uint RegisterWindowMessage(string msg)
```

Parameters

msg [string](#) 

The message to be registered.

Returns

[uint](#) 

If the message is successfully registered, the return value is a message identifier in the range **0xC000** through **0xFFFF**.

If the function fails, the return value is zero. To get extended error information, call [GetLastError](#) .

Class VisualStyle

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Provides methods for managing and drawing with visual styles.

```
public class VisualStyle : IDisposable, IHandle
```

Inheritance

[object](#)  ← VisualStyle

Implements

[IDisposable](#) , [IHandle](#)

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Properties

Handle

Gets the handle (**H**THEME) of the opened theme data.

```
public nint Handle { get; }
```

Property Value

[nint](#) 

Methods

Dispose()

Releases the theme data for this [VisualStyle](#) instance.

```
public void Dispose()
```

DrawText(Graphics, string, ref Rectangle, ThemeTextOptions)

Draws formatted text in the specified rectangle without visual styles. Formats the text according to the specified method (expanding tabs, justifying characters, breaking lines, and so forth).

```
public static int DrawText(Graphics gh, string text, ref Rectangle rect,
    ThemeTextOptions options)
```

Parameters

gh [Graphics](#)

The [Graphics](#) surface to draw on.

text [string](#)

A string that specifies the text to be drawn.

rect [Rectangle](#)

A [Rectangle](#) structure that contains the rectangle (in logical coordinates) in which the text is to be formatted.

options [ThemeTextOptions](#)

The method of formatting the text.

Returns

[int](#)

If the function succeeds, the return value is the height of the text in logical units. If [AlignMiddle](#) or [AlignBottom](#) is specified, the return value is the offset from `lpRect->top` to the bottom of the drawn text. If the function fails, the return value is zero.

Remarks

Unless the [NoClip](#) option is used, `DrawText` clips the text so that it does not appear outside the specified rectangle. Note that text with significant overhang may be clipped, for example, an initial "W" in the text

string or text that is in italics. All formatting is assumed to have multiple lines unless the [SingleLine](#) option is specified.

If the selected font is too large for the specified rectangle, the function does not attempt to substitute a smaller font.

DrawText(Graphics, string, ref Rectangle, ThemeTextOptions, ref DrawTextAdvancedOptions)

Draws formatted text in the specified rectangle without visual styles. Formats the text according to the specified method (expanding tabs, justifying characters, breaking lines, and so forth).

```
public static int DrawText(Graphics gh, string text, ref Rectangle rect, ThemeTextOptions options, ref DrawTextAdvancedOptions adv)
```

Parameters

gh [Graphics](#)

The [Graphics](#) surface to draw on.

text [string](#)

A string that specifies the text to be drawn.

rect [Rectangle](#)

A [Rectangle](#) structure that contains the rectangle (in logical coordinates) in which the text is to be formatted.

options [ThemeTextOptions](#)

The method of formatting the text.

adv [DrawTextAdvancedOptions](#)

A [DrawTextAdvancedOptions](#) structure that specifies additional formatting options.

Returns

[int](#)

If the function succeeds, the return value is the height of the text in logical units. If [AlignMiddle](#) or [AlignBottom](#) is specified, the return value is the offset from lpRect->top to the bottom of the drawn text. If the function fails, the return value is zero.

Remarks

Unless the [NoClip](#) option is used, DrawText clips the text so that it does not appear outside the specified rectangle. Note that text with significant overhang may be clipped, for example, an initial "W" in the text string or text that is in italics. All formatting is assumed to have multiple lines unless the DT_SINGLELINE format is specified.

If the selected font is too large for the specified rectangle, the function does not attempt to substitute a smaller font.

DrawThemeText(Graphics, int, int, string, int, ThemeTextOptions, Rectangle)

Draws text using the color and font defined by the visual style.

```
public void DrawThemeText(Graphics gh, int iPartId, int iStateId, string text, int cchText, ThemeTextOptions options, Rectangle rect)
```

Parameters

gh [Graphics](#)

The [Graphics](#) surface to draw the text on.

iPartId [int](#)

The control part that has the desired text appearance. See [Aero Parts and States](#) for more info. If this value is 0, the text is drawn in the default font or a font selected into the device context.

iStateId [int](#)

The control state that has the desired text appearance. See [Aero Parts and States](#) for more info.

text [string](#)

A string containing the text to draw.

cchText [int](#)

An integer containing the number of characters to draw. If this parameter is set to -1, all characters are drawn.

options [ThemeTextOptions](#)

A combination of [ThemeTextOptions](#) values specifying options for the string's formatting.

rect [Rectangle](#)

A [Rectangle](#) that contains in logical coordinates, a rectangle in which the text is to be drawn. It is recommended to use `extentRect` from `GetThemeTextExtent` to retrieve the correct coordinates.

Remarks

OpenThemeData(Window, string)

Opens the theme data for a window and its associated class.

```
public static VisualStyle OpenThemeData(Window wnd, string classList)
```

Parameters

wnd [Window](#)

The window for which theme data is requested.

classList [string](#)

A semicolon-separated list of classes. See the [Visual Styles](#) article for more info.

Returns

[VisualStyle](#)

The visual style opened.

Remarks

[OpenThemeData\(Window, string\)](#) tries to match each class, one at a time, to a class data section in the active theme. If a match is found, an associated [VisualStyle](#) is returned. If no match is found, a [FileNotFoundException](#) is thrown. You should guard against this in a `try-catch` block, and fall back to the

static non-visual-style dependent functions in the `catch` block instead, such as [DrawText\(Graphics, string, ref Rectangle, ThemeTextOptions\)](#).

The `classList` parameter contains a list, not just a single name, to provide the class an opportunity to get the best match between the class and the current visual style. For example, a button may pass `"OkButton;Button"` if its ID is `ID_OK`. If the current visual style has an entry for `OkButton`, that is used; otherwise no visual style is applied.

Exceptions

[FileNotFoundException](#) 

A matching class data section was not found.

Enum Win32ErrorCode

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Defines Win32 error codes.

```
public enum Win32ErrorCode
```

Extension Methods

[Macros.HResultFromWin32\(Win32ErrorCode\)](#), [EnumExtensions.GetValues\(Enum\)](#)

Fields

DNS_ERROR_AUTOZONE_ALREADY_EXISTS = 9610

DNS automatic zone already exists.

DNS_ERROR_AXFR = 9752

DNS zone transfer failed.

DNS_ERROR_BACKGROUND_LOADING = 9568

This operation is not allowed while the DNS server is loading zones in the background. Try again later.

DNS_ERROR_BAD_PACKET = 9502

Bad DNS packet.

DNS_ERROR_CANNOT_FIND_ROOT_HINTS = 9564

The DNS server could not find a set of root hints.

DNS_ERROR_CNAME_COLLISION = 9709

A CNAME record already exists for the given name.

DNS_ERROR_CNAME_LOOP = 9707

CNAME loop detected.

DNS_ERROR_DATAFILE_OPEN_FAILURE = 9653

Failed to open the data file for the DNS zone.

DNS_ERROR_DATAFILE_PARSING = 9655

Failure while reading datafile for DNS zone.

DNS_ERROR_DP_ALREADY_ENLISTED = 9904

This DNS server is already enlisted in the specified directory partition.

DNS_ERROR_DP_ALREADY_EXISTS = 9902

The specified directory partition already exists.

DNS_ERROR_DP_DOES_NOT_EXIST = 9901

The specified directory partition does not exist.

DNS_ERROR_DP_FSMO_ERROR = 9906

The application directory partition operation failed. The domain controller holding the domain naming master role is down or unable to service the request or is not running Windows Server 2003.

DNS_ERROR_DP_NOT_AVAILABLE = 9905

The directory partition is not available at this time. Wait a few minutes and try again.

DNS_ERROR_DP_NOT_ENLISTED = 9903

This DNS server is not enlisted in the specified directory partition.

DNS_ERROR_DS_UNAVAILABLE = 9717

The directory service is unavailable.

DNS_ERROR_DS_ZONE_ALREADY_EXISTS = 9718

DNS zone already exists in the directory service.

DNS_ERROR_DWORD_VALUE_TOO_LARGE = 9567

The specified value is too large for this parameter.

DNS_ERROR_DWORD_VALUE_TOO_SMALL = 9566

The specified value is too small for this parameter.

DNS_ERROR_FILE_WRITEBACK_FAILED = 9654

Failed to write the data file for the DNS zone.

`DNS_ERROR_FORWARDER_ALREADY_EXISTS = 9619`

A conditional forwarding zone already exists for that name.

`DNS_ERROR_INCONSISTENT_ROOT_HINTS = 9565`

The DNS server found root hints but they were not consistent across all adapters.

`DNS_ERROR_INVALID_DATAFILE_NAME = 9652`

Invalid data file name for the DNS zone.

`DNS_ERROR_INVALID_IP_ADDRESS = 9552`

Invalid IP address.

`DNS_ERROR_INVALID_NAME_CHAR = 9560`

DNS name contains an invalid character.

`DNS_ERROR_INVALID_PROPERTY = 9553`

Invalid property.

`DNS_ERROR_INVALID_TYPE = 9551`

Invalid DNS type.

`DNS_ERROR_INVALID_ZONE_OPERATION = 9603`

Invalid operation for DNS zone.

`DNS_ERROR_INVALID_ZONE_TYPE = 9611`

Invalid DNS zone type.

`DNS_ERROR_NAME_DOES_NOT_EXIST = 9714`

DNS name does not exist.

`DNS_ERROR_NAME_NOT_IN_ZONE = 9706`

Name not in DNS zone.

`DNS_ERROR_NBSTAT_INIT_FAILED = 9617`

NBTSTAT initialization call failed.

`DNS_ERROR_NEED_SECONDARY_ADDRESSES = 9614`

Need secondary IP address.

`DNS_ERROR_NEED_WINS_SERVERS = 9616`

Need WINS servers.

`DNS_ERROR_NODE_CREATION_FAILED = 9703`

Node creation failure in DNS.

`DNS_ERROR_NODE_IS_CNAME = 9708`

Node is a CNAME DNS record.

`DNS_ERROR_NON_RFC_NAME = 9556`

DNS name does not comply with RFC specifications.

`DNS_ERROR_NOT_ALLOWED_ON_RODC = 9569`

The operation requested is not permitted on against a DNS server running on a read-only DC.

`DNS_ERROR_NOT_ALLOWED_ON_ROOT_SERVER = 9562`

The operation requested is not permitted on a DNS root server.

`DNS_ERROR_NOT_ALLOWED_UNDER_DELEGATION = 9563`

The record could not be created because this part of the DNS namespace has been delegated to another server.

`DNS_ERROR_NOT_UNIQUE = 9555`

Record for given name and type is not unique.

`DNS_ERROR_NO_BOOTFILE_IF_DS_ZONE = 9719`

DNS server not creating or reading the boot file for the directory service integrated DNS zone.

`DNS_ERROR_NO_CREATE_CACHE_DATA = 9713`

Could not create DNS cache data.

`DNS_ERROR_NO_DNS_SERVERS = 9852`

No DNS servers configured for local system.

`DNS_ERROR_NO_PACKET = 9503`

No DNS packet.

`DNS_ERROR_NO_TCPIP = 9851`

TCP/IP network protocol not installed.

`DNS_ERROR_NO_ZONE_INFO = 9602`

DNS zone information not available.

`DNS_ERROR_NUMERIC_NAME = 9561`

DNS name is entirely numeric.

`DNS_ERROR_PRIMARY_REQUIRES_DATAFILE = 9651`

The primary DNS zone requires a data file.

`DNS_ERROR_RCODE = 9504`

DNS error, check rcode.

`DNS_ERROR_RCODE_BADKEY = 9017`

DNS bad key.

`DNS_ERROR_RCODE_BADSIG = 9016`

DNS signature failed to verify.

`DNS_ERROR_RCODE_BADTIME = 9018`

DNS signature validity expired.

`DNS_ERROR_RCODE_FORMAT_ERROR = 9001`

DNS server unable to interpret format.

`DNS_ERROR_RCODE_NAME_ERROR = 9003`

DNS name does not exist.

`DNS_ERROR_RCODE_NOTAUTH = 9009`

DNS server not authoritative for zone.

`DNS_ERROR_RCODE_NOTZONE = 9010`

DNS name in update or prereq is not in zone.

`DNS_ERROR_RCODE_NOT_IMPLEMENTED = 9004`

DNS request not supported by name server.

`DNS_ERROR_RCODE_NXRRSET = 9008`

DNS RR set that should to exist, does not exist.

`DNS_ERROR_RCODE_REFUSED = 9005`

DNS operation refused.

`DNS_ERROR_RCODE_SERVER_FAILURE = 9002`

DNS server failure.

`DNS_ERROR_RCODE_YXDOMAIN = 9006`

DNS name that should not exist, does exist.

`DNS_ERROR_RCODE_YXRRSET = 9007`

DNS resource record (RR) set that should not exist, does exist.

`DNS_ERROR_RECORD_ALREADY_EXISTS = 9711`

DNS record already exists.

`DNS_ERROR_RECORD_DOES_NOT_EXIST = 9701`

DNS record does not exist.

`DNS_ERROR_RECORD_FORMAT = 9702`

DNS record format error.

`DNS_ERROR_RECORD_ONLY_AT_ZONE_ROOT = 9710`

Record is only at DNS zone root.

`DNS_ERROR_RECORD_TIMED_OUT = 9705`

DNS record timed out.

DNS_ERROR_SECONDARY_DATA = 9712

Secondary DNS zone data error.

DNS_ERROR_SECONDARY_REQUIRES_MASTER_IP = 9612

Secondary DNS zone requires master IP address.

DNS_ERROR_SOA_DELETE_INVALID = 9618

Invalid delete of SOA.

DNS_ERROR_TRY_AGAIN_LATER = 9554

Try DNS operation again later.

DNS_ERROR_UNKNOWN_RECORD_TYPE = 9704

Unknown DNS record type.

DNS_ERROR_UNSECURE_PACKET = 9505

Unsecured DNS packet.

DNS_ERROR_WINS_INIT_FAILED = 9615

WINS initialization failed.

DNS_ERROR_ZONE_ALREADY_EXISTS = 9609

DNS zone already exists.

DNS_ERROR_ZONE_CONFIGURATION_ERROR = 9604

Invalid DNS zone configuration.

DNS_ERROR_ZONE_CREATION_FAILED = 9608

DNS zone creation failed.

DNS_ERROR_ZONE_DOES_NOT_EXIST = 9601

DNS zone does not exist.

DNS_ERROR_ZONE_HAS_NO_NS_RECORDS = 9606

DNS zone has no Name Server (NS) record.

`DNS_ERROR_ZONE_HAS_NO_SOA_RECORD = 9605`

DNS zone has no start of authority (SOA) record.

`DNS_ERROR_ZONE_IS_SHUTDOWN = 9621`

The operation cannot be performed because this zone is shut down.

`DNS_ERROR_ZONE_LOCKED = 9607`

DNS zone is locked.

`DNS_ERROR_ZONE_NOT_SECONDARY = 9613`

DNS zone not secondary.

`DNS_ERROR_ZONE_REQUIRES_MASTER_IP = 9620`

This zone must be configured with one or more master DNS server IP addresses.

`DNS_INFO_ADDED_LOCAL_WINS = 9753`

Added local WINS server.

`DNS_INFO_AXFR_COMPLETE = 9751`

DNS AXFR (zone transfer) complete.

`DNS_INFO_NO_RECORDS = 9501`

No records found for given DNS query.

`DNS_STATUS_CONTINUE_NEEDED = 9801`

Secure update call needs to continue update request.

`DNS_STATUS_DOTTED_NAME = 9558`

DNS name is dotted (multilabel).

`DNS_STATUS_FQDN = 9557`

DNS name is a fully qualified DNS name.

`DNS_STATUS_SINGLE_PART_NAME = 9559`

DNS name is a single-part name.

`DNS_WARNING_DOMAIN_UNDELETED = 9716`

DNS domain was undeleted.

`DNS_WARNING_PTR_CREATE_FAILED = 9715`

Could not create pointer (PTR) record.

`EPT_S_CANT_CREATE = 1899`

The endpoint mapper database entry could not be created.

`EPT_S_CANT_PERFORM_OP = 1752`

The server endpoint cannot perform the operation.

`EPT_S_INVALID_ENTRY = 1751`

The entry is invalid.

`EPT_S_NOT_REGISTERED = 1753`

There are no more endpoints available from the endpoint mapper.

`ERROR_ABANDONED_WAIT_0 = 735`

The call failed because the handle associated with it was closed.

`ERROR_ABANDON_HIBERFILE = 787`

A valid hibernation file has been invalidated and should be abandoned.

`ERROR_ABIOS_ERROR = 538`

An error occurred in the ABIOS subsystem.

`ERROR_ACCESS_AUDIT_BY_POLICY = 785`

Access to %1 is monitored by policy rule %2.

`ERROR_ACCESS_DENIED = 5`

Access is denied.

`ERROR_ACCESS_DISABLED_BY_POLICY = 1260`

This program is blocked by Group Policy. For more information, contact your system administrator.

ERROR_ACCESS_DISABLED_NO_SAFER_UI_BY_POLICY = 786

Access to %1 has been restricted by your administrator by policy rule %2.

ERROR_ACCESS_DISABLED_WEBBLADE = 1277

Windows cannot open this program because it has been disabled.

ERROR_ACCESS_DISABLED_WEBBLADE_TAMPER = 1278

Windows cannot open this program because the license enforcement system has been tampered with or become corrupted.

ERROR_ACCOUNT_DISABLED = 1331

Logon failure: Account currently disabled.

ERROR_ACCOUNT_EXPIRED = 1793

The user's account has expired.

ERROR_ACCOUNT_LOCKED_OUT = 1909

The referenced account is currently locked out and cannot be logged on to.

ERROR_ACCOUNT_RESTRICTION = 1327

Logon failure: User account restriction. Possible reasons are blank passwords not allowed, logon hour restrictions, or a policy restriction has been enforced.

ERROR_ACPI_ERROR = 669

An error occurred in the Advanced Configuration and Power Interface (ACPI) subsystem.

ERROR_ACTIVATION_COUNT_EXCEEDED = 7059

Activation has already been reset the maximum number of times for this installation. Your activation timer will not be cleared.

ERROR_ADAP_HDW_ERR = 57

A network adapter hardware error occurred.

ERROR_ADDRESS_ALREADY_ASSOCIATED = 1227

The network transport endpoint already has an address associated with it.

ERROR_ADDRESS_NOT_ASSOCIATED = 1228

An address has not yet been associated with the network endpoint.

ERROR_ADVANCED_INSTALLER_FAILED = 14099

An advanced installer failed during setup or servicing.

ERROR_ALIAS_EXISTS = 1379

The specified local group already exists.

ERROR_ALLOCATE_BUCKET = 602

The bucket array must be grown. Retry transaction after doing so.

ERROR_ALLOTTED_SPACE_EXCEEDED = 1344

No more memory is available for security information updates.

ERROR_ALL_NODES_NOT_AVAILABLE = 5037

All cluster nodes must be running to perform this operation.

ERROR_ALL_USER_TRUST_QUOTA_EXCEEDED = 1933

The total delegated trust creation quota has been exceeded.

ERROR_ALREADY_ASSIGNED = 85

The local device name is already in use.

ERROR_ALREADY_EXISTS = 183

Cannot create a file when that file already exists.

ERROR_ALREADY_FIBER = 1280

The current thread has already been converted to a fiber.

ERROR_ALREADY_INITIALIZED = 1247

An attempt was made to perform an initialization operation when initialization has already been completed.

ERROR_ALREADY_REGISTERED = 1242

The service is already registered.

ERROR_ALREADY_RUNNING_LKG = 1074

The system is currently running with the last-known-good configuration.

ERROR_ALREADY_THREAD = 1281

The current thread has already been converted from a fiber.

ERROR_ALREADY_WAITING = 1904

The specified printer handle is already being waited on.

ERROR_ALREADY_WIN32 = 719

The value already corresponds with a Win 32 error code.

ERROR_AMBIGUOUS_SYSTEM_DEVICE = 15250

The requested system device cannot be identified due to multiple indistinguishable devices potentially matching the identification criteria.

ERROR_APPHELP_BLOCK = 1259

AppHelp dialog canceled, thus preventing the application from starting.

ERROR_APP_INIT_FAILURE = 575

{Application Error} The application failed to initialize properly (0x%lx). Click OK to terminate the application.

ERROR_APP_WRONG_OS = 1151

The specified program is not a Windows or MS-DOS program.

ERROR_ARBITRATION_UNHANDLED = 723

The arbiter has deferred arbitration of these resources to its parent.

ERROR_ARENA_TRASHED = 7

The storage control blocks were destroyed.

ERROR_ARITHMETIC_OVERFLOW = 534

Arithmetic result exceeded 32 bits.

ERROR_ASSERTION_FAILURE = 668

An assertion failure has occurred.

ERROR_ATOMIC_LOCKS_NOT_SUPPORTED = 174

The file system does not support atomic changes to the lock type.

ERROR_AUDIT_FAILED = 606

{Audit Failed} An attempt to generate a security audit failed.

ERROR_AUTHENTICATION_FIREWALL_FAILED = 1935

Logon failure: The machine you are logging on to is protected by an authentication firewall. The specified account is not allowed to authenticate to the machine.

ERROR_AUTODATASEG_EXCEEDS_64k = 199

The operating system cannot run this application program.

ERROR_BACKUP_CONTROLLER = 586

This operation is only allowed for the primary domain controller (PDC) of the domain.

ERROR_BADDB = 1009

The configuration registry database is corrupt.

ERROR_BADKEY = 1010

The configuration registry key is invalid.

ERROR_BADSTARTPOSITION = 778

The iterator's start position is invalid.

ERROR_BAD_ACCESSOR_FLAGS = 773

The data provider was not able to interpret the flags set for a column binding in an accessor.

ERROR_BAD_ARGUMENTS = 160

One or more arguments are not correct.

ERROR_BAD_COMMAND = 22

The device does not recognize the command.

ERROR_BAD_COMPRESSION_BUFFER = 605

The specified buffer contains ill-formed data.

ERROR_BAD_CONFIGURATION = 1610

The configuration data for this product is corrupt. Contact your support personnel.

ERROR_BAD_CURRENT_DIRECTORY = 703

{Invalid Current Directory} The process cannot switch to the startup current directory %hs. Select OK to set current directory to %hs, or select CANCEL to exit.

ERROR_BAD_DESCRIPTOR_FORMAT = 1361

A security descriptor is not in the right format (absolute or self-relative).

ERROR_BAD_DEVICE = 1200

The specified device name is invalid.

ERROR_BAD_DEV_TYPE = 66

The network resource type is not correct.

ERROR_BAD_DLL_ENTRYPOINT = 609

{Invalid DLL Entrypoint} The dynamic link library %hs is not written correctly. The stack pointer has been left in an inconsistent state. The entry point should be declared as WINAPI or STDCALL. Select YES to fail the DLL load. Select NO to continue execution. Selecting NO can cause the application to operate incorrectly.

ERROR_BAD_DRIVER = 2001

The specified driver is invalid.

ERROR_BAD_DRIVER_LEVEL = 119

The system does not support the command requested.

ERROR_BAD_ENVIRONMENT = 10

The environment is incorrect.

ERROR_BAD_EXE_FORMAT = 193

%1 is not a valid Win32 application.

ERROR_BAD_FILE_TYPE = 222

The file type being saved or retrieved has been blocked.

ERROR_BAD_FORMAT = 11

An attempt was made to load a program with an incorrect format.

ERROR_BAD_FUNCTION_TABLE = 559

A malformed function table was encountered during an unwind operation.

ERROR_BAD_IMPERSONATION_LEVEL = 1346

Either a required impersonation level was not provided, or the provided impersonation level is invalid.

ERROR_BAD_INHERITANCE_ACL = 1340

The inherited ACL or ACE could not be built.

ERROR_BAD_LENGTH = 24

The program issued a command but the command length is incorrect.

ERROR_BAD_LOGON_SESSION_STATE = 1365

The logon session is not in a state that is consistent with the requested operation.

ERROR_BAD_NETPATH = 53

The network path was not found.

ERROR_BAD_NET_NAME = 67

The network name cannot be found.

ERROR_BAD_NET_RESP = 58

The specified server cannot perform the requested operation.

ERROR_BAD_PATHNAME = 161

The specified path is invalid.

ERROR_BAD_PIPE = 230

The pipe state is invalid.

ERROR_BAD_PROFILE = 1206

The network connection profile is corrupted.

ERROR_BAD_PROVIDER = 1204

The specified network provider name is invalid.

ERROR_BAD_QUERY_SYNTAX = 1615

SQL query syntax invalid or unsupported.

ERROR_BAD_RECOVERY_POLICY = 6012

Recovery policy configured for this system contains invalid recovery certificate.

ERROR_BAD_REM_ADAP = 60

The remote adapter is not compatible.

ERROR_BAD_SERVICE_ENTRYPOINT = 610

{Invalid Service Callback Entrypoint} The %hs service is not written correctly. The stack pointer has been left in an inconsistent state. The callback entry point should be declared as WINAPI or STDCALL. Selecting OK will cause the service to continue operation. However, the service process might operate incorrectly.

ERROR_BAD_STACK = 543

An invalid or unaligned stack was encountered during an unwind operation.

ERROR_BAD_THREADID_ADDR = 159

The address for the thread ID is not correct.

ERROR_BAD_TOKEN_TYPE = 1349

The type of the token is inappropriate for its attempted use.

ERROR_BAD_UNIT = 20

The system cannot find the device specified.

ERROR_BAD_USER_PROFILE = 1253

The specified user does not have a valid profile.

ERROR_BAD_VALIDATION_CLASS = 1348

The validation information class requested was invalid.

ERROR_BEGINNING_OF_MEDIA = 1102

The beginning of the tape or a partition was encountered.

ERROR_BEYOND_VDL = 1289

The operation occurred beyond the valid data length of the file.

ERROR_BIOS_FAILED_TO_CONNECT_INTERRUPT = 585

The system bios failed to connect a system interrupt to the device or bus for which the device is connected.

ERROR_BOOT_ALREADY_ACCEPTED = 1076

The current boot has already been accepted for use as the last-known-good control set.

ERROR_BROKEN_PIPE = 109

The pipe has been ended.

ERROR_BUFFER_ALL_ZEROS = 754

Specified buffer contains all zeros.

ERROR_BUFFER_OVERFLOW = 111

The file name is too long.

ERROR_BUSY = 170

The requested resource is in use.

ERROR_BUSY_DRIVE = 142

The system cannot perform aJOINorSUBSTat this time.

ERROR_BUS_RESET = 1111

The I/O bus was reset.

ERROR_CACHE_PAGE_LOCKED = 752

Cached page was locked during operation.

ERROR_CALLBACK_POP_STACK = 768

An exception occurred in a user mode callback and the kernel callback frame should be removed.

ERROR_CALLBACK_SUPPLIED_INVALID_DATA = 1273

An application-defined callback gave invalid data when called.

ERROR_CALL_NOT_IMPLEMENTED = 120

This function is not supported on this system.

ERROR_CANCELLED = 1223

The operation was canceled by the user.

ERROR_CANCEL_VIOLATION = 173

A lock request was not outstanding for the supplied cancel region.

ERROR_CANNOT_ABORT_TRANSACTIONS = 6848

The transactional resource manager had too many transactions outstanding that could not be aborted.
The transactional resource manager has been shut down.

ERROR_CANNOT_ACCEPT_TRANSACTED_WORK = 6847

The transactional resource manager cannot currently accept transacted work due to a transient condition, such as low resources.

ERROR_CANNOT_COPY = 266

The copy functions cannot be used.

ERROR_CANNOT_DETECT_DRIVER_FAILURE = 1080

Failure actions can only be set for Win32 services, not for drivers.

ERROR_CANNOT_DETECT_PROCESS_ABORT = 1081

This service runs in the same process as the service control manager. Therefore, the service control manager cannot take action if this service's process terminates unexpectedly.

ERROR_CANNOT_EXECUTE_FILE_IN_TRANSACTION = 6838

This file is open for modification in an unresolved transaction and can be opened for execution only by a transacted reader.

ERROR_CANNOT_FIND_WND_CLASS = 1407

Cannot find window class.

ERROR_CANNOT_IMPERSONATE = 1368

Unable to impersonate using a named pipe until data has been read from that pipe.

ERROR_CANNOT_LOAD_REGISTRY_FILE = 589

{Registry File Failure} The registry cannot load the hive (file): %hs or its log or alternate. It is corrupt, absent, or not writable.

ERROR_CANNOT_MAKE = 82

The directory or file cannot be created.

ERROR_CANNOT_OPEN_PROFILE = 1205

Unable to open the network connection profile.

ERROR_CANTFETCHBACKWARDS = 770

The data provider cannot fetch backward through a result set.

ERROR_CANTOPEN = 1011

The configuration registry key could not be opened.

ERROR_CANTREAD = 1012

The configuration registry key could not be read.

ERROR_CANTSCROLLBACKWARDS = 771

The data provider cannot scroll backward through a result set.

ERROR_CANTWRITE = 1013

The configuration registry key could not be written.

ERROR_CANT_ACCESS_DOMAIN_INFO = 1351

Configuration information could not be read from the domain controller, either because the machine is unavailable, or access has been denied.

ERROR_CANT_ACCESS_FILE = 1920

The file cannot be accessed by the system.

ERROR_CANT_BREAK_TRANSACTIONAL_DEPENDENCY = 6824

The operation cannot be performed because another transaction is depending on the fact that this property will not change.

ERROR_CANT_CREATE_MORE_STREAM_MINIVERSIONS = 6812

It is not possible to create any more miniversions for this stream.

ERROR_CANT_CROSS_RM_BOUNDARY = 6825

The operation would involve a single file with two transactional resource managers and is therefore not allowed.

ERROR_CANT_DELETE_LAST_ITEM = 4335

The last remaining item in this group or resource cannot be deleted.

ERROR_CANT_DISABLE_MANDATORY = 1310

The group cannot be disabled.

ERROR_CANT_ENABLE_DENY_ONLY = 629

A group marked use for deny only cannot be enabled.

ERROR_CANT_EVICT_ACTIVE_NODE = 5009

A cluster node cannot be evicted from the cluster unless the node is down or it is the last node.

ERROR_CANT_OPEN_ANONYMOUS = 1347

Cannot open an anonymous level security token.

ERROR_CANT_OPEN_MINIVERSION_WITH_MODIFY_INTENT = 6811

It is not possible to open a miniversion with modify access.

ERROR_CANT_RECOVER_WITH_HANDLE_OPEN = 6818

The file cannot be recovered because a handle is still open on it.

ERROR_CANT_RESOLVE_FILENAME = 1921

The name of the file cannot be resolved by the system.

ERROR_CANT_TERMINATE_SELF = 555

Indicates that a thread attempted to terminate itself by default (called NtTerminateThread with NULL) and it was the last thread in the current process.

ERROR_CANT_WAIT = 554

Used to indicate that an operation cannot continue without blocking for I/O.

ERROR_CAN_NOT_COMPLETE = 1003

Cannot complete this function.

ERROR_CAN_NOT_DEL_LOCAL_WINS = 4001

The local WINS cannot be deleted.

ERROR_CARDBUS_NOT_SUPPORTED = 724

The inserted CardBus device cannot be started because of a configuration error on %hs".

ERROR_CHECKING_FILE_SYSTEM = 712

Checking file system on %wZ.

ERROR_CHECKOUT_REQUIRED = 221

The file must be checked out before saving changes.

ERROR_CHILD_MUST_BE_VOLATILE = 1021

Cannot create a stable subkey under a volatile parent key.

ERROR_CHILD_NOT_COMPLETE = 129

The %1 application cannot be run in Win32 mode.

ERROR_CHILD_WINDOW_MENU = 1436

Child windows cannot have menus.

ERROR_CIRCULAR_DEPENDENCY = 1059

Circular service dependency was specified.

ERROR_CLASS_ALREADY_EXISTS = 1410

Class already exists.

ERROR_CLASS_DOES_NOT_EXIST = 1411

Class does not exist.

ERROR_CLASS_HAS_WINDOWS = 1412

Class still has open windows.

ERROR_CLENER_CARTRIDGE_INSTALLED = 4340

A cleaner cartridge is present in the tape library.

ERROR_CLENER_CARTRIDGE_SPENT = 4333

The cleaner cartridge has performed the maximum number of drive cleanings.

ERROR_CLENER_SLOT_NOT_SET = 4332

A cleaner slot is not reserved.

ERROR_CLENER_SLOT_SET = 4331

A cleaner slot is already reserved.

ERROR_CLIENT_SERVER_PARAMETERS_INVALID = 597

The parameters passed to the server in the client/server shared memory window were invalid. Too much data might have been put in the shared memory window.

ERROR_CLIPBOARD_NOT_OPEN = 1418

Thread does not have a clipboard open.

ERROR_CLIPPING_NOT_SUPPORTED = 2005

The requested clipping operation is not supported.

ERROR_CLUSCFG_ALREADY_COMMITTED = 5901

The cluster configuration action has already been committed.

ERROR_CLUSCFG_ROLLBACK_FAILED = 5902

The cluster configuration action could not be rolled back.

ERROR_CLUSCFG_SYSTEM_DISK_DRIVE_LETTER_CONFLICT = 5903

The drive letter assigned to a system disk on one node conflicted with the drive letter assigned to a disk on another node.

ERROR_CLUSTERLOG_CHKPOINT_NOT_FOUND = 5032

No checkpoint record was found in the cluster log.

ERROR_CLUSTERLOG_CORRUPT = 5029

The cluster log is corrupt.

ERROR_CLUSTERLOG_EXCEEDS_MAXSIZE = 5031

The cluster log exceeds its maximum size.

ERROR_CLUSTERLOG_NOT_ENOUGH_SPACE = 5033

The minimum required disk space needed for logging is not available.

ERROR_CLUSTERLOG_RECORD_EXCEEDS_MAXSIZE = 5030

The record could not be written to the cluster log because it exceeds the maximum size.

ERROR_CLUSTER_CANT_CREATE_DUP_CLUSTER_NAME = 5900

A cluster cannot be created with the specified cluster name because that cluster name is already in use. Specify a different name for the cluster.

ERROR_CLUSTER_CANT_DESERIALIZE_DATA = 5923

An error occurred while deserializing a stream of cluster data.

ERROR_CLUSTER_DATABASE_SEQMISMATCH = 5083

The join operation failed because the cluster database sequence number has changed or is incompatible with the locker node. This can happen during a join operation if the cluster database was changing during the join.

ERROR_CLUSTER_DATABASE_TRANSACTION_IN_PROGRESS = 5918

An internal cluster error occurred. A cluster database transaction was attempted while a transaction was already in progress.

ERROR_CLUSTER_DATABASE_TRANSACTION_NOT_IN_PROGRESS = 5919

An internal cluster error occurred. There was an attempt to commit a cluster database transaction while no transaction was in progress.

ERROR_CLUSTER_EVICT_WITHOUT_CLEANUP = 5896

The cluster node was evicted from the cluster successfully, but the node was not cleaned up. To determine what clean-up steps failed and how to recover, see the Failover Clustering application event log using Event Viewer.

ERROR_CLUSTER_GROUP_MOVING = 5908

The group is unable to accept the request because it is moving to another node.

ERROR_CLUSTER_GUM_NOT_LOCKER = 5085

A non-locker code received a request to reserve the lock for making global updates.

ERROR_CLUSTER_INCOMPATIBLE_VERSIONS = 5075

The cluster join operation failed due to incompatible software versions between the joining node and its sponsor.

ERROR_CLUSTER_INSTANCE_ID_MISMATCH = 5893

The join operation failed because the cluster instance ID of the joining node does not match the cluster instance ID of the sponsor node.

ERROR_CLUSTER_INTERNAL_INVALID_FUNCTION = 5912

An internal cluster error occurred. A call to an invalid function was attempted.

ERROR_CLUSTER_INVALID_IPV6_NETWORK = 5926

The cluster network is not valid for an IPv6 address resource, or it does not match the configured address.

ERROR_CLUSTER_INVALID_IPV6_TUNNEL_NETWORK = 5927

The cluster network is not valid for an IPv6 tunnel resource. Check the configuration of the IP Address resource on which the IPv6 tunnel resource depends.

ERROR_CLUSTER_INVALID_NETWORK = 5054

The cluster network is not valid.

ERROR_CLUSTER_INVALID_NETWORK_PROVIDER = 5049

The cluster network provider is not valid.

ERROR_CLUSTER_INVALID_NODE = 5039

The cluster node is not valid.

ERROR_CLUSTER_INVALID_REQUEST = 5048

The cluster request is not valid for this object.

ERROR_CLUSTER_INVALID_STRING_FORMAT = 5917

An input string of characters is not in a valid format for the data it represents.

ERROR_CLUSTER_INVALID_STRING_TERMINATION = 5916

An input string of characters is not properly terminated.

ERROR_CLUSTER_IPADDR_IN_USE = 5057

The cluster IP address is already in use.

ERROR_CLUSTER_JOIN_ABORTED = 5074

The cluster join operation was aborted.

ERROR_CLUSTER_JOIN_IN_PROGRESS = 5041

A node is in the process of joining the cluster.

ERROR_CLUSTER_JOIN_NOT_IN_PROGRESS = 5053

A cluster join operation is not in progress.

ERROR_CLUSTER_LAST_INTERNAL_NETWORK = 5066

The cluster network is the only one configured for internal cluster communication between two or more active cluster nodes. The internal communication capability cannot be removed from the network.

ERROR_CLUSTER_LOCAL_NODE_NOT_FOUND = 5043

The cluster local node information was not found.

ERROR_CLUSTER_MAXNUM_OF_RESOURCES_EXCEEDED = 5076

This resource cannot be created because the cluster has reached the limit on the number of resources it can monitor.

ERROR_CLUSTER_MEMBERSHIP_HALT = 5892

The membership engine requested shutdown of the cluster service on this node.

ERROR_CLUSTER_MEMBERSHIP_INVALID_STATE = 5890

An operation was attempted that is incompatible with the current membership state of the node.

ERROR_CLUSTER_MISMATCHED_COMPUTER_ACCT_NAME = 5905

The name of the corresponding computer account does not match the network name for this resource.

ERROR_CLUSTER_NETINTERFACE_EXISTS = 5046

The cluster network interface already exists.

ERROR_CLUSTER_NETINTERFACE_NOT_FOUND = 5047

The cluster network interface was not found.

ERROR_CLUSTER_NETWORK_ALREADY_OFFLINE = 5064

The cluster network is already offline.

ERROR_CLUSTER_NETWORK_ALREADY_ONLINE = 5063

The cluster network is already online.

ERROR_CLUSTER_NETWORK_EXISTS = 5044

The cluster network already exists.

ERROR_CLUSTER_NETWORK_HAS_DEPENDENTS = 5067

One or more cluster resources depend on the network to provide service to clients. The client access capability cannot be removed from the network.

ERROR_CLUSTER_NETWORK_NOT_FOUND = 5045

The cluster network was not found.

ERROR_CLUSTER_NETWORK_NOT_FOUND_FOR_IP = 5894

A matching cluster network for the specified IP address could not be found.

ERROR_CLUSTER_NETWORK_NOT_INTERNAL = 5060

The cluster network is not configured for internal cluster communication.

ERROR_CLUSTER_NODE_ALREADY_DOWN = 5062

The cluster node is already down.

ERROR_CLUSTER_NODE_ALREADY_HAS_DFS_ROOT = 5088

A DFS root already exists in this cluster node.

ERROR_CLUSTER_NODE_ALREADY_MEMBER = 5065

The cluster node is already a member of the cluster.

ERROR_CLUSTER_NODE_ALREADY_UP = 5061

The cluster node is already up.

ERROR_CLUSTER_NODE_DOWN = 5050

The cluster node is down.

ERROR_CLUSTER_NODE_EXISTS = 5040

The cluster node already exists.

ERROR_CLUSTER_NODE_NOT_FOUND = 5042

The cluster node was not found.

ERROR_CLUSTER_NODE_NOT_MEMBER = 5052

The cluster node is not a member of the cluster.

ERROR_CLUSTER_NODE_NOT_PAUSED = 5058

The cluster node is not paused.

ERROR_CLUSTER_NODE_NOT_READY = 5072

The cluster node is not ready to perform the requested operation.

ERROR_CLUSTER_NODE_PAUSED = 5070

The cluster node is paused.

ERROR_CLUSTER_NODE_SHUTTING_DOWN = 5073

The cluster node is shutting down.

ERROR_CLUSTER_NODE_UNREACHABLE = 5051

The cluster node is not reachable.

ERROR_CLUSTER_NODE_UP = 5056

The cluster node is up.

ERROR_CLUSTER_NO_NET_ADAPTERS = 5906

No network adapters are available.

ERROR_CLUSTER_NO_QUORUM = 5925

A quorum of cluster nodes was not present to form a cluster.

ERROR_CLUSTER_NO_RPC_PACKAGES_REGISTERED = 5081

No authentication package could be registered with the RPC server.

ERROR_CLUSTER_NO_SECURITY_CONTEXT = 5059

No cluster security context is available.

ERROR_CLUSTER_NULL_DATA = 5920

An internal cluster error occurred. Data was not properly initialized.

ERROR_CLUSTER_OLD_VERSION = 5904

One or more nodes in the cluster are running a version of Windows that does not support this operation.

ERROR_CLUSTER_OWNER_NOT_IN_PREFLIST = 5082

You cannot bring the group online because the owner of the group is not in the preferred list for the group. To change the owner node for the group, move the group.

ERROR_CLUSTER_PARAMETER_MISMATCH = 5897

Two or more parameter values specified for a resource's properties are in conflict.

ERROR_CLUSTER_PARAMETER_OUT_OF_BOUNDS = 5913

A parameter value is out of acceptable range.

ERROR_CLUSTER_PARTIAL_READ = 5921

An error occurred while reading from a stream of data. An unexpected number of bytes was returned.

ERROR_CLUSTER_PARTIAL_SEND = 5914

A network error occurred while sending data to another node in the cluster. The number of bytes transmitted was less than required.

ERROR_CLUSTER_PARTIAL_WRITE = 5922

An error occurred while writing to a stream of data. The required number of bytes could not be written.

ERROR_CLUSTER_POISONED = 5907

The cluster node has been poisoned.

ERROR_CLUSTER_PROPERTY_DATA_TYPE_MISMATCH = 5895

The actual data type of the property did not match the expected data type of the property.

ERROR_CLUSTER_QUORUMLOG_NOT_FOUND = 5891

The quorum resource does not contain the quorum log.

ERROR_CLUSTER_REGISTRY_INVALID_FUNCTION = 5915

An invalid cluster registry operation was attempted.

ERROR_CLUSTER_RESNAME_NOT_FOUND = 5080

The specified resource name is not supported by this resource DLL. This might be due to a bad (or changed) name supplied to the resource DLL.

ERROR_CLUSTER_RESOURCE_TYPE_BUSY = 5909

The resource type cannot accept the request because it is too busy performing another operation.

ERROR_CLUSTER_RESOURCE_TYPE_NOT_FOUND = 5078

The specified resource type was not found.

ERROR_CLUSTER_RESTYPE_NOT_SUPPORTED = 5079

The specified node does not support a resource of this type. This might be due to version inconsistencies or due to the absence of the resource DLL on this node.

ERROR_CLUSTER_SHUTTING_DOWN = 5022

The cluster software is shutting down.

ERROR_CLUSTER_SYSTEM_CONFIG_CHANGED = 5077

The system configuration changed during the cluster join or form operation. The join or form operation was aborted.

ERROR_CLUSTER_WRONG_OS_VERSION = 5899

This computer cannot be made a member of a cluster because it does not have the correct version of Windows installed.

ERROR_COLORSPACE_MISMATCH = 2021

The specified transform does not match the bitmap's color space.

ERROR_COMMITMENT_LIMIT = 1455

The paging file is too small for this operation to complete.

ERROR_COMMITMENT_MINIMUM = 635

{Virtual Memory Minimum Too Low} Your system is low on virtual memory. Windows is increasing the size of your virtual memory paging file. During this process, memory requests for some applications might be denied. For more information, see Help.

ERROR_COMPRESSION_DISABLED = 769

Compression is disabled for this volume.

ERROR_CONNECTED_OTHER_PASSWORD = 2108

The network connection was made successfully, but the user had to be prompted for a password other than the one originally specified.

ERROR_CONNECTED_OTHER_PASSWORD_DEFAULT = 2109

The network connection was made successfully using default credentials.

ERROR_CONNECTION_ABORTED = 1236

The network connection was aborted by the local system.

ERROR_CONNECTION_ACTIVE = 1230

An invalid operation was attempted on an active network connection.

ERROR_CONNECTION_COUNT_LIMIT = 1238

A connection to the server could not be made because the limit on the number of concurrent connections for this account has been reached.

ERROR_CONNECTION_INVALID = 1229

An operation was attempted on a nonexistent network connection.

ERROR_CONNECTION_REFUSED = 1225

The remote system refused the network connection.

ERROR_CONNECTION_UNAVAIL = 1201

The device is not currently connected but it is a remembered connection.

ERROR_CONTENT_BLOCKED = 1296

An operation failed because the content was blocked.

ERROR_CONTEXT_EXPIRED = 1931

The context has expired and can no longer be used.

ERROR_CONTINUE = 1246

Continue with work in progress.

ERROR_CONTROLLING_IEPORT = 4329

Unable to open or close the inject/eject port.

ERROR_CONTROL_C_EXIT = 572

{Application Exit by CTRL+C} The application terminated as a result of a CTRL+C.

ERROR_CONTROL_ID_NOT_FOUND = 1421

Control ID not found.

ERROR_CONVERT_TO_LARGE = 600

Internal OFS status codes indicating how an allocation operation is handled. Either it is retried after the containing onode is moved or the extent stream is converted to a large stream.

ERROR_CORE_DRIVER_PACKAGE_NOT_FOUND = 3016

Unable to find a core driver package that is required by the printer driver package.

ERROR_CORE_RESOURCE = 5026

The cluster resource could not be deleted because it is a core resource.

ERROR_CORRUPT_SYSTEM_FILE = 634

The system file %1 has become corrupt and has been replaced.

ERROR_COULD_NOT_INTERPRET = 552

The passed ACL did not contain the minimum required information.

ERROR_COULD_NOT_RESIZE_LOG = 6629

The log could not be set to the requested size.

ERROR_COUNTER_TIMEOUT = 1121

A serial I/O operation completed because the time-out period expired. (The IOCTL_SERIAL_XOFF_COUNTER did not reach zero.)

ERROR_CRASH_DUMP = 753

Crash dump exists in paging file.

ERROR_CRC = 23

Data error (cyclic redundancy check).

ERROR_CREATE_FAILED = 1631

The Windows Installer service failed to start. Contact your support personnel.

ERROR_CRM_PROTOCOL_ALREADY_EXISTS = 6710

The resource manager tried to register a protocol that already exists.

ERROR_CRM_PROTOCOL_NOT_FOUND = 6712

The requested propagation protocol was not registered as a CRM.

ERROR_CSCSHARE_OFFLINE = 1262

The share is currently offline or does not exist.

ERROR_CS_ENCRYPTION_EXISTING_ENCRYPTED_FILE = 6019

File is encrypted and should be opened in client-side encryption mode.

ERROR_CS_ENCRYPTION_FILE_NOT_CSE = 6021

The SMB client requested a client-side extension (CSE) file system control (FSCTL) on a non-CSE file.

ERROR_CS_ENCRYPTION_INVALID_SERVER_RESPONSE = 6017

The remote server sent an invalid response for a file being opened with client-side encryption.

ERROR_CS_ENCRYPTION_NEW_ENCRYPTED_FILE = 6020

A new encrypted file is being created and a \$EFS needs to be provided.

ERROR_CS_ENCRYPTION_UNSUPPORTED_SERVER = 6018

Client-side encryption is not supported by the remote server even though it claims to support it.

ERROR_CTX_ACCOUNT_RESTRICTION = 7064

Unable to log you on because of an account restriction.

ERROR_CTX_BAD_VIDEO_MODE = 7025

An attempt has been made to connect to a session whose video mode is not supported by the current client.

ERROR_CTX_CANNOT_MAKE_EVENTLOG_ENTRY = 7005

A registry key for event logging could not be created for this session.

ERROR_CTX_CDM_CONNECT = 7066

The Client Drive Mapping Service has connected on terminal connection.

ERROR_CTX_CDM_DISCONNECT = 7067

The Client Drive Mapping Service has disconnected on terminal connection.

ERROR_CTX_CLIENT_LICENSE_IN_USE = 7052

Your request to connect to this terminal server has been rejected. Your terminal server client license number is currently being used by another user. Call your system administrator to obtain a unique license number.

ERROR_CTX_CLIENT_LICENSE_NOT_SET = 7053

Your request to connect to this terminal server has been rejected. Your terminal server client license number has not been entered for this copy of the terminal server client. Contact your system administrator.

ERROR_CTX_CLIENT_QUERY_TIMEOUT = 7040

The client failed to respond to the server connect message.

ERROR_CTX_CLOSE_PENDING = 7007

A close operation is pending on the session.

ERROR_CTX_CONSOLE_CONNECT = 7042

Reconnecting a disconnected session to the console is not supported.

ERROR_CTX_CONSOLE_DISCONNECT = 7041

Disconnecting the console session is not supported.

ERROR_CTX_ENCRYPTION_LEVEL_REQUIRED = 7061

You do not have the proper encryption level to access this session.

ERROR_CTX_GRAPHICS_INVALID = 7035

The application attempted to enable DOS graphics mode. DOS graphics mode is not supported.

ERROR_CTX_INVALID_MODEMNAME = 7010

The modem name was not found in the MODEM.INF file.

ERROR_CTX_INVALID_PD = 7002

The specified protocol driver is invalid.

ERROR_CTX_INVALID_WD = 7049

The specified terminal connection driver is invalid.

ERROR_CTX_LICENSE_CLIENT_INVALID = 7055

The client you are using is not licensed to use this system. Your logon request is denied.

ERROR_CTX_LICENSE_EXPIRED = 7056

The system license has expired. Your logon request is denied.

ERROR_CTX_LICENSE_NOT_AVAILABLE = 7054

The number of connections to this computer is limited and all connections are in use right now. Try connecting later or contact your system administrator.

ERROR_CTX_LOGON_DISABLED = 7037

Your interactive logon privilege has been disabled. Contact your administrator.

ERROR_CTX_MODEM_INF_NOT_FOUND = 7009

The MODEM.INF file was not found.

ERROR_CTX_MODEM_RESPONSE_BUSY = 7015

Busy signal detected at remote site on callback.

ERROR_CTX_MODEM_RESPONSE_ERROR = 7011

The modem did not accept the command sent to it. Verify that the configured modem name matches the attached modem.

ERROR_CTX_MODEM_RESPONSE_NO_CARRIER = 7013

Carrier detect has failed or carrier has been dropped due to disconnect.

ERROR_CTX_MODEM_RESPONSE_NO_DIALTONE = 7014

Dial tone not detected within the required time. Verify that the phone cable is properly attached and functional.

ERROR_CTX_MODEM_RESPONSE_TIMEOUT = 7012

The modem did not respond to the command sent to it. Verify that the modem is properly cabled and turned on.

ERROR_CTX_MODEM_RESPONSE_VOICE = 7016

Voice detected at remote site on callback.

ERROR_CTX_NOT_CONSOLE = 7038

The requested operation can be performed only on the system console. This is most often the result of a driver or system DLL requiring direct console access.

ERROR_CTX_NO_FORCE_LOGOFF = 7063

The user %s\%s is already logged on to the console of this computer. You do not have permission to log in at this time. To resolve this issue, contact %s\%s and have them log off.

ERROR_CTX_NO_OUTBUF = 7008

There are no free output buffers available.

ERROR_CTX_PD_NOT_FOUND = 7003

The specified protocol driver was not found in the system path.

ERROR_CTX_SECURITY_LAYER_ERROR = 7068

The terminal server security layer detected an error in the protocol stream and has disconnected the client.

ERROR_CTX_SERVICE_NAME_COLLISION = 7006

A service with the same name already exists on the system.

ERROR_CTX_SESSION_IN_USE = 7062

The user %s\%s is currently logged on to this computer. Only the current user or an administrator can log on to this computer.

ERROR_CTX_SHADOW_DENIED = 7044

The request to control another session remotely was denied.

ERROR_CTX_SHADOW_DISABLED = 7051

The requested session is not configured to allow remote control.

ERROR_CTX_SHADOW_ENDED_BY_MODE_CHANGE = 7058

The remote control of the console was terminated because the display mode was changed. Changing the display mode in a remote control session is not supported.

ERROR_CTX_SHADOW_INVALID = 7050

The requested session cannot be controlled remotely. This might be because the session is disconnected or does not currently have a user logged on.

ERROR_CTX_SHADOW_NOT_RUNNING = 7057

Remote control could not be terminated because the specified session is not currently being remotely controlled.

ERROR_CTX_TD_ERROR = 7017

Transport driver error.

ERROR_CTX_WD_NOT_FOUND = 7004

The specified terminal connection driver was not found in the system path.

ERROR_CTX_WINSTATIONS_DISABLED = 7060

Remote logons are currently disabled.

ERROR_CTX_WINSTATION_ACCESS_DENIED = 7045

The requested session access is denied.

ERROR_CTX_WINSTATION_ALREADY_EXISTS = 7023

The specified session name is already in use.

ERROR_CTX_WINSTATION_BUSY = 7024

The requested operation cannot be completed because the terminal connection is currently busy processing a connect, disconnect, reset, or delete operation.

ERROR_CTX_WINSTATION_NAME_INVALID = 7001

The specified session name is invalid.

ERROR_CTX_WINSTATION_NOT_FOUND = 7022

The specified session cannot be found.

ERROR_CURRENT_DIRECTORY = 16

The directory cannot be removed.

ERROR_CURRENT_DOMAIN_NOT_ALLOWED = 1399

This operation cannot be performed on the current domain.

ERROR_CURRENT_TRANSACTION_NOT_VALID = 6714

The current transaction context associated with the thread is not a valid handle to a transaction object.

ERROR_DATABASE_BACKUP_CORRUPT = 5087

The backed-up cluster database is possibly corrupt.

ERROR_DATABASE_DOES_NOT_EXIST = 1065

The database specified does not exist.

ERROR_DATABASE_FAILURE = 4313

Unable to read from or write to the database.

ERROR_DATABASE_FULL = 4314

The database is full.

ERROR_DATATYPE_MISMATCH = 1629

Data supplied is of wrong type.

ERROR_DATA_LOST_REPAIR = 6843

Windows has discovered corruption in a file, and that file has since been repaired. Data loss might have occurred.

ERROR_DATA_NOT_ACCEPTED = 592

{Data Not Accepted} The transport driver interface (TDI) client could not handle the data received during an indication.

ERROR_DBG_COMMAND_EXCEPTION = 697

Debugger command communication exception.

ERROR_DBG_CONTINUE = 767

Debugger continued.

ERROR_DBG_CONTROL_BREAK = 696

Debugger received control break.

ERROR_DBG_CONTROL_C = 693

Debugger got control C.

ERROR_DBG_EXCEPTION_HANDLED = 766

Debugger handled exception.

ERROR_DBG_EXCEPTION_NOT_HANDLED = 688

Debugger did not handle the exception.

`ERROR_DBG_PRINTEXCEPTION_C = 694`

Debugger printed exception on control C.

`ERROR_DBG_REPLY_LATER = 689`

Debugger will reply later.

`ERROR_DBG_RIPEXCEPTION = 695`

Debugger received Routing Information Protocol (RIP) exception.

`ERROR_DBG_TERMINATE_PROCESS = 692`

Debugger terminated process.

`ERROR_DBG_TERMINATE_THREAD = 691`

Debugger terminated thread.

`ERROR_DBG_UNABLE_TO_PROVIDE_HANDLE = 690`

Debugger cannot provide handle.

`ERROR_DC_NOT_FOUND = 1425`

Invalid device context (DC) handle.

`ERROR_DDE_FAIL = 1156`

An error occurred in sending the command to the application.

`ERROR_DEBUGGER_INACTIVE = 1284`

An attempt to perform an operation on a debug object failed because the object is in the process of being deleted.

`ERROR_DEBUG_ATTACH_FAILED = 590`

{Unexpected Failure in DebugActiveProcess} An unexpected failure occurred while processing a DebugActiveProcess API request. Choosing OK will terminate the process, and choosing Cancel will ignore the error.

`ERROR_DECRYPTION_FAILED = 6001`

The specified file could not be decrypted.

ERROR_DELAY_LOAD_FAILED = 1285

An attempt to delay-load a .dll or get a function address in a delay-loaded .dll failed.

ERROR_DELETE_PENDING = 303

The file cannot be opened because it is in the process of being deleted.

ERROR_DELETING_ICM_XFORM = 2019

There was an error while deleting the color transform.

ERROR_DEPENDENCY_ALREADY_EXISTS = 5003

The cluster resource cannot be made dependent on the specified resource because it is already dependent.

ERROR_DEPENDENCY_NOT_ALLOWED = 5069

The cluster quorum resource is not allowed to have any dependencies.

ERROR_DEPENDENCY_NOT_FOUND = 5002

The cluster resource dependency cannot be found.

ERROR_DEPENDENT_RESOURCE_EXISTS = 5001

The operation cannot be completed because other resources depend on this resource.

ERROR_DEPENDENT_RESOURCE_PROPERTY_CONFLICT = 5924

One or more property values for this resource are in conflict with one or more property values associated with its dependent resources.

ERROR_DEPENDENT_SERVICES_RUNNING = 1051

A stop control has been sent to a service that other running services are dependent on.

ERROR_DESTINATION_ELEMENT_FULL = 1161

The indicated destination element already contains media.

ERROR_DESTROY_OBJECT_OF_OTHER_THREAD = 1435

Cannot destroy object created by another thread.

ERROR_DEVICE_ALREADY_ATTACHED = 548

An attempt was made to attach to a device that was already attached to another device.

ERROR_DEVICE_ALREADY_REMEMBERED = 1202

The local device name has a remembered connection to another network resource.

ERROR_DEVICE_DOOR_OPEN = 1166

The device has indicated that its door is open.

ERROR_DEVICE_ENUMERATION_ERROR = 648

The device encountered an error while applying power or reading the device configuration. This might be caused by a failure of your hardware or by a poor connection.

ERROR_DEVICE_NOT_AVAILABLE = 4319

The device is not ready for use.

ERROR_DEVICE_NOT_CONNECTED = 1167

The device is not connected.

ERROR_DEVICE_NOT_PARTITIONED = 1107

Tape partition information could not be found when loading a tape.

ERROR_DEVICE_REINITIALIZATION_NEEDED = 1164

The indicated device requires re-initialization due to hardware errors.

ERROR_DEVICE_REMOVED = 1617

The device has been removed.

ERROR_DEVICE_REQUIRES_CLEANING = 1165

The device has indicated that cleaning is required before further operations are attempted.

ERROR_DEV_NOT_EXIST = 55

The specified network resource or device is no longer available.

ERROR_DHCP_ADDRESS_CONFLICT = 4100

The Dynamic Host Configuration Protocol (DHCP) client has obtained an IP address that is already in use on the network. The local interface will be disabled until the DHCP client can obtain a new address.

ERROR_DIFFERENT_SERVICE_ACCOUNT = 1079

The account specified for this service is different from the account specified for other services running in the same process.

ERROR_DIRECTORY = 267

The directory name is invalid.

ERROR_DIRECTORY_NOT_RM = 6803

The specified directory does not contain a resource manager.

ERROR_DIRECT_ACCESS_HANDLE = 130

Attempt to use a file handle to an open disk partition for an operation other than raw disk I/O.

ERROR_DIR_EFS_DISALLOWED = 6010

The directory has been disabled for encryption.

ERROR_DIR_NOT_EMPTY = 145

The directory is not empty.

ERROR_DIR_NOT_ROOT = 144

The directory is not a subdirectory of the root directory.

ERROR_DISCARDED = 157

The segment is already discarded and cannot be locked.

ERROR_DISK_CHANGE = 107

The program stopped because an alternate disk was not inserted.

ERROR_DISK_CORRUPT = 1393

The disk structure is corrupted and unreadable.

ERROR_DISK_FULL = 112

There is not enough space on the disk.

ERROR_DISK_OPERATION_FAILED = 1127

While accessing the hard disk, a disk operation failed even after retries.

ERROR_DISK_QUOTA_EXCEEDED = 1295

An operation failed because the storage quota was exceeded.

ERROR_DISK_RECALIBRATE_FAILED = 1126

While accessing the hard disk, a recalibrate operation failed, even after retries.

ERROR_DISK_REPAIR_DISABLED = 780

The attempted operation required self-healing to be enabled.

ERROR_DISK_RESET_FAILED = 1128

While accessing the hard disk, a disk controller reset was needed, but that also failed.

ERROR_DISK_TOO_FRAGMENTED = 302

The volume is too fragmented to complete this operation.

ERROR_DLL_INIT_FAILED = 1114

A DLL initialization routine failed.

ERROR_DLL_INIT_FAILED_LOGOFF = 624

{DLL Initialization Failed} The application failed to initialize because the window station is shutting down.

ERROR_DLL_MIGHT_BE_INCOMPATIBLE = 687

The application is loading executable code from the module %hs. This is secure, but might be incompatible with previous releases of the operating system. An alternative, %hs, is available. Should the application use the secure module %hs?

ERROR_DLL_MIGHT_BE_INSECURE = 686

The application is attempting to run executable code from the module %hs. This might be insecure. An alternative, %hs, is available. Should the application use the secure module %hs?

ERROR_DLL_NOT_FOUND = 1157

One of the library files needed to run this application cannot be found.

ERROR_DOMAIN_CONTROLLER_EXISTS = 1250

A domain controller with the specified name already exists.

ERROR_DOMAIN_CONTROLLER_NOT_FOUND = 1908

Could not find the domain controller for this domain.

ERROR_DOMAIN_CTRLR_CONFIG_ERROR = 581

A domain server has an incorrect configuration.

ERROR_DOMAIN_EXISTS = 1356

The specified domain already exists.

ERROR_DOMAIN_LIMIT_EXCEEDED = 1357

An attempt was made to exceed the limit on the number of domains per server.

ERROR_DOMAIN_TRUST_INCONSISTENT = 1810

The name or SID of the domain specified is inconsistent with the trust information for that domain.

ERROR_DOWNGRADE_DETECTED = 1265

The system detected a possible attempt to compromise security. Ensure that you can contact the server that authenticated you.

ERROR_DRIVERS_LEAKING_LOCKED_PAGES = 729

A device driver is leaking locked I/O pages, causing system degradation. The system has automatically enabled a tracking code to try and catch the culprit.

ERROR_DRIVER_BLOCKED = 1275

This driver has been blocked from loading.

ERROR_DRIVER_CANCEL_TIMEOUT = 594

{Cancel Timeout} The driver %hs failed to complete a canceled I/O request in the allotted time.

ERROR_DRIVER_DATABASE_ERROR = 652

There was an error [%2] processing the driver database.

ERROR_DRIVER_FAILED_PRIOR_UNLOAD = 654

The driver could not be loaded because a previous version of the driver is still in memory.

ERROR_DRIVER_FAILED_SLEEP = 633

{System Standby Failed} The driver %hs does not support standby mode. Updating this driver might allow the system to go to standby mode.

ERROR_DRIVER_PROCESS_TERMINATED = 1291

The process hosting the driver for this device has been terminated.

ERROR_DRIVE_LOCKED = 108

The disk is in use or locked by another process.

ERROR_DRIVE_MEDIA_MISMATCH = 4303

The drive and medium are not compatible, or they exist in different libraries.

ERROR_DS_ADD_REPLICA_INHIBITED = 8302

The add replica operation cannot be performed. The naming context must be writable to create the replica.

ERROR_DS_ADMIN_LIMIT_EXCEEDED = 8228

The administrative limit for this request was exceeded.

ERROR_DS_AFFECTS_MULTIPLE_DSAS = 8249

The operation affects multiple DSAs.

ERROR_DS_AG_CANT_HAVE_UNIVERSAL_MEMBER = 8578

An account group cannot have a universal group as a member.

ERROR_DS_ALIASED_OBJ_MISSING = 8334

The aliased object is missing.

ERROR_DS_ALIAS_DEREF_PROBLEM = 8244

There is an alias dereferencing problem.

ERROR_DS_ALIAS_POINTS_TO_ALIAS = 8336

An alias is not permitted to refer to another alias.

ERROR_DS_ALIAS_PROBLEM = 8241

There is an alias problem.

ERROR_DS_ATTRIBUTE_OR_VALUE_EXISTS = 8205

The specified directory service attribute or value already exists.

ERROR_DS_ATTRIBUTE_OWNED_BY_SAM = 8346

Access to the attribute is not permitted because the attribute is owned by the SAM.

ERROR_DS_ATTRIBUTE_TYPE_UNDEFINED = 8204

The attribute type specified to the directory service is not defined.

ERROR_DS_ATT_ALREADY_EXISTS = 8318

The specified attribute is already present on the object.

ERROR_DS_ATT_IS_NOT_ON_OBJ = 8310

The attribute specified in the operation is not present on the object.

ERROR_DS_ATT_NOT_DEF_FOR_CLASS = 8317

An attempt was made to modify an object to include an attribute that is not legal for its class.

ERROR_DS_ATT_NOT_DEF_IN_SCHEMA = 8303

A reference to an attribute that is not defined in the schema occurred.

ERROR_DS_ATT_SCHEMA_REQ_ID = 8399

Cannot read the governs class identifier for the schema record.

ERROR_DS_ATT_SCHEMA_REQ_SYNTAX = 8416

Failed to obtain the required syntax for the attribute schema.

ERROR_DS_ATT_VAL_ALREADY_EXISTS = 8323

The specified value already exists.

ERROR_DS_AUDIT_FAILURE = 8625

A required audit event could not be generated for the operation.

ERROR_DS_AUTHORIZATION_FAILED = 8599

The directory service failed to authorize the request.

ERROR_DS_AUTH_METHOD_NOT_SUPPORTED = 8231

The requested authentication method is not supported by the server.

ERROR_DS_AUTH_UNKNOWN = 8234

The authentication mechanism is unknown.

ERROR_DS_AUX_CLS_TEST_FAIL = 8389

Schema update failed: Class in the Aux Class list does not exist or is not an auxiliary class.

ERROR_DS_BACKLINK_WITHOUT_LINK = 8482

The schema update operation tried to add a backward link attribute that has no corresponding forward link.

ERROR_DS_BAD_ATT_SCHEMA_SYNTAX = 8400

The attribute schema has bad syntax.

ERROR_DS_BAD_HIERARCHY_FILE = 8425

The hierarchy file is invalid.

ERROR_DS_BAD_INSTANCE_TYPE = 8313

The specified instance type is not valid.

ERROR_DS_BAD_NAME_SYNTAX = 8335

The object name has bad syntax.

ERROR_DS_BAD_RDN_ATT_ID_SYNTAX = 8392

Schema update failed: Rdn-Att-Id has wrong syntax.

ERROR_DS_BUILD_HIERARCHY_TABLE_FAILED = 8426

The attempt to build the hierarchy table failed.

ERROR_DS_BUSY = 8206

The directory service is busy.

ERROR_DS_CANT_ACCESS_REMOTE_PART_OF_AD = 8585

Could not access a partition of the directory service located on a remote server. Make sure at least one server is running for the partition in question.

ERROR_DS_CANT_ADD_ATT_VALUES = 8320

The specified attribute is not present, or has no values.

ERROR_DS_CANT_ADD_SYSTEM_ONLY = 8358

It is not permitted to add an attribute that is owned by the system.

ERROR_DS_CANT_ADD_TO_GC = 8550

The attribute is not allowed to be replicated to the GC because of security reasons.

ERROR_DS_CANT_CACHE_ATT = 8401

The attribute could not be cached.

ERROR_DS_CANT_CACHE_CLASS = 8402

The class could not be cached.

ERROR_DS_CANT_CREATE_IN_NONDOMAIN_NC = 8553

Security principal objects can only be created inside domain naming contexts.

ERROR_DS_CANT_CREATE_UNDER_SCHEMA = 8510

An object of this class cannot be created under the schema container. You can only create Attribute-Schema and Class-Schema objects under the schema container.

ERROR_DS_CANT_DELETE = 8398

The requested delete operation could not be performed.

ERROR_DS_CANT_DELETE_DSA_OBJ = 8340

The DSA object cannot be deleted.

ERROR_DS_CANT_DEL_MASTER_CROSSREF = 8375

It is not permitted to delete a master cross-reference.

ERROR_DS_CANT_DEMOTE_WITH_WRITEABLE_NC = 8604

Writable NCs prevent this DC from demoting.

ERROR_DS_CANT_DEREF_ALIAS = 8337

The alias cannot be dereferenced.

ERROR_DS_CANT_DERIVE_SPN_FOR_DELETED_DOMAIN = 8603

The directory service cannot derive an SPN with which to mutually authenticate the target server because the server's domain has been deleted from the forest.

ERROR_DS_CANT_DERIVE_SPN_WITHOUT_SERVER_REF = 8589

The directory service cannot derive an SPN with which to mutually authenticate the target server because the corresponding server object in the local DS database has no serverReference attribute.

ERROR_DS_CANT_FIND_DC_FOR_SRC_DOMAIN = 8537

The operation could not locate a DC for the source domain.

ERROR_DS_CANT_FIND_DSA_OBJ = 8419

The DSA object could not be found.

ERROR_DS_CANT_FIND_EXPECTED_NC = 8420

The naming context could not be found.

ERROR_DS_CANT_FIND_NC_IN_CACHE = 8421

The naming context could not be found in the cache.

ERROR_DS_CANT_MIX_MASTER_AND_REPS = 8331

The object and parent must be of the same type, either both masters or both replicas.

ERROR_DS_CANT_MOD_OBJ_CLASS = 8215

The directory service detected an attempt to modify the object class of an object.

ERROR_DS_CANT_MOD_PRIMARYGROUPID = 8506

Cannot change the primary group ID of a domain controller account.

ERROR_DS_CANT_MOD_SYSTEM_ONLY = 8369

The attribute cannot be modified because it is owned by the system.

ERROR_DS_CANT_MOVE_ACCOUNT_GROUP = 8498

Cross-domain moves of nonempty account groups is not allowed.

ERROR_DS_CANT_MOVE_APP_BASIC_GROUP = 8608

Cross-domain moves of nonempty basic application groups is not allowed.

ERROR_DS_CANT_MOVE_APP_QUERY_GROUP = 8609

Cross-domain moves of nonempty query-based application groups is not allowed.

ERROR_DS_CANT_MOVE_DELETED_OBJECT = 8489

The object being moved across domains is already known to be deleted by the destination server. The source server does not have the latest version of the source object.

ERROR_DS_CANT_MOVE_RESOURCE_GROUP = 8499

Cross-domain moves of nonempty resource groups is not allowed.

ERROR_DS_CANT_ON_NON_LEAF = 8213

The directory service can perform the requested operation only on a leaf object.

ERROR_DS_CANT_ON_RDN = 8214

The directory service cannot perform the requested operation on the relative distinguished name (RDN) attribute of an object.

ERROR_DS_CANT_REMOVE_ATT_CACHE = 8403

The attribute could not be removed from the cache.

ERROR_DS_CANT_REMOVE_CLASS_CACHE = 8404

The class could not be removed from the cache.

ERROR_DS_CANT_REM_MISSING_ATT = 8324

The attribute cannot be removed because it is not present on the object.

ERROR_DS_CANT_REM_MISSING_ATT_VAL = 8325

The attribute value cannot be removed because it is not present on the object.

ERROR_DS_CANT_REPLACE_HIDDEN_REC = 8424

The operation cannot replace the hidden record.

ERROR_DS_CANT_RETRIEVE_ATTS = 8481

The search failed to retrieve attributes from the database.

ERROR_DS_CANT_RETRIEVE_CHILD = 8422

The child object could not be retrieved.

ERROR_DS_CANT_RETRIEVE_DN = 8405

The distinguished name attribute could not be read.

ERROR_DS_CANT_RETRIEVE_INSTANCE = 8407

The instance type attribute could not be retrieved.

ERROR_DS_CANT_RETRIEVE_SD = 8526

The Security Descriptor attribute could not be read.

ERROR_DS_CANT_START = 8531

Directory Service cannot start.

ERROR_DS_CANT_TREE_DELETE_CRITICAL_OBJ = 8560

Critical directory service system objects cannot be deleted during tree deletion operations. The tree deletion might have been partially performed.

ERROR_DS_CANT_WITH_ACCT_GROUP_MEMBERSHPS = 8493

Cannot move objects with memberships across domain boundaries because, once moved, this violates the membership conditions of the account group. Remove the object from any account group memberships and retry.

ERROR_DS_CHILDREN_EXIST = 8332

The operation cannot be performed because child objects exist. This operation can only be performed on a leaf object.

ERROR_DS_CLASS_MUST_BE_CONCRETE = 8359

The class of the object must be structural; you cannot instantiate an abstract class.

ERROR_DS_CLASS_NOT_DSA = 8343

The object must be of class DSA.

ERROR_DS_CLIENT_LOOP = 8259

A referral loop was detected by the client.

ERROR_DS_CODE_INCONSISTENCY = 8408

An internal error has occurred.

ERROR_DS_COMPARE_FALSE = 8229

The compare response was false.

ERROR_DS_COMPARE_TRUE = 8230

The compare response was true.

ERROR_DS_CONFIDENTIALITY_REQUIRED = 8237

This request requires a secure connection.

ERROR_DS_CONFIG_PARAM_MISSING = 8427

The directory configuration parameter is missing from the registry.

ERROR_DS_CONSTRAINT_VIOLATION = 8239

A constraint violation occurred.

ERROR_DS_CONSTRUCTED_ATT_MOD = 8475

Modification of a constructed attribute is not allowed.

ERROR_DS_CONTROL_NOT_FOUND = 8258

The specified control is not supported by the server.

ERROR_DS_COULDNT_CONTACT_FSMO = 8367

The requested Flexible Single Master Operations (FSMO) operation failed. The current FSMO holder could not be contacted.

ERROR_DS_COULDNT_IDENTIFY_OBJECTS_FOR_TREE_DELETE = 8503

The directory service failed to identify the list of objects to delete while attempting a tree deletion.

ERROR_DS_COULDNT_LOCK_TREE_FOR_DELETE = 8502

The directory service failed to lock a tree in preparation for a tree deletion because the tree was in use.

ERROR_DS_COULDNT_UPDATE_SPNS = 8525

While processing a change to the DNS host name for an object, the SPN values could not be kept in sync.

ERROR_DS_COUNTING_AB_INDICES_FAILED = 8428

The attempt to count the address book indices failed.

ERROR_DS_CROSS_DOMAIN_CLEANUP_REQD = 8491

A cross-domain move operation failed because two versions of the moved object exist—one each in the source and destination domains. The destination object needs to be removed to restore the system to a consistent state.

ERROR_DS_CROSS_DOM_MOVE_ERROR = 8216

The requested cross-domain move operation could not be performed.

ERROR_DS_CROSS_NC_DN_RENAME = 8368

Modification of a distinguished name across a naming context is not permitted.

ERROR_DS_CROSS_REF_BUSY = 8602

A cross-reference is in use locally with the same name.

ERROR_DS_CROSS_REF_EXISTS = 8374

A cross-reference already exists.

ERROR_DS_CR_IMPOSSIBLE_TO_VALIDATE = 8495

The directory cannot validate the proposed naming context name because it does not hold a replica of the naming context above the proposed naming context. Ensure that the domain naming master role is held by a server that is configured as a GC server, and that the server is up-to-date with its replication partners. (Applies only to Windows 2000 operating system domain naming masters.)

ERROR_DS_CR_IMPOSSIBLE_TO_VALIDATE_V2 = 8586

The directory cannot validate the proposed naming context (or partition) name because it does not hold a replica, nor can it contact a replica of the naming context above the proposed naming context. Ensure that the parent naming context is properly registered in the DNS, and at least one replica of this naming context is reachable by the domain naming master.

ERROR_DS_DATABASE_ERROR = 8409

A database error has occurred.

ERROR_DS_DECODING_ERROR = 8253

A decoding error has occurred.

ERROR_DS_DESTINATION_AUDITING_NOT_ENABLED = 8536

The operation requires that destination domain auditing be enabled.

ERROR_DS_DESTINATION_DOMAIN_NOT_IN_FOREST = 8535

The destination domain MUST be in the forest.

ERROR_DS_DIFFERENT_REPL_EPOCHS = 8593

The directory service cannot perform the requested operation because the servers involved are of different replication epochs (which is usually related to a domain rename that is in progress).

ERROR_DS_DISALLOWED_IN_SYSTEM_CONTAINER = 8615

The requested operation is not allowed on an object under the system container.

ERROR_DS_DNS_LOOKUP_FAILURE = 8524

The DSA operation is unable to proceed because of a DNS lookup failure.

ERROR_DS_DOMAIN_RENAME_IN_PROGRESS = 8612

The directory service cannot perform the requested operation because a domain rename operation is in progress.

ERROR_DS_DOMAIN_VERSION_TOO_HIGH = 8564

The version of the operating system installed is incompatible with the current domain functional level. You must upgrade to a new version of the operating system before this server can become a domain controller in this domain.

ERROR_DS_DOMAIN_VERSION_TOO_LOW = 8566

The version of the operating system installed on this server no longer supports the current domain functional level. You must raise the domain functional level before this server can become a domain controller in this domain.

ERROR_DS_DRA_ABANDON_SYNC = 8462

The replication synchronization attempt was abandoned because of a lack of updates.

ERROR_DS_DRA_ACCESS_DENIED = 8453

Replication access was denied.

ERROR_DS_DRA_BAD_DN = 8439

The DN specified for this replication operation is invalid.

ERROR_DS_DRA_BAD_INSTANCE_TYPE = 8445

The replication operation encountered an object with an invalid instance type.

ERROR_DS_DRA_BAD_NC = 8440

The naming context specified for this replication operation is invalid.

ERROR_DS_DRA_BUSY = 8438

The directory service is too busy to complete the replication operation at this time.

ERROR_DS_DRA_CONNECTION_FAILED = 8444

The server specified for this replication operation could not be contacted.

ERROR_DS_DRA_DB_ERROR = 8451

The replication operation encountered a database error.

ERROR_DS_DRA_DN_EXISTS = 8441

The DN specified for this replication operation already exists.

ERROR_DS_DRA_EARLIER_SCHEMA_CONFLICT = 8544

The replication operation could not be completed due to a previous schema incompatibility.

ERROR_DS_DRA_EXTN_CONNECTION_FAILED = 8466

The server specified for this replication operation was contacted, but that server was unable to contact an additional server needed to complete the operation.

ERROR_DS_DRA_GENERIC = 8436

The replication operation failed.

ERROR_DS_DRA_INCOMPATIBLE_PARTIAL_SET = 8464

A synchronization attempt failed because the destination DC is currently waiting to synchronize new partial attributes from the source. This condition is normal if a recent schema change modified the partial attribute set. The destination partial attribute set is not a subset of the source partial attribute set.

ERROR_DS_DRA_INCONSISTENT_DIT = 8443

The replication operation encountered a database inconsistency.

ERROR_DS_DRA_INTERNAL_ERROR = 8442

The replication system encountered an internal error.

ERROR_DS_DRA_INVALID_PARAMETER = 8437

An invalid parameter was specified for this replication operation.

ERROR_DS_DRA_MAIL_PROBLEM = 8447

The replication operation encountered an error with the mail system.

ERROR_DS_DRA_MISSING_PARENT = 8460

The replication operation failed because a required parent object is missing.

ERROR_DS_DRA_NAME_COLLISION = 8458

The replication operation failed due to a collision of object names.

ERROR_DS_DRA_NOT_SUPPORTED = 8454

The requested operation is not supported by this version of the directory service.

ERROR_DS_DRA_NO_REPLICA = 8452

The naming context is in the process of being removed or is not replicated from the specified server.

ERROR_DS_DRA_OBJ_IS_REP_SOURCE = 8450

The naming context cannot be removed because it is replicated to another server.

ERROR_DS_DRA_OBJ_NC_MISMATCH = 8545

The replication update could not be applied because either the source or the destination has not yet received information regarding a recent cross-domain move operation.

ERROR_DS_DRA_OUT_OF_MEM = 8446

The replication operation failed to allocate memory.

ERROR_DS_DRA_OUT_SCHEDULE_WINDOW = 8617

The scheduled replication did not take place because the system was too busy to execute the request within the schedule window. The replication queue is overloaded. Consider reducing the number of partners or decreasing the scheduled replication frequency.

ERROR_DS_DRA_PREEMPTED = 8461

The replication operation was preempted.

ERROR_DS_DRA_RECYCLED_TARGET = 8639

The replication operation failed because the target object referenced by a link value is recycled.

ERROR_DS_DRA_REF_ALREADY_EXISTS = 8448

The replication reference information for the target server already exists.

ERROR_DS_DRA_REF_NOT_FOUND = 8449

The replication reference information for the target server does not exist.

ERROR_DS_DRA_REPL_PENDING = 8477

The replication request has been posted; waiting for a reply.

ERROR_DS_DRA_RPC_CANCELLED = 8455

The replication RPC was canceled.

ERROR_DS_DRA_SCHEMA_CONFLICT = 8543

The replication operation could not be completed due to a schema incompatibility.

ERROR_DS_DRA_SCHEMA_INFO_SHIP = 8542

Schema information could not be included in the replication request.

ERROR_DS_DRA_SCHEMA_MISMATCH = 8418

The replication operation failed because of a schema mismatch between the servers involved.

ERROR_DS_DRA_SHUTDOWN = 8463

The replication operation was terminated because the system is shutting down.

ERROR_DS_DRA_SINK_DISABLED = 8457

The destination server is currently rejecting replication requests.

ERROR_DS_DRA_SOURCE_DISABLED = 8456

The source server is currently rejecting replication requests.

ERROR_DS_DRA_SOURCE_IS_PARTIAL_REPLICA = 8465

The replication synchronization attempt failed because a master replica attempted to sync from a partial replica.

ERROR_DS_DRA_SOURCE_REINSTALLED = 8459

The replication source has been reinstalled.

ERROR_DS_DRS_EXTENSIONS_CHANGED = 8594

The directory service binding must be renegotiated due to a change in the server extensions information.

ERROR_DS_DSA_MUST_BE_INT_MASTER = 8342

The operation can only be performed on an internal master DSA object.

ERROR_DS_DST_DOMAIN_NOT_NATIVE = 8496

Destination domain must be in native mode.

ERROR_DS_DST_NC_MISMATCH = 8486

The source and destination for a cross-domain move do not agree on the naming contexts in the forest. Either the source or the destination does not have the latest version of the Partitions container.

ERROR_DS_DS_REQUIRED = 8478

The requested operation requires a directory service, and none was available.

ERROR_DS_DUPLICATE_ID_FOUND = 8605

The requested object has a nonunique identifier and cannot be retrieved.

ERROR_DS_DUP_LDAP_DISPLAY_NAME = 8382

Schema update failed: Duplicate LDAP display name.

ERROR_DS_DUP_LINK_ID = 8468

Schema update failed: An attribute with the same link identifier already exists.

ERROR_DS_DUP_MAPI_ID = 8380

Schema update failed: Duplicate Message Application Programming Interface (MAPI) identifier.

ERROR_DS_DUP_MSDS_INTID = 8597

Schema update failed: Duplicate msDS-IntId. Retry the operation.

ERROR_DS_DUP_OID = 8379

Schema update failed: Duplicate OID.

ERROR_DS_DUP_RDN = 8378

Schema update failed: Duplicate RDN.

ERROR_DS_DUP_SCHEMA_ID_GUID = 8381

Schema update failed: Duplicate schema ID GUID.

ERROR_DS_ENCODING_ERROR = 8252

An encoding error has occurred.

ERROR_DS_EPOCH_MISMATCH = 8483

The source and destination of a cross-domain move do not agree on the object's epoch number. Either the source or the destination does not have the latest version of the object.

ERROR_DS_EXISTING_AD_CHILD_NC = 8613

The directory service detected a child partition below the requested partition name. The partition hierarchy must be created in a top down method.

ERROR_DS_EXISTS_IN_AUX_CLS = 8393

Schema deletion failed: Class is used as an auxiliary class.

ERROR_DS_EXISTS_IN_MAY_HAVE = 8386

Schema deletion failed: Attribute is used in the May-Contain list.

ERROR_DS_EXISTS_IN_MUST_HAVE = 8385

Schema deletion failed: Attribute is used in the Must-Contain list.

ERROR_DS_EXISTS_IN_POSS_SUP = 8395

Schema deletion failed: Class is used as a Poss-Superior.

ERROR_DS_EXISTS_IN_RDNATTID = 8598

Schema deletion failed: Attribute is used in rDNAttID.

ERROR_DS_EXISTS_IN_SUB_CLS = 8394

Schema deletion failed: Class is used as a subclass.

ERROR_DS_FILTER_UNKNOWN = 8254

The search filter cannot be recognized.

ERROR_DS_FILTER_USES_CONSTRUCTED_ATTRS = 8555

A filter was passed that uses constructed attributes.

ERROR_DS_FOREST_VERSION_TOO_HIGH = 8563

The version of the operating system installed is incompatible with the current forest functional level. You must upgrade to a new version of the operating system before this server can become a domain controller in this forest.

ERROR_DS_FOREST_VERSION_TOO_LOW = 8565

The version of the operating system installed on this server no longer supports the current forest functional level. You must raise the forest functional level before this server can become a domain controller in this forest.

ERROR_DS_GCVERIFY_ERROR = 8417

The GC verification failed. The GC is not available or does not support the operation. Some part of the directory is currently not available.

ERROR_DS_GC_NOT_AVAILABLE = 8217

Unable to contact the global catalog (GC) server.

ERROR_DS_GC_REQUIRED = 8547

The requested operation can be performed only on a GC server.

ERROR_DS_GENERIC_ERROR = 8341

A directory service error has occurred.

ERROR_DS_GLOBAL_CANT_HAVE_CROSSDOMAIN_MEMBER = 8519

A global group cannot have a cross-domain member.

ERROR_DS_GLOBAL_CANT_HAVE_LOCAL_MEMBER = 8516

A global group cannot have a local group as a member.

ERROR_DS_GLOBAL_CANT_HAVE_UNIVERSAL_MEMBER = 8517

A global group cannot have a universal group as a member.

ERROR_DS_GOVERNSID_MISSING = 8410

ThegovernsIDattribute is missing.

ERROR_DS_GROUP_CONVERSION_ERROR = 8607

The group cannot be converted due to attribute restrictions on the requested group type.

ERROR_DS_HAVE_PRIMARY_MEMBERS = 8521

A group with primary members cannot change to a security-disabled group.

ERROR_DS_HIERARCHY_TABLE_MALLOC_FAILED = 8429

The allocation of the hierarchy table failed.

ERROR_DS_HIGH_DSA_VERSION = 8642

The functional level of the domain (or forest) cannot be lowered to the requested value.

ERROR_DS_ILLEGAL_BASE_SCHEMA_MOD = 8507

An attempt was made to modify the base schema.

ERROR_DS_ILLEGAL_MOD_OPERATION = 8311

Illegal modify operation. Some aspect of the modification is not permitted.

ERROR_DS_ILLEGAL_SUPERIOR = 8345

The object cannot be added because the parent is not on the list of possible superiors.

ERROR_DS_ILLEGAL_XDOM_MOVE_OPERATION = 8492

This object cannot be moved across domain boundaries either because cross-domain moves for this class are not allowed, or the object has some special characteristics, for example, a trust account or a restricted relative identifier (RID), that prevent its move.

ERROR_DS_INAPPROPRIATE_AUTH = 8233

Inappropriate authentication.

ERROR_DS_INAPPROPRIATE_MATCHING = 8238

Inappropriate matching.

ERROR_DS_INCOMPATIBLE_CONTROLS_USED = 8574

Unable to continue operation because multiple conflicting controls were used.

ERROR_DS_INCOMPATIBLE_VERSION = 8567

The version of the operating system installed on this server is incompatible with the functional level of the domain or forest.

ERROR_DS_INCORRECT_ROLE_OWNER = 8210

The requested operation could not be performed because the directory service is not the master for that type of operation.

ERROR_DS_INIT_FAILURE = 8532

Directory Services could not start.

ERROR_DS_INIT_FAILURE_CONSOLE = 8561

Directory Services could not start because of the following error: %1. Error Status: 0x%2. Click OK to shut down the system. You can use the Recovery Console to further diagnose the system.

ERROR_DS_INSTALL_NO_SCH_VERSION_IN_INIFILE = 8512

The replica or child install failed to read theobjectVersionattribute in the SCHEMA section of the file schema.ini in the System32 directory.

ERROR_DS_INSTALL_NO_SRC_SCH_VERSION = 8511

The replica or child install failed to get theobjectVersionattribute on the schema container on the source DC. Either the attribute is missing on the schema container or the credentials supplied do not

have permission to read it.

ERROR_DS_INSTALL_SCHEMA_MISMATCH = 8467

The version of the directory service schema of the source forest is not compatible with the version of the directory service on this computer.

ERROR_DS_INSUFFICIENT_ATTR_TO_CREATE_OBJECT = 8606

Insufficient attributes were given to create an object. This object might not exist because it might have been deleted and the garbage already collected.

ERROR_DS_INSUFF_ACCESS_RIGHTS = 8344

Insufficient access rights to perform the operation.

ERROR_DS_INTERNAL_FAILURE = 8430

The directory service encountered an internal failure.

ERROR_DS_INVALID_ATTRIBUTE_SYNTAX = 8203

The attribute syntax specified to the directory service is invalid.

ERROR_DS_INVALID_DMD = 8360

The schema object could not be found.

ERROR_DS_INVALID_DN_SYNTAX = 8242

An invalid dn syntax has been specified.

ERROR_DS_INVALID_GROUP_TYPE = 8513

The specified group type is invalid.

ERROR_DS_INVALID_LDAP_DISPLAY_NAME = 8479

The LDAP display name of the class or attribute contains non-ASCII characters.

ERROR_DS_INVALID_NAME_FOR_SPN = 8554

An SPN could not be constructed because the provided host name is not in the necessary format.

ERROR_DS_INVALID_ROLE_OWNER = 8366

The role owner attribute could not be read.

ERROR_DS_INVALID_SCRIPT = 8600

The directory service cannot process the script because it is invalid.

ERROR_DS_INVALID_SEARCH_FLAG = 8500

The search flags for the attribute are invalid. The ambiguous name resolution (ANR) bit is valid only on attributes of Unicode or Teletex strings.

ERROR_DS_INVALID_SEARCH_FLAG_SUBTREE = 8626

The search flags for the attribute are invalid. The subtree index bit is valid only on single-valued attributes.

ERROR_DS_INVALID_SEARCH_FLAG_TUPLE = 8627

The search flags for the attribute are invalid. The tuple index bit is valid only on attributes of Unicode strings.

ERROR_DS_IS_LEAF = 8243

The object is a leaf object.

ERROR_DS_KEY_NOT_UNIQUE = 8527

The object requested was not found, but an object with that key was found.

ERROR_DS_LDAP_SEND_QUEUE_FULL = 8616

The LDAP server's network send queue has filled up because the client is not processing the results of its requests fast enough. No more requests will be processed until the client catches up. If the client does not catch up then it will be disconnected.

ERROR_DS_LINK_ID_NOT_AVAILABLE = 8577

Schema update failed: There are no link identifiers available.

ERROR_DS_LOCAL_CANT_HAVE_CROSSDOMAIN_LOCAL_MEMBER = 8520

A local group cannot have another cross domain local group as a member.

ERROR_DS_LOCAL_ERROR = 8251

A local error has occurred.

ERROR_DS_LOCAL_MEMBER_OF_LOCAL_ONLY = 8548

A local group can only be a member of other local groups in the same domain.

ERROR_DS_LOOP_DETECT = 8246

A loop has been detected.

ERROR_DS_LOW_DSA_VERSION = 8568

The functional level of the domain (or forest) cannot be raised to the requested value because one or more domain controllers in the domain (or forest) are at a lower, incompatible functional level.

ERROR_DS_MACHINE_ACCOUNT_CREATED_PRENT4 = 8572

The machine account was created before Windows NT 4.0. The account needs to be re-created.

ERROR_DS_MACHINE_ACCOUNT_QUOTA_EXCEEDED = 8557

Your computer could not be joined to the domain. You have exceeded the maximum number of computer accounts you are allowed to create in this domain. Contact your system administrator to have this limit reset or increased.

ERROR_DS_MASTERDSA_REQUIRED = 8314

The operation must be performed at a master DSA.

ERROR_DS_MAX_OBJ_SIZE_EXCEEDED = 8304

The maximum size of an object has been exceeded.

ERROR_DS_MEMBERSHIP_EVALUATED_LOCALLY = 8201

The directory service evaluated group memberships locally.

ERROR_DS_MISSING_EXPECTED_ATT = 8411

An expected attribute is missing.

ERROR_DS_MISSING_FSMO_SETTINGS = 8434

The directory service is missing mandatory configuration information and is unable to determine the ownership of floating single-master operation roles.

ERROR_DS_MISSING_INFRASTRUCTURE_CONTAINER = 8497

The operation cannot be performed because the server does not have an infrastructure container in the domain of interest.

ERROR_DS_MISSING_REQUIRED_ATT = 8316

A required attribute is missing.

ERROR_DS_MISSING_SUPREF = 8406

No superior reference has been configured for the directory service. The directory service is, therefore, unable to issue referrals to objects outside this forest.

ERROR_DS_MODIFYDN_DISALLOWED_BY_FLAG = 8581

A system flag has been set on the object that does not allow the object to be moved or renamed.

ERROR_DS_MODIFYDN_DISALLOWED_BY_INSTANCE_TYPE = 8579

Rename or move operations on naming context heads or read-only objects are not allowed.

ERROR_DS_MODIFYDN_WRONG_GRANDPARENT = 8582

This object is not allowed to change its grandparent container. Moves are not forbidden on this object, but are restricted to sibling containers.

ERROR_DS_MUST_BE_RUN_ON_DST_DC = 8558

For security reasons, the operation must be run on the destination DC.

ERROR_DS_NAME_ERROR_DOMAIN_ONLY = 8473

Name translation: Unable to resolve completely, only the domain was found.

ERROR_DS_NAME_ERROR_NOT_FOUND = 8470

Name translation: Could not find the name or insufficient right to see name.

ERROR_DS_NAME_ERROR_NOT_UNIQUE = 8471

Name translation: Input name mapped to more than one output name.

ERROR_DS_NAME_ERROR_NO_MAPPING = 8472

Name translation: The input name was found but not the associated output format.

ERROR_DS_NAME_ERROR_NO_SYNTACTICAL_MAPPING = 8474

Name translation: Unable to perform purely syntactical mapping at the client without going out to the wire.

ERROR_DS_NAME_ERROR_RESOLVING = 8469

Name translation: Generic processing error.

ERROR_DS_NAME_ERROR_TRUST_REFERRAL = 8583

Unable to resolve completely; a referral to another forest was generated.

ERROR_DS_NAME_NOT_UNIQUE = 8571

The requested name already exists as a unique identifier.

ERROR_DS_NAME_REFERENCE_INVALID = 8373

The name reference is invalid.

ERROR_DS_NAME_TOO_LONG = 8348

The name is too long.

ERROR_DS_NAME_TOO_MANY_PARTS = 8347

The name has too many parts.

ERROR_DS_NAME_TYPE_UNKNOWN = 8351

The directory service cannot get the attribute type for a name.

ERROR_DS_NAME_UNPARSEABLE = 8350

The directory service encountered an error parsing a name.

ERROR_DS_NAME_VALUE_TOO_LONG = 8349

The name value is too long.

ERROR_DS_NAMING_MASTER_GC = 8523

Only DSAs configured to be GC servers should be allowed to hold the domain naming master FSMO role. (Applies only to Windows 2000 servers.)

ERROR_DS_NAMING_VIOLATION = 8247

There is a naming violation.

ERROR_DS_NCNAME_MISSING_CR_REF = 8412

The specified naming context is missing a cross-reference.

ERROR_DS_NCNAME_MUST_BE_NC = 8357

The object must be a naming context.

ERROR_DS_NC_MUST_HAVE_NC_PARENT = 8494

A naming context head must be the immediate child of another naming context head, not of an interior node.

ERROR_DS_NC_STILL_HAS_DSAS = 8546

The requested domain could not be deleted because there exist domain controllers that still host this domain.

ERROR_DS_NONEXISTENT_MAY_HAVE = 8387

Schema update failed: Attribute in May-Contain list does not exist.

ERROR_DS_NONEXISTENT_MUST_HAVE = 8388

Schema update failed: Attribute in the Must-Contain list does not exist.

ERROR_DS_NONEXISTENT_POSS_SUP = 8390

Schema update failed: Class in the Poss-Superiors list does not exist.

ERROR_DS_NONSAFE_SCHEMA_CHANGE = 8508

Adding a new mandatory attribute to an existing class, deleting a mandatory attribute from an existing class, or adding an optional attribute to the special class Top that is not a backlink attribute (directly or through inheritance, for example, by adding or deleting an auxiliary class) is not allowed.

ERROR_DS_NON_ASQ_SEARCH = 8624

The requested search operation is not supported for attribute scoped query (ASQ) searches.

ERROR_DS_NON_BASE_SEARCH = 8480

The requested search operation is only supported for base searches.

ERROR_DS_NOTIFY_FILTER_TOO_COMPLEX = 8377

Notification filter is too complex.

ERROR_DS_NOT_AN_OBJECT = 8352

The name does not identify an object; the name identifies a phantom.

ERROR_DS_NOT_AUTHORITY_FOR_DST_NC = 8487

The destination of a cross-domain move is not authoritative for the destination naming context.

ERROR_DS_NOT_CLOSEST = 8588

The GC server is not in the closest site.

ERROR_DS_NOT_INSTALLED = 8200

An error occurred while installing the directory service. For more information, see the event log.

ERROR_DS_NOT_ON_BACKLINK = 8362

The operation cannot be performed on a back link.

ERROR_DS_NOT_SUPPORTED = 8256

The specified method is not supported.

ERROR_DS_NOT_SUPPORTED_SORT_ORDER = 8570

The sort order requested is not supported.

ERROR_DS_NO_ATTRIBUTE_OR_VALUE = 8202

The specified directory service attribute or value does not exist.

ERROR_DS_NO_BEHAVIOR_VERSION_IN_MIXEDDOMAIN = 8569

The forest functional level cannot be raised to the requested value because one or more domains are still in mixed-domain mode. All domains in the forest must be in native mode for you to raise the forest functional level.

ERROR_DS_NO_CHAINED_EVAL = 8328

Chained evaluation is not permitted.

ERROR_DS_NO_CHAINING = 8327

Chaining is not permitted.

ERROR_DS_NO_CHECKPOINT_WITH_PDC = 8551

The checkpoint with the PDC could not be taken because too many modifications are currently being processed.

ERROR_DS_NO_CROSSREF_FOR_NC = 8363

The cross-reference for the specified naming context could not be found.

ERROR_DS_NO_DELETED_NAME = 8355

Failed to create name for deleted object.

ERROR_DS_NO_FPO_IN_UNIVERSAL_GROUPS = 8549

Foreign security principals cannot be members of universal groups.

ERROR_DS_NO_MORE_RIDS = 8209

The directory service has exhausted the pool of relative identifiers.

ERROR_DS_NO_MSDS_INTID = 8596

Schema update failed: No values for msDS-IntId are available.

ERROR_DS_NO_NEST_GLOBALGROUP_IN_MIXEDDOMAIN = 8514

You cannot nest global groups in a mixed domain if the group is security-enabled.

ERROR_DS_NO_NEST_LOCALGROUP_IN_MIXEDDOMAIN = 8515

You cannot nest local groups in a mixed domain if the group is security-enabled.

ERROR_DS_NO_NTDSA_OBJECT = 8623

The NTDS Settings object for the domain controller does not exist.

ERROR_DS_NO_OBJECT_MOVE_IN_SCHEMA_NC = 8580

Move operations on objects in the schema naming context are not allowed.

ERROR_DS_NO_PARENT_OBJECT = 8329

The operation could not be performed because the object's parent is either uninstantiated or deleted.

ERROR_DS_NO_PKT_PRIVACY_ON_CONNECTION = 8533

The connection between client and server requires packet privacy or better.

ERROR_DS_NO_RDN_DEFINED_IN_SCHEMA = 8306

An attempt was made to add an object of a class that does not have an RDN defined in the schema.

ERROR_DS_NO_REF_DOMAIN = 8575

Unable to find a valid security descriptor reference domain for this partition.

ERROR_DS_NO_REQUESTED_ATTS_FOUND = 8308

None of the requested attributes were found on the objects.

ERROR_DS_NO_RESULTS_RETURNED = 8257

No results were returned.

ERROR_DS_NO_RIDS_ALLOCATED = 8208

The directory service was unable to allocate a relative identifier.

ERROR_DS_NO_SERVER_OBJECT = 8622

The server object for the domain controller does not exist.

ERROR_DS_NO_SUCH_OBJECT = 8240

There is no such object on the server.

ERROR_DS_NO_TREE_DELETE_ABOVE_NC = 8501

Tree deletions starting at an object that has an NC head as a descendant are not allowed.

ERROR_DS_NTDSRIPT_PROCESS_ERROR = 8592

The directory service cannot process the script because of an error.

ERROR_DS_NTDSRIPT_SYNTAX_ERROR = 8591

The directory service cannot parse the script because of a syntax error.

ERROR_DS_OBJECT_BEING_REMOVED = 8339

The operation cannot continue because the object is in the process of being removed.

ERROR_DS_OBJECT_CLASS_REQUIRED = 8315

The object class attribute must be specified.

ERROR_DS_OBJECT_RESULTS_TOO_LARGE = 8248

The result set is too large.

ERROR_DS_OBJ_CLASS_NOT_DEFINED = 8371

The specified class is not defined.

ERROR_DS_OBJ_CLASS_NOT_SUBCLASS = 8372

The specified class is not a subclass.

ERROR_DS_OBJ_CLASS_VIOLATION = 8212

The requested operation did not satisfy one or more constraints associated with the class of the object.

ERROR_DS_OBJ_GUID_EXISTS = 8361

A local object with this GUID (dead or alive) already exists.

ERROR_DS_OBJ_NOT_FOUND = 8333

Directory object not found.

ERROR_DS_OBJ_STRING_NAME_EXISTS = 8305

An attempt was made to add an object to the directory with a name that is already in use.

ERROR_DS_OBJ_TOO_LARGE = 8312

The specified object is too large.

ERROR_DS_OFFSET_RANGE_ERROR = 8262

The search results exceed the offset range specified.

ERROR_DS_OPERATIONS_ERROR = 8224

An operations error occurred.

ERROR_DS_OUT_OF_SCOPE = 8338

The operation is out of scope.

ERROR_DS_OUT_OF_VERSION_STORE = 8573

The database is out of version store.

ERROR_DS_PARAM_ERROR = 8255

One or more parameters are illegal.

ERROR_DS_PARENT_IS_AN_ALIAS = 8330

Having a parent that is an alias is not permitted. Aliases are leaf objects.

ERROR_DS_PDC_OPERATION_IN_PROGRESS = 8490

Another operation that requires exclusive access to the PDC FSMO is already in progress.

ERROR_DS_POLICY_NOT_KNOWN = 8618

At this time, it cannot be determined if the branch replication policy is available on the hub domain controller. Retry at a later time to account for replication latencies.

ERROR_DS_PROTOCOL_ERROR = 8225

A protocol error occurred.

ERROR_DS_RANGE_CONSTRAINT = 8322

A value for the attribute was not in the acceptable range of values.

ERROR_DS_RDN_DOESNT_MATCH_SCHEMA = 8307

An attempt was made to add an object using an RDN that is not the RDN defined in the schema.

ERROR_DS_RECALCSHEMA_FAILED = 8396

Schema update failed in recalculating validation cache.

ERROR_DS_REFERRAL = 8235

A referral was returned from the server.

ERROR_DS_REFERRAL_LIMIT_EXCEEDED = 8260

The preset referral limit was exceeded.

ERROR_DS_REFUSING_FSMO_ROLES = 8433

This directory server is shutting down, and cannot take ownership of new floating single-master operation roles.

ERROR_DS_REMOTE_CROSSREF_OP_FAILED = 8601

The remote create cross-reference operation failed on the domain naming master FSMO. The operation's error is in the extended data.

ERROR_DS_REPLICATOR_ONLY = 8370

Only the replicator can perform this function.

ERROR_DS_REPLICA_SET_CHANGE_NOT_ALLOWED_ON_DISABLED_CR = 8595

The operation is not allowed on a disabled cross-reference.

ERROR_DS_REPL_LIFETIME_EXCEEDED = 8614

The directory service cannot replicate with this server because the time since the last replication with this server has exceeded the tombstone lifetime.

ERROR_DS_RESERVED_LINK_ID = 8576

Schema update failed: The link identifier is reserved.

ERROR_DS_RIDMGR_INIT_ERROR = 8211

The directory service was unable to initialize the subsystem that allocates relative identifiers.

ERROR_DS_ROLE_NOT_VERIFIED = 8610

The FSMO role ownership could not be verified because its directory partition did not replicate successfully with at least one replication partner.

ERROR_DS_ROOT_CANT_BE_SUBREF = 8326

The specified root object cannot be a subreference.

ERROR_DS_ROOT_MUST_BE_NC = 8301

The root object must be the head of a naming context. The root object cannot have an instantiated parent.

ERROR_DS_ROOT_REQUIRES_CLASS_TOP = 8432

A root object requires a class of "top".

ERROR_DS_SAM_INIT_FAILURE = 8504

SAM initialization failed because of the following error: %1. Error Status: 0x%2. Click OK to shut down the system and reboot into Directory Services Restore Mode. Check the event log for detailed information.

ERROR_DS_SAM_INIT_FAILURE_CONSOLE = 8562

SAM initialization failed because of the following error: %1. Error Status: 0x%2. Click OK to shut down the system. You can use the Recovery Console to further diagnose the system.

ERROR_DS_SAM_NEED_BOOTKEY_FLOPPY = 8530

SAM needs to get the boot key from the floppy disk.

ERROR_DS_SAM_NEED_BOOTKEY_PASSWORD = 8529

SAM needs to get the boot password.

ERROR_DS_SCHEMA_ALLOC_FAILED = 8415

Schema allocation failed. Check if the machine is running low on memory.

ERROR_DS_SCHEMA_NOT_LOADED = 8414

The schema is not loaded.

ERROR_DS_SCHEMA_UPDATE_DISALLOWED = 8509

Schema update is not allowed on this DC because the DC is not the schema FSMO role owner.

ERROR_DS_SECURITY_CHECKING_ERROR = 8413

A security checking error has occurred.

ERROR_DS_SECURITY_ILLEGAL_MODIFY = 8423

The modification was not permitted for security reasons.

ERROR_DS_SEC_DESC_INVALID = 8354

The security descriptor is invalid.

ERROR_DS_SEC_DESC_TOO_SHORT = 8353

The security descriptor is too short.

ERROR_DS_SEMANTIC_ATT_TEST = 8383

Schema update failed: Range-Lower less than Range-Upper.

ERROR_DS_SENSITIVE_GROUP_VIOLATION = 8505

Only an administrator can modify the membership list of an administrative group.

ERROR_DS_SERVER_DOWN = 8250

The server is not operational.

ERROR_DS_SHUTTING_DOWN = 8364

The operation could not be performed because the directory service is shutting down.

ERROR_DS_SINGLE_USER_MODE_FAILED = 8590

The directory service failed to enter single-user mode.

ERROR_DS_SINGLE_VALUE_CONSTRAINT = 8321

Multiple values were specified for an attribute that can have only one value.

ERROR_DS_SIZELIMIT_EXCEEDED = 8227

The size limit for this request was exceeded.

ERROR_DS_SORT_CONTROL_MISSING = 8261

The search requires a SORT control.

ERROR_DS_SOURCE_AUDITING_NOT_ENABLED = 8552

The operation requires that source domain auditing be enabled.

ERROR_DS_SOURCE_DOMAIN_IN_FOREST = 8534

The source domain cannot be in the same forest as the destination.

ERROR_DS_SPN_VALUE_NOT_UNIQUE_IN_FOREST = 8647

The operation failed because the SPN value provided for addition/modification is not unique forest-wide.

ERROR_DS_SRC_AND_DST_NC_IDENTICAL = 8485

The source and destination for the cross-domain move operation are identical. The caller should use a local move operation instead of a cross-domain move operation.

ERROR_DS_SRC_AND_DST_OBJECT_CLASS_MISMATCH = 8540

The source and destination object must be of the same type.

ERROR_DS_SRC_DC_MUST_BE_SP4_OR_GREATER = 8559

For security reasons, the source DC must be NT4SP4 or greater.

ERROR_DS_SRC_GUID_MISMATCH = 8488

The source and destination of a cross-domain move do not agree on the identity of the source object. Either the source or the destination does not have the latest version of the source object.

ERROR_DS_SRC_NAME_MISMATCH = 8484

The source and destination of a cross-domain move do not agree on the object's current name. Either the source or the destination does not have the latest version of the object.

ERROR_DS_SRC_OBJ_NOT_GROUP_OR_USER = 8538

The source object must be a group or user.

ERROR_DS_SRC_SID_EXISTS_IN_FOREST = 8539

The source object's SID already exists in the destination forest.

ERROR_DS_STRING_SD_CONVERSION_FAILED = 8522

The schema cache load failed to convert the string default security descriptor (SD) on a class-schema object.

ERROR_DS_STRONG_AUTH_REQUIRED = 8232

A more secure authentication method is required for this server.

ERROR_DS_SUBREF_MUST_HAVE_PARENT = 8356

The parent of a new subreference must exist.

ERROR_DS_SUBTREE_NOTIFY_NOT_NC_HEAD = 8376

Subtree notifications are only supported on naming context (NC) heads.

ERROR_DS_SUB_CLS_TEST_FAIL = 8391

Schema update failed: Class in the subclass of the list does not exist or does not satisfy hierarchy rules.

ERROR_DS_SYNTAX_MISMATCH = 8384

Schema update failed: Syntax mismatch.

ERROR_DS_THREAD_LIMIT_EXCEEDED = 8587

The thread limit for this request was exceeded.

ERROR_DS_TIMELIMIT_EXCEEDED = 8226

The time limit for this request was exceeded.

ERROR_DS_TREE_DELETE_NOT_FINISHED = 8397

The tree deletion is not finished. The request must be made again to continue deleting the tree.

ERROR_DS_UNABLE_TO_SURRENDER_ROLES = 8435

The directory service was unable to transfer ownership of one or more floating single-master operation roles to other servers.

ERROR_DS_UNAVAILABLE = 8207

The directory service is unavailable.

ERROR_DS_UNAVAILABLE_CRIT_EXTENSION = 8236

The server does not support the requested critical extension.

ERROR_DS_UNICODEPWD_NOT_IN_QUOTES = 8556

TheunicodePwdattribute value must be enclosed in quotation marks.

ERROR_DS_UNIVERSAL_CANT_HAVE_LOCAL_MEMBER = 8518

A universal group cannot have a local group as a member.

ERROR_DS_UNKNOWN_ERROR = 8431

The directory service encountered an unknown failure.

ERROR_DS_UNKNOWN_OPERATION = 8365

The directory service request is invalid.

ERROR_DS_UNWILLING_TO_PERFORM = 8245

The server is unwilling to process the request.

ERROR_DS_UPN_VALUE_NOT_UNIQUE_IN_FOREST = 8648

The operation failed because the UPN value provided for addition/modification is not unique forest-wide.

ERROR_DS_USER_BUFFER_TO_SMALL = 8309

The user buffer is too small.

ERROR_DS_VERSION_CHECK_FAILURE = 643

This version of Windows is not compatible with the behavior version of directory forest, domain, or domain controller.

ERROR_DS_WKO_CONTAINER_CANNOT_BE_SPECIAL = 8611

The target container for a redirection of a well-known object container cannot already be a special container.

ERROR_DS_WRONG_LINKED_ATT_SYNTAX = 8528

The syntax of the linked attribute being added is incorrect. Forward links can only have syntax 2.5.5.1, 2.5.5.7, and 2.5.5.14, and backlinks can only have syntax 2.5.5.1.

ERROR_DS_WRONG_OM_OBJ_CLASS = 8476

The OM-Object-Class specified is incorrect for an attribute with the specified syntax.

ERROR_DUPLICATE_SERVICE_NAME = 1078

The name is already in use as either a service name or a service display name.

ERROR_DUPLICATE_TAG = 2014

The specified tag is already present.

ERROR_DUP_DOMAINNAME = 1221

The workgroup or domain name is already in use by another computer on the network.

ERROR_DUP_NAME = 52

You were not connected because a duplicate name exists on the network. Go to System in Control Panel to change the computer name, and then try again.

ERROR_DYNLINK_FROM_INVALID_RING = 196

The operating system cannot run this application program.

ERROR_EAS_DIDNT_FIT = 275

The extended attributes did not fit in the buffer.

ERROR_EAS_NOT_SUPPORTED = 282

The mounted file system does not support extended attributes.

ERROR_EA_ACCESS_DENIED = 994

Access to the extended attribute was denied.

ERROR_EA_FILE_CORRUPT = 276

The extended attribute file on the mounted file system is corrupt.

ERROR_EA_LIST_INCONSISTENT = 255

The extended attributes are inconsistent.

ERROR_EA_TABLE_FULL = 277

The extended attribute table file is full.

ERROR_EC_LOG_DISABLED = 15081

The log of the subscription is in a disabled state and events cannot be forwarded to it. The log must first be enabled before the subscription can be activated.

ERROR_EC_SUBSCRIPTION_CANNOT_ACTIVATE = 15080

The subscription fails to activate.

ERROR_EFS_ALG_BLOB_TOO_BIG = 6013

The encryption algorithm used on the source file needs a bigger key buffer than the one on the destination file.

ERROR_EFS_DISABLED = 6015

This machine is disabled for file encryption.

ERROR_EFS_NOT_ALLOWED_IN_TRANSACTION = 6831

The encryption operation could not be completed because a transaction is active.

ERROR_EFS_SERVER_NOT_TRUSTED = 6011

The server is not trusted for remote encryption operation.

ERROR_EFS_VERSION_NOT_SUPPORT = 6016

A newer system is required to decrypt this encrypted file.

ERROR_ELEVATION_REQUIRED = 740

The requested operation requires elevation.

ERROR_EMPTY = 4306

The library, drive, or media pool is empty.

ERROR_ENCRYPTION_FAILED = 6000

The specified file could not be encrypted.

ERROR_END_OF_MEDIA = 1100

The physical end of the tape has been reached.

ERROR_ENLISTMENT_NOT_FOUND = 6717

The specified enlistment object could not be opened because it was not found.

ERROR_ENLISTMENT_NOT_SUPERIOR = 6820

The request was rejected because the enlistment in question is not a superior enlistment.

ERROR_ENVVAR_NOT_FOUND = 203

The system could not find the environment option that was entered.

ERROR_EOM_OVERFLOW = 1129

Physical end of tape encountered.

ERROR_ERRORS_ENCOUNTERED = 774

One or more errors occurred while processing the request.

ERROR_EVALUATION_EXPIRATION = 622

{Windows Evaluation Notification} The evaluation period for this installation of Windows has expired. This system will shut down in 1 hour. To restore access to this installation of Windows, upgrade this installation using a licensed distribution of this product.

ERROR_EVENTLOG_CANT_START = 1501

No event log file could be opened, so the event logging service did not start.

ERROR_EVENTLOG_FILE_CHANGED = 1503

The event log file has changed between read operations.

ERROR_EVENTLOG_FILE_CORRUPT = 1500

The event log file is corrupted.

ERROR_EVENT_DONE = 710

{TDI Event Done} The TDI indication has completed successfully.

ERROR_EVENT_PENDING = 711

{TDI Event Pending} The TDI indication has entered the pending state.

ERROR_EVT_CANNOT_OPEN_CHANNEL_OF_QUERY = 15036

The channel at index %1 of the query cannot be opened.

ERROR_EVT_CHANNEL_CANNOT_ACTIVATE = 15025

The channel fails to activate.

ERROR_EVT_CHANNEL_NOT_FOUND = 15007

The specified channel could not be found. Check channel configuration.

ERROR_EVT_CONFIGURATION_ERROR = 15010

Configuration error.

ERROR_EVT_EVENT_DEFINITION_NOT_FOUND = 15032

The event definition could not be found for the event ID (%1).

ERROR_EVT_EVENT_TEMPLATE_NOT_FOUND = 15003

The template for an event definition cannot be found in the resource (error = %1).

ERROR_EVT_FILTER_ALREADYSCOPED = 15014

An expression can only be followed by a change-of-scope operation if it itself evaluates to a node set and is not already part of some other change-of-scope operation.

ERROR_EVT_FILTER_INVARG = 15016

Left side arguments to binary operators must be either attributes, nodes, or variables and right side arguments must be constants.

ERROR_EVT_FILTER_INVTEST = 15017

A step operation must involve either a node test or, in the case of a predicate, an algebraic expression against which to test each node in the node set identified by the preceding node set can be evaluated.

ERROR_EVT_FILTER_INVTYPE = 15018

This data type is currently unsupported.

ERROR_EVT_FILTER_NOTELTSET = 15015

Cannot perform a step operation from a term that does not represent an element set.

ERROR_EVT_FILTER_PARSEERR = 15019

A syntax error occurred at position %1!d!

ERROR_EVT_FILTER_TOO_COMPLEX = 15026

The xpath expression exceeded supported complexity. Simplify it or split it into two or more simple expressions.

ERROR_EVT_FILTER_UNEXPECTEDTOKEN = 15021

The token encountered was unexpected.

ERROR_EVT_FILTER_UNSUPPORTEDOP = 15020

This operator is unsupported by this implementation of the filter.

ERROR_EVT_INVALID_CHANNEL_PATH = 15000

The specified channel path is invalid.

ERROR_EVT_INVALID_CHANNEL_PROPERTY_VALUE = 15023

Channel property %1!s! contains an invalid value. The value has an invalid type, is outside the valid range, cannot be updated, or is not supported by this type of channel.

ERROR_EVT_INVALID_EVENT_DATA = 15005

The event data raised by the publisher is not compatible with the event template definition in the publisher's manifest.

ERROR_EVT_INVALID_OPERATION_OVER_ENABLED_DIRECT_CHANNEL = 15022

The requested operation cannot be performed over an enabled direct channel. The channel must first be disabled before performing the requested operation.

ERROR_EVT_INVALID_PUBLISHER_NAME = 15004

The specified publisher name is invalid.

ERROR_EVT_INVALID_PUBLISHER_PROPERTY_VALUE = 15024

Publisher property %1!s! contains an invalid value. The value has an invalid type, is outside the valid range, cannot be updated, or is not supported by this type of publisher.

ERROR_EVT_INVALID_QUERY = 15001

The specified query is invalid.

ERROR_EVT_MALFORMED_XML_TEXT = 15008

The specified XML text was not well-formed. See extended error for more details.

ERROR_EVT_MAX_INSERTS_REACHED = 15031

The maximum number of replacements has been reached.

ERROR_EVT_MESSAGE_ID_NOT_FOUND = 15028

The message ID for the desired message could not be found.

ERROR_EVT_MESSAGE_LOCALE_NOT_FOUND = 15033

The locale-specific resource for the desired message is not present.

ERROR_EVT_MESSAGE_NOT_FOUND = 15027

The message resource is present but the message is not found in the string or message table.

ERROR_EVT_NON_VALIDATING_MSXML = 15013

Registered Microsoft XML (MSXML) does not support validation.

ERROR_EVT_PUBLISHER_DISABLED = 15037

The publisher has been disabled and its resource is not available. This usually occurs when the publisher is in the process of being uninstalled or upgraded.

ERROR_EVT_PUBLISHER_METADATA_NOT_FOUND = 15002

The publisher metadata cannot be found in the resource.

ERROR_EVT_QUERY_RESULT_INVALID_POSITION = 15012

Query result is currently at an invalid position.

ERROR_EVT_QUERY_RESULT_STALE = 15011

The query result is stale or invalid. This might be due to the log being cleared or rolling over after the query result was created. Users should handle this code by releasing the query result object and reissuing the query.

ERROR_EVT_SUBSCRIPTION_TO_DIRECT_CHANNEL = 15009

The caller is trying to subscribe to a direct channel which is not allowed. The events for a direct channel go directly to a log file and cannot be subscribed to.

ERROR_EVT_UNRESOLVED_PARAMETER_INSERT = 15030

The description string for the parameter reference (%1) could not be found.

ERROR_EVT_UNRESOLVED_VALUE_INSERT = 15029

The substitution string for the insert index (%1) could not be found.

ERROR_EVT_VERSION_TOO_NEW = 15035

The resource is too new to be compatible.

ERROR_EVT_VERSION_TOO_OLD = 15034

The resource is too old to be compatible.

ERROR_EXCEPTION_IN_SERVICE = 1064

An exception occurred in the service when handling the control request.

ERROR_EXCL_SEM_ALREADY_OWNED = 101

The exclusive semaphore is owned by another process.

ERROR_EXE_CANNOT_MODIFY_SIGNED_BINARY = 217

The image file %1 is signed, unable to modify.

ERROR_EXE_CANNOT_MODIFY_STRONG_SIGNED_BINARY = 218

The image file %1 is strong signed, unable to modify.

ERROR_EXE_MACHINE_TYPE_MISMATCH = 216

This version of %1 is not compatible with the version of Windows you're running. Check your computer's system information to see whether you need an x86 (32-bit) or x64 (64-bit) version of the program, and then contact the software publisher.

ERROR_EXE_MARKED_INVALID = 192

The operating system cannot run %1.

ERROR_EXTENDED_ERROR = 1208

An extended error has occurred.

ERROR_EXTRANEIOUS_INFORMATION = 677

{Too Much Information} The specified ACL contained more information than was expected.

ERROR_FAILED_DRIVER_ENTRY = 647

The driver was not loaded because it failed its initialization call.

ERROR_FAILED_SERVICE_CONTROLLER_CONNECT = 1063

The service process could not connect to the service controller.

ERROR_FAIL_I24 = 83

Fail on INT 24.

ERROR_FAIL_NOACTION_REBOOT = 350

No action was taken because a system reboot is required.

ERROR_FAIL_REBOOT_INITIATED = 3018

The requested operation failed. A system reboot has been initiated to roll back changes made.

ERROR_FAIL_REBOOT_REQUIRED = 3017

The requested operation failed. A system reboot is required to roll back changes made.

ERROR_FAIL_RESTART = 352

The restart operation failed.

ERROR_FAIL_SHUTDOWN = 351

The shutdown operation failed.

ERROR_FATAL_APP_EXIT = 713

{Fatal Application Exit} %hs.

ERROR_FILEMARK_DETECTED = 1101

A tape access reached a filemark.

ERROR_FILENAME_EXCED_RANGE = 206

The file name or extension is too long.

ERROR_FILE_CHECKED_OUT = 220

This file is checked out or locked for editing by another user.

ERROR_FILE_CORRUPT = 1392

The file or directory is corrupted and unreadable.

ERROR_FILE_ENCRYPTED = 6002

The specified file is encrypted and the user does not have the ability to decrypt it.

ERROR_FILE_EXISTS = 80

The file exists.

ERROR_FILE_IDENTITY_NOT_PERSISTENT = 6823

The file cannot be opened in a transaction because its identity depends on the outcome of an unresolved transaction.

ERROR_FILE_INVALID = 1006

The volume for a file has been externally altered so that the opened file is no longer valid.

ERROR_FILE_NOT_ENCRYPTED = 6007

The specified file is not encrypted.

ERROR_FILE_NOT_FOUND = 2

The system cannot find the file specified.

ERROR_FILE_OFFLINE = 4350

The remote storage service was not able to recall the file.

ERROR_FILE_READ_ONLY = 6009

The specified file is read-only.

ERROR_FILE_SYSTEM_LIMITATION = 665

The requested operation could not be completed due to a file system limitation.

ERROR_FILE_TOO_LARGE = 223

The file size exceeds the limit allowed and cannot be saved.

ERROR_FIRMWARE_UPDATED = 728

Windows has detected that the system firmware (BIOS) was updated (previous firmware date = %2, current firmware date %3).

ERROR_FLOATED_SECTION = 6846

I/O was attempted on a section object that has been floated as a result of a transaction ending. There is no valid data.

ERROR_FLOAT_MULTIPLE_FAULTS = 630

{EXCEPTION} Multiple floating point faults.

ERROR_FLOAT_MULTIPLE_TRAPS = 631

{EXCEPTION} Multiple floating point traps.

ERROR_FLOPPY_BAD_REGISTERS = 1125

The floppy disk controller returned inconsistent results in its registers.

ERROR_FLOPPY_ID_MARK_NOT_FOUND = 1122

No ID address mark was found on the floppy disk.

ERROR_FLOPPY_UNKNOWN_ERROR = 1124

The floppy disk controller reported an error that is not recognized by the floppy disk driver.

ERROR_FLOPPY_VOLUME = 584

The paging file cannot be created on a floppy disk.

ERROR_FLOPPY_WRONG_CYLINDER = 1123

Mismatch between the floppy disk sector ID field and the floppy disk controller track address.

ERROR_FORMS_AUTH_REQUIRED = 224

Access denied. Before opening files in this location, you must first browse to the website and select the option to sign in automatically.

ERROR_FOUND_OUT_OF_SCOPE = 601

The attempt to find the object found an object matching by ID on the volume but it is out of the scope of the handle used for the operation.

ERROR_FSFILTER_OP_COMPLETED_SUCCESSFULLY = 762

A file system or file system filter driver has successfully completed an FsFilter operation.

ERROR_FS_DRIVER_REQUIRED = 588

A volume has been accessed for which a file system driver is required that has not yet been loaded.

ERROR_FT_READ_RECOVERY_FROM_BACKUP = 704

{Redundant Read} To satisfy a read request, the NT fault-tolerant file system successfully read the requested data from a redundant copy. This was done because the file system encountered a failure on a member of the fault-tolerant volume, but it was unable to reassign the failing area of the device.

ERROR_FT_WRITE_RECOVERY = 705

{Redundant Write} To satisfy a write request, the Windows NT operating system fault-tolerant file system successfully wrote a redundant copy of the information. This was done because the file system encountered a failure on a member of the fault-tolerant volume, but it was not able to reassign the failing area of the device.

ERROR_FULLSCREEN_MODE = 1007

The requested operation cannot be performed in full-screen mode.

ERROR_FULL_BACKUP = 4004

The backup failed. Check the directory to which you are backing the database.

ERROR_FUNCTION_FAILED = 1627

Function failed during execution.

ERROR_FUNCTION_NOT_CALLED = 1626

Function could not be executed.

ERROR_GENERIC_NOT_MAPPED = 1360

Generic access types were contained in an access mask that should already be mapped to nongeneric types.

ERROR_GEN_FAILURE = 31

A device attached to the system is not functioning.

ERROR_GLOBAL_ONLY_HOOK = 1429

This hook procedure can only be set globally.

ERROR_GRACEFUL_DISCONNECT = 1226

The network connection was gracefully closed.

ERROR_GROUP_EXISTS = 1318

The specified group already exists.

ERROR_GROUP_NOT_AVAILABLE = 5012

The cluster group is not available for any new requests.

ERROR_GROUP_NOT_FOUND = 5013

The cluster group could not be found.

ERROR_GROUP_NOT_ONLINE = 5014

The operation could not be completed because the cluster group is not online.

ERROR_GUID_SUBSTITUTION_MADE = 680

{GUID Substitution} During the translation of a GUID to a Windows SID, no administratively defined GUID prefix was found. A substitute prefix was used, which will not compromise system security. However, this might provide more restrictive access than intended.

ERROR_HANDLES_CLOSED = 676

{Handles Closed} Handles to objects have been automatically closed because of the requested operation.

ERROR_HANDLE_DISK_FULL = 39

The disk is full.

ERROR_HANDLE_EOF = 38

Reached the end of the file.

ERROR_HANDLE_NO_LONGER_VALID = 6815

The handle has been invalidated by a transaction. The most likely cause is the presence of memory mapping on a file, or an open handle when the transaction ended or rolled back to savepoint.

ERROR_HIBERNATED = 726

The system was put into hibernation.

ERROR_HIBERNATION_FAILURE = 656

The system has failed to hibernate (the error code is %hs). Hibernation will be disabled until the system is restarted.

ERROR_HOOK_NEEDS_HMOD = 1428

Cannot set nonlocal hook without a module handle.

ERROR_HOOK_NOT_INSTALLED = 1431

The hook procedure is not installed.

ERROR_HOOK_TYPE_NOT_ALLOWED = 1458

Hook type not allowed.

ERROR_HOST_DOWN = 1256

The remote system is not available. For information about network troubleshooting, see Windows Help.

ERROR_HOST_NODE_NOT_AVAILABLE = 5005

A cluster node is not available for this operation.

ERROR_HOST_NODE_NOT_GROUP_OWNER = 5016

The operation failed because either the specified cluster node is not the owner of the group, or the node is not a possible owner of the group.

ERROR_HOST_NODE_NOT_RESOURCE_OWNER = 5015

The operation failed because either the specified cluster node is not the owner of the resource, or the node is not a possible owner of the resource.

ERROR_HOST_UNREACHABLE = 1232

The network location cannot be reached. For information about network troubleshooting, see Windows Help.

ERROR_HOTKEY_ALREADY_REGISTERED = 1409

Hot key is already registered.

ERROR_HOTKEY_NOT_REGISTERED = 1419

Hot key is not registered.

ERROR_HWNDS_HAVE_DIFF_PARENT = 1441

All handles to windows in a multiple-window position structure must have the same parent.

ERROR_ICM_NOT_ENABLED = 2018

Image Color Management is not enabled.

ERROR_IEPORT_FULL = 4341

Cannot use the IEport because it is not empty.

ERROR_ILLEGAL_CHARACTER = 582

An illegal character was encountered. For a multibyte character set, this includes a lead byte without a succeeding trail byte. For the Unicode character set, this includes the characters 0xFFFF and 0xFFFE.

ERROR_ILLEGAL_DLL_RELOCATION = 623

{Illegal System DLL Relocation} The system DLL %hs was relocated in memory. The application will not run properly. The relocation occurred because the DLL %hs occupied an address range reserved for Windows system DLLs. The vendor supplying the DLL should be contacted for a new DLL.

ERROR_ILLEGAL_ELEMENT_ADDRESS = 1162

The indicated element does not exist.

ERROR_ILLEGAL_FLOAT_CONTEXT = 579

{EXCEPTION} A real-mode application issued a floating-point instruction, and floating-point hardware is not present.

ERROR_ILL_FORMED_PASSWORD = 1324

Unable to update the password. The value provided for the new password contains values that are not allowed in passwords.

ERROR_IMAGE_MACHINE_TYPE_MISMATCH = 706

{Machine Type Mismatch} The image file %hs is valid, but is for a machine type other than the current machine. Select OK to continue, or CANCEL to fail the DLL load.

ERROR_IMAGE_MACHINE_TYPE_MISMATCH_EXE = 720

{Machine Type Mismatch} The image file %hs is valid, but is for a machine type other than the current machine.

ERROR_IMAGE_NOT_AT_BASE = 700

{Image Relocated} An image file could not be mapped at the address specified in the image file. Local fixes must be performed on this image.

ERROR_IMPLEMENTATION_LIMIT = 1292

An operation attempted to exceed an implementation-defined limit.

ERROR_INCOMPATIBLE_SERVICE_PRIVILEGE = 1297

A privilege that the service requires to function properly does not exist in the service account configuration. The Services Microsoft Management Console (MMC) snap-in (Services.msc) and the Local Security Settings MMC snap-in (Secpol.msc) can be used to view the service configuration and the account configuration.

ERROR_INCOMPATIBLE_SERVICE_SID_TYPE = 1290

The service start failed because one or more services in the same process have an incompatible service SID type setting. A service with a restricted service SID type can only coexist in the same process with other services with a restricted SID type.

ERROR_INCORRECT_ADDRESS = 1241

The network address could not be used for the operation requested.

ERROR_INCORRECT_SIZE = 1462

Incorrect size argument.

ERROR_INC_BACKUP = 4003

The backup failed. Was a full backup done before?

ERROR_INDEX_ABSENT = 1611

Component qualifier not present.

ERROR_INDIGENOUS_TYPE = 4338

The media type cannot be removed from this library because at least one drive in the library reports it can support this media type.

ERROR_INDOUBT_TRANSACTIONS_EXIST = 6827

The operation would leave a transactional resource manager in an inconsistent state and is, therefore, not allowed.

ERROR_INFLOOP_IN_RELOC_CHAIN = 202

The operating system cannot run %1.

ERROR_INSTALL_ALREADY_RUNNING = 1618

Another installation is already in progress. Complete that installation before proceeding with this install.

ERROR_INSTALL_FAILURE = 1603

Fatal error during installation.

ERROR_INSTALL_LANGUAGE_UNSUPPORTED = 1623

The language of this installation package is not supported by your system.

ERROR_INSTALL_LOG_FAILURE = 1622

Error opening installation log file. Verify that the specified log file location exists and that you can write to it.

ERROR_INSTALL_NOTUSED = 1634

Component not used on this computer.

ERROR_INSTALL_PACKAGE_INVALID = 1620

This installation package could not be opened. Contact the application vendor to verify that this is a valid Windows Installer package.

ERROR_INSTALL_PACKAGE_OPEN_FAILED = 1619

This installation package could not be opened. Verify that the package exists and that you can access it, or contact the application vendor to verify that this is a valid Windows Installer package.

ERROR_INSTALL_PACKAGE_REJECTED = 1625

This installation is forbidden by system policy. Contact your system administrator.

ERROR_INSTALL_PACKAGE_VERSION = 1613

This installation package cannot be installed by the Windows Installer service. You must install a Windows service pack that contains a newer version of the Windows Installer service.

ERROR_INSTALL_PLATFORM_UNSUPPORTED = 1633

This installation package is not supported by this processor type. Contact your product vendor.

ERROR_INSTALL_REMOTE_DISALLOWED = 1640

Only administrators have permission to add, remove, or configure server software during a Terminal Services remote session. If you want to install or configure software on the server, contact your network administrator.

ERROR_INSTALL_REMOTE_PROHIBITED = 1645

The Windows Installer does not permit installation from a Remote Desktop Connection.

ERROR_INSTALL_SERVICE_FAILURE = 1601

The Windows Installer service could not be accessed. This can occur if the Windows Installer is not correctly installed. Contact your support personnel for assistance.

ERROR_INSTALL_SERVICE_SAFEBOOT = 1652

The Windows Installer service is not accessible in Safe Mode. Try again when your computer is not in Safe Mode or you can use System Restore to return your machine to a previous good state.

ERROR_INSTALL_SOURCE_ABSENT = 1612

The installation source for this product is not available. Verify that the source exists and that you can access it.

ERROR_INSTALL_SUSPEND = 1604

Installation suspended, incomplete.

ERROR_INSTALL_TEMP_UNWRITABLE = 1632

The Temp folder is on a drive that is full or is inaccessible. Free up space on the drive or verify that you have write permission on the Temp folder.

ERROR_INSTALL_TRANSFORM_FAILURE = 1624

Error applying transforms. Verify that the specified transform paths are valid.

ERROR_INSTALL_TRANSFORM_REJECTED = 1644

One or more customizations are not permitted by a software restriction policy.

ERROR_INSTALL_UI_FAILURE = 1621

There was an error starting the Windows Installer service user interface. Contact your support personnel.

ERROR_INSTALL_USEREXIT = 1602

User canceled installation.

ERROR_INSTRUCTION_MISALIGNMENT = 549

An attempt was made to execute an instruction at an unaligned address, and the host system does not support unaligned instruction references.

ERROR_INSUFFICIENT_BUFFER = 122

The data area passed to a system call is too small.

ERROR_INSUFFICIENT_LOGON_INFO = 608

There is insufficient account information to log you on.

ERROR_INSUFFICIENT_POWER = 639

There is not enough power to complete the requested operation.

ERROR_INSUFFICIENT_RESOURCE_FOR_SPECIFIED_SHARED_SECTION_SIZE = 781

The Desktop heap encountered an error while allocating session memory. There is more information in the system event log.

ERROR_INTERNAL_DB_CORRUPTION = 1358

Unable to complete the requested operation because of either a catastrophic media failure or a data structure corruption on the disk.

ERROR_INTERNAL_DB_ERROR = 1383

The local security authority database contains an internal inconsistency.

ERROR_INTERNAL_ERROR = 1359

An internal error occurred.

ERROR_INTERRUPT_STILL_CONNECTED = 764

The specified interrupt vector is still connected.

ERROR_INTERRUPT_VECTOR_ALREADY_CONNECTED = 763

The specified interrupt vector was already connected.

ERROR_INVALID_ACCEL_HANDLE = 1403

Invalid accelerator table handle.

ERROR_INVALID_ACCESS = 12

The access code is invalid.

ERROR_INVALID_ACCOUNT_NAME = 1315

The name provided is not a properly formed account name.

ERROR_INVALID_ACL = 1336

The ACL structure is invalid.

ERROR_INVALID_ADDRESS = 487

Attempt to access invalid address.

ERROR_INVALID_AT_INTERRUPT_TIME = 104

Cannot request exclusive semaphores at interrupt time.

ERROR_INVALID_BLOCK = 9

The storage control block address is invalid.

ERROR_INVALID_BLOCK_LENGTH = 1106

When accessing a new tape of a multivolume partition, the current block size is incorrect.

ERROR_INVALID_CATEGORY = 117

The Input Output Control (IOCTL) call made by the application program is not correct.

ERROR_INVALID_CLEAER = 4310

The media identifier does not represent a valid cleaner.

ERROR_INVALID_CLUSTER_IPV6_ADDRESS = 5911

The address is not valid for an IPv6 Address resource. A global IPv6 address is required, and it must match a cluster network. Compatibility addresses are not permitted.

ERROR_INVALID_CMM = 2010

The specified color management module is invalid.

ERROR_INVALID_COLORINDEX = 2022

The specified named color index is not present in the profile.

ERROR_INVALID_COLORSPACE = 2017

The specified color space is invalid.

ERROR_INVALID_COMBOBOX_MESSAGE = 1422

Invalid message for a combo box because it does not have an edit control.

ERROR_INVALID_COMMAND_LINE = 1639

Invalid command-line argument. Consult the Windows Installer SDK for detailed command line help.

ERROR_INVALID_COMPUTERNAME = 1210

The format of the specified computer name is invalid.

ERROR_INVALID_CRUNTIME_PARAMETER = 1288

The parameter passed to aCruntime function is incorrect.

ERROR_INVALID_CURSOR_HANDLE = 1402

Invalid cursor handle.

ERROR_INVALID_DATA = 13

The data is invalid.

ERROR_INVALID_DATATYPE = 1804

The specified data type is invalid.

ERROR_INVALID_DEVICE_OBJECT_PARAMETER = 650

The device object parameter is either not a valid device object or is not attached to the volume specified by the file name.

ERROR_INVALID_DLL = 1154

One of the library files needed to run this application is damaged.

ERROR_INVALID_DOMAINNAME = 1212

The format of the specified domain name is invalid.

ERROR_INVALID_DOMAIN_ROLE = 1354

This operation is only allowed for the PDC of the domain.

ERROR_INVALID_DOMAIN_STATE = 1353

The domain was in the wrong state to perform the security operation.

ERROR_INVALID_DRIVE = 15

The system cannot find the drive specified.

ERROR_INVALID_DRIVE_OBJECT = 4321

The drive identifier does not represent a valid drive.

ERROR_INVALID_DWP_HANDLE = 1405

Invalid handle to a multiple-window position structure.

ERROR_INVALID_EA_HANDLE = 278

The specified extended attribute handle is invalid.

ERROR_INVALID_EA_NAME = 254

The specified extended attribute name was invalid.

ERROR_INVALID_EDIT_HEIGHT = 1424

Height must be less than 256.

ERROR_INVALID_ENVIRONMENT = 1805

The environment specified is invalid.

ERROR_INVALID_EVENTNAME = 1211

The format of the specified event name is invalid.

ERROR_INVALID_EVENT_COUNT = 151

The number of specified semaphore events for DosMuxSemWait is not correct.

ERROR_INVALID_EXE_SIGNATURE = 191

Cannot run %1 in Win32 mode.

ERROR_INVALID_FIELD = 1616

Record field does not exist.

ERROR_INVALID_FILTER_PROC = 1427

Invalid hook procedure.

ERROR_INVALID_FLAGS = 1004

Invalid flags.

ERROR_INVALID_FLAG_NUMBER = 186

The flag passed is not correct.

ERROR_INVALID_FORM_NAME = 1902

The specified form name is invalid.

ERROR_INVALID_FORM_SIZE = 1903

The specified form size is invalid.

ERROR_INVALID_FUNCTION = 1

Incorrect function.

ERROR_INVALID_GROUPNAME = 1209

The format of the specified group name is invalid.

ERROR_INVALID_GROUP_ATTRIBUTES = 1345

The specified attributes are invalid, or incompatible with the attributes for the group as a whole.

ERROR_INVALID_GW_COMMAND = 1443

Invalid GW_* command.

ERROR_INVALID_HANDLE = 6

The handle is invalid.

ERROR_INVALID_HANDLE_STATE = 1609

Handle is in an invalid state.

ERROR_INVALID_HOOK_FILTER = 1426

Invalid hook procedure type.

ERROR_INVALID_HOOK_HANDLE = 1404

Invalid hook handle.

ERROR_INVALID_HW_PROFILE = 619

The specified hardware profile configuration is invalid.

ERROR_INVALID_ICON_HANDLE = 1414

Invalid icon handle.

ERROR_INVALID_ID_AUTHORITY = 1343

The value provided was an invalid value for an identifier authority.

ERROR_INVALID_IMAGE_HASH = 577

The hash for the image cannot be found in the system catalogs. The image is likely corrupt or the victim of tampering.

ERROR_INVALID_IMPORT_OF_NON_DLL = 1276

A DLL referenced a module that was neither a DLL nor the process's executable image.

ERROR_INVALID_INDEX = 1413

Invalid index.

ERROR_INVALID_KEYBOARD_HANDLE = 1457

Invalid keyboard layout handle.

ERROR_INVALID_LABEL = 1299

Indicates a particular SID cannot be assigned as the label of an object.

ERROR_INVALID_LB_MESSAGE = 1432

Invalid message for single-selection list box.

ERROR_INVALID_LDT_DESCRIPTOR = 564

Indicates that the user supplied an invalid descriptor when trying to set up LDT descriptors.

ERROR_INVALID_LDT_OFFSET = 563

Indicates that the starting value for the LDT information was not an integral multiple of the selector size.

ERROR_INVALID_LDT_SIZE = 561

Indicates that an attempt was made to grow a local domain table (LDT) by setting its size, or that the size was not an even number of selectors.

ERROR_INVALID_LEVEL = 124

The system call level is not correct.

ERROR_INVALID_LIBRARY = 4301

The library identifier does not represent a valid library.

ERROR_INVALID_LIST_FORMAT = 153

The DosMuxSemWait list is not correct.

ERROR_INVALID_LOGON_HOURS = 1328

Logon failure: Account logon time restriction violation.

ERROR_INVALID_LOGON_TYPE = 1367

A logon request contained an invalid logon type value.

ERROR_INVALID_MEDIA = 4300

The media identifier does not represent a valid medium.

ERROR_INVALID_MEDIA_POOL = 4302

The media pool identifier does not represent a valid media pool.

ERROR_INVALID_MEMBER = 1388

A new member could not be added to a local group because the member has the wrong account type.

ERROR_INVALID_MENU_HANDLE = 1401

Invalid menu handle.

ERROR_INVALID_MESSAGE = 1002

The window cannot act on the sent message.

ERROR_INVALID_MESSAGEDEST = 1218

The format of the specified message destination is invalid.

ERROR_INVALID_MESSAGE_NAME = 1217

The format of the specified message name is invalid.

ERROR_INVALID_MINALLOC_SIZE = 195

The operating system cannot run %1.

ERROR_INVALID_MODULETYPE = 190

The operating system cannot run %1.

ERROR_INVALID_MONITOR_HANDLE = 1461

Invalid monitor handle.

ERROR_INVALID_MSGBOX_STYLE = 1438

Invalid message box style.

ERROR_INVALID_NAME = 123

The file name, directory name, or volume label syntax is incorrect.

ERROR_INVALID_NETNAME = 1214

The format of the specified network name is invalid.

ERROR_INVALID_OPERATION = 4317

The operation identifier is not valid.

ERROR_INVALID_OPERATION_ON_QUORUM = 5068

This operation cannot be performed on the cluster resource because it is the quorum resource. This quorum resource cannot be brought offline and its possible owners list cannot be modified.

ERROR_INVALID_OPLOCK_PROTOCOL = 301

An invalid oplock acknowledgment was received by the system.

ERROR_INVALID_ORDINAL = 182

The operating system cannot run %1.

ERROR_INVALID_OWNER = 1307

This SID cannot be assigned as the owner of this object.

ERROR_INVALID_PARAMETER = 87

The parameter is incorrect.

ERROR_INVALID_PASSWORD = 86

The specified network password is not correct.

ERROR_INVALID_PASSWORDNAME = 1216

The format of the specified password is invalid.

ERROR_INVALID_PATCH_XML = 1650

The XML update data is invalid.

ERROR_INVALID_PIXEL_FORMAT = 2000

The pixel format is invalid.

ERROR_INVALID_PLUGPLAY_DEVICE_PATH = 620

The specified Plug and Play registry device path is invalid.

ERROR_INVALID_PORT_ATTRIBUTES = 545

Invalid object attributes specified to NtCreatePort or invalid port attributes specified to NtConnectPort.

ERROR_INVALID_PRIMARY_GROUP = 1308

This SID cannot be assigned as the primary group of an object.

ERROR_INVALID_PRINTER_COMMAND = 1803

The printer command is invalid.

ERROR_INVALID_PRINTER_NAME = 1801

The printer name is invalid.

ERROR_INVALID_PRINTER_STATE = 1906

The state of the printer is invalid.

ERROR_INVALID_PRINT_MONITOR = 3007

The specified print monitor does not have the required functions.

ERROR_INVALID_PRIORITY = 1800

The specified priority is invalid.

ERROR_INVALID_PROFILE = 2011

The specified color profile is invalid.

ERROR_INVALID_QUOTA_LOWER = 547

An attempt was made to lower a quota limit below the current usage.

ERROR_INVALID_REPARSE_DATA = 4392

The data present in the reparse point buffer is invalid.

ERROR_INVALID_SCROLLBAR_RANGE = 1448

Scroll bar range cannot be greater than MAXLONG.

ERROR_INVALID_SECURITY_DESCR = 1338

The security descriptor structure is invalid.

ERROR_INVALID_SEGDPL = 198

The operating system cannot run %1.

ERROR_INVALID_SEGMENT_NUMBER = 180

The system detected a segment number that was not correct.

ERROR_INVALID_SEPARATOR_FILE = 1799

The specified separator file is invalid.

ERROR_INVALID_SERVER_STATE = 1352

The SAM or local security authority (LSA) server was in the wrong state to perform the security operation.

ERROR_INVALID_SERVICENAME = 1213

The format of the specified service name is invalid.

ERROR_INVALID_SERVICE_ACCOUNT = 1057

The account name is invalid or does not exist, or the password is invalid for the account name specified.

ERROR_INVALID_SERVICE_CONTROL = 1052

The requested control is not valid for this service.

ERROR_INVALID_SERVICE_LOCK = 1071

The specified service database lock is invalid.

ERROR_INVALID_SHARENAME = 1215

The format of the specified share name is invalid.

ERROR_INVALID_SHOWWIN_COMMAND = 1449

Cannot show or remove the window in the way specified.

ERROR_INVALID_SID = 1337

The SID structure is invalid.

ERROR_INVALID_SIGNAL_NUMBER = 209

The signal being posted is not correct.

ERROR_INVALID_SPI_VALUE = 1439

Invalid system-wide (SPI_*) parameter.

ERROR_INVALID_STACKSEG = 189

The operating system cannot run %1.

ERROR_INVALID_STARTING_CODESEG = 188

The operating system cannot run %1.

ERROR_INVALID_STATE = 5023

The group or resource is not in the correct state to perform the requested operation.

ERROR_INVALID_SUB_AUTHORITY = 1335

The sub-authority part of an SID is invalid for this particular use.

ERROR_INVALID_TABLE = 1628

Invalid or unknown table specified.

ERROR_INVALID_TARGET_HANDLE = 114

The target internal file identifier is incorrect.

ERROR_INVALID_TASK_INDEX = 1551

The specified task index is invalid.

ERROR_INVALID_TASK_NAME = 1550

The specified task name is invalid.

ERROR_INVALID_THREAD_ID = 1444

Invalid thread identifier.

ERROR_INVALID_TIME = 1901

The specified time is invalid.

ERROR_INVALID_TRANSACTION = 6700

The transaction handle associated with this operation is not valid.

ERROR_INVALID_TRANSFORM = 2020

The specified color transform is invalid.

ERROR_INVALID_UNWIND_TARGET = 544

An invalid unwind target was encountered during an unwind operation.

ERROR_INVALID_USER_BUFFER = 1784

The supplied user buffer is not valid for the requested operation.

ERROR_INVALID_VARIANT = 604

The supplied variant structure contains invalid data.

ERROR_INVALID_VERIFY_SWITCH = 118

The verify-on-write switch parameter value is not correct.

ERROR_INVALID_WINDOW_HANDLE = 1400

Invalid window handle.

ERROR_INVALID_WINDOW_STYLE = 2002

The window style or class attribute is invalid for this operation.

ERROR_INVALID_WORKSTATION = 1329

Logon failure: User not allowed to log on to this computer.

ERROR_IOPL_NOT_ENABLED = 197

The operating system is not presently configured to run this application.

ERROR_IO_DEVICE = 1117

The request could not be performed because of an I/O device error.

ERROR_IO_INCOMPLETE = 996

Overlapped I/O event is not in a signaled state.

ERROR_IO_PENDING = 997

Overlapped I/O operation is in progress.

ERROR_IO_PRIVILEGE_FAILED = 571

{Privilege Failed} The I/O permissions for the process could not be changed.

ERROR_IO_REISSUE_AS_CACHED = 3950

Reissue the given operation as a cached I/O operation.

ERROR_IPSEC_DEFAULT_MM_AUTH_NOT_FOUND = 13014

The specified default main mode authentication list was not found.

ERROR_IPSEC_DEFAULT_MM_POLICY_NOT_FOUND = 13013

The specified default main mode policy was not found.

ERROR_IPSEC_DEFAULT_QM_POLICY_NOT_FOUND = 13015

The specified default quick mode policy was not found.

ERROR_IPSEC_IKE_ADD_UPDATE_KEY_FAILED = 13860

Failed to add an SA to the IPsec driver. The most common cause for this is if the IKE negotiation took too long to complete. If the problem persists, reduce the load on the faulting machine.

ERROR_IPSEC_IKE_ATTRIB_FAIL = 13802

The IKE security attributes are unacceptable.

ERROR_IPSEC_IKE_AUTH_FAIL = 13801

The IKE authentication credentials are unacceptable.

ERROR_IPSEC_IKE_BENIGN_REINIT = 13878

A temporary state was created to perform reinitialization. This is not a real failure.

ERROR_IPSEC_IKE_CERT_CHAIN_POLICY_MISMATCH = 13887

Certificate does not chain to a trusted root in IPsec policy.

ERROR_IPSEC_IKE_CGA_AUTH_FAILED = 13892

Could not verify the binding between the color graphics adapter (CGA) address and the certificate.

ERROR_IPSEC_IKE_CRL_FAILED = 13817

The certificate revocation check failed.

ERROR_IPSEC_IKE_DECRYPT = 13867

Error decrypting payload.

ERROR_IPSEC_IKE_DH_FAIL = 13822

There was a failure in the Diffie-Hellman computation.

ERROR_IPSEC_IKE_DH_FAILURE = 13864

Diffie-Hellman failure.

ERROR_IPSEC_IKE_DOS_COOKIE_SENT = 13890

Sent DOS cookie notify to initiator.

ERROR_IPSEC_IKE_DROP_NO_RESPONSE = 13813

There was no response from a peer.

ERROR_IPSEC_IKE_ENCRYPT = 13866

Error encrypting payload.

ERROR_IPSEC_IKE_ERROR = 13816

An unknown error occurred.

ERROR_IPSEC_IKE_FAILQUERYSSP = 13854

Failed to query the Kerberos package to obtain the max token size.

ERROR_IPSEC_IKE_FAILSSPINIT = 13853

Failed to obtain the security function table dispatch address from the SSPI.

ERROR_IPSEC_IKE_GENERAL_PROCESSING_ERROR = 13804

General processing error.

ERROR_IPSEC_IKE_GETSPIFAIL = 13857

Failed to obtain a new service provider interface (SPI) for the inbound SA from the IPsec driver. The most common cause for this is that the driver does not have the correct filter. Check your policy to verify the filters.

ERROR_IPSEC_IKE_INVALID_AUTH_ALG = 13874

Invalid authentication algorithm.

ERROR_IPSEC_IKE_INVALID_CERT_KEYLEN = 13881

Key length in the certificate is too small for configured security requirements.

ERROR_IPSEC_IKE_INVALID_CERT_TYPE = 13819

Invalid certificate type.

ERROR_IPSEC_IKE_INVALID_COOKIE = 13846

Invalid cookie received.

ERROR_IPSEC_IKE_INVALID_ENCRYPT_ALG = 13873

Invalid encryption algorithm.

ERROR_IPSEC_IKE_INVALID_FILTER = 13858

Given filter is invalid.

ERROR_IPSEC_IKE_INVALID_GROUP = 13865

Invalid Diffie-Hellman group.

ERROR_IPSEC_IKE_INVALID_HASH = 13870

Hash verification failed.

ERROR_IPSEC_IKE_INVALID_HASH_ALG = 13871

Invalid hash algorithm.

ERROR_IPSEC_IKE_INVALID_HASH_SIZE = 13872

Invalid hash size.

ERROR_IPSEC_IKE_INVALID_HEADER = 13824

Invalid header.

ERROR_IPSEC_IKE_INVALID_KEY_USAGE = 13818

Invalid certificate key usage.

ERROR_IPSEC_IKE_INVALID_MM_FOR_QM = 13894

The parameters of the main mode are invalid for this quick mode.

ERROR_IPSEC_IKE_INVALID_PAYLOAD = 13843

Invalid payload received.

ERROR_IPSEC_IKE_INVALID_POLICY = 13861

Invalid policy.

ERROR_IPSEC_IKE_INVALID_RESPONDER_LIFETIME_NOTIFY = 13879

The lifetime value received in the Responder Lifetime Notify is below the Windows 2000 configured minimum value. Fix the policy on the peer machine.

ERROR_IPSEC_IKE_INVALID_SIG = 13875

Invalid certificate signature.

ERROR_IPSEC_IKE_INVALID_SIGNATURE = 13826

Failed to verify signature.

ERROR_IPSEC_IKE_INVALID_SITUATION = 13863

Invalid situation.

ERROR_IPSEC_IKE_INVALID_UMATTS = 13889

Received invalid AuthIP user mode attributes.

ERROR_IPSEC_IKE_KERBEROS_ERROR = 13827

Failed to authenticate using Kerberos.

ERROR_IPSEC_IKE_LOAD_FAILED = 13876

Load failed.

ERROR_IPSEC_IKE_LOAD_SOFT_SA = 13844

Soft SA loaded.

ERROR_IPSEC_IKE_MM_ACQUIRE_DROP = 13809

The negotiation request sat in the queue too long.

ERROR_IPSEC_IKE_MM_DELAY_DROP = 13814

The negotiation took too long.

ERROR_IPSEC_IKE_MM_EXPIRED = 13885

Main mode SA lifetime expired or the peer sent a main mode delete.

ERROR_IPSEC_IKE_MM_LIMIT = 13882

Maximum number of established MM SAs to peer exceeded.

ERROR_IPSEC_IKE_NEGOTIATION_DISABLED = 13883

The IKE received a policy that disables negotiation.

ERROR_IPSEC_IKE_NEGOTIATION_PENDING = 13803

The IKE negotiation is in progress.

ERROR_IPSEC_IKE_NEG_STATUS_BEGIN = 13800

Starts the list of frequencies of various IKE Win32 error codes encountered during negotiations.

ERROR_IPSEC_IKE_NEG_STATUS_END = 13897

Ends the list of frequencies of various IKE Win32 error codes encountered during negotiations.

ERROR_IPSEC_IKE_NOTCBPRIV = 13851

Failed to enable trusted computer base (TCB) privilege.

ERROR_IPSEC_IKE_NO_CERT = 13806

The IKE failed to find a valid machine certificate. Contact your network security administrator about installing a valid certificate in the appropriate certificate store.

ERROR_IPSEC_IKE_NO_MM_POLICY = 13850

There is no available main mode IKE policy.

ERROR_IPSEC_IKE_NO_PEER_CERT = 13847

Peer failed to send valid machine certificate.

ERROR_IPSEC_IKE_NO_POLICY = 13825

No policy configured.

ERROR_IPSEC_IKE_NO_PRIVATE_KEY = 13820

The IKE negotiation failed because the machine certificate used does not have a private key. IPsec certificates require a private key. Contact your network security administrator about a certificate that has a private key.

ERROR_IPSEC_IKE_NO_PUBLIC_KEY = 13828

The peer's certificate did not have a public key.

ERROR_IPSEC_IKE_OUT_OF_MEMORY = 13859

Memory allocation failed.

ERROR_IPSEC_IKE_PEER_CRL_FAILED = 13848

Certification revocation check of peer's certificate failed.

ERROR_IPSEC_IKE_PEER_MM_ASSUMED_INVALID = 13886

Main mode SA assumed to be invalid because peer stopped responding.

ERROR_IPSEC_IKE_POLICY_CHANGE = 13849

New policy invalidated SAs formed with the old policy.

ERROR_IPSEC_IKE_POLICY_MATCH = 13868

Policy match error.

ERROR_IPSEC_IKE_PROCESS_ERR = 13829

Error processing the error payload.

ERROR_IPSEC_IKE_PROCESS_ERR_CERT = 13835

Error processing the certification payload.

ERROR_IPSEC_IKE_PROCESS_ERR_CERT_REQ = 13836

Error processing the certificate request payload.

ERROR_IPSEC_IKE_PROCESS_ERR_DELETE = 13841

Error processing the delete payload.

ERROR_IPSEC_IKE_PROCESS_ERR_HASH = 13837

Error processing the hash payload.

ERROR_IPSEC_IKE_PROCESS_ERR_ID = 13834

Error processing the ID payload.

ERROR_IPSEC_IKE_PROCESS_ERR_KEY = 13833

Error processing the key exchange payload.

ERROR_IPSEC_IKE_PROCESS_ERR_NATOA = 13893

Error processing the NatOA payload.

ERROR_IPSEC_IKE_PROCESS_ERR_NONCE = 13839

Error processing the nonce payload.

ERROR_IPSEC_IKE_PROCESS_ERR_NOTIFY = 13840

Error processing the notify payload.

ERROR_IPSEC_IKE_PROCESS_ERR_PROP = 13831

Error processing the proposal payload.

ERROR_IPSEC_IKE_PROCESS_ERR_SA = 13830

Error processing the SA payload.

ERROR_IPSEC_IKE_PROCESS_ERR_SIG = 13838

Error processing the signature payload.

ERROR_IPSEC_IKE_PROCESS_ERR_TRANS = 13832

Error processing the transform payload.

ERROR_IPSEC_IKE_PROCESS_ERR_VENDOR = 13842

Error processing the VendorId payload.

ERROR_IPSEC_IKE_QM_ACQUIRE_DROP = 13810

The negotiation request sat in the queue too long.

ERROR_IPSEC_IKE_QM_DELAY_DROP = 13815

The negotiation took too long.

ERROR_IPSEC_IKE_QM_EXPIRED = 13895

The quick mode SA was expired by the IPsec driver.

ERROR_IPSEC_IKE_QM_LIMIT = 13884

Reached maximum quick mode limit for the main mode. New main mode will be started.

ERROR_IPSEC_IKE_QUEUE_DROP_MM = 13811

The negotiation request sat in the queue too long.

ERROR_IPSEC_IKE_QUEUE_DROP_NO_MM = 13812

The negotiation request sat in the queue too long.

ERROR_IPSEC_IKE_RPC_DELETE = 13877

Deleted by using an RPC call.

ERROR_IPSEC_IKE_SA_DELETED = 13807

The IKE security association (SA) was deleted by a peer before it was completely established.

ERROR_IPSEC_IKE_SA_REAPED = 13808

The IKE SA was deleted before it was completely established.

ERROR_IPSEC_IKE_SECLoadFAIL = 13852

Failed to load SECURITY.DLL.

ERROR_IPSEC_IKE_SHUTTING_DOWN = 13891

The IKE service is shutting down.

ERROR_IPSEC_IKE_SOFT_SA_TORN_DOWN = 13845

Soft SA torn down.

ERROR_IPSEC_IKE_SRVACQFAIL = 13855

Failed to obtain the Kerberos server credentials for the Internet Security Association and Key Management Protocol (ISAKMP)/ERROR_IPSEC_IKE service. Kerberos authentication will not function. The most likely reason for this is lack of domain membership. This is normal if your computer is a member of a workgroup.

ERROR_IPSEC_IKE_SRVQUERYCRED = 13856

Failed to determine the SSPI principal name for ISAKMP/ERROR_IPSEC_IKE service (QueryCredentialsAttributes).

ERROR_IPSEC_IKE_TIMED_OUT = 13805

Negotiation timed out.

ERROR_IPSEC_IKE_TOO_MANY_FILTERS = 13896

Too many dynamically added IKEEXT filters were detected.

ERROR_IPSEC_IKE_UNEXPECTED_MESSAGE_ID = 13888

Received unexpected message ID.

ERROR_IPSEC_IKE_UNKNOWN_DOI = 13862

Invalid digital object identifier (DOI).

ERROR_IPSEC_IKE_UNSUPPORTED_ID = 13869

Unsupported ID.

ERROR_IPSEC_MM_AUTH_EXISTS = 13010

The specified main mode authentication list exists.

ERROR_IPSEC_MM_AUTH_IN_USE = 13012

The specified main mode authentication list is being used.

ERROR_IPSEC_MM_AUTH_NOT_FOUND = 13011

The specified main mode authentication list was not found.

ERROR_IPSEC_MM_AUTH_PENDING_DELETION = 13022

The main mode authentication bundle is pending deletion.

ERROR_IPSEC_MM_FILTER_EXISTS = 13006

The specified main mode filter already exists.

ERROR_IPSEC_MM_FILTER_NOT_FOUND = 13007

The specified main mode filter was not found.

ERROR_IPSEC_MM_FILTER_PENDING_DELETION = 13018

The main mode filter is pending deletion.

ERROR_IPSEC_MM_POLICY_EXISTS = 13003

The specified main mode policy already exists.

ERROR_IPSEC_MM_POLICY_IN_USE = 13005

The specified main mode policy is being used.

ERROR_IPSEC_MM_POLICY_NOT_FOUND = 13004

The specified main mode policy was not found.

ERROR_IPSEC_MM_POLICY_PENDING_DELETION = 13021

The main mode policy is pending deletion.

ERROR_IPSEC_QM_POLICY_EXISTS = 13000

The specified quick mode policy already exists.

ERROR_IPSEC_QM_POLICY_IN_USE = 13002

The specified quick mode policy is being used.

ERROR_IPSEC_QM_POLICY_NOT_FOUND = 13001

The specified quick mode policy was not found.

ERROR_IPSEC_QM_POLICY_PENDING_DELETION = 13023

The quick mode policy is pending deletion.

ERROR_IPSEC_TRANSPORT_FILTER_PENDING_DELETION = 13019

The transport filter is pending deletion.

ERROR_IPSEC_TRANSPORT_FILTER_EXISTS = 13008

The specified transport mode filter already exists.

ERROR_IPSEC_TRANSPORT_FILTER_NOT_FOUND = 13009

The specified transport mode filter does not exist.

ERROR_IPSEC_TUNNEL_FILTER_EXISTS = 13016

The specified tunnel mode filter exists.

ERROR_IPSEC_TUNNEL_FILTER_NOT_FOUND = 13017

The specified tunnel mode filter was not found.

ERROR_IPSEC_TUNNEL_FILTER_PENDING_DELETION = 13020

The tunnel filter is pending deletion.

ERROR_IP_ADDRESS_CONFLICT1 = 611

There is an IP address conflict with another system on the network.

ERROR_IP_ADDRESS_CONFLICT2 = 612

There is an IP address conflict with another system on the network.

ERROR_IRQ_BUSY = 1119

Unable to open a device that was sharing an IRQ with other devices. At least one other device that uses that IRQ was already opened.

ERROR_IS_JOINED = 134

An attempt was made to use aJOINorSUBSTcommand on a drive that has already been joined.

ERROR_IS_JOIN_PATH = 147

Not enough resources are available to process this command.

ERROR_IS_JOIN_TARGET = 133

AJOINorSUBSTcommand cannot be used for a drive that contains previously joined drives.

ERROR_IS_SUBSTED = 135

An attempt was made to use aJOINorSUBSTcommand on a drive that has already been substituted.

ERROR_IS_SUBST_PATH = 146

The path specified is being used in a substitute.

ERROR_IS_SUBST_TARGET = 149

An attempt was made to join or substitute a drive for which a directory on the drive is the target of a previous substitute.

ERROR_ITERATED_DATA_EXCEEDS_64k = 194

The operating system cannot run %1.

ERROR_JOIN_TO_JOIN = 138

The system tried to join a drive to a directory on a joined drive.

ERROR_JOIN_TO_SUBST = 140

The system tried to join a drive to a directory on a substituted drive.

ERROR_JOURNAL_DELETE_IN_PROGRESS = 1178

The volume change journal is being deleted.

ERROR_JOURNAL_ENTRY_DELETED = 1181

The journal entry has been deleted from the journal.

ERROR_JOURNAL_HOOK_SET = 1430

The journal hook procedure is already installed.

ERROR_JOURNAL_NOT_ACTIVE = 1179

The volume change journal is not active.

ERROR_KEY_DELETED = 1018

Illegal operation attempted on a registry key that has been marked for deletion.

ERROR_KEY_HAS_CHILDREN = 1020

Cannot create a symbolic link in a registry key that already has subkeys or values.

ERROR_KM_DRIVER_BLOCKED = 1930

The printer driver is not compatible with a policy enabled on your computer that blocks Windows NT 4.0 operating system drivers.

ERROR_LABEL_TOO_LONG = 154

The volume label you entered exceeds the label character limit of the destination file system.

ERROR_LAST_ADMIN = 1322

The last remaining administration account cannot be disabled or deleted.

ERROR_LB_WITHOUT_TABSTOPS = 1434

This list box does not support tab stops.

ERROR_LIBRARY_FULL = 4322

Library is full. No slot is available for use.

ERROR_LIBRARY_OFFLINE = 4305

The operation cannot be performed on an offline library.

ERROR_LICENSE_QUOTA_EXCEEDED = 1395

The service being accessed is licensed for a particular number of connections. No more connections can be made to the service at this time because the service has accepted the maximum number of connections.

ERROR_LISTBOX_ID_NOT_FOUND = 1416

The list box identifier was not found.

ERROR_LM_CROSS_ENCRYPTION_REQUIRED = 1390

A cross-encrypted password is necessary to change this user password.

ERROR_LOCAL_USER_SESSION_KEY = 1303

No encryption key is available. A well-known encryption key was returned.

ERROR_LOCKED = 212

The segment is locked and cannot be reallocated.

ERROR_LOCK_FAILED = 167

Unable to lock a region of a file.

ERROR_LOCK_VIOLATION = 33

The process cannot access the file because another process has locked a portion of the file.

ERROR_LOGIN_TIME_RESTRICTION = 1239

Attempting to log on during an unauthorized time of day for this account.

ERROR_LOGIN_WKSTA_RESTRICTION = 1240

The account is not authorized to log on from this station.

ERROR_LOGON_FAILURE = 1326

Logon failure: Unknown user name or bad password.

ERROR_LOGON_NOT_GRANTED = 1380

Logon failure: The user has not been granted the requested logon type at this computer.

ERROR_LOGON_SERVER_CONFLICT = 568

The Netlogon service cannot start because another Netlogon service running in the domain conflicts with the specified role.

ERROR_LOGON_SESSION_COLLISION = 1366

The logon session ID is already in use.

ERROR_LOGON_SESSION_EXISTS = 1363

Cannot start a new logon session with an ID that is already in use.

ERROR_LOGON_TYPE_NOT_GRANTED = 1385

Logon failure: The user has not been granted the requested logon type at this computer.

ERROR_LOG_APPENDED_FLUSH_FAILED = 6647

Records were appended to the log or reservation changes were made, but the log could not be flushed.

ERROR_LOG_ARCHIVE_IN_PROGRESS = 6633

Log archival is in progress.

ERROR_LOG_ARCHIVE_NOT_IN_PROGRESS = 6632

The operation requires an archive context.

ERROR_LOG_BLOCKS_EXHAUSTED = 6605

The log service user marshaling buffers are exhausted.

ERROR_LOG_BLOCK_INCOMPLETE = 6603

The log service encountered a partial or incomplete log block.

ERROR_LOG_BLOCK_INVALID = 6609

The log service encountered an invalid log block.

ERROR_LOG_BLOCK_VERSION = 6608

The log service encountered an invalid log block version.

ERROR_LOG_CANT_DELETE = 6616

The log service cannot delete a log file or file system container.

ERROR_LOG_CLIENT_ALREADY_REGISTERED = 6636

A log client has already registered on the stream.

ERROR_LOG_CLIENT_NOT_REGISTERED = 6637

A log client has not been registered on the stream.

ERROR_LOG_CONTAINER_LIMIT_EXCEEDED = 6617

The log service has reached the maximum allowable containers allocated to a log file.

ERROR_LOG_CONTAINER_OPEN_FAILED = 6641

The log service encountered an error when attempting to open a log container.

ERROR_LOG_CONTAINER_READ_FAILED = 6639

The log service encountered an error when attempting to read from a log container.

ERROR_LOG_CONTAINER_STATE_INVALID = 6642

The log service encountered an invalid container state when attempting a requested action.

ERROR_LOG_CONTAINER_WRITE_FAILED = 6640

The log service encountered an error when attempting to write to a log container.

ERROR_LOG_CORRUPTION_DETECTED = 6817

The log data is corrupt.

ERROR_LOG_DEDICATED = 6631

The operation failed because the log is a dedicated log.

ERROR_LOG_EPHEMERAL = 6634

The operation requires a non-ephemeral log, but the log is ephemeral.

ERROR_LOG_FILE_FULL = 1502

The event log file is full.

ERROR_LOG_FULL = 6628

The log space is exhausted.

ERROR_LOG_FULL_HANDLER_IN_PROGRESS = 6638

A request has already been made to handle the log full condition.

ERROR_LOG_GROWTH_FAILED = 6833

An attempt to create space in the transactional resource manager's log failed. The failure status has been recorded in the event log.

ERROR_LOG_INCONSISTENT_SECURITY = 6646

Security on the log and its containers is inconsistent.

ERROR_LOG_INVALID_RANGE = 6604

The log service encountered an attempt to access data outside the active log range.

ERROR_LOG_METADATA_CORRUPT = 6612

The log service encountered a corrupted metadata file.

ERROR_LOG_METADATA_FLUSH_FAILED = 6645

The log metadata flush failed.

ERROR_LOG_METADATA_INCONSISTENT = 6614

The log service encountered a metadata file with inconsistent data.

ERROR_LOG_METADATA_INVALID = 6613

The log service encountered a metadata file that could not be created by the log file system.

ERROR_LOG_MULTIPLEXED = 6630

The log is multiplexed; no direct writes to the physical log are allowed.

ERROR_LOG_NOT_ENOUGH_CONTAINERS = 6635

The log must have at least two containers before it can be read from or written to.

ERROR_LOG_NO_RESTART = 6611

The log service encountered a log stream with no restart area.

ERROR_LOG_PINNED = 6644

The log space cannot be reclaimed because the log is pinned.

ERROR_LOG_PINNED_ARCHIVE_TAIL = 6623

Log space cannot be reclaimed because the log is pinned by the archive tail.

ERROR_LOG_PINNED_RESERVATION = 6648

The log is pinned due to reservation consuming most of the log space. Free some reserved records to make space available.

ERROR_LOG_POLICY_ALREADY_INSTALLED = 6619

The log policy could not be installed because a policy of the same type is already present.

ERROR_LOG_POLICY_CONFLICT = 6622

A policy on the log in question prevented the operation from completing.

ERROR_LOG_POLICY_INVALID = 6621

The installed set of policies on the log is invalid.

ERROR_LOG_POLICY_NOT_INSTALLED = 6620

The log policy in question was not installed at the time of the request.

ERROR_LOG_READ_CONTEXT_INVALID = 6606

The log service encountered an attempt to read from a marshaling area with an invalid read context.

ERROR_LOG_READ_MODE_INVALID = 6610

The log service encountered an attempt to read the log with an invalid read mode.

ERROR_LOG_RECORDS_RESERVED_INVALID = 6625

The number of reserved log records or the adjustment of the number of reserved log records is invalid.

ERROR_LOG_RECORD_NONEXISTENT = 6624

The log record is not a record in the log file.

ERROR_LOG_RESERVATION_INVALID = 6615

The log service encountered an attempt to erroneous allocate or dispose reservation space.

ERROR_LOG_RESIZE_INVALID_SIZE = 6806

The requested log size is invalid.

ERROR_LOG_RESTART_INVALID = 6607

The log service encountered an invalid log restart area.

ERROR_LOG_SECTOR_INVALID = 6600

The log service encountered an invalid log sector.

ERROR_LOG_SECTOR_PARITY_INVALID = 6601

The log service encountered a log sector with invalid block parity.

ERROR_LOG_SECTOR_REMAPPED = 6602

The log service encountered a remapped log sector.

ERROR_LOG_SPACE_RESERVED_INVALID = 6626

The reserved log space or the adjustment of the log space is invalid.

ERROR_LOG_START_OF_LOG = 6618

The log service has attempted to read or write backward past the start of the log.

ERROR_LOG_STATE_INVALID = 6643

The log service is not in the correct state to perform a requested action.

ERROR_LOG_TAIL_INVALID = 6627

A new or existing archive tail or base of the active log is invalid.

ERROR_LONGJUMP = 682

A long jump has been executed.

ERROR_LOST_WRITEBEHIND_DATA = 596

{Delayed Write Failed} Windows was unable to save all the data for the file %hs. The data has been lost. This error might be caused by a failure of your computer hardware or network connection. Try to save this file elsewhere.

ERROR_LOST_WRITEBEHIND_DATA_LOCAL_DISK_ERROR = 790

{Delayed Write Failed} Windows was unable to save all the data for the file %hs; the data has been lost. This error can be caused if the device has been removed or the media is write-protected.

ERROR_LOST_WRITEBEHIND_DATA_NETWORK_DISCONNECTED = 788

{Delayed Write Failed} Windows was unable to save all the data for the file %hs; the data has been lost. This error can be caused by network connectivity issues. Try to save this file elsewhere.

ERROR_LOST_WRITEBEHIND_DATA_NETWORK_SERVER_ERROR = 789

{Delayed Write Failed} Windows was unable to save all the data for the file %hs; the data has been lost. This error was returned by the server on which the file exists. Try to save this file elsewhere.

ERROR_LUIDS_EXHAUSTED = 1334

No more LUIDs are available.

ERROR_MACHINE_LOCKED = 1271

The machine is locked and cannot be shut down without the force option.

ERROR_MAGAZINE_NOT_PRESENT = 1163

The indicated element is part of a magazine that is not present.

ERROR_MALFORMED_SUBSTITUTION_STRING = 14094

A string containing localized substitutable content was malformed. Either a dollar sign (\$) was followed by something other than a left parenthesis or another dollar sign, or a substitution's right parenthesis was not found.

ERROR_MAPPED_ALIGNMENT = 1132

The base address or the file offset specified does not have the proper alignment.

ERROR_MARSHALL_OVERFLOW = 603

The user/kernel marshaling buffer has overflowed.

ERROR_MAX_SESSIONS_REACHED = 353

The maximum number of sessions has been reached.

ERROR_MAX_THRDS_REACHED = 164

No more threads can be created in the system.

ERROR_MCA_EXCEPTION = 784

A thread is getting dispatched with MCA EXCEPTION because of MCA.

ERROR_MCA_INTERNAL_ERROR = 15205

An internal monitor configuration API error occurred.

ERROR_MCA_INVALID_CAPABILITIES_STRING = 15200

The monitor returned a DDC/CI capabilities string that did not comply with the ACCESS.bus 3.0, DDC/CI 1.1, or MCCS 2 Revision 1 specification.

ERROR_MCA_INVALID_TECHNOLOGY_TYPE_RETURNED = 15206

The monitor returned an invalid monitor technology type. CRT, plasma, and LCD (TFT) are examples of monitor technology types. This error implies that the monitor violated the MCCS 2.0 or MCCS 2.0 Revision 1 specification.

ERROR_MCA_INVALID_VCP_VERSION = 15201

The monitor's VCP version (0xDF) VCP code returned an invalid version value.

ERROR_MCA_MCCS_VERSION_MISMATCH = 15203

The MCCS version in a monitor's mcs_ver capability does not match the MCCS version the monitor reports when the VCP version (0xDF) VCP code is used.

ERROR_MCA_MONITOR_VIOLATES_MCCS_SPECIFICATION = 15202

The monitor does not comply with the MCCS specification it claims to support.

ERROR_MCA_OCCURED = 651

A machine check error has occurred. Check the system event log for additional information.

ERROR_MCA_UNSUPPORTED_COLOR_TEMPERATURE = 15207

The SetMonitorColorTemperature() caller passed a color temperature to it that the current monitor did not support. CRT, plasma, and LCD (TFT) are examples of monitor technology types. This error implies that the monitor violated the MCCS 2.0 or MCCS 2.0 Revision 1 specification.

ERROR_MCA_UNSUPPORTED_MCCS_VERSION = 15204

The monitor configuration API works only with monitors that support the MCCS 1.0, MCCS 2.0, or MCCS 2.0 Revision 1 specifications.

ERROR_MEDIA_CHANGED = 1110

The media in the drive might have changed.

ERROR_MEDIA_CHECK = 679

{Media Changed} The media might have changed.

ERROR_MEDIA_INCOMPATIBLE = 4315

The medium is not compatible with the device or media pool.

ERROR_MEDIA_NOT_AVAILABLE = 4318

The media is not mounted or ready for use.

ERROR_MEDIA_OFFLINE = 4304

The medium currently exists in an offline library and must be online to perform this operation.

ERROR_MEDIA_UNAVAILABLE = 4308

No media is currently available in this media pool or library.

ERROR_MEDIUM_NOT_ACCESSIBLE = 4323

The transport cannot access the medium.

ERROR_MEMBERS_PRIMARY_GROUP = 1374

The user cannot be removed from a group because the group is currently the user's primary group.

ERROR_MEMBER_IN_ALIAS = 1378

The specified account name is already a member of the group.

ERROR_MEMBER_IN_GROUP = 1320

Either the specified user account is already a member of the specified group, or the specified group cannot be deleted because it contains a member.

ERROR_MEMBER_NOT_IN_ALIAS = 1377

The specified account name is not a member of the group.

ERROR_MEMBER_NOT_IN_GROUP = 1321

The specified user account is not a member of the specified group account.

ERROR_MEMORY_HARDWARE = 779

The hardware has reported an uncorrectable memory error.

ERROR_MENU_ITEM_NOT_FOUND = 1456

A menu item was not found.

ERROR_MESSAGE_EXCEEDS_MAX_SIZE = 4336

The message provided exceeds the maximum size allowed for this parameter.

ERROR_MESSAGE_SYNC_ONLY = 1159

The message can be used only with synchronous operations.

ERROR_METAFILE_NOT_SUPPORTED = 2003

The requested metafile operation is not supported.

ERROR_META_EXPANSION_TOO_LONG = 208

The asterisk (*) or question mark (?) global file name characters are entered incorrectly, or too many global file name characters are specified.

ERROR_MINIVERSION_INACCESSIBLE_FROM_SPECIFIED_TRANSACTION = 6810

A miniversion can only be opened in the context of the transaction that created it.

ERROR_MISSING_SYSTEMFILE = 573

{Missing System File} The required system file %hs is bad or missing.

ERROR_MOD_NOT_FOUND = 126

The specified module could not be found.

ERROR_MORE_DATA = 234

More data is available.

ERROR_MORE_WRITES = 1120

A serial I/O operation was completed by another write to the serial port. (The IOCTL_SERIAL_XOFF_COUNTER reached zero.)

ERROR_MOUNT_POINT_NOT_RESOLVED = 649

The create operation failed because the name contained at least one mount point that resolves to a volume to which the specified device object is not attached.

ERROR_MP_PROCESSOR_MISMATCH = 725

The CPUs in this multiprocessor system are not all the same revision level. To use all processors the operating system restricts itself to the features of the least capable processor in the system. If problems occur with this system, contact the CPU manufacturer to see if this mix of processors is supported.

ERROR_MR_MID_NOT_FOUND = 317

The system cannot find message text for message number 0x%1 in the message file for %2.

ERROR_MUI_FILE_NOT_FOUND = 15100

The resource loader failed to find the Multilingual User Interface (MUI) file.

ERROR_MUI_FILE_NOT_LOADED = 15105

The resource loader cache does not have a loaded MUI entry.

ERROR_MUI_INTLSETTINGS_INVALID_LOCALE_NAME = 15108

Locale installation failed.

ERROR_MUI_INTLSETTINGS_UILANG_NOT_INSTALLED = 15107

User interface language installation failed.

ERROR_MUI_INVALID_FILE = 15101

The resource loader failed to load the MUI file because the file failed to pass validation.

ERROR_MUI_INVALID_LOCALE_NAME = 15103

The RC manifest has an invalid culture name.

ERROR_MUI_INVALID_RC_CONFIG = 15102

The release candidate (RC) manifest is corrupted with garbage data, is an unsupported version, or is missing a required item.

ERROR_MUI_INVALID_ULTIMATEFALLBACK_NAME = 15104

The RC Manifest has an invalid ultimate fallback name.

ERROR_MUTANT_LIMIT_EXCEEDED = 587

An attempt was made to acquire a mutant such that its maximum count would have been exceeded.

ERROR_MUTUAL_AUTH_FAILED = 1397

Mutual authentication failed. The server's password is out of date at the domain controller.

ERROR_NEGATIVE_SEEK = 131

An attempt was made to move the file pointer before the beginning of the file.

ERROR_NESTING_NOT_ALLOWED = 215

Cannot nest calls to LoadModule.

ERROR_NETLOGON_NOT_STARTED = 1792

An attempt was made to log on, but the network logon service was not started.

ERROR_NETNAME_DELETED = 64

The specified network name is no longer available.

ERROR_NETWORK_ACCESS_DENIED = 65

Network access is denied.

ERROR_NETWORK_BUSY = 54

The network is busy.

ERROR_NETWORK_NOT_AVAILABLE = 5035

A cluster network is not available for this operation.

ERROR_NETWORK_UNREACHABLE = 1231

The network location cannot be reached. For information about network troubleshooting, see Windows Help.

ERROR_NET_OPEN_FAILED = 570

The NtCreateFile API failed. This error should never be returned to an application, it is a place holder for the Windows LAN Manager Redirector to use in its internal error mapping routines.

ERROR_NET_WRITE_FAULT = 88

A write fault occurred on the network.

ERROR_NOACCESS = 998

Invalid access to memory location.

ERROR_NODE_CANNOT_BE_CLUSTERED = 5898

This computer cannot be made a member of a cluster.

ERROR_NODE_CANT_HOST_RESOURCE = 5071

The cluster resource cannot be brought online. The owner node cannot run this resource.

ERROR_NODE_NOT_AVAILABLE = 5036

A cluster node is not available for this operation.

ERROR_NOINTERFACE = 632

The requested interface is not supported.

ERROR_NOLOGON_INTERDOMAIN_TRUST_ACCOUNT = 1807

The account used is an interdomain trust account. Use your global user account or local user account to access this server.

ERROR_NOLOGON_SERVER_TRUST_ACCOUNT = 1809

The account used is a server trust account. Use your global user account or local user account to access this server.

ERROR_NOLOGON_WORKSTATION_TRUST_ACCOUNT = 1808

The account used is a computer account. Use your global user account or local user account to access this server.

ERROR_NONE_MAPPED = 1332

No mapping between account names and SIDs was done.

ERROR_NONPAGED_SYSTEM_RESOURCES = 1451

Insufficient system resources exist to complete the requested service.

ERROR_NON_ACCOUNT_SID = 1257

The security identifier provided is not from an account domain.

ERROR_NON_DOMAIN_SID = 1258

The security identifier provided does not have a domain component.

ERROR_NON_MDICHILD_WINDOW = 1445

Cannot process a message from a window that is not a multiple document interface (MDI) window.

ERROR_NOTHING_TO_TERMINATE = 758

A process being terminated has no threads to terminate.

ERROR_NOTIFY_CLEANUP = 745

This indicates that a notify change request has been completed due to closing the handle which made the notify change request.

ERROR_NOTIFY_ENUM_DIR = 1022

A notify change request is being completed and the information is not being returned in the caller's buffer. The caller now needs to enumerate the files to find the changes.

ERROR_NOT_ALL_ASSIGNED = 1300

Not all privileges or groups referenced are assigned to the caller.

ERROR_NOT_AUTHENTICATED = 1244

The operation being requested was not performed because the user has not been authenticated.

ERROR_NOT_A_REPARSE_POINT = 4390

The file or directory is not a reparse point.

ERROR_NOT_CAPABLE = 775

The implementation is not capable of performing the request.

ERROR_NOT_CHILD_WINDOW = 1442

The window is not a child window.

ERROR_NOT_CONTAINER = 1207

Cannot enumerate a noncontainer.

ERROR_NOT_DOS_DISK = 26

The specified disk cannot be accessed.

ERROR_NOT_EMPTY = 4307

The library, drive, or media pool must be empty to perform this operation.

ERROR_NOT_ENOUGH_MEMORY = 8

Not enough storage is available to process this command.

ERROR_NOT_ENOUGH_QUOTA = 1816

Not enough quota is available to process this command.

ERROR_NOT_ENOUGH_SERVER_MEMORY = 1130

Not enough server storage is available to process this command.

ERROR_NOT_EXPORT_FORMAT = 6008

The specified file is not in the defined EFS export format.

ERROR_NOT_FOUND = 1168

Element not found.

ERROR_NOT_JOINED = 136

The system tried to delete the JOIN of a drive that is not joined.

ERROR_NOT_LOCKED = 158

The segment is already unlocked.

ERROR_NOT_LOGGED_ON = 1245

The operation being requested was not performed because the user has not logged on to the network. The specified service does not exist.

ERROR_NOT_LOGON_PROCESS = 1362

The requested action is restricted for use by logon processes only. The calling process has not registered as a logon process.

ERROR_NOT_OWNER = 288

Attempt to release mutex not owned by caller.

ERROR_NOT_QUORUM_CAPABLE = 5021

The cluster could not make the specified resource a quorum resource because it is not capable of being a quorum resource.

ERROR_NOT_QUORUM_CLASS = 5025

The cluster could not make the specified resource a quorum resource because it does not belong to a shared storage class.

ERROR_NOT_READY = 21

The device is not ready.

ERROR_NOT_REGISTRY_FILE = 1017

The system attempted to load or restore a file into the registry, but the specified file is not in a registry file format.

ERROR_NOT_SAFEBOOT_SERVICE = 1084

This service cannot be started in Safe Mode.

ERROR_NOT_SAFE_MODE_DRIVER = 646

The driver was not loaded because the system is booting into safe mode.

ERROR_NOT_SAME_DEVICE = 17

The system cannot move the file to a different disk drive.

ERROR_NOT_SNAPSHOT_VOLUME = 6841

The target volume is not a snapshot volume. This operation is only valid on a volume mounted as a snapshot.

ERROR_NOT_SUBSTED = 137

The system tried to delete the substitution of a drive that is not substituted.

ERROR_NOT_SUPPORTED = 50

The request is not supported.

ERROR_NOT_SUPPORTED_ON_SBS = 1254

This operation is not supported on a computer running Windows Server 2003 operating system for Small Business Server.

ERROR_NOT_SUPPORTED_ON_STANDARD_SERVER = 8584

The requested action is not supported on a standard server.

ERROR_NOT_TINY_STREAM = 598

The stream is not a tiny stream.

ERROR_NO_ASSOCIATION = 1155

No application is associated with the specified file for this operation.

ERROR_NO_BROWSER_SERVERS_FOUND = 6118

The list of servers for this workgroup is not currently available

ERROR_NO_CALLBACK_ACTIVE = 614

A callback return system service cannot be executed when no callback is active.

ERROR_NO_DATA = 232

The pipe is being closed.

ERROR_NO_DATA_DETECTED = 1104

No more data is on the tape.

ERROR_NO_EFS = 6004

The required encryption driver is not loaded for this system.

ERROR_NO_EVENT_PAIR = 580

An event pair synchronization operation was performed using the thread-specific client/server event pair object, but no event pair object was associated with the thread.

ERROR_NO_GUID_TRANSLATION = 560

Indicates that an attempt was made to assign protection to a file system file or directory and one of the SIDs in the security descriptor could not be translated into a GUID that could be stored by the file system. This causes the protection attempt to fail, which might cause a file creation attempt to fail.

ERROR_NO_IMPERSONATION_TOKEN = 1309

An attempt has been made to operate on an impersonation token by a thread that is not currently impersonating a client.

ERROR_NO_INHERITANCE = 1391

Indicates an ACL contains no inheritable components.

ERROR_NO_LOGON_SERVERS = 1311

There are currently no logon servers available to service the logon request.

ERROR_NO_LOG_SPACE = 1019

System could not allocate the required space in a registry log.

ERROR_NO_MATCH = 1169

There was no match for the specified key in the index.

ERROR_NO_MEDIA_IN_DRIVE = 1112

No media in drive.

ERROR_NO_MORE_DEVICES = 1248

No more local devices.

ERROR_NO_MORE_FILES = 18

There are no more files.

ERROR_NO_MORE_ITEMS = 259

No more data is available.

ERROR_NO_MORE_MATCHES = 626

There are no more matches for the current index enumeration.

ERROR_NO_MORE_SEARCH_HANDLES = 113

No more internal file identifiers are available.

ERROR_NO_MORE_USER_HANDLES = 1158

The current process has used all of its system allowance of handles for Windows manager objects.

ERROR_NO_NETWORK = 1222

The network is not present or not started.

ERROR_NO_NET_OR_BAD_PATH = 1203

The network path was either typed incorrectly, does not exist, or the network provider is not currently available. Try retyping the path or contact your network administrator.

ERROR_NO_PAGEFILE = 578

{No Paging File Specified} No paging file was specified in the system configuration.

ERROR_NO_PROC_SLOTS = 89

The system cannot start another process at this time.

ERROR_NO_PROMOTION_ACTIVE = 8222

A domain controller promotion is not currently active.

ERROR_NO_QUOTAS_FOR_ACCOUNT = 1302

No system quota limits are specifically set for this account.

ERROR_NO_RECOVERY_POLICY = 6003

There is no valid encryption recovery policy configured for this system.

ERROR_NO_RECOVERY_PROGRAM = 1082

No recovery program has been configured for this service.

ERROR_NO_SAVEPOINT_WITH_OPEN_FILES = 6842

The savepoint operation failed because files are open on the transaction. This is not permitted.

ERROR_NO_SCROLLBARS = 1447

The window does not have scroll bars.

ERROR_NO_SECRETS = 8620

The local account store does not contain secret material for the specified account.

ERROR_NO_SECURITY_ON_OBJECT = 1350

Unable to perform a security operation on an object that has no associated security.

ERROR_NO_SHUTDOWN_IN_PROGRESS = 1116

Unable to abort the system shutdown because no shutdown was in progress.

ERROR_NO_SIGNAL_SENT = 205

No process in the command subtree has a signal handler.

ERROR_NO_SITENAME = 1919

No site name is available for this machine.

ERROR_NO_SITE_SETTINGS_OBJECT = 8619

The site settings object for the specified site does not exist.

ERROR_NO_SPOOL_SPACE = 62

Space to store the file waiting to be printed is not available on the server.

ERROR_NO_SUCH_ALIAS = 1376

The specified local group does not exist.

ERROR_NO_SUCH_DOMAIN = 1355

The specified domain either does not exist or could not be contacted.

ERROR_NO_SUCH_GROUP = 1319

The specified group does not exist.

ERROR_NO_SUCH_LOGON_SESSION = 1312

A specified logon session does not exist. It might already have been terminated.

ERROR_NO_SUCH_MEMBER = 1387

A member could not be added to or removed from the local group because the member does not exist.

ERROR_NO_SUCH_PACKAGE = 1364

A specified authentication package is unknown.

ERROR_NO_SUCH_PRIVILEGE = 1313

A specified privilege does not exist.

ERROR_NO_SUCH_SITE = 1249

The specified site does not exist.

ERROR_NO_SUCH_USER = 1317

The specified account does not exist.

ERROR_NO_SUPPORTING_DRIVES = 4339

This offline media cannot be mounted on this system because no enabled drives are present that can be used.

ERROR_NO_SYSTEM_MENU = 1437

The window does not have a system menu.

ERROR_NO_SYSTEM_RESOURCES = 1450

Insufficient system resources exist to complete the requested service.

ERROR_NO_TOKEN = 1008

An attempt was made to reference a token that does not exist.

ERROR_NO_TRACKING_SERVICE = 1172

The tracking (workstation) service is not running.

ERROR_NO_TRUST_LSA_SECRET = 1786

The workstation does not have a trust secret.

ERROR_NO_TRUST_SAM_ACCOUNT = 1787

The security database on the server does not have a computer account for this workstation trust relationship.

ERROR_NO_TXF_METADATA = 6816

There is no transaction metadata on the file.

ERROR_NO_UNICODE_TRANSLATION = 1113

No mapping for the Unicode character exists in the target multibyte code page.

ERROR_NO_USER_KEYS = 6006

There are no Encrypting File System (EFS) keys defined for the user.

ERROR_NO_USER_SESSION_KEY = 1394

There is no user session key for the specified logon session.

ERROR_NO_VOLUME_ID = 1173

The volume ID could not be found.

ERROR_NO_VOLUME_LABEL = 125

The disk has no volume label.

ERROR_NO_WILDCARD_CHARACTERS = 1417

No wildcards were found.

ERROR_NO_WRITABLE_DC_FOUND = 8621

Could not find a writable domain controller in the domain.

ERROR_NO_YIELD_PERFORMED = 721

A yield execution was performed and no thread was available to run.

ERROR_NT_CROSS_ENCRYPTION_REQUIRED = 1386

A cross-encrypted password is necessary to change a user password.

ERROR_NULL_LM_PASSWORD = 1304

The password is too complex to be converted to a LAN Manager password. The LAN Manager password returned is a null string.

ERROR_OBJECT_ALREADY_EXISTS = 5010

The object already exists.

ERROR_OBJECT_IN_LIST = 5011

The object is already in the list.

ERROR_OBJECT_NAME_EXISTS = 698

{Object Exists} An attempt was made to create an object and the object name already existed.

ERROR_OBJECT_NOT_FOUND = 4312

The object identifier does not represent a valid object.

ERROR_OBJECT_NO_LONGER_EXISTS = 6807

The object (file, stream, link) corresponding to the handle has been deleted by a transaction savepoint rollback.

ERROR_OLD_WIN_VERSION = 1150

The specified program requires a newer version of Windows.

ERROR_ONLY_IF_CONNECTED = 1251

This operation is supported only when you are connected to the server.

ERROR_OPEN_FAILED = 110

The system cannot open the device or file specified.

ERROR_OPERATION_ABORTED = 995

The I/O operation has been aborted because of either a thread exit or an application request.

ERROR_OPLOCK_BREAK_IN_PROGRESS = 742

An open/create operation completed while an oplock break is underway.

ERROR_OPLOCK_NOT_GRANTED = 300

The oplock request is denied.

ERROR_OUTOFMEMORY = 14

Not enough storage is available to complete this operation.

ERROR_OUT_OF_PAPER = 28

The printer is out of paper.

ERROR_OUT_OF_STRUCTURES = 84

Storage to process this request is not available.

ERROR_OVERRIDE_NOCHANGES = 1252

The group policy framework should call the extension even if there are no changes.

ERROR_PAGED_SYSTEM_RESOURCES = 1452

Insufficient system resources exist to complete the requested service.

ERROR_PAGEFILE_CREATE_FAILED = 576

{Unable to Create Paging File} The creation of the paging file %hs failed (%lx). The requested size was %ld.

ERROR_PAGEFILE_QUOTA = 1454

Insufficient quota to complete the requested service.

ERROR_PAGEFILE_QUOTA_EXCEEDED = 567

Page file quota was exceeded.

ERROR_PAGE_FAULT_COPY_ON_WRITE = 749

Page fault was a demand zero fault.

ERROR_PAGE_FAULT_DEMAND_ZERO = 748

Page fault was a demand zero fault.

ERROR_PAGE_FAULT_GUARD_PAGE = 750

Page fault was a demand zero fault.

ERROR_PAGE_FAULT_PAGING_FILE = 751

Page fault was satisfied by reading from a secondary storage device.

ERROR_PAGE_FAULT_TRANSITION = 747

Page fault was a transition fault.

ERROR_PARAMETER_QUOTA_EXCEEDED = 1283

Data present in one of the parameters is more than the function can operate on.

ERROR_PARTIAL_COPY = 299

Only part of a ReadProcessMemory or WriteProcessMemory request was completed.

ERROR_PARTITION_FAILURE = 1105

Tape could not be partitioned.

ERROR_PASSWORD_EXPIRED = 1330

Logon failure: The specified account password has expired.

ERROR_PASSWORD_MUST_CHANGE = 1907

The user's password must be changed before logging on the first time.

ERROR_PASSWORD_RESTRICTION = 1325

Unable to update the password. The value provided for the new password does not meet the length, complexity, or history requirements of the domain.

ERROR_PATCH_MANAGED_ADVERTISED_PRODUCT = 1651

Windows Installer does not permit updating of managed advertised products. At least one feature of the product must be installed before applying the update.

ERROR_PATCH_NO_SEQUENCE = 1648

No valid sequence could be found for the set of updates.

ERROR_PATCH_PACKAGE_INVALID = 1636

This update package could not be opened. Contact the application vendor to verify that this is a valid Windows Installer update package.

ERROR_PATCH_PACKAGE_OPEN_FAILED = 1635

This update package could not be opened. Verify that the update package exists and that you can access it, or contact the application vendor to verify that this is a valid Windows Installer update package.

ERROR_PATCH_PACKAGE_REJECTED = 1643

The update package is not permitted by a software restriction policy.

ERROR_PATCH_PACKAGE_UNSUPPORTED = 1637

This update package cannot be processed by the Windows Installer service. You must install a Windows service pack that contains a newer version of the Windows Installer service.

ERROR_PATCH_REMOVAL_DISALLOWED = 1649

Update removal was disallowed by policy.

ERROR_PATCH_REMOVAL_UNSUPPORTED = 1646

Uninstallation of the update package is not supported.

ERROR_PATCH_TARGET_NOT_FOUND = 1642

The upgrade cannot be installed by the Windows Installer service because the program to be upgraded might be missing, or the upgrade might update a different version of the program. Verify that the program to be upgraded exists on your computer and that you have the correct upgrade.

ERROR_PATH_BUSY = 148

The path specified cannot be used at this time.

ERROR_PATH_NOT_FOUND = 3

The system cannot find the path specified.

ERROR_PER_USER_TRUST_QUOTA_EXCEEDED = 1932

The current user's delegated trust creation quota has been exceeded.

ERROR_PIPE_BUSY = 231

All pipe instances are busy.

ERROR_PIPE_CONNECTED = 535

There is a process on the other end of the pipe.

ERROR_PIPE_LISTENING = 536

Waiting for a process to open the other end of the pipe.

ERROR_PIPE_LOCAL = 229

The pipe is local.

ERROR_PIPE_NOT_CONNECTED = 233

No process is on the other end of the pipe.

ERROR_PKINIT_FAILURE = 1263

The Kerberos protocol encountered an error while validating the KDC certificate during smartcard logon. There is more information in the system event log.

ERROR_PLUGPLAY_QUERY_VETOED = 683

The Plug and Play query operation was not successful.

ERROR_PNP_BAD_MPS_TABLE = 671

A device is missing in the system BIOS MultiProcessor Specification (MPS) table. This device will not be used. Contact your system vendor for system BIOS update.

ERROR_PNP_INVALID_ID = 674

Driver %2 returned invalid ID for a child device (%3).

ERROR_PNP_IRQ_TRANSLATION_FAILED = 673

An interrupt request (IRQ) translator failed to translate resources.

ERROR_PNP_REBOOT_REQUIRED = 638

Device will not start without a reboot.

ERROR_PNP_RESTART_ENUMERATION = 636

A device was removed so enumeration must be restarted.

ERROR_PNP_TRANSLATION_FAILED = 672

A translator failed to translate resources.

ERROR_POINT_NOT_FOUND = 1171

The point passed to GetMouseMovePoints is not in the buffer.

ERROR_POLICY_OBJECT_NOT_FOUND = 8219

The policy object does not exist.

ERROR_POLICY_ONLY_IN_DS = 8220

The requested policy information is only in the directory service.

ERROR_POPUP_ALREADY_ACTIVE = 1446

Pop-up menu already active.

ERROR_PORT_MESSAGE_TOO_LONG = 546

Length of message passed to NtRequestPort or NtRequestWaitReplyPort was longer than the maximum message allowed by the port.

ERROR_PORT_NOT_SET = 642

An attempt to remove a process DebugPort was made, but a port was not already associated with the process.

ERROR_PORT_UNREACHABLE = 1234

No service is operating at the destination network endpoint on the remote system.

ERROR_POSSIBLE_DEADLOCK = 1131

A potential deadlock condition has been detected.

ERROR_POTENTIAL_FILE_FOUND = 1180

A file was found, but it might not be the correct file.

ERROR_PREDEFINED_HANDLE = 714

The specified registry key is referenced by a predefined handle.

ERROR_PRIMARY_TRANSPORT_CONNECT_FAILED = 746

{Connect Failure on Primary Transport} An attempt was made to connect to the remote server %hs on the primary transport, but the connection failed. The computer was able to connect on a secondary transport.

ERROR_PRINTER_ALREADY_EXISTS = 1802

The printer already exists.

ERROR_PRINTER_DELETED = 1905

The specified printer has been deleted.

ERROR_PRINTER_DRIVER_ALREADY_INSTALLED = 1795

The specified printer driver is already installed.

ERROR_PRINTER_DRIVER_BLOCKED = 3014

The printer driver is known to harm the system.

ERROR_PRINTER_DRIVER_DOWNLOAD_NEEDED = 3019

The specified printer driver was not found on the system and needs to be downloaded.

ERROR_PRINTER_DRIVER_IN_USE = 3001

The specified printer driver is currently in use.

ERROR_PRINTER_DRIVER_PACKAGE_IN_USE = 3015

The specified printer driver package is currently in use.

ERROR_PRINTER_DRIVER_WARNED = 3013

The printer driver is known to be unreliable.

ERROR_PRINTER_HAS_JOBS_QUEUED = 3009

The requested operation is not allowed when there are jobs queued to the printer.

ERROR_PRINTER_NOT_FOUND = 3012

No printers were found.

ERROR_PRINTER_NOT_SHAREABLE = 3022

The specified printer cannot be shared.

ERROR_PRINTQ_FULL = 61

The print queue is full.

ERROR_PRINT_CANCELLED = 63

Your file waiting to be printed was deleted.

ERROR_PRINT_MONITOR_ALREADY_INSTALLED = 3006

The specified print monitor has already been installed.

ERROR_PRINT_MONITOR_IN_USE = 3008

The specified print monitor is currently in use.

ERROR_PRINT_PROCESSOR_ALREADY_INSTALLED = 3005

The specified print processor has already been installed.

ERROR_PRIVATE_DIALOG_INDEX = 1415

Using private DIALOG window words.

ERROR_PRIVILEGE_NOT_HELD = 1314

A required privilege is not held by the client.

ERROR_PROCESS_ABORTED = 1067

The process terminated unexpectedly.

ERROR_PROCESS_IN_JOB = 760

The specified process is part of a job.

ERROR_PROCESS_IS_PROTECTED = 1293

Either the target process, or the target thread's containing process, is a protected process.

ERROR_PROCESS_MODE_ALREADY_BACKGROUND = 402

The process is already in background processing mode.

ERROR_PROCESS_MODE_NOT_BACKGROUND = 403

The process is not in background processing mode.

ERROR_PROCESS_NOT_IN_JOB = 759

The specified process is not part of a job.

ERROR_PROC_NOT_FOUND = 127

The specified procedure could not be found.

ERROR_PRODUCT_UNINSTALLED = 1614

Product is uninstalled.

ERROR_PRODUCT_VERSION = 1638

Another version of this product is already installed. Installation of this version cannot continue. To configure or remove the existing version of this product, use Add/Remove Programs in Control Panel.

ERROR_PROFILE_DOES_NOT_MATCH_DEVICE = 2023

The specified profile is intended for a device of a different type than the specified device.

ERROR_PROFILE_NOT_ASSOCIATED_WITH_DEVICE = 2015

The specified color profile is not associated with any device.

ERROR_PROFILE_NOT_FOUND = 2016

The specified color profile was not found.

ERROR_PROFILING_AT_LIMIT = 553

The number of active profiling objects is at the maximum and no more can be started.

ERROR_PROFILING_NOT_STARTED = 550

Profiling not started.

ERROR_PROFILING_NOT_STOPPED = 551

Profiling not stopped.

ERROR_PROMOTION_ACTIVE = 8221

A domain controller promotion is currently active.

ERROR_PROTOCOL_UNREACHABLE = 1233

The network location cannot be reached. For information about network troubleshooting, see Windows Help.

ERROR_PWD_HISTORY_CONFLICT = 617

You have attempted to change your password to one that you have used in the past. The policy of your user account does not allow this. Select a password that you have not previously used.

ERROR_PWD_TOO_RECENT = 616

The policy of your user account does not allow you to change passwords too frequently. This is done to prevent users from changing back to a familiar, but potentially discovered, password. If you feel your password has been compromised, contact your administrator immediately to have a new one assigned.

ERROR_PWD_TOO_SHORT = 615

The password provided is too short to meet the policy of your user account. Choose a longer password.

ERROR_QUORUMLOG_OPEN_FAILED = 5028

The quorum log could not be created or mounted successfully.

ERROR_QUORUM_DISK_NOT_FOUND = 5086

The quorum disk could not be located by the cluster service.

ERROR_QUORUM_NOT_ALLOWED_IN_THIS_GROUP = 5928

Quorum resource cannot reside in the available storage group.

ERROR_QUORUM_OWNER_ALIVE = 5034

The cluster node failed to take control of the quorum resource because the resource is owned by another active node.

ERROR_QUORUM_RESOURCE = 5020

The cluster resource could not be deleted or brought offline because it is the quorum resource.

ERROR_QUORUM_RESOURCE_ONLINE_FAILED = 5027

The quorum resource failed to come online.

ERROR_QUOTA_LIST_INCONSISTENT = 621

The specified quota list is internally inconsistent with its descriptor.

ERROR_RANGE_LIST_CONFLICT = 627

The range could not be added to the range list because of a conflict.

ERROR_RANGE_NOT_FOUND = 644

The specified range could not be found in the range list.

ERROR_RDP_PROTOCOL_ERROR = 7065

The RDP component %2 detected an error in the protocol stream and has disconnected the client.

ERROR_READ_FAULT = 30

The system cannot read from the specified device.

ERROR_RECEIVE_EXPEDITED = 708

{Expedited Data Received} The network transport returned data to its client that was marked as expedited by the remote system.

ERROR_RECEIVE_PARTIAL = 707

{Partial Data Received} The network transport returned partial data to its client. The remaining data will be sent later.

ERROR_RECEIVE_PARTIAL_EXPEDITED = 709

{Partial Expedited Data Received} The network transport returned partial data to its client and this data was marked as expedited by the remote system. The remaining data will be sent later.

ERROR_RECOVERY_FAILURE = 1279

A transaction recover failed.

ERROR_RECOVERY_NOT_NEEDED = 6821

The transactional resource manager is already consistent. Recovery is not needed.

ERROR_REC_NON_EXISTENT = 4005

The name does not exist in the WINS database.

ERROR_REDIRECTOR_HAS_OPEN_HANDLES = 1794

The redirector is in use and cannot be unloaded.

ERROR_REDIR_PAUSED = 72

The specified printer or disk device has been paused.

ERROR_REGISTRY_CORRUPT = 1015

The registry is corrupted. The structure of one of the files containing registry data is corrupted, or the system's memory image of the file is corrupted, or the file could not be recovered because the alternate copy or log was absent or corrupted.

ERROR_REGISTRY_HIVE_RECOVERED = 685

{Registry Hive Recovered} Registry hive (file): %hs was corrupted and it has been recovered. Some data might have been lost.

ERROR_REGISTRY_IO_FAILED = 1016

An I/O operation initiated by the registry failed and cannot be recovered. The registry could not read in, write out, or flush one of the files that contain the system's image of the registry.

ERROR_REGISTRY_QUOTA_LIMIT = 613

{Low On Registry Space} The system has reached the maximum size allowed for the system part of the registry. Additional storage requests will be ignored.

ERROR_REGISTRY_RECOVERED = 1014

One of the files in the registry database had to be recovered by use of a log or alternate copy. The recovery was successful.

ERROR_REG_NAT_CONSUMPTION = 1261

A program attempt to use an invalid register value. Normally caused by an uninitialized register. This error is Itanium specific.

ERROR_RELOC_CHAIN_XEEDS_SEGLIM = 201

The operating system cannot run %1.

ERROR_REMOTE_FILE_VERSION_MISMATCH = 6814

The remote server sent mismatching version numbers or FID for a file opened with transactions.

ERROR_REMOTE_PRINT_CONNECTIONS_BLOCKED = 1936

Remote connections to the Print Spooler are blocked by a policy set on your machine.

ERROR_REMOTE_SESSION_LIMIT_EXCEEDED = 1220

An attempt was made to establish a session to a network server, but there are already too many sessions established to that server.

ERROR_REMOTE_STORAGE_MEDIA_ERROR = 4352

The remote storage service encountered a media error.

ERROR_REMOTE_STORAGE_NOT_ACTIVE = 4351

The remote storage service is not operational at this time.

ERROR_REM_NOT_LIST = 51

Windows cannot find the network path. Verify that the network path is correct and the destination computer is not busy or turned off. If Windows still cannot find the network path, contact your network administrator.

ERROR_REPARSE = 741

A reparse should be performed by the object manager because the name of the file resulted in a symbolic link.

ERROR_REPARSE_ATTRIBUTE_CONFLICT = 4391

The reparse point attribute cannot be set because it conflicts with an existing attribute.

ERROR_REPARSE_OBJECT = 755

A reparse should be performed by the object manager because the name of the file resulted in a symbolic link.

ERROR_REPARSE_TAG_INVALID = 4393

The tag present in the reparse point buffer is invalid.

ERROR_REPARSE_TAG_MISMATCH = 4394

There is a mismatch between the tag specified in the request and the tag present in the reparse point.

ERROR_REPLY_MESSAGE_MISMATCH = 595

{Reply Message Mismatch} An attempt was made to reply to a local procedure call (LPC) message, but the thread specified by the client ID in the message was not waiting on that message.

ERROR_REQUEST_ABORTED = 1235

The request was aborted.

ERROR_REQUEST_OUT_OF_SEQUENCE = 776

The client of a component requested an operation that is not valid given the state of the component instance.

ERROR_REQUEST_REFUSED = 4320

The operator or administrator has refused the request.

ERROR_REQUIRES_INTERACTIVE_WINDOWSTATION = 1459

This operation requires an interactive window station.

ERROR_REQ_NOT_ACCEP = 71

No more connections can be made to this remote computer at this time because the computer has accepted the maximum number of connections.

ERROR_RESMON_CREATE_FAILED = 5017

The cluster resource could not be created in the specified resource monitor.

ERROR_RESMON_INVALID_STATE = 5084

The resource monitor will not allow the fail operation to be performed while the resource is in its current state. This can happen if the resource is in a pending state.

ERROR_RESMON_ONLINE_FAILED = 5018

The cluster resource could not be brought online by the resource monitor.

ERROR_RESOURCEMANAGER_NOT_FOUND = 6716

The specified resource manager object could not be opened because it was not found.

ERROR_RESOURCEMANAGER_READ_ONLY = 6707

The specified resource manager made no changes or updates to the resource under this transaction.

ERROR_RESOURCE_CALL_TIMED_OUT = 5910

The call to the cluster resource DLL timed out.

ERROR_RESOURCE_DATA_NOT_FOUND = 1812

The specified image file did not contain a resource section.

ERROR_RESOURCE_DISABLED = 4309

A resource required for this operation is disabled.

ERROR_RESOURCE_ENUM_USER_STOP = 15106

The user stopped resource enumeration.

ERROR_RESOURCE_FAILED = 5038

A cluster resource failed.

ERROR_RESOURCE_LANG_NOT_FOUND = 1815

The specified resource language ID cannot be found in the image file.

ERROR_RESOURCE_NAME_NOT_FOUND = 1814

The specified resource name cannot be found in the image file.

ERROR_RESOURCE_NOT_AVAILABLE = 5006

The cluster resource is not available.

ERROR_RESOURCE_NOT_FOUND = 5007

The cluster resource could not be found.

ERROR_RESOURCE_NOT_ONLINE = 5004

The cluster resource is not online.

ERROR_RESOURCE_NOT_PRESENT = 4316

The resource required for this operation does not exist.

ERROR_RESOURCE_ONLINE = 5019

The operation could not be completed because the cluster resource is online.

ERROR_RESOURCE_PROPERTIES_STORED = 5024

The properties were stored but not all changes will take effect until the next time the resource is brought online.

ERROR_RESOURCE_PROPERTY_UNCHANGEABLE = 5089

An attempt to modify a resource property failed because it conflicts with another existing property.

ERROR_RESOURCE_REQUIREMENTS_CHANGED = 756

The device has succeeded a query-stop and its resource requirements have changed.

ERROR_RESOURCE_TYPE_NOT_FOUND = 1813

The specified resource type cannot be found in the image file.

ERROR_RESUME_HIBERNATION = 727

The system was resumed from hibernation.

ERROR_RETRY = 1237

The operation could not be completed. A retry should be performed.

ERROR_REVISION_MISMATCH = 1306

Indicates two revision levels are incompatible.

ERROR_RING2SEG_MUST_BE_MOVABLE = 200

The code segment cannot be greater than or equal to 64 KB.

ERROR_RING2_STACK_IN_USE = 207

The ring 2 stack is in use.

ERROR_RMODE_APP = 1153

The specified program was written for an earlier version of Windows.

ERROR_RM_ALREADY_STARTED = 6822

The transactional resource manager has already been started.

ERROR_RM_DISCONNECTED = 6819

The transaction outcome is unavailable because the resource manager responsible for it is disconnected.

ERROR_RM_METADATA_CORRUPT = 6802

The metadata of the resource manager has been corrupted. The resource manager will not function.

ERROR_RM_NOT_ACTIVE = 6801

Transaction support within the specified file system resource manager is not started or was shut down due to an error.

ERROR_ROLLBACK_TIMER_EXPIRED = 6829

A rollback could not be scheduled because a previously scheduled rollback has already been executed or is queued for execution.

ERROR_ROWSNOTRELEASED = 772

The data provider requires that previously fetched data is released before asking for more data.

ERROR_RPL_NOT_ALLOWED = 4006

Replication with a nonconfigured partner is not allowed.

ERROR_RXACT_COMMITTED = 744

This success level status indicates that the transaction state already exists for the registry subtree, but that a transaction commit was previously aborted. The commit has now been completed.

ERROR_RXACT_COMMIT_FAILURE = 1370

An internal security database corruption has been encountered.

ERROR_RXACT_COMMIT_NECESSARY = 678

This warning level status indicates that the transaction state already exists for the registry subtree, but that a transaction commit was previously aborted. The commit has NOT been completed, but it has not been rolled back either (so it can still be committed if desired).

ERROR_RXACT_INVALID_STATE = 1369

The transaction state of a registry subtree is incompatible with the requested operation.

ERROR_RXACT_STATE_CREATED = 701

This informational level status indicates that a specified registry subtree transaction state did not yet exist and had to be created.

ERROR_SAME_DRIVE = 143

The system cannot join or substitute a drive to or for a directory on the same drive.

ERROR_SAM_INIT_FAILURE = 8541

SAM initialization failed because of the following error: %1. Error Status: 0x%2. Click OK to shut down the system and reboot into Safe Mode. Check the event log for detailed information.

ERROR_SCOPE_NOT_FOUND = 318

The scope specified was not found.

ERROR_SCREEN_ALREADY_LOCKED = 1440

Screen already locked.

ERROR_SECRET_TOO_LONG = 1382

The length of a secret exceeds the maximum length allowed.

ERROR_SECTOR_NOT_FOUND = 27

The drive cannot find the sector requested.

ERROR_SEEK = 25

The drive cannot locate a specific area or track on the disk.

ERROR_SEEK_ON_DEVICE = 132

The file pointer cannot be set on the specified device or file.

ERROR_SEGMENT_NOTIFICATION = 702

{Segment Load} A virtual DOS machine (VDM) is loading, unloading, or moving an MS-DOS or Win16 program segment image. An exception is raised so a debugger can load, unload, or track symbols and breakpoints within these 16-bit segments.

ERROR_SEM_IS_SET = 102

The semaphore is set and cannot be closed.

ERROR_SEM_NOT_FOUND = 187

The specified system semaphore name was not found.

ERROR_SEM_OWNER_DIED = 105

The previous ownership of this semaphore has ended.

ERROR_SEM_TIMEOUT = 121

The semaphore time-out period has expired.

ERROR_SEM_USER_LIMIT = 106

Insert the disk for drive %1.

ERROR_SERIAL_NO_DEVICE = 1118

No serial device was successfully initialized. The serial driver will unload.

ERROR_SERVER_DISABLED = 1341

The server is currently disabled.

ERROR_SERVER_HAS_OPEN_HANDLES = 1811

The server is in use and cannot be unloaded.

ERROR_SERVER_NOT_DISABLED = 1342

The server is currently enabled.

ERROR_SERVER_SHUTDOWN_IN_PROGRESS = 1255

The server machine is shutting down.

ERROR_SERVER_SID_MISMATCH = 628

The server process is running under a SID different than that required by the client.

ERROR_SERVICE_ALREADY_RUNNING = 1056

An instance of the service is already running.

ERROR_SERVICE_CANNOT_ACCEPT_CTRL = 1061

The service cannot accept control messages at this time.

ERROR_SERVICE_DATABASE_LOCKED = 1055

The service database is locked.

ERROR_SERVICE_DEPENDENCY_DELETED = 1075

The dependency service does not exist or has been marked for deletion.

ERROR_SERVICE_DEPENDENCY_FAIL = 1068

The dependency service or group failed to start.

ERROR_SERVICE_DISABLED = 1058

The service cannot be started, either because it is disabled or because it has no enabled devices associated with it.

ERROR_SERVICE_DOES_NOT_EXIST = 1060

The specified service does not exist as an installed service.

ERROR_SERVICE_EXISTS = 1073

The specified service already exists.

ERROR_SERVICE_LOGON_FAILED = 1069

The service did not start due to a logon failure.

ERROR_SERVICE_MARKED_FOR_DELETE = 1072

The specified service has been marked for deletion.

ERROR_SERVICE_NEVER_STARTED = 1077

No attempts to start the service have been made since the last boot.

ERROR_SERVICE_NOTIFY_CLIENT_LAGGING = 1294

The service notification client is lagging too far behind the current state of services in the machine.

ERROR_SERVICE_NOT_ACTIVE = 1062

The service has not been started.

ERROR_SERVICE_NOT_FOUND = 1243

The specified service does not exist.

ERROR_SERVICE_NOT_IN_EXE = 1083

The executable program that this service is configured to run in does not implement the service.

ERROR_SERVICE_NO_THREAD = 1054

A thread could not be created for the service.

ERROR_SERVICE_REQUEST_TIMEOUT = 1053

The service did not respond to the start or control request in a timely fashion.

ERROR_SERVICE_SPECIFIC_ERROR = 1066

The service has returned a service-specific error code.

ERROR_SERVICE_START_HANG = 1070

After starting, the service stopped responding in a start-pending state.

ERROR_SESSION_CREDENTIAL_CONFLICT = 1219

Multiple connections to a server or shared resource by the same user, using more than one user name, are not allowed. Disconnect all previous connections to the server or shared resource and try again.

ERROR_SETCOUNT_ON_BAD_LB = 1433

LB_SETCOUNT sent to non-lazy list box.

ERROR_SETMARK_DETECTED = 1103

A tape access reached the end of a set of files.

ERROR_SET_NOT_FOUND = 1170

The property set specified does not exist on the object.

ERROR_SET_POWER_STATE_FAILED = 1141

The system BIOS failed an attempt to change the system power state.

ERROR_SET_POWER_STATE_VETOED = 1140

An attempt to change the system power state was vetoed by another application or driver.

ERROR_SHARED_POLICY = 8218

The policy object is shared and can only be modified at the root.

ERROR_SHARING_BUFFER_EXCEEDED = 36

Too many files opened for sharing.

ERROR_SHARING_PAUSED = 70

The remote server has been paused or is in the process of being started.

ERROR_SHARING_VIOLATION = 32

The process cannot access the file because it is being used by another process.

ERROR_SHUTDOWN_CLUSTER = 5008

The cluster is being shut down.

ERROR_SHUTDOWN_IN_PROGRESS = 1115

A system shutdown is in progress.

ERROR_SHUTDOWN_IS_SCHEDULED = 1190

A system shutdown has already been scheduled.

ERROR_SHUTDOWN_USERS_LOGGED_ON = 1191

The system shutdown cannot be initiated because there are other users logged on to the computer.

ERROR_SIGNAL_PENDING = 162

A signal is already pending.

ERROR_SIGNAL_REFUSED = 156

The recipient process has refused the signal.

ERROR_SINGLE_INSTANCE_APP = 1152

Cannot start more than one instance of the specified program.

ERROR_SMARTCARD_SUBSYSTEM_FAILURE = 1264

The Kerberos protocol encountered an error while attempting to utilize the smartcard subsystem.

ERROR_SOME_NOT_MAPPED = 1301

Some mapping between account names and SIDs was not done.

ERROR_SOURCE_ELEMENT_EMPTY = 1160

The indicated source element has no media.

ERROR_SPARSE_NOT_ALLOWED_IN_TRANSACTION = 6844

The sparse operation could not be completed because a transaction is active on the file.

ERROR_SPECIAL_ACCOUNT = 1371

Cannot perform this operation on built-in accounts.

ERROR_SPECIAL_GROUP = 1372

Cannot perform this operation on this built-in special group.

ERROR_SPECIAL_USER = 1373

Cannot perform this operation on this built-in special user.

ERROR_SPL_NO_ADDJOB = 3004

An AddJob call was not issued.

ERROR_SPL_NO_STARTDOC = 3003

A StartDocPrinter call was not issued.

ERROR_SPOOL_FILE_NOT_FOUND = 3002

The spool file was not found.

ERROR_STACK_BUFFER_OVERRUN = 1282

The system detected an overrun of a stack-based buffer in this application. This overrun could potentially allow a malicious user to gain control of this application.

ERROR_STACK_OVERFLOW = 1001

Recursion too deep; the stack overflowed.

ERROR_STACK_OVERFLOW_READ = 599

The request must be handled by the stack overflow code.

ERROR_STATIC_INIT = 4002

The importation from the file failed.

ERROR_STOPPED_ON_SYMLINK = 681

The create operation stopped after reaching a symbolic link.

ERROR_STREAM_MINIVERSION_NOT_FOUND = 6808

The specified file miniversion was not found for this transacted file open.

ERROR_STREAM_MINIVERSION_NOT_VALID = 6809

The specified file miniversion was found but has been invalidated. The most likely cause is a transaction savepoint rollback.

ERROR_SUBST_TO_JOIN = 141

The system tried toSUBSTa drive to a directory on a joined drive.

ERROR_SUBST_TO_SUBST = 139

The system tried to substitute a drive to a directory on a substituted drive.

ERROR_SUCCESS_REBOOT_INITIATED = 1641

The requested operation completed successfully. The system will be restarted so the changes can take effect.

ERROR_SUCCESS_REBOOT_REQUIRED = 3010

The requested operation is successful. Changes will not be effective until the system is rebooted.

ERROR_SUCCESS_RESTART_REQUIRED = 3011

The requested operation is successful. Changes will not be effective until the service is restarted.

ERROR_SWAPERROR = 999

Error performing in-page operation.

ERROR_SXS_ACTIVATION_CONTEXT_DISABLED = 14006

The application attempted to activate a disabled activation context.

ERROR_SXS_ASSEMBLY_IS_NOT_A_DEPLOYMENT = 14103

The assembly is not a deployment.

ERROR_SXS_ASSEMBLY_MISSING = 14081

The referenced assembly could not be found.

ERROR_SXS_ASSEMBLY_NOT_FOUND = 14003

The referenced assembly is not installed on your system.

ERROR_SXS_ASSEMBLY_NOT_LOCKED = 14097

The component must be locked before making the request.

ERROR_SXS_CANT_GEN_ACTCTX = 14001

The application has failed to start because its side-by-side configuration is incorrect. See the application event log for more detail.

ERROR_SXS_COMPONENT_STORE_CORRUPT = 14098

The component store has been corrupted.

ERROR_SXS_CORRUPTION = 14083

The application isolation metadata for this process or thread has become corrupt.

ERROR_SXS_CORRUPT_ACTIVATION_STACK = 14082

The activation context activation stack for the running thread of execution is corrupt.

ERROR_SXS_DUPLICATE_ASSEMBLY_NAME = 14027

Two or more components referenced directly or indirectly by the application manifest are different versions of the same component, which is not permitted.

ERROR_SXS_DUPLICATE_CLSID = 14023

Two or more components referenced directly or indirectly by the application manifest have the same COM server CLSIDs.

ERROR_SXS_DUPLICATE_DLL_NAME = 14021

Two or more components referenced directly or indirectly by the application manifest have files by the same name.

ERROR_SXS_DUPLICATE_IID = 14024

Two or more components referenced directly or indirectly by the application manifest have proxies for the same COM interface IIDs.

ERROR_SXS_DUPLICATE_PROGID = 14026

Two or more components referenced directly or indirectly by the application manifest have the same COM ProgIDs.

ERROR_SXS_DUPLICATE_TLBID = 14025

Two or more components referenced directly or indirectly by the application manifest have the same COM type library TLBIDs.

ERROR_SXS_DUPLICATE_WINDOWCLASS_NAME = 14022

Two or more components referenced directly or indirectly by the application manifest have window classes with the same name.

ERROR_SXS_EARLY_DEACTIVATION = 14084

The activation context being deactivated is not the most recently activated one.

ERROR_SXS_FILE_HASH_MISMATCH = 14028

A component's file does not match the verification information present in the component manifest.

ERROR_SXS_FILE_NOT_PART_OF_ASSEMBLY = 14104

The file is not a part of the assembly.

ERROR_SXS_IDENTITIES_DIFFERENT = 14102

The component identities are different.

ERROR_SXS_IDENTITY_DUPLICATE_ATTRIBUTE = 14092

An identity contains two definitions for the same attribute.

ERROR_SXS_IDENTITY_PARSE_ERROR = 14093

The identity string is malformed. This might be due to a trailing comma, more than two unnamed attributes, a missing attribute name, or a missing attribute value.

ERROR_SXS_INCORRECT_PUBLIC_KEY_OKEN = 14095

The public key token does not correspond to the public key specified.

ERROR_SXS_INVALID_ACTCTXDATA_FORMAT = 14002

The application binding data format is invalid.

ERROR_SXS_INVALID_ASSEMBLY_IDENTITY_ATTRIBUTE = 14017

The manifest contains an attribute for the assembly identity that is not valid.

ERROR_SXS_INVALID_ASSEMBLY_IDENTITY_ATTRIBUTE_NAME = 14080

The supplied assembly identity has one or more attribute names that contain characters not permitted in XML names.

ERROR_SXS_INVALID_DEACTIVATION = 14085

The activation context being deactivated is not active for the current thread of execution.

ERROR_SXS_INVALID_IDENTITY_ATTRIBUTE_NAME = 14091

The name of an attribute in an identity is not within the legal range.

ERROR_SXS_INVALID_IDENTITY_ATTRIBUTE_VALUE = 14090

The value of an attribute in an identity is not within the legal range.

ERROR_SXS_INVALID_XML_NAMESPACE_URI = 14014

The manifest contains a reference to an invalid URI.

ERROR_SXS_KEY_NOT_FOUND = 14007

The requested lookup key was not found in any active activation context.

ERROR_SXS_LEAF_MANIFEST_DEPENDENCY_NOT_INSTALLED = 14016

The manifest for an assembly used by the application has a reference to a dependent assembly that is not installed.

ERROR_SXS_MANIFEST_FORMAT_ERROR = 14004

The manifest file does not begin with the required tag and format information.

ERROR_SXS_MANIFEST_IDENTITY_SAME_BUT_CONTENTS_DIFFERENT = 14101

The identities of the manifests are identical, but the contents are different.

ERROR_SXS_MANIFEST_INVALID_REQUIRED_DEFAULT_NAMESPACE = 14019

The manifest has a default namespace specified on the assembly element but its value is not urn:schemas-microsoft-com:asm.v1".

ERROR_SXS_MANIFEST_MISSING_REQUIRED_DEFAULT_NAMESPACE = 14018

The manifest is missing the required default namespace specification on the assembly element.

ERROR_SXS_MANIFEST_PARSE_ERROR = 14005

The manifest file contains one or more syntax errors.

ERROR_SXS_MANIFEST_TOO_BIG = 14105

The size of the manifest exceeds the maximum allowed.

ERROR_SXS_MISSING_ASSEMBLY_IDENTITY_ATTRIBUTE = 14079

The supplied assembly identity is missing one or more attributes that must be present in this context.

ERROR_SXS_MULTIPLE_DEACTIVATION = 14086

The activation context being deactivated has already been deactivated.

ERROR_SXS_POLICY_PARSE_ERROR = 14029

The policy manifest contains one or more syntax errors.

ERROR_SXS_PRIVATE_MANIFEST_CROSS_PATH_WITH_REPARSE_POINT = 14020

The private manifest probed has crossed the reparse-point-associated path.

ERROR_SXS_PROCESS_DEFAULT_ALREADY_SET = 14011

An attempt to set the process default activation context failed because the process default activation context was already set.

ERROR_SXS_PROCESS_TERMINATION_REQUESTED = 14087

A component used by the isolation facility has requested to terminate the process.

ERROR_SXS_PROTECTION_CATALOG_FILE_MISSING = 14078

Assembly Protection Error: The catalog for an assembly is missing.

ERROR_SXS_PROTECTION_CATALOG_NOT_VALID = 14076

Assembly Protection Error: The catalog for an assembly is not valid, or does not match the assembly's manifest.

ERROR_SXS_PROTECTION_PUBLIC_KEY_TOO_SHORT = 14075

Assembly Protection Error: The public key for an assembly was too short to be allowed.

ERROR_SXS_PROTECTION_RECOVERY_FAILED = 14074

Assembly Protection Error: Unable to recover the specified assembly.

ERROR_SXS_RELEASE_ACTIVATION_CONTEXT = 14088

A kernel mode component is releasing a reference on an activation context.

ERROR_SXS_ROOT_MANIFEST_DEPENDENCY_OT_INSTALLED = 14015

The application manifest contains a reference to a dependent assembly that is not installed.

ERROR_SXS_SECTION_NOT_FOUND = 14000

The requested section was not present in the activation context.

ERROR_SXS_SETTING_NOT_REGISTERED = 14106

The setting is not registered.

ERROR_SXS_SYSTEM_DEFAULT_ACTIVATION_CONTEXT_EMPTY = 14089

The activation context of the system default assembly could not be generated.

ERROR_SXS_THREAD_QUERIES_DISABLED = 14010

Lack of system resources has required isolated activation to be disabled for the current thread of execution.

ERROR_SXS_TRANSACTION_CLOSURE_INCOMPLETE = 14107

One or more required members of the transaction are not present.

ERROR_SXS_UNKNOWN_ENCODING = 14013

The encoding requested is not recognized.

ERROR_SXS_UNKNOWN_ENCODING_GROUP = 14012

The encoding group identifier specified is not recognized.

ERROR_SXS_UNTRANSLATABLE_HResult = 14077

An HRESULT could not be translated to a corresponding Win32 error code.

ERROR_SXS_VERSION_CONFLICT = 14008

A component version required by the application conflicts with another active component version.

ERROR_SXS_WRONG_SECTION_TYPE = 14009

The type requested activation context section does not match the query API used.

ERROR_SXS_XML_E_BADCHARDATA = 14036

Manifest Parse Error: An Invalid character was found in text content.

ERROR_SXS_XML_E_BADCHARINSTRING = 14034

Manifest Parse Error: A string literal contained an invalid character.

ERROR_SXS_XML_E_BADNAMECHAR = 14033

Manifest Parse Error: A name contained an invalid character.

ERROR_SXS_XML_E_BADPEREFINSUBSET = 14059

Manifest Parse Error: Parameter entities cannot be used inside markup declarations in an internal subset.

ERROR_SXS_XML_E_BADSTARTNAMECHAR = 14032

Manifest Parse Error: A name started with an invalid character.

ERROR_SXS_XML_E_BADXMLCASE = 14069

Manifest Parse Error: The name "xml" is reserved and must be lowercase.

ERROR_SXS_XML_E_BADXMLDECL = 14056

Manifest Parse Error: Invalid XML declaration.

ERROR_SXS_XML_E_COMMENTSYNTAX = 14031

Manifest Parse Error: Incorrect syntax was used in a comment.

ERROR_SXS_XML_E_DUPLICATEATTRIBUTE = 14053

Manifest Parse Error: Duplicate attribute.

ERROR_SXS_XML_E_EXPECTINGCLOSEQUOTE = 14045

Manifest Parse Error: A single (') or double (") quotation mark is missing.

ERROR_SXS_XML_E_EXPECTINGTAGEND = 14038

Manifest Parse Error: The angle bracket (>) character was expected.

ERROR_SXS_XML_E_INCOMPLETE_ENCODING = 14043

Manifest Parse Error: End of file reached in invalid state for current encoding.

ERROR_SXS_XML_E_INTERNALERROR = 14041

Manifest Parse Error: Internal error.

ERROR_SXS_XML_E_INVALIDATROOTLEVEL = 14055

Manifest Parse Error: Invalid at the top level of the document.

ERROR_SXS_XML_E_INVALIDENCODING = 14067

Manifest Parse Error: System does not support the specified encoding.

ERROR_SXS_XML_E_INVALIDSWITCH = 14068

Manifest Parse Error: Switch from current encoding to specified encoding not supported.

ERROR_SXS_XML_E_INVALID_DECIMAL = 14047

Manifest Parse Error: Invalid character for decimal digit.

ERROR_SXS_XML_E_INVALID_HEXIDECIMAL = 14048

Manifest Parse Error: Invalid character for hexadecimal digit.

ERROR_SXS_XML_E_INVALID_STANDALONE = 14070

Manifest Parse Error: The stand-alone attribute must have the value "yes" or "no".

ERROR_SXS_XML_E_INVALID_UNICODE = 14049

Manifest Parse Error: Invalid Unicode character value for this platform.

ERROR_SXS_XML_E_INVALID_VERSION = 14072

Manifest Parse Error: Invalid version number.

ERROR_SXS_XML_E_MISSINGEQUALS = 14073

Manifest Parse Error: Missing equal sign (=) between the attribute and the attribute value.

ERROR_SXS_XML_E_MISSINGQUOTE = 14030

Manifest Parse Error: A string literal was expected, but no opening quotation mark was found.

ERROR_SXS_XML_E_MISSINGROOT = 14057

Manifest Parse Error: XML document must have a top-level element.

ERROR_SXS_XML_E_MISSINGSEMICOLON = 14039

Manifest Parse Error: A semicolon (;) was expected.

ERROR_SXS_XML_E_MISSINGWHITESPACE = 14037

Manifest Parse Error: Required white space was missing.

ERROR_SXS_XML_E_MISSING_PAREN = 14044

Manifest Parse Error: Missing parenthesis.

ERROR_SXS_XML_E_MULTIPLEROOTS = 14054

Manifest Parse Error: Only one top-level element is allowed in an XML document.

ERROR_SXS_XML_E_MULTIPLE_COLONS = 14046

Manifest Parse Error: Multiple colons are not allowed in a name.

ERROR_SXS_XML_E_RESERVEDNAMESPACE = 14066

Manifest Parse Error: The namespace prefix is not allowed to start with the reserved string xml".

ERROR_SXS_XML_E_UNBALANCEDPAREN = 14040

Manifest Parse Error: Unbalanced parentheses.

ERROR_SXS_XML_E_UNCLOSEDCDATA = 14065

Manifest Parse Error: A CDATA section was not closed.

ERROR_SXS_XML_E_UNCLOSEDCOMMENT = 14063

Manifest Parse Error: A comment was not closed.

ERROR_SXS_XML_E_UNCLOSEDDECL = 14064

Manifest Parse Error: A declaration was not closed.

ERROR_SXS_XML_E_UNCLOSEDENDTAG = 14061

Manifest Parse Error: End element was missing the angle bracket (>) character.

ERROR_SXS_XML_E_UNCLOSEDSTARTTAG = 14060

Manifest Parse Error: Element was not closed.

ERROR_SXS_XML_E_UNCLOSEDSTRING = 14062

Manifest Parse Error: A string literal was not closed.

ERROR_SXS_XML_E_UNCLOSEDTAG = 14052

Manifest Parse Error: The following tags were not closed: %1.

ERROR_SXS_XML_E_UNEXPECTEDENDTAG = 14051

Manifest Parse Error: End tag was not expected at this location.

ERROR_SXS_XML_E_UNEXPECTEDEOF = 14058

Manifest Parse Error: Unexpected end of file.

ERROR_SXS_XML_E_UNEXPECTED_STANDALONE = 14071

Manifest Parse Error: The stand-alone attribute cannot be used in external entities.

ERROR_SXS_XML_E_UNEXPECTED_WHITESPACE = 14042

Manifest Parse Error: Whitespace is not allowed at this location.

ERROR_SXS_XML_E_WHITESPACEORQUESTIONMARK = 14050

Manifest Parse Error: Expecting whitespace or question mark (?).

ERROR_SXS_XML_E_XMLDECLSYNTAX = 14035

Manifest Parse Error: Invalid syntax for an XML declaration.

ERROR_SYMLINK_CLASS_DISABLED = 1463

The symbolic link cannot be followed because its type is disabled.

ERROR_SYMLINK_NOT_SUPPORTED = 1464

This application does not support the current operation on symbolic links.

ERROR_SYNCHRONIZATION_REQUIRED = 569

On applicable Windows Server releases, the Security Accounts Manager (SAM) database is significantly out of synchronization with the copy on the domain controller. A complete synchronization is required.

ERROR_SYNC_FOREGROUND_REFRESH_REQUIRED = 1274

The Group Policy framework should call the extension in the synchronous foreground policy refresh.

ERROR_SYSTEM_DEVICE_NOT_FOUND = 15299

The requested system device cannot be found.

ERROR_SYSTEM_HIVE_TOO_LARGE = 653

The system hive size has exceeded its limit.

ERROR_SYSTEM_IMAGE_BAD_SIGNATURE = 637

{Fatal System Error} The system image %s is not properly signed. The file has been replaced with the signed file. The system has been shut down.

ERROR_SYSTEM_POWERSTATE_COMPLEX_TRANSITION = 783

The system power state is transitioning from %2 to %3 but could enter %4.

ERROR_SYSTEM_POWERSTATE_TRANSITION = 782

The system power state is transitioning from %2 to %3.

ERROR_SYSTEM_PROCESS_TERMINATED = 591

{Fatal System Error} The %hs system process terminated unexpectedly with a status of 0x%08x (0x%08x 0x%08x). The system has been shut down.

ERROR_SYSTEM_SHUTDOWN = 641

The system is in the process of shutting down.

ERROR_SYSTEM_TRACE = 150

System trace information was not specified in your CONFIG.SYS file, or tracing is disallowed.

ERROR_TAG_NOT_FOUND = 2012

The specified tag was not found.

ERROR_TAG_NOT_PRESENT = 2013

A required tag is not present.

ERROR_THREAD_1_INACTIVE = 210

The signal handler cannot be set.

ERROR_THREAD_ALREADY_IN_TASK = 1552

The specified thread is already joining a task.

ERROR_THREAD_MODE_ALREADY_BACKGROUND = 400

The thread is already in background processing mode.

ERROR_THREAD_MODE_NOT_BACKGROUND = 401

The thread is not in background processing mode.

ERROR_THREAD_NOT_IN_PROCESS = 566

An attempt was made to operate on a thread within a specific process, but the thread specified is not in the process specified.

ERROR_THREAD_WAS_SUSPENDED = 699

{Thread Suspended} A thread termination occurred while the thread was suspended. The thread was resumed and termination proceeded.

ERROR_TIMEOUT = 1460

This operation returned because the time-out period expired.

ERROR_TIMER_NOT_CANCELED = 541

An attempt was made to cancel or set a timer that has an associated asynchronous procedure call (APC) and the subject thread is not the thread that originally set the timer with an associated APC routine.

ERROR_TIMER_RESOLUTION_NOT_SET = 607

The timer resolution was not previously set by the current process.

ERROR_TIMER_RESUME_IGNORED = 722

The resume flag to a timer API was ignored.

ERROR_TIME_SKEW = 1398

There is a time and/or date difference between the client and server.

ERROR_TLW_WITH_WSCHILD = 1406

Cannot create a top-level child window.

ERROR_TM_IDENTITY_MISMATCH = 6845

The call to create a transaction manager object failed because the Tm Identity stored in the logfile does not match the Tm Identity that was passed in as an argument.

ERROR_TM_INITIALIZATION_FAILED = 6706

The transaction manager was unable to be successfully initialized. Transacted operations are not supported.

ERROR_TM_VOLATILE = 6828

The operation could not be completed because the transaction manager does not have a log.

ERROR_TOKEN_ALREADY_IN_USE = 1375

The token is already in use as a primary token.

ERROR_TOO_MANY_CMDS = 56

The network BIOS command limit has been reached.

ERROR_TOO_MANY_CONTEXT_IDS = 1384

During a logon attempt, the user's security context accumulated too many SIDs.

ERROR_TOO_MANY_LINKS = 1142

An attempt was made to create more links on a file than the file system supports.

ERROR_TOO_MANY_LUIDS_REQUESTED = 1333

Too many local user identifiers (LUIDs) were requested at one time.

ERROR_TOO_MANY_MODULES = 214

Too many dynamic-link modules are attached to this program or dynamic-link module.

ERROR_TOO_MANY_MUXWAITERS = 152

DosMuxSemWait did not execute; too many semaphores are already set.

ERROR_TOO_MANY_NAMES = 68

The name limit for the local computer network adapter card was exceeded.

ERROR_TOO_MANY_OPEN_FILES = 4

The system cannot open the file.

ERROR_TOO_MANY_POSTS = 298

Too many posts were made to a semaphore.

ERROR_TOO_MANY_SECRETS = 1381

The maximum number of secrets that can be stored in a single system has been exceeded.

ERROR_TOO_MANY_SEMAPHORES = 100

Cannot create another system semaphore.

ERROR_TOO_MANY_SEM_REQUESTS = 103

The semaphore cannot be set again.

ERROR_TOO_MANY_SESS = 69

The network BIOS session limit was exceeded.

ERROR_TOO_MANY_SIDS = 1389

Too many SIDs have been specified.

ERROR_TOO_MANY_TCBS = 155

Cannot create another thread.

ERROR_TOO_MANY_THREADS = 565

Indicates a process has too many threads to perform the requested action. For example, assignment of a primary token can be performed only when a process has zero or one threads.

ERROR_TRANSACTED_MAPPING_UNSUPPORTED_REMOTE = 6834

Memory mapping (creating a mapped section) to a remote file under a transaction is not supported.

ERROR_TRANSACTIONAL_CONFLICT = 6800

The function attempted to use a name that is reserved for use by another transaction.

ERROR_TRANSACTIONAL_OPEN_NOT_ALLOWED = 6832

This object is not allowed to be opened in a transaction.

ERROR_TRANSACTIONMANAGER_NOT_FOUND = 6718

The specified transaction manager object could not be opened because it was not found.

ERROR_TRANSACTIONMANAGER_NOT_ONLINE = 6719

The specified resource manager was unable to create an enlistment because its associated transaction manager is not online.

ERROR_TRANSACTIONMANAGER_RECOVERY_NAME_COLLISION = 6720

The specified transaction manager was unable to create the objects contained in its log file in the ObjectB namespace. Therefore, the transaction manager was unable to recover.

ERROR_TRANSACTIONS_NOT_FROZEN = 6839

The request to thaw frozen transactions was ignored because transactions were not previously frozen.

ERROR_TRANSACTIONS_UNSUPPORTED_REMOTE = 6805

The remote server or share does not support transacted file operations.

ERROR_TRANSACTION_ALREADY_ABORTED = 6704

It is too late to perform the requested operation because the transaction has already been aborted.

ERROR_TRANSACTION_ALREADY_COMMITTED = 6705

It is too late to perform the requested operation because the transaction has already been committed.

ERROR_TRANSACTION_FREEZE_IN_PROGRESS = 6840

Transactions cannot be frozen because a freeze is already in progress.

ERROR_TRANSACTION_INVALID_MARSHALL_BUFFER = 6713

The buffer passed in to PushTransaction or PullTransaction is not in a valid format.

ERROR_TRANSACTION_NOT_ACTIVE = 6701

The requested operation was made in the context of a transaction that is no longer active.

ERROR_TRANSACTION_NOT_FOUND = 6715

The specified transaction object could not be opened because it was not found.

ERROR_TRANSACTION_NOT_JOINED = 6708

The resource manager has attempted to prepare a transaction that it has not successfully joined.

ERROR_TRANSACTION_NOT_REQUESTED = 6703

The caller has called a response API, but the response is not expected because the transaction manager did not issue the corresponding request to the caller.

ERROR_TRANSACTION_PROPAGATION_FAILED = 6711

The attempt to propagate the transaction failed.

ERROR_TRANSACTION_REQUEST_NOT_VALID = 6702

The requested operation is not valid on the transaction object in its current state.

ERROR_TRANSACTION_REQUIRED_PROMOTION = 6837

Promotion was required to allow the resource manager to enlist, but the transaction was set to disallow it.

ERROR_TRANSACTION_SCOPE_CALLBACKS_NOT_SET = 6836

A transaction scope could not be entered because the scope handler has not been initialized.

ERROR_TRANSACTION_SUPERIOR_EXISTS = 6709

The transaction object already has a superior enlistment, and the caller attempted an operation that would have created a new superior. Only a single superior enlistment is allowed.

ERROR_TRANSFORM_NOT_SUPPORTED = 2004

The requested transformation operation is not supported.

ERROR_TRANSLATION_COMPLETE = 757

The translator has translated these resources into the global space and no further translations should be performed.

ERROR_TRANSPORT_FULL = 4328

Cannot use the transport because it is already in use.

ERROR_TRUSTED_DOMAIN_FAILURE = 1788

The trust relationship between the primary domain and the trusted domain failed.

ERROR_TRUSTED_RELATIONSHIP_FAILURE = 1789

The trust relationship between this workstation and the primary domain failed.

ERROR_TRUST_FAILURE = 1790

The network logon failed.

ERROR_TS_INCOMPATIBLE_SESSIONS = 7069

The target session is incompatible with the current session.

ERROR_TXF_ATTRIBUTE_CORRUPT = 6830

The transactional metadata attribute on the file or directory is corrupt and unreadable.

ERROR_TXF_DIR_NOT_EMPTY = 6826

The \$Txf directory must be empty for this operation to succeed.

ERROR_TXF_METADATA_ALREADY_PRESENT = 6835

Transaction metadata is already present on this file and cannot be superseded.

ERROR_UNABLE_TO_CLEAN = 4311

The drive cannot be cleaned or does not support cleaning.

ERROR_UNABLE_TO_EJECT_MOUNTED_MEDIA = 4330

Unable to eject the medium because it is in a drive.

ERROR_UNABLE_TO_INVENTORY_DRIVE = 4325

Unable to retrieve the drive status.

ERROR_UNABLE_TO_INVENTORY_SLOT = 4326

Unable to retrieve the slot status.

ERROR_UNABLE_TO_INVENTORY_TRANSPORT = 4327

Unable to retrieve status about the transport.

ERROR_UNABLE_TO_LOAD_MEDIUM = 4324

Unable to load the medium into the drive.

ERROR_UNABLE_TO_LOCK_MEDIA = 1108

Unable to lock the media eject mechanism.

ERROR_UNABLE_TO_MOVE_REPLACEMENT = 1176

Unable to move the replacement file to the file to be replaced. The file to be replaced has retained its original name.

ERROR_UNABLE_TO_MOVE_REPLACEMENT_2 = 1177

Unable to move the replacement file to the file to be replaced. The file to be replaced has been renamed using the backup name.

ERROR_UNABLE_TO_REMOVE_REPLACED = 1175

Unable to remove the file to be replaced.

ERROR_UNABLE_TO_UNLOAD_MEDIA = 1109

Unable to unload the media.

ERROR_UNDEFINED_CHARACTER = 583

The Unicode character is not defined in the Unicode character set installed on the system.

ERROR_UNEXPECTED_MM_CREATE_ERR = 556

If an MM error is returned that is not defined in the standard FsRtl filter, it is converted to one of the following errors that is guaranteed to be in the filter. In this case, information is lost; however, the filter correctly handles the exception.

ERROR_UNEXPECTED_MM_EXTEND_ERR = 558

If an MM error is returned that is not defined in the standard FsRtl filter, it is converted to one of the following errors that is guaranteed to be in the filter. In this case, information is lost; however, the filter correctly handles the exception.

ERROR_UNEXPECTED_MM_MAP_ERROR = 557

If an MM error is returned that is not defined in the standard FsRtl filter, it is converted to one of the following errors that is guaranteed to be in the filter. In this case, information is lost; however, the filter correctly handles the exception.

ERROR_UNEXPECTED_OMID = 4334

Unexpected on-medium identifier.

ERROR_UNEXP_NET_ERR = 59

An unexpected network error occurred.

ERROR_UNHANDLED_EXCEPTION = 574

{Application Error} The exception %s (0x%08lx) occurred in the application at location 0x%08lx.

ERROR_UNIDENTIFIED_ERROR = 1287

Insufficient information exists to identify the cause of failure.

ERROR_UNKNOWN_COMPONENT = 1607

Component ID not registered.

ERROR_UNKNOWN_FEATURE = 1606

Feature ID not registered.

ERROR_UNKNOWN_PATCH = 1647

The update is not applied to this product.

ERROR_UNKNOWN_PORT = 1796

The specified port is unknown.

ERROR_UNKNOWN_PRINTER_DRIVER = 1797

The printer driver is unknown.

ERROR_UNKNOWN_PRINTPROCESSOR = 1798

The print processor is unknown.

ERROR_UNKNOWN_PRINT_MONITOR = 3000

The specified print monitor is unknown.

ERROR_UNKNOWN_PRODUCT = 1605

This action is valid only for products that are currently installed.

ERROR_UNKNOWN_PROPERTY = 1608

Unknown property.

ERROR_UNKNOWN_REVISION = 1305

The revision level is unknown.

ERROR_UNMAPPED_SUBSTITUTION_STRING = 14096

A substitution string had no mapping.

ERROR_UNRECOGNIZED_MEDIA = 1785

The disk media is not recognized. It might not be formatted.

ERROR_UNRECOGNIZED_VOLUME = 1005

The volume does not contain a recognized file system. Be sure that all required file system drivers are loaded and that the volume is not corrupted.

ERROR_UNSUPPORTED_COMPRESSION = 618

The specified compression format is unsupported.

ERROR_UNSUPPORTED_TYPE = 1630

Data of this type is not supported.

ERROR_UNWIND = 542

Unwind exception code.

ERROR_UNWIND_CONSOLIDATE = 684

A frame consolidation has been executed.

ERROR_USER_DELETE_TRUST_QUOTA_EXCEEDED = 1934

The current user's delegated trust deletion quota has been exceeded.

ERROR_USER_EXISTS = 1316

The specified account already exists.

ERROR_USER_MAPPED_FILE = 1224

The requested operation cannot be performed on a file with a user-mapped section open.

ERROR_USER_PROFILE_LOAD = 500

User profile cannot be loaded.

ERROR_VALIDATE_CONTINUE = 625

The validation process needs to continue on to the next step.

ERROR_VC_DISCONNECTED = 240

The session was canceled.

ERROR_VDM_DISALLOWED = 1286

%1 is a 16-bit application. You do not have permissions to execute 16-bit applications. Check your permissions with your system administrator.

ERROR_VDM_HARD_ERROR = 593

The NT Virtual DOS Machine (NTVDM) encountered a hard error.

ERROR_VERIFIER_STOP = 537

Application verifier has found an error in the current process.

ERROR_VERSION_PARSE_ERROR = 777

A version number could not be parsed.

ERROR_VIRUS_DELETED = 226

This file contains a virus and cannot be opened. Due to the nature of this virus, the file has been removed from this location.

ERROR_VIRUS_INFECTED = 225

Operation did not complete successfully because the file contains a virus.

ERROR_VOLSnap_HIBERNATE_READY = 761

{Volume Shadow Copy Service} The system is now ready for hibernation.

ERROR_VOLSnap_PREPARE_HIBERNATE = 655

{Volume Shadow Copy Service} Wait while the Volume Shadow Copy Service prepares volume %hs for hibernation.

ERROR_VOLUME_CONTAINS_SYS_FILES = 4337

The volume contains system or paging files.

ERROR_VOLUME_MOUNTED = 743

A new volume has been mounted by a file system.

ERROR_VOLUME_NOT_SIS_ENABLED = 4500

Single Instance Storage (SIS) is not available on this volume.

ERROR_VOLUME_NOT_SUPPORT_EFS = 6014

The disk partition does not support file encryption.

ERROR_WAIT_FOR_OPLOCK = 765

An operation is blocked waiting for an oplock.

ERROR_WAIT_NO_CHILDREN = 128

There are no child processes to wait for.

ERROR_WAKE_SYSTEM = 730

The system has awoken.

ERROR_WAKE_SYSTEM_DEBUGGER = 675

{Kernel Debugger Awakened} the system debugger was awakened by an interrupt.

ERROR_WAS_LOCKED = 717

{Page Locked} One of the pages to lock was already locked.

ERROR_WAS_UNLOCKED = 715

{Page Unlocked} The page protection of a locked page was changed to 'No Access' and the page was unlocked from memory and from the process.

ERROR_WINDOW_NOT_COMBOBOX = 1423

The window is not a combo box.

ERROR_WINDOW_NOT_DIALOG = 1420

The window is not a valid dialog window.

ERROR_WINDOW_OF_OTHER_THREAD = 1408

Invalid window; it belongs to other thread.

ERROR_WINS_INTERNAL = 4000

Windows Internet Name Service (WINS) encountered an error while processing the command.

ERROR_WMI_ALREADY_DISABLED = 4212

The WMI data block or event notification has already been disabled.

ERROR_WMI_ALREADY_ENABLED = 4206

The WMI data block or event notification has already been enabled.

ERROR_WMI_DP_FAILED = 4209

The WMI data provider failed to carry out the request.

ERROR_WMI_DP_NOT_FOUND = 4204

The WMI data provider could not be located.

ERROR_WMI_GUID_DISCONNECTED = 4207

The WMI data block is no longer available.

ERROR_WMI_GUID_NOT_FOUND = 4200

The GUID passed was not recognized as valid by a WMI data provider.

ERROR_WMI_INSTANCE_NOT_FOUND = 4201

The instance name passed was not recognized as valid by a WMI data provider.

ERROR_WMI_INVALID_MOF = 4210

The WMI Managed Object Format (MOF) information is not valid.

ERROR_WMI_INVALID_REGINFO = 4211

The WMI registration information is not valid.

ERROR_WMI_ITEMID_NOT_FOUND = 4202

The data item ID passed was not recognized as valid by a WMI data provider.

ERROR_WMI_READ_ONLY = 4213

The WMI data item or data block is read-only.

ERROR_WMI_SERVER_UNAVAILABLE = 4208

The WMI data service is not available.

ERROR_WMI_SET_FAILURE = 4214

The WMI data item or data block could not be changed.

ERROR_WMI_TRY_AGAIN = 4203

The WMI request could not be completed and should be retried.

ERROR_WMI_UNRESOLVED_INSTANCE_REF = 4205

The WMI data provider references an instance set that has not been registered.

ERROR_WORKING_SET_QUOTA = 1453

Insufficient quota to complete the requested service.

ERROR_WOW_ASSERTION = 670

WOW assertion error.

ERROR_WRITE_FAULT = 29

The system cannot write to the specified device.

ERROR_WRITE_PROTECT = 19

The media is write-protected.

ERROR_WRONG_DISK = 34

The wrong disk is in the drive. Insert %2 (Volume Serial Number: %3) into drive %1.

ERROR_WRONG_EFS = 6005

The file was encrypted with a different encryption driver than is currently loaded.

ERROR_WRONG_PASSWORD = 1323

Unable to update the password. The value provided as the current password is incorrect.

ERROR_WRONG_TARGET_NAME = 1396

Logon failure: The target account name is incorrect.

ERROR_WX86_ERROR = 540

An error occurred in the WX86 subsystem.

ERROR_WX86_WARNING = 539

A warning occurred in the WX86 subsystem.

ERROR_XML_ENCODING_MISMATCH = 14100

The character encoding in the XML declaration did not match the encoding used in the document.

FRS_ERR_AUTHENTICATION = 8008

The file replication service cannot satisfy the request because authenticated RPC is not available. The event log might contain more information.

FRS_ERR_CHILD_TO_PARENT_COMM = 8011

The file replication service cannot communicate with the file replication service on the domain controller. The event log might contain more information.

FRS_ERR_INSUFFICIENT_PRIV = 8007

The file replication service cannot satisfy the request because the user has insufficient privileges. The event log might contain more information.

FRS_ERR_INTERNAL = 8005

The file replication service terminated the request. The event log might contain more information.

FRS_ERR_INTERNAL_API = 8004

The file replication service API terminated the request. The event log might contain more information.

FRS_ERR_INVALID_API_SEQUENCE = 8001

The file replication service API was called incorrectly.

FRS_ERR_INVALID_SERVICE_PARAMETER = 8017

The file replication service detected an invalid parameter.

FRS_ERR_PARENT_AUTHENTICATION = 8010

The file replication service cannot satisfy the request because authenticated RPC is not available on the domain controller. The event log might contain more information.

FRS_ERR_PARENT_INSUFFICIENT_PRIV = 8009

The file replication service cannot satisfy the request because the user has insufficient privileges on the domain controller. The event log might contain more information.

FRS_ERR_PARENT_TO_CHILD_COMM = 8012

The file replication service on the domain controller cannot communicate with the file replication service on this computer. The event log might contain more information.

FRS_ERR_SERVICE_COMM = 8006

The file replication service cannot be contacted. The event log might contain more information.

FRS_ERR_STARTING_SERVICE = 8002

The file replication service cannot be started.

FRS_ERR_STOPPING_SERVICE = 8003

The file replication service cannot be stopped.

FRS_ERR_SYSVOL_DEMOTE = 8016

The file replication service cannot stop replicating the system volume because of an internal error. The event log might contain more information.

FRS_ERR_SYSVOL_IS_BUSY = 8015

The file replication service cannot process the request. The system volume is busy with a previous request.

FRS_ERR_SYSVOL_POPULATE = 8013

The file replication service cannot populate the system volume because of an internal error. The event log might contain more information.

FRS_ERR_SYSVOL_POPULATE_TIMEOUT = 8014

The file replication service cannot populate the system volume because of an internal time-out. The event log might contain more information.

NERR_ACFFileIOFail = 2229

A disk I/O failure occurred.

NERR_ACFNoParent = 2232

The parent directory could not be located.

NERR_ACFNoRoom = 2228

There are too many names in the user accounts database.

NERR_ACFNotFound = 2219

The security database could not be found.

NERR_ACFNotLoaded = 2227

The security database has not been started.

NERR_ACFTooManyLists = 2230

The limit of 64 entries per resource was exceeded.

NERR_AccountExpired = 2239

This user account has expired.

NERR_AccountLockedOut = 2702

The account is locked out.

NERR_AccountUndefined = 2238

This user account is undefined.

NERR_AcctLimitExceeded = 2434

A user's session with this server has been deleted.

NERR_ActiveConns = 2402

Active connections still exist.

NERR_AddForwarded = 2275

This message alias has been added but is still forwarded.

NERR_AlertExists = 2430

The specified client is already registered for the specified event.

NERR_AlreadyExists = 2276

This message alias already exists locally.

NERR_AlreadyForwarded = 2274

This message alias has already been forwarded.

NERR_AlreadyLoggedIn = 2200

This workstation is already logged on to the LAN.

NERR_BadAsgType = 2251

This asg_type is invalid.

NERR_BadComponent = 2356

This path component is invalid.

NERR_BadControlRecv = 2193

The service control dispatcher pipe read failed.

NERR_BadDest = 2382

The destination path is illegal.

NERR_BadDev = 2341

The requested device is invalid.

NERR_BadDevString = 2340

This list of devices is invalid.

NERR_BadDosFunction = 2502

The program below called an unsupported MS-DOS function.

NERR_BadDosRetCode = 2500

The program below returned an MS-DOS error code.

NERR_BadEventName = 2143

The event name is invalid.

NERR_BadFileChecksum = 2504

The file below is corrupt.

NERR_BadPassword = 2203

The password parameter is invalid.

NERR_BadPasswordCore = 2403

This share name or password is invalid.

NERR_BadQueueDevString = 2334

This device name list is invalid.

NERR_BadQueuePriority = 2335

The queue priority is invalid.

NERR_BadReceive = 2282

The message was sent but not received.

NERR_BadRecipient = 2433

The alert recipient is invalid.

NERR_BadServiceName = 2185

The service name is invalid.

NERR_BadServiceProgName = 2188

The configuration file contains an invalid service program name.

NERR_BadSource = 2381

The source path is illegal.

NERR_BadTransactConfig = 2141

The server is not configured for transactions.

NERR_BadUasConfig = 2450

The user accounts database is not configured correctly.

NERR_BadUsername = 2202

The user name or group name parameter is invalid.

NERR_BrowserConfiguredToNotRun = 2550

The browser service was configured with MaintainServerList=No.

NERR_BrowserNotStarted = 2139

The requested information is not available.

NERR_BrowserTableIncomplete = 2319

The information in the list of servers might be incorrect.

NERR_BufTooSmall = 2123

The API return buffer is too small.

NERR_CallingRplSrvr = 2515

Connecting to the Remoteboot server.

NERR_CanNotGrowSegment = 2233

Unable to add to the security database session cache segment.

NERR_CanNotGrowUASFile = 2456

Unable to add to the user accounts database.

NERR_CantConnectRplSrvr = 2513

Cannot connect to the Remoteboot server.

NERR_CantOpenImageFile = 2514

Cannot open image file on the Remoteboot server.

NERR_CantType = 2357

Could not determine the type of input.

NERR_CfgCompNotFound = 2146

The specified component could not be found in the configuration information.

NERR_CfgParamNotFound = 2147

The specified parameter could not be found in the configuration information.

NERR_ClientNameNotFound = 2312

A session does not exist with that computer name.

NERR_CommDevInUse = 2343

This device is already in use as a communication device.

NERR_ComputerAccountNotFound = 2697

The specified computer account could not be found.

NERR_DCNotFound = 2453

Could not find the domain controller for this domain.

NERR_DataTypeInvalid = 2167

The data type is not supported by the print processor.

NERR_DatabaseUpToDate = 2248

No updates are necessary to this replicant network or local security database.

NERR_DefaultJoinRequired = 2694

The destination domain controller does not support creating machine accounts in organizational units (OUs).

NERR_DelComputerName = 2278

The computer name could not be deleted.

NERR_DeleteLater = 2298

This message alias will be deleted later.

NERR_DestExists = 2153

The printer destination already exists.

NERR_DestIdle = 2158

This printer destination is idle and cannot accept control operations.

NERR_DestInvalidOp = 2159

This printer destination request contains an invalid control function.

NERR_DestInvalidState = 2162

This operation cannot be performed on the print destination in its current state.

NERR_DestNoRoom = 2157

No more printer destinations can be added.

NERR_DestNotFound = 2152

The printer destination cannot be found.

NERR_DevInUse = 2404

The device is being accessed by an active process.

NERR_DevInvalidOpCode = 2331

The operation is invalid for this device.

NERR_DevNotFound = 2332

This device cannot be shared.

NERR_DevNotOpen = 2333

This device was not open.

NERR_DevNotRedirected = 2107

The device is not connected.

NERR_DeviceIsShared = 2252

This device is currently being shared.

NERR_DeviceNotShared = 2311

This device is not shared.

NERR_DeviceShareConflict = 2318

This device cannot be shared as both a spooled and a nonspooled resource.

NERR_DfsAlreadyShared = 2664

The server share specified is already shared in the DFS.

NERR_DfsBadRenamePath = 2671

The specified rename target path is invalid.

NERR_DfsCantCreateJunctionPoint = 2669

Unable to create a link.

NERR_DfsCantRemoveDfsRoot = 2682

The DFS root cannot be deleted. Uninstall DFS if required.

NERR_DfsCantRemoveLastServerShare = 2677

Cannot remove the last server share supporting this root or link.

NERR_DfsChildOrParentInDfs = 2683

A child or parent directory of the share is already in a DFS.

NERR_DfsCyclicalName = 2674

A cycle in the DFS name was detected.

NERR_DfsDataIsIdentical = 2681

The DFS data being reconciled is identical.

NERR_DfsDuplicateService = 2676

This link is already supported by the specified server share.

NERR_DfsInconsistent = 2679

The internal state of the DFS Service has become inconsistent.

NERR_DfsInternalCorruption = 2660

The internal database maintained by the DFS service is corrupt.

NERR_DfsInternalError = 2690

DFS internal error.

NERR_DfsLeafVolume = 2667

The operation is not valid in this portion of the namespace.

NERR_DfsNoSuchServer = 2673

The specified server is not a server for this link.

NERR_DfsNoSuchShare = 2665

The indicated server share does not support the indicated DFS namespace.

NERR_DfsNoSuchVolume = 2662

There is no DFS name whose entry path matches the input entry path.

NERR_DfsNotALeafVolume = 2666

The operation is not valid in this portion of the namespace.

NERR_DfsNotSupportedInServerDfs = 2675

The operation is not supported on a server-based DFS.

NERR_DfsServerNotDfsAware = 2670

The server is not DFS-aware.

NERR_DfsServerUpgraded = 2680

The DFS Service has been installed on the specified server.

NERR_DfsVolumeAlreadyExists = 2663

A root or link with the given name already exists.

NERR_DfsVolumeDataCorrupt = 2661

One of the records in the internal DFS database is corrupt.

NERR_DfsVolumeHasMultipleServers = 2668

The operation is ambiguous because the link has multiple servers.

NERR_DfsVolumeIsInterDfs = 2678

The operation is not supported for an inter-DFS link.

NERR_DfsVolumeIsOffline = 2672

The specified DFS link is offline.

NERR_DifferentServers = 2383

The source and destination paths are on different servers.

NERR_DriverNotFound = 2166

The device driver does not exist.

NERR_DupNameReboot = 2144

The computer name already exists on the network. Change it and reboot the computer.

NERR_DuplicateName = 2297

A duplicate message alias exists on the network.

NERR_DuplicateShare = 2118

The name has already been shared.

NERR_ErrCommRunSrv = 2389

An error occurred when communicating with a Run server.

NERR_ErrorExecingGhost = 2391

An error occurred when starting a background process.

NERR_ExecFailure = 2315

A failure occurred when executing a remote administration command.

NERR_FileIdNotFound = 2314

There is not an open file with that identification number.

NERR_GroupExists = 2223

The group already exists.

NERR_GroupNotFound = 2220

The group name could not be found.

NERR_GrpMsgProcessor = 2280

An error occurred in the domain message processor.

NERR_ImageParamErr = 2508

Image parameter substitution failed.

NERR_InUseBySpooler = 2342

This device is already in use by the spooler.

NERR_IncompleteDel = 2299

The message alias was not successfully deleted from all networks.

NERR_InternalError = 2140

An internal error occurred.

NERR_InvalidAPI = 2142

The requested API is not supported on the remote server.

NERR_InvalidComputer = 2351

This computer name is invalid.

NERR_InvalidDatabase = 2247

The security database is corrupted.

NERR_InvalidDevice = 2294

This is an invalid device name.

NERR_InvalidLana = 2400

The LAN adapter number is invalid.

NERR_InvalidLogSeek = 2440

The log file does not contain the requested record number.

NERR_InvalidLogonHours = 2241

The user is not allowed to log on at this time.

NERR_InvalidMaxUsers = 2122

The Peer service supports only two simultaneous users.

NERR_InvalidUASOp = 2451

This operation is not permitted when the Net Logon service is running.

NERR_InvalidWorkgroupName = 2695

The specified workgroup name is invalid.

NERR_InvalidWorkstation = 2240

The user is not allowed to log on from this workstation.

NERR_IsDfsShare = 2321

The share must be removed from the Distributed File System (DFS) before it can be deleted.

NERR_ItemNotFound = 2115

The queue is empty.

NERR_JobInvalidState = 2164

This operation cannot be performed on the print job in its current state.

NERR_JobNoRoom = 2156

No more print jobs can be added.

NERR_JobNotFound = 2151

The print job does not exist.

NERR_LanmanIniError = 2131

An error occurred when opening or reading the configuration file.

NERR_LastAdmin = 2452

This operation is not allowed on the last administrative account.

NERR_LineTooLong = 2149

A line in the configuration file is too long.

NERR_LocalDrive = 2405

The drive letter is in use locally.

NERR_LocalForward = 2279

Messages cannot be forwarded back to the same workstation.

NERR_LogFileChanged = 2378

This log file has changed between reads.

NERR_LogFileCorrupt = 2379

Log file %1 is corrupt.

NERR_LogOverflow = 2377

Log file %1 is full.

NERR_LogonDomainExists = 2216

There is already a logon domain for this computer.

NERR_LogonNoUserPath = 2211

The server is configured without a valid user path.

NERR_LogonScriptError = 2212

An error occurred while loading or running the logon script.

NERR_LogonServerConflict = 2210

A centralized logon server conflict occurred.

NERR_LogonServerNotFound = 2215

The logon server could not be found.

NERR_LogonTrackingError = 2454

Could not set logon information for this user.

NERR_LogonsPaused = 2209

Network logons are paused.

NERR_MaxLenExceeded = 2354

The string and prefix specified are too long.

NERR_MsgAlreadyStarted = 2271

The Messenger service is already started.

NERR_MsgInitFailed = 2272

The Messenger service failed to start.

NERR_MsgNotStarted = 2284

The Messenger service has not been started.

NERR_MultipleNets = 2300

This operation is not supported on computers with multiple networks.

NERR_NameInUse = 2283

The message alias is currently in use. Try again later.

NERR_NameNotForwarded = 2288

Messages for this alias are not currently being forwarded.

NERR_NameNotFound = 2273

The message alias could not be found on the network.

NERR_NameUsesIncompatibleCodePage = 2696

The specified computer name is incompatible with the default language used on the domain controller.

NERR_NetNameNotFound = 2310

This shared resource does not exist.

NERR_NetNotStarted = 2102

The workstation driver is not installed.

NERR_NetlogonNotStarted = 2455

The Net Logon service has not been started.

NERR_NetworkError = 2136

A general network error occurred.

NERR_NoAlternateServers = 2467

There are no alternate servers registered on this server.

NERR_NoCommDevs = 2337

There are no shared communication devices.

NERR_NoComputerName = 2270

The computer name could not be added as a message alias. The name might already exist on the network.

NERR_NoForwardName = 2286

The forwarded message alias could not be found on the network.

NERR_NoNetworkResource = 2105

A network resource shortage occurred.

NERR_NoRoom = 2119

The server is currently out of the requested resource.

NERR_NoRplBootSystem = 2505

No loader is specified in the boot-block definition file.

NERR_NoSuchAlert = 2432

An invalid or nonexistent alert name was raised.

NERR_NoSuchConnection = 2462

The connection identification does not specify a valid connection.

NERR_NoSuchServer = 2460

The server identification does not specify a valid server.

NERR_NoSuchSession = 2461

The session identification does not specify a valid session.

NERR_NonDosFloppyUsed = 2510

The image was not generated from an MS-DOS disk formatted with /S.

NERR_NonValidatedLogon = 2217

The logon server could not validate the logon.

NERR_NotInCache = 2235

This user is not cached in the user accounts database session cache.

NERR_NotInDispatchTbl = 2192

The service control dispatcher could not find the service name in the dispatch table.

NERR_NotLocalDomain = 2320

The computer is not active in this domain.

NERR_NotLocalName = 2285

The name is not on the local computer.

NERR_NotLoggedOn = 2201

The workstation is not logged on to the LAN.

NERR_NotPrimary = 2226

This operation is allowed only on the PDC of the domain.

NERR_OpenFiles = 2401

There are open files on the connection.

NERR_PasswordCantChange = 2243

The password of this user cannot change.

NERR_PasswordExpired = 2242

The password of this user has expired.

NERR_PasswordFilterError = 2705

The password does not meet the requirements of the password filter DLLs.

NERR_PasswordHistConflict = 2244

This password cannot be used now.

NERR_PasswordMismatch = 2458

A password mismatch has been detected.

NERR_PasswordMustChange = 2701

The password must change at the next logon.

NERR_PasswordNotComplexEnough = 2704

The password does not meet the complexity policy.

NERR_PasswordTooLong = 2703

The password is too long.

NERR_PasswordTooRecent = 2246

The password of this user is too recent to change.

NERR_PasswordTooShort = 2245

The password does not meet the password policy requirements. Check the minimum password length, password complexity, and password history requirements.

NERR_PausedRemote = 2281

The message was sent, but the recipient has paused the Messenger service.

NERR_PersonalSku = 2698

This version of Windows cannot be joined to a domain.

NERR_ProcNoRespond = 2160

The print processor is not responding.

NERR_ProcNotFound = 2168

The print processor is not installed.

NERR_ProfileCleanup = 2372

The system cannot delete current connections to network resources.

NERR_ProfileFileTooBig = 2370

Profile files cannot exceed 64 KB.

NERR_ProfileLoadErr = 2374

An error occurred while loading the profile file.

NERR_ProfileOffset = 2371

The start offset is out of range.

NERR_ProfileSaveErr = 2375

Errors occurred while saving the profile file. The profile was partially saved.

NERR_ProfileUnknownCmd = 2373

The system was unable to parse the command line in this file.

NERR_ProgNeedsExtraMem = 2501

The program below needs more memory.

NERR_QExists = 2154

The print queue already exists.

NERR_QNoRoom = 2155

No more printers can be added.

NERR_QNotFound = 2150

The printer does not exist.

NERR_QinvalidState = 2163

This operation cannot be performed on the print queue in its current state.

NERR_QueueNotFound = 2338

The queue you specified does not exist.

NERR_RPL_CONNECTED = 2519

A second connection to a Remoteboot resource is not allowed.

NERR_RedirectedPath = 2117

The operation is invalid on a redirected resource.

NERR_RemoteBootFailed = 2503

The workstation failed to boot.

NERR_RemoteErr = 2127

A remote API error occurred.

NERR_RemoteFull = 2287

The message alias table on the remote station is full.

NERR_RemoteOnly = 2106

This operation is not supported on workstations.

NERR_ResourceExists = 2225

The resource permission list already exists.

NERR_ResourceNotFound = 2222

The resource name could not be found.

NERR_RplAdapterInfoCorrupted = 2625

Adapter ID record information has been corrupted.

NERR_RplAdapterNameUnavailable = 2633

Chosen network adapter ID is in use by some other workstation.

NERR_RplAdapterNotFound = 2637

Adapter record was not found.

NERR_RplBackupDatabase = 2636

Service failed to back up the Remoteboot database.

NERR_RplBadDatabase = 2612

Service failed to start because its database is absent or corrupt.

NERR_RplBadRegistry = 2611

Service failed to start due to bad startup information in the registry.

NERR_RplBootInUse = 2635

There are workstations, profiles, or configurations using this boot block.

NERR_RplBootInfoCorrupted = 2628

Boot block record information has been corrupted.

NERR_RplBootNameUnavailable = 2640

The boot name or vendor ID is in use by some other boot block record.

NERR_RplBootNotFound = 2631

Boot block record was not found.

NERR_RplBootRestart = 2511

Remote boot will be restarted later.

NERR_RplBootServiceTerm = 2517

Remote boot service was stopped, check the error log for the cause of the problem.

NERR_RplBootStartFailed = 2518

Remote boot startup failed; check the error log for the cause of the problem.

NERR_RplCannotEnum = 2615

Cannot enumerate service records.

NERR_RplConfigInfoCorrupted = 2623

Configuration record information has been corrupted.

NERR_RplConfigNameUnavailable = 2641

The configuration name is in use by some other configuration.

NERR_RplConfigNotEmpty = 2634

There are profiles using this configuration.

NERR_RplConfigNotFound = 2624

Configuration record was not found.

NERR_RplIncompatibleProfile = 2632

Chosen profile is incompatible with this workstation.

NERR_RplInternal = 2626

An internal service error has occurred.

NERR_RplLoadrDiskErr = 2507

A disk I/O error occurred.

NERR_RplLoadrNetBiosErr = 2506

NetBIOS returned an error: The network control blocks (NCBs) and Server Message Block (SMB) are dumped above.

NERR_RplNeedsRPLUSERAcct = 2630

The RPLUSER local group could not be found.

NERR_RplNoAdaptersStarted = 2610

Service failed to start because none of the network adapters started with this service.

NERR_RplNotRplServer = 2614

Service failed to start because the RPLUSER group is absent.

NERR_RplProfileInfoCorrupted = 2619

Profile record information has been corrupted.

NERR_RplProfileNameUnavailable = 2621

Profile name is in use by some other profile.

NERR_RplProfileNotEmpty = 2622

There are workstations using this profile.

NERR_RplProfileNotFound = 2620

Profile record was not found.

NERR_RplRplfilesShare = 2613

Service failed to start because the RPLFILES share is absent.

NERR_RplSrvrCallFailed = 2512

The call to the Remoteboot server failed.

NERR_RplVendorInfoCorrupted = 2627

Vendor ID record information has been corrupted.

NERR_RplVendorNameUnavailable = 2639

Vendor name is in use by some other vendor record.

NERR_RplVendorNotFound = 2638

Vendor record was not found.

NERR_RplWkstaInfoCorrupted = 2616

Workstation record information has been corrupted.

NERR_RplWkstaNameUnavailable = 2618

Workstation name is in use by some other workstation.

NERR_RplWkstaNeedsUserAcct = 2629

The user account for this workstation record is missing.

NERR_RplWkstaNotFound = 2617

Workstation record was not found.

NERR_RunSrvPaused = 2385

The Run server you requested is paused.

NERR_ServerNotStarted = 2114

The Server service is not started.

NERR_ServiceCtlBusy = 2187

The service control is busy.

NERR_ServiceCtlNotValid = 2191

The requested pause or stop is not valid for this service.

NERR_ServiceCtlTimeout = 2186

The service is not responding to the control function.

NERR_ServiceEntryLocked = 2183

The service does not respond to control actions.

NERR_ServiceInstalled = 2182

The requested service has already been started.

NERR_ServiceKillProc = 2190

The service ended abnormally.

NERR_ServiceNotCtrl = 2189

The service could not be controlled in its present state.

NERR_ServiceNotInstalled = 2184

The service has not been started.

NERR_ServiceNotStarting = 2194

A thread for the new service could not be created.

NERR_ServiceTableFull = 2181

The service table is full.

NERR_ServiceTableLocked = 2180

The service database is locked.

NERR_SetupAlreadyJoined = 2691

This machine is already joined to a domain.

NERR_SetupDomainController = 2693

This machine is a domain controller and cannot be unjoined from a domain.

NERR_SetupNotJoined = 2692

This machine is not currently joined to a domain.

NERR_ShareMem = 2104

An internal error occurred. The network cannot access a shared memory segment.

NERR_ShareNotFound = 2392

The shared resource you are connected to could not be found.

NERR_SourceIsDir = 2380

The source path cannot be a directory.

NERR_SpeGroupOp = 2234

This operation is not allowed on this special group.

NERR_SpoolNoMemory = 2165

A spooler memory allocation failure occurred.

NERR_SpoolerNotLoaded = 2161

The spooler is not running.

NERR_StandaloneLogon = 2214

The logon server was not specified. The computer will be logged on as STANDALONE.

NERR_StartingRplBoot = 2516

Connecting to the Remoteboot server.

NERR_Success = 0

The operation completed successfully.

NERR_SyncRequired = 2249

This replicant database is outdated; synchronization is required.

NERR_TimeDiffAtDC = 2457

This server's clock is not synchronized with the PDC's clock.

NERR_TmpFile = 2316

A failure occurred when opening a remote temporary file.

NERR_TooManyAlerts = 2431

The alert table is full.

NERR_TooManyConnections = 2465

The server has reached the maximum number of connections it supports.

NERR_TooManyEntries = 2362

The buffer for types is not big enough.

NERR_TooManyFiles = 2466

The server cannot open more files because it has reached its maximum number.

NERR_TooManyImageParams = 2509

Too many image parameters cross disk sector boundaries.

NERR_TooManyItems = 2121

Requested addition of items exceeds the maximum allowed.

NERR_TooManyNames = 2277

The maximum number of added message aliases has been exceeded.

NERR_TooManyServers = 2463

There is no space for another entry in the table of available servers.

NERR_TooManySessions = 2464

The server has reached the maximum number of sessions it supports.

NERR_TooMuchData = 2317

The data returned from a remote administration command has been truncated to 64 KB.

NERR_TruncatedBroadcast = 2289

The broadcast message was truncated.

NERR_TryDownLevel = 2470

Try the down-level (remote admin protocol) version of API instead.

NERR_UPSDriverNotStarted = 2480

The uninterruptible power supply (UPS) driver could not be accessed by the UPS service.

NERR_UPSInvalidCommPort = 2482

The UPS service could not access the specified Comm Port.

NERR_UPSInvalidConfig = 2481

The UPS service is not configured correctly.

NERR_UPSShutdownFailed = 2484

The UPS service failed to perform a system shut down.

NERR_UPSSignalAsserted = 2483

The UPS indicated a line fail or low battery situation. Service not started.

NERR_UnableToAddName_F = 2205

The logon processor did not add the message alias.

NERR_UnableToAddName_W = 2204

The logon processor did not add the message alias.

NERR_UnableToDelName_F = 2207

The logoff processor did not delete the message alias.

NERR_UnableToDelName_W = 2206

The logoff processor did not delete the message alias.

NERR_UnknownDevDir = 2116

The device or directory does not exist.

NERR_UnknownServer = 2103

The server could not be located.

NERR_UseNotFound = 2250

The network connection could not be found.

NERR_UserExists = 2224

The user account already exists.

NERR_UserInGroup = 2236

The user already belongs to this group.

NERR_UserLogon = 2231

Deleting a user with a session is not allowed.

NERR_UserNotFound = 2221

The user name could not be found.

NERR_UserNotInGroup = 2237

The user does not belong to this group.

NERR_WkstaInconsistentState = 2137

The Workstation service is in an inconsistent state. Restart the computer before restarting the Workstation service.

NERR_WkstaNotStarted = 2138

The Workstation service has not been started.

NERR_WriteFault = 2295

A write fault occurred.

OR_INVALID_OID = 1911

The object specified was not found.

OR_INVALID_OXID = 1910

The object exporter specified was not found.

OR_INVALID_SET = 1912

The object set specified was not found.

PEERDIST_ERROR_ALREADY_COMPLETED = 4060

Cannot perform the requested operation because it has already been carried out.

PEERDIST_ERROR_ALREADY_EXISTS = 4058

An element already exists and was not replaced.

PEERDIST_ERROR_ALREADY_INITIALIZED = 4055

The supplied object has already been initialized.

PEERDIST_ERROR_CANNOT_PARSE_CONTENTINFO = 4051

The supplied content information is malformed.

PEERDIST_ERROR_CONTENTINFO_VERSION_UNSUPPORTED = 4050

The version of the supplied content information is not supported.

PEERDIST_ERROR_INVALIDATED = 4057

The supplied object has already been invalidated.

PEERDIST_ERROR_INVALID_CONFIGURATION = 4063

A configuration value is invalid.

PEERDIST_ERROR_MISSING_DATA = 4052

The requested data cannot be found in local or peer caches.

PEERDIST_ERROR_NOT_INITIALIZED = 4054

The supplied object has not been initialized.

PEERDIST_ERROR_NOT_LICENSED = 4064

The SKU is not licensed.

PEERDIST_ERROR_NO_MORE = 4053

No more data is available or required.

PEERDIST_ERROR_OPERATION_NOTFOUND = 4059

Cannot cancel the requested operation as it has already been completed.

PEERDIST_ERROR_OUT_OF_BOUNDS = 4061

An operation accessed data beyond the bounds of valid data.

PEERDIST_ERROR_SERVICE_UNAVAILABLE = 4065

PeerDist Service is still initializing and will be available shortly.

PEERDIST_ERROR_SHUTDOWN_IN_PROGRESS = 4056

A shutdown operation is already in progress.

PEERDIST_ERROR_VERSION_UNSUPPORTED = 4062

The requested version is not supported.

RPC_S_ADDRESS_ERROR = 1768

An addressing error occurred in the RPC server.

RPC_S_ALREADY_LISTENING = 1713

The RPC server is already listening.

RPC_S_ALREADY_REGISTERED = 1711

The object UUID) has already been registered.

RPC_S_BINDING_HAS_NO_AUTH = 1746

The binding does not contain any authentication information.

RPC_S_BINDING_INCOMPLETE = 1819

The binding handle does not contain all the required information.

RPC_S_CALL_CANCELLED = 1818

The RPC was canceled.

RPC_S_CALL_FAILED = 1726

The RPC failed.

RPC_S_CALL_FAILED_DNE = 1727

The RPC failed and did not execute.

RPC_S_CALL_IN_PROGRESS = 1791

An RPC is already in progress for this thread.

RPC_S_CANNOT_SUPPORT = 1764

The requested operation is not supported.

RPC_S_CANT_CREATE_ENDPOINT = 1720

The endpoint cannot be created.

RPC_S_COMM_FAILURE = 1820

A communications failure occurred during an RPC.

RPC_S_DUPLICATE_ENDPOINT = 1740

The endpoint is a duplicate.

RPC_S_ENTRY_ALREADY_EXISTS = 1760

The entry already exists.

RPC_S_ENTRY_NOT_FOUND = 1761

The entry is not found.

RPC_S_ENTRY_TYPE_MISMATCH = 1922

The entry is not of the expected type.

RPC_S_FP_DIV_ZERO = 1769

A floating-point operation at the RPC server caused a division by zero.

RPC_S_FP_OVERFLOW = 1771

A floating-point overflow occurred at the RPC server.

RPC_S_FP_UNDERFLOW = 1770

A floating-point underflow occurred at the RPC server.

RPC_S_GROUP_MEMBER_NOT_FOUND = 1898

The group member was not found.

RPC_S_GRP_ELT_NOT_ADDED = 1928

The group element could not be added.

`RPC_S_GRP_ELT_NOT_REMOVED = 1929`

The group element could not be removed.

`RPC_S_INCOMPLETE_NAME = 1755`

The entry name is incomplete.

`RPC_S_INTERFACE_NOT_EXPORTED = 1924`

The interface could not be exported to the specified entry.

`RPC_S_INTERFACE_NOT_FOUND = 1759`

The interface was not found.

`RPC_S_INTERNAL_ERROR = 1766`

An internal error occurred in an RPC.

`RPC_S_INVALID_ASYNC_CALL = 1915`

Invalid asynchronous RPC call handle for this operation.

`RPC_S_INVALID_ASYNC_HANDLE = 1914`

Invalid asynchronous RPC handle.

`RPC_S_INVALID_AUTH_IDENTITY = 1749`

The security context is invalid.

`RPC_S_INVALID_BINDING = 1702`

The binding handle is invalid.

`RPC_S_INVALID_BOUND = 1734`

The array bounds are invalid.

`RPC_S_INVALID_ENDPOINT_FORMAT = 1706`

The endpoint format is invalid.

`RPC_S_INVALID_NAF_ID = 1763`

The network address family is invalid.

`RPC_S_INVALID_NAME_SYNTAX = 1736`

The name syntax is invalid.

`RPC_S_INVALID_NETWORK_OPTIONS = 1724`

The network options are invalid.

`RPC_S_INVALID_NET_ADDR = 1707`

The network address is invalid.

`RPC_S_INVALID_OBJECT = 1900`

The object UUID is the nil UUID.

`RPC_S_INVALID_RPC_PROTSEQ = 1704`

The RPC protocol sequence is invalid.

`RPC_S_INVALID_STRING_BINDING = 1700`

The string binding is invalid.

`RPC_S_INVALID_STRING_UUID = 1705`

The stringUUID is invalid.

`RPC_S_INVALID_TAG = 1733`

The tag is invalid.

`RPC_S_INVALID_TIMEOUT = 1709`

The time-out value is invalid.

`RPC_S_INVALID_VERS_OPTION = 1756`

The version option is invalid.

`RPC_S_MAX_CALLS_TOO_SMALL = 1742`

The maximum number of calls is too small.

`RPC_S_NAME_SERVICE_UNAVAILABLE = 1762`

The name service is unavailable.

RPC_S_NOTHING_TO_EXPORT = 1754

No interfaces have been exported.

RPC_S_NOT_ALL_OBJS_EXPORTED = 1923

Not all object UUIDs could be exported to the specified entry.

RPC_S_NOT_ALL_OBJS_UNEXPORTED = 1758

There is nothing to unexport.

RPC_S_NOT_CANCELLED = 1826

The thread is not canceled.

RPC_S_NOT_LISTENING = 1715

The RPC server is not listening.

RPC_S_NOT_RPC_ERROR = 1823

The error specified is not a valid Windows RPC error code.

RPC_S_NO_BINDINGS = 1718

There are no bindings.

RPC_S_NO_CALL_ACTIVE = 1725

There are no RPCs active on this thread.

RPC_S_NO_CONTEXT_AVAILABLE = 1765

No security context is available to allow impersonation.

RPC_S_NO_ENDPOINT_FOUND = 1708

No endpoint was found.

RPC_S_NO_ENTRY_NAME = 1735

The binding does not contain an entry name.

RPC_S_NO_INTERFACES = 1817

No interfaces have been registered.

`RPC_S_NO_MORE_BINDINGS = 1806`

There are no more bindings.

`RPC_S_NO_MORE_MEMBERS = 1757`

There are no more members.

`RPC_S_NO_PRINC_NAME = 1822`

No principal name is registered.

`RPC_S_NO_PROTSEQS = 1719`

There are no protocol sequences.

`RPC_S_NO_PROTSEQS_REGISTERED = 1714`

No protocol sequences have been registered.

`RPC_S_OBJECT_NOT_FOUND = 1710`

The object UUID) was not found.

`RPC_S_OUT_OF_RESOURCES = 1721`

Not enough resources are available to complete this operation.

`RPC_S_PRF_ELT_NOT_ADDED = 1926`

The specified profile element could not be added.

`RPC_S_PRF_ELT_NOT_REMOVED = 1927`

The specified profile element could not be removed.

`RPC_S_PROCNUM_OUT_OF_RANGE = 1745`

The procedure number is out of range.

`RPC_S_PROFILE_NOT_ADDED = 1925`

The specified profile entry could not be added.

`RPC_S_PROTOCOL_ERROR = 1728`

An RPC protocol error occurred.

RPC_S_PROTSEQ_NOT_FOUND = 1744

The RPC protocol sequence was not found.

RPC_S_PROTSEQ_NOT_SUPPORTED = 1703

The RPC protocol sequence is not supported.

RPC_S_PROXY_ACCESS_DENIED = 1729

Access to the HTTP proxy is denied.

RPC_S_SEC_PKG_ERROR = 1825

A security package-specific error occurred.

RPC_S_SEND_INCOMPLETE = 1913

Some data remains to be sent in the request buffer.

RPC_S_SERVER_TOO_BUSY = 1723

The RPC server is too busy to complete this operation.

RPC_S_SERVER_UNAVAILABLE = 1722

The RPC server is unavailable.

RPC_S_STRING_TOO_LONG = 1743

The string is too long.

RPC_S_TYPE_ALREADY_REGISTERED = 1712

The type UUID has already been registered.

RPC_S_UNKNOWN_AUTHN_LEVEL = 1748

The authentication level is unknown.

RPC_S_UNKNOWN_AUTHN_SERVICE = 1747

The authentication service is unknown.

RPC_S_UNKNOWN_AUTHN_TYPE = 1741

The authentication type is unknown.

`RPC_S_UNKNOWN_AUTHZ_SERVICE = 1750`

The authorization service is unknown.

`RPC_S_UNKNOWN_IF = 1717`

The interface is unknown.

`RPC_S_UNKNOWN_MGR_TYPE = 1716`

The manager type is unknown.

`RPC_S_UNSUPPORTED_AUTHN_LEVEL = 1821`

The requested authentication level is not supported.

`RPC_S_UNSUPPORTED_NAME_SYNTAX = 1737`

The name syntax is not supported.

`RPC_S_UNSUPPORTED_TRANS_SYN = 1730`

The transfer syntax is not supported by the RPC server.

`RPC_S_UNSUPPORTED_TYPE = 1732`

The UUID type is not supported.

`RPC_S_UUID_LOCAL_ONLY = 1824`

A UUID that is valid only on this computer has been allocated.

`RPC_S_UUID_NO_ADDRESS = 1739`

No network address is available to use to construct a UUID.

`RPC_S_WRONG_KIND_OF_BINDING = 1701`

The binding handle is not the correct type.

`RPC_S_ZERO_DIVIDE = 1767`

The RPC server attempted an integer division by zero.

`RPC_X_BAD_STUB_DATA = 1783`

The stub received bad data.

RPC_X_BYTE_COUNT_TOO_SMALL = 1782

The byte count is too small.

RPC_X_ENUM_VALUE_OUT_OF_RANGE = 1781

The enumeration value is out of range.

RPC_X_INVALID_ES_ACTION = 1827

Invalid operation on the encoding/decoding handle.

RPC_X_INVALID_PIPE_OBJECT = 1830

The RPC pipe object is invalid or corrupted.

RPC_X_NO_MORE_ENTRIES = 1772

The list of RPC servers available for the binding of auto handles has been exhausted.

RPC_X_NULL_REF_POINTER = 1780

A null reference pointer was passed to the stub.

RPC_X_PIPE_CLOSED = 1916

The RPC pipe object has already been closed.

RPC_X_PIPE_DISCIPLINE_ERROR = 1917

The RPC call completed before all pipes were processed.

RPC_X_PIPE_EMPTY = 1918

No more data is available from the RPC pipe.

RPC_X_SS_CANNOT_GET_CALL_HANDLE = 1779

The stub is unable to get the RPC handle.

RPC_X_SS_CHAR_TRANS_OPEN_FAIL = 1773

Unable to open the character translation table file.

RPC_X_SS_CHAR_TRANS_SHORT_FILE = 1774

The file containing the character translation table has fewer than 512 bytes.

RPC_X_SS_CONTEXT_DAMAGED = 1777

The context handle changed during an RPC.

RPC_X_SS_HANDLES_MISMATCH = 1778

The binding handles passed to an RPC do not match.

RPC_X_SS_IN_NULL_CONTEXT = 1775

A null context handle was passed from the client to the host during an RPC.

RPC_X_WRONG_ES_VERSION = 1828

Incompatible version of the serializing package.

RPC_X_WRONG_PIPE_ORDER = 1831

An invalid operation was attempted on an RPC pipe object.

RPC_X_WRONG_PIPE_VERSION = 1832

Unsupported RPC pipe version.

RPC_X_WRONG_STUB_VERSION = 1829

Incompatible version of the RPC stub.

SCHED_E_SERVICE_NOT_LOCALSYSTEM = 6200

The Task Scheduler service must be configured to run in the System account to function properly. Individual tasks can be configured to run in other accounts.

WAIT_TIMEOUT = 258

The wait operation timed out.

WARNING_IPSEC_MM_POLICY_PRUNED = 13024

The main mode policy was successfully added, but some of the requested offers are not supported.

WARNING_IPSEC_QM_POLICY_PRUNED = 13025

The quick mode policy was successfully added, but some of the requested offers are not supported.

WSAEACCES = 10013

An attempt was made to access a socket in a way forbidden by its access permissions.

WSAEADDRINUSE = 10048

Only one usage of each socket address (protocol/network address/port) is normally permitted.

WSAEADDRNOTAVAIL = 10049

The requested address is not valid in its context.

WSAEAFNOSUPPORT = 10047

An address incompatible with the requested protocol was used.

WSAEALREADY = 10037

An operation was attempted on a nonblocking socket that already had an operation in progress.

WSAEBADF = 10009

The file handle supplied is not valid.

WSAECANCELLED = 10103

A call to WSALookupServiceEnd was made while this call was still processing. The call has been canceled.

WSAECONNABORTED = 10053

An established connection was aborted by the software in your host machine.

WSAECONNREFUSED = 10061

No connection could be made because the target machine actively refused it.

WSAECONNRESET = 10054

An existing connection was forcibly closed by the remote host.

WSAEDESTADDRREQ = 10039

A required address was omitted from an operation on a socket.

WSAEDISCON = 10101

Returned by WSAREcv or WSAREcvFrom to indicate that the remote party has initiated a graceful shutdown sequence.

WSAEDQUOT = 10069

Ran out of disk quota.

WSAEFAULT = 10014

The system detected an invalid pointer address in attempting to use a pointer argument in a call.

WSAEHOSTDOWN = 10064

A socket operation failed because the destination host was down.

WSAEHOSTUNREACH = 10065

A socket operation was attempted to an unreachable host.

WSAEINPROGRESS = 10036

A blocking operation is currently executing.

WSAEINTR = 10004

A blocking operation was interrupted by a call to WSACancelBlockingCall.

WSAEINVAL = 10022

An invalid argument was supplied.

WSAEINVALIDPROCTABLE = 10104

The procedure call table is invalid.

WSAEINVALIDPROVIDER = 10105

The requested service provider is invalid.

WSAEISCONN = 10056

A connect request was made on an already connected socket.

WSAELOOP = 10062

Cannot translate name.

WSAEMFILE = 10024

Too many open sockets.

WSAEMSGSIZE = 10040

A message sent on a datagram socket was larger than the internal message buffer or some other network limit, or the buffer used to receive a datagram into was smaller than the datagram itself.

WSAENAMETOOLONG = 10063

Name or name component was too long.

WSAENETDOWN = 10050

A socket operation encountered a dead network.

WSAENETRESET = 10052

The connection has been broken due to keep-alive activity detecting a failure while the operation was in progress.

WSAENETUNREACH = 10051

A socket operation was attempted to an unreachable network.

WSAENOBUFS = 10055

An operation on a socket could not be performed because the system lacked sufficient buffer space or because a queue was full.

WSAENOMORE = 10102

No more results can be returned by WSALookupServiceNext.

WSAENOPROTOPT = 10042

An unknown, invalid, or unsupported option or level was specified in a getsockopt or setsockopt call.

WSAENOTCONN = 10057

A request to send or receive data was disallowed because the socket is not connected and (when sending on a datagram socket using a sendto call) no address was supplied.

WSAENOTEMPTY = 10066

Cannot remove a directory that is not empty.

WSAENOTSOCK = 10038

An operation was attempted on something that is not a socket.

WSAEOPNOTSUPP = 10045

The attempted operation is not supported for the type of object referenced.

WSAEPFNOSUPPORT = 10046

The protocol family has not been configured into the system or no implementation for it exists.

WSAEPROCLIM = 10067

A Windows Sockets implementation might have a limit on the number of applications that can use it simultaneously.

WSAEPROTONOSUPPORT = 10043

The requested protocol has not been configured into the system, or no implementation for it exists.

WSAEPROTOTYPE = 10041

A protocol was specified in the socket function call that does not support the semantics of the socket type requested.

WSAEPROVIDERFAILEDINIT = 10106

The requested service provider could not be loaded or initialized.

WSAEREFUSED = 10112

A database query failed because it was actively refused.

WSAEREMOTE = 10071

Item is not available locally.

WSAESHUTDOWN = 10058

A request to send or receive data was disallowed because the socket had already been shut down in that direction with a previous shutdown call.

WSAESOCKTNOSUPPORT = 10044

The support for the specified socket type does not exist in this address family.

WSAESTALE = 10070

File handle reference is no longer available.

WSAETIMEDOUT = 10060

A connection attempt failed because the connected party did not properly respond after a period of time, or the established connection failed because the connected host failed to respond.

WSAETOOMANYREFS = 10059

Too many references to a kernel object.

WSAEUSERS = 10068

Ran out of quota.

WSAEWOULDBLOCK = 10035

A nonblocking socket operation could not be completed immediately.

WSAHOST_NOT_FOUND = 11001

No such host is known.

WSANOTINITIALISED = 10093

Either the application has not called WSStartup, or WSStartup failed.

WSANO_DATA = 11004

The requested name is valid, but no data of the requested type was found.

WSANO_RECOVERY = 11003

A nonrecoverable error occurred during a database lookup.

WSASERVICE_NOT_FOUND = 10108

No such service is known. The service cannot be found in the specified namespace.

WSASYSCALLFAILURE = 10107

A system call that should never fail has failed.

WSASYSNOTREADY = 10091

WSStartup cannot function at this time because the underlying system it uses to provide network services is currently unavailable.

WSATRY_AGAIN = 11002

This is usually a temporary error during host name resolution and means that the local server did not receive a response from an authoritative server.

WSATYPE_NOT_FOUND = 10109

The specified class was not found.

WSAVERNOTSUPPORTED = 10092

The Windows Sockets version requested is not supported.

WSA_E_CANCELLED = 10111

A call to WSALookupServiceEnd was made while this call was still processing. The call has been canceled.

WSA_E_NO_MORE = 10110

No more results can be returned by WSALookupServiceNext.

WSA_QOS_ADMISSION_FAILURE = 11010

Error due to lack of resources.

WSA_QOS_BAD_OBJECT = 11013

There is a problem with some part of the filterspec or provider-specific buffer in general.

WSA_QOS_BAD_STYLE = 11012

Unknown or conflicting style.

WSA_QOS_EFILTERCOUNT = 11021

An incorrect number of QOS FILTERSPECS were specified in the FLOWDESCRIPTOR.

WSA_QOS_EFILTERSTYLE = 11019

An invalid QOS filter style was used.

WSA_QOS_EFILTERTYPE = 11020

An invalid QOS filter type was used.

WSA_QOS_EFLOWCOUNT = 11023

An incorrect number of flow descriptors was specified in the QOS structure.

WSA_QOS_EFLOWDESC = 11026

An invalid QOS flow descriptor was found in the flow descriptor list.

WSA_QOS_EFLOWSPEC = 11017

An invalid or inconsistent flowspec was found in the QOS structure.

WSA_QOS_EOBJLENGTH = 11022

An object with an invalidObjectLengthfield was specified in the QOS provider-specific buffer.

WSA_QOS_EPOLICYOBJ = 11025

An invalid policy object was found in the QOS provider-specific buffer.

WSA_QOS_EPROVSPECBUF = 11018

Invalid QOS provider-specific buffer.

WSA_QOS_EPSFILTERSPEC = 11028

An invalid FILTERSPEC was found in the QOS provider-specific buffer.

WSA_QOS_EPSFLOWSPEC = 11027

An invalid or inconsistent flowspec was found in the QOS provider-specific buffer.

WSA_QOS_ESDMODEOBJ = 11029

An invalid shape discard mode object was found in the QOS provider-specific buffer.

WSA_QOS_ESERVICETYPE = 11016

An invalid or unrecognized service type was found in the flowspec.

WSA_QOS_ESHAPERATEOBJ = 11030

An invalid shaping rate object was found in the QOS provider-specific buffer.

WSA_QOS_EUNKNOWNPSOBJ = 11024

An unrecognized object was found in the QOS provider-specific buffer.

WSA_QOS_GENERIC_ERROR = 11015

General quality of serve (QOS) error.

WSA_QOS_NO_RECEIVERS = 11008

There are no receivers.

WSA_QOS_NO_SENDERS = 11007

There are no senders.

WSA_QOS_POLICY_FAILURE = 11011

Rejected for administrative reasons—bad credentials.

WSA_QOS_RECEIVERS = 11005

At least one reserve has arrived.

WSA_QOS_REQUEST_CONFIRMED = 11009

Reserve has been confirmed.

WSA_QOS_RESERVED_PETYPE = 11031

A reserved policy element was found in the QOS provider-specific buffer.

WSA_QOS_SENDERS = 11006

At least one path has arrived.

WSA_QOS_TRAFFIC_CTRL_ERROR = 11014

There is a problem with some part of the flowspec.

Class WinConstants

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Defines common Win32 constants.

```
public static class WinConstants
```

Inheritance

[object](#)  ← WinConstants

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Fields

FACILITY_WIN32

Defines the **FACILITY_WIN32** macro from **winerror.h**.

```
public const int FACILITY_WIN32 = 7
```

Field Value

[int](#) 

MAX_PATH

Defines the **MAX_PATH** constant, being the maximum length of a file path for applications that don't have long file path support (unless overridden with the `\?</code> UNC path prefix).`

```
public const int MAX_PATH = 260
```

Field Value

Struct WinMessage

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Provides information about a Windows message.

```
public struct WinMessage
```

Inherited Members

[ValueType.Equals\(object\)](#) , [ValueType.GetHashCode\(\)](#) , [ValueType.ToString\(\)](#) ,
[object.Equals\(object, object\)](#) , [object.GetType\(\)](#) , [object.ReferenceEquals\(object, object\)](#)

Properties

CursorPos

The cursor position, in screen coordinates, when the message was posted.

```
public Point CursorPos { get; }
```

Property Value

[Point](#)

LParam

Additional information about the message. The exact meaning depends on the value of the [MessageId](#) member.

```
public nint LParam { get; }
```

Property Value

[nint](#)

Remarks

This value represents the LPARAM before the framework processes it as a managed object when received in your window procedure.

MessageId

The message identifier. Applications can only use the low word; the high word is reserved by the system.

```
public uint MessageId { get; }
```

Property Value

[uint](#)

Time

The time at which the message was posted.

```
public DateTime Time { get; }
```

Property Value

[DateTime](#)

WParam

Additional information about the message. The exact meaning depends on the value of the [MessageId](#) member.

```
public nuint WParam { get; }
```

Property Value

[nuint](#)

Window

The window whose window procedure receives the message. This member is [null](#)  when the message is a thread message.

```
public Window? Window { get; }
```

Property Value

[Window](#)

Class Window

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Provides methods for managing Win32 windows.

```
public class Window : IHandle
```

Inheritance

[object](#)  ← Window

Implements

[IHandle](#)

Inherited Members

[object.Equals\(object, object\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  ,
[object.ReferenceEquals\(object, object\)](#) 

Constructors

Window(ExWindowStyles, string, string, uint, Point, Size, Window?, nint?, object?)

Creates an overlapped, pop-up, or child window with an extended window style.

```
public Window(ExWindowStyles exStyle, string className, string windowName, uint style, Point location, Size size, Window? parent = null, nint? hMenu = null, object? param = null)
```

Parameters

exStyle [ExWindowStyles](#)

The extended window style of the window being created.

className [string](#) 

A class name string or a class atom.

windowName [string](#)

The window name. If the window style specifies a title bar, the window title specified in **windowName** is displayed in the title bar. When using the function to create controls, such as buttons, check boxes, and static controls, use **windowName** to specify the text of the control. When creating a static control with the **SS_ICON** style, use **windowName** to specify the icon name or identifier. To specify an identifier, use the syntax "#num".

style [uint](#)

The style of the window being created. This parameter can be a bitwise OR (|) combination of constants from the normal [WindowStyles](#), combined with any additional control style values.

location [Point](#)

The initial position of the window. For an overlapped or pop-up window, the location is relative to the top-left corner of the screen. For a child window, the location is relative to the top-left corner of the parent window's client area. If this parameter is set to [WindowDefaultLocation](#), Windows automatically selects the position.

size [Size](#)

The initial size of the window, in device units. If this parameter is set to [WindowDefaultSize](#), Windows automatically selects a size.

parent [Window](#)

The parent or owner window of the window being created. To create a child window or an owned window, supply a valid window. This parameter is optional for pop-up windows.

hMenu [hMenu](#)?

A handle to a menu, or specifies a child-window identifier, depending on the window style. For an overlapped or pop-up window, **hMenu** identifies the menu to be used with the window; it can be [null](#) if the class menu is to be used. For a child window, **hMenu** specifies the child-window identifier, an integer value used by a dialog box control to notify its parent about events. The application determines the child-window identifier; it must be unique for all child windows with the same parent window.

param [object](#)

Pointer to a value to be passed to the window through the **CREATESTRUCT** structure (lpCreateParams member) pointed to by the **lParam** parameter of the **WM_CREATE** message. This message is sent to the

created window by this function before it returns. See the Memory Notes section in Remarks for more info.

Remarks

The function sends `WM_NCCREATE`, `WM_NCCALCSIZE`, and `WM_CREATE` messages to the window being created.

If the created window is a child window, its default position is at the bottom of the Z-order. If the created window is a top-level window, its default position is at the top of the Z-order (but beneath all topmost windows unless the created window is itself topmost).

For information on controlling whether the Taskbar displays a button for the created window, see [Managing Taskbar Buttons](#).

The `NoActivate` value for `exStyle` prevents foreground activation by the system. To prevent queue activation when the user clicks on the window, you must process the `WM_MOUSEACTIVATE` message appropriately. To bring the window to the foreground or to activate it programmatically, use [MoveToForeground\(\)](#) or [SetActive\(\)](#). Returning `FALSE` (1) to `WM_NCACTIVATE` prevents the window from losing queue activation. However, the return value is ignored at activation time.

Memory Notes

If you set the `param` parameter to a non-[null](#) value, the `lParam` parameter inside the `WM_CREATE` message will point to a

Fields

WindowDefaultLocation

Specifies the default location for a window.

```
public static readonly Point WindowDefaultLocation
```

Field Value

[Point](#)

WindowDefaultSize

Specifies the default size for a window.

```
public static readonly Size WindowDefaultSize
```

Field Value

[Size](#)↗

Properties

Enabled

Gets or sets the enabled state of the window.

```
public bool Enabled { get; set; }
```

Property Value

[bool](#)↗

Handle

Gets the handle (HWND) of the window.

```
public nint Handle { get; }
```

Property Value

[nint](#)↗

Parent

Gets the parent window of this [Window](#).

```
public Window? Parent { get; }
```

Property Value

[Window](#)

Remarks

To obtain a window's owner window, use `GetWindow` with the `GW_OWNER` flag. To obtain the parent window and not the owner, use `GetAncestor` with the `GA_PARENT` flag.

State

Gets the state of the window.

```
public WindowState State { get; }
```

Property Value

[WindowState](#)

The state of the window.

Text

Gets or sets the text of the window. For top-level windows, this is usually the window title. For controls, this is the text displayed on the control.

```
public string Text { get; set; }
```

Property Value

[string](#)[↗]

Visible

Gets or sets the visibility of the window. For more options, use the [Show\(WindowShowCommand\)](#) or [ShowAsync\(WindowShowCommand\)](#) functions.

```
public bool Visible { get; set; }
```

Property Value

[bool](#) 

Methods

Close()

Closes the window.

```
public void Close()
```

Remarks

There is a difference between closing a window and destroying a window. Closing a window sends the window a `WM_CLOSE` message, which the app can then handle to prompt the user to, for example, save their changes. If the app handles `WM_CLOSE` the app can call the [Destroy\(\)](#) function, which actually destroys the window. The [DefaultWndProc\(Window, uint, uint, object\)](#) automatically does this (without a prompt). Destroying the window is when the actual exiting happens. On destroy, the window receives a `WM_DESTROY` message.

For more details, see [Closing the Window](#) .

To destroy a window, an application must use the [Destroy\(\)](#) function.

Destroy()

Destroys the window.

```
public bool Destroy()
```

Returns

[bool](#) 

[true](#)  if the function succeeded; otherwise, [false](#) .

Remarks

The function sends `WM_DESTROY` and `WM_NCDESTROY` messages to the window to deactivate it and remove the keyboard focus from it. The function also destroys the window's menu, destroys timers, removes clipboard ownership, and breaks the clipboard viewer chain (if the window is at the top of the viewer chain).

If the specified window is a parent or owner window, the [Destroy\(\)](#) function automatically destroys the associated child or owned windows when it destroys the parent or owner window. The function first destroys child or owned windows, and then it destroys the parent or owner window.

[Destroy\(\)](#) also destroys modeless dialog boxes.

A thread cannot use [Destroy\(\)](#) to destroy a window created by a different thread.

If the window being destroyed is a child window that does not have the [NoParentNotify](#) style, a `WM_PARENTNOTIFY` message is sent to the parent.

Equals(object?)

Determines whether two [Window](#) instances represent the same window.

```
public override bool Equals(object? obj)
```

Parameters

`obj` [object](#)

The [Window](#) to compare.

Returns

[bool](#)

[true](#) if the specified [Window](#) represents the same [Window](#); otherwise, [false](#).

Exists(nint)

Determines whether the specified window handle identifies an existing window.

```
public static bool Exists(nint hwnd)
```

Parameters

hwnd [nint](#)

A handle to the window to be tested.

Returns

[bool](#)

[true](#) if the window handle identifies an existing window; otherwise, [false](#).

Remarks

A thread should not use [Exists\(nint\)](#) for a window that it did not create because the window could be destroyed after this function was called. Further, because window handles are recycled the handle could even point to a different window.

FromHandle(nint)

Creates a [Window](#) from an existing handle.

```
public static Window? FromHandle(nint hwnd)
```

Parameters

hwnd [nint](#)

The handle of the window.

Returns

[Window](#)

A [Window](#) corresponding to the handle specified by **hwnd**, or [null](#) if the window does not exist.

FromPoint(Point)

Retrieves the window that is under the specified point.

```
public static Window? FromPoint(Point pt)
```

Parameters

pt [Point](#)

The point to be checked.

Returns

[Window](#)

The window that is under the point. If no window exists at the given point, the return value is [null](#). If the point is over a static text control, the return value is the window under the static text control.

GetHashCode()

Gets the hash code for the current [Window](#).

```
public override int GetHashCode()
```

Returns

[int](#)

The hash code for the current [Window](#).

GetWindowLongPtr(int)

Retrieves information about the specified window. The function also retrieves the value at a specified offset into the extra window memory.

```
public nint GetWindowLongPtr(int nIndex)
```

Parameters

nIndex [int](#)

The zero-based offset to the value to be retrieved. Valid values are in the range zero through the number of bytes of extra window memory, minus the size of a LONG_PTR. To retrieve any other value, specify one of the [WindowLongPtr](#).

Returns

[nint](#)

If the function succeeds, the return value is the requested value.

If the function fails, the return value is zero. To get extended error information, call `<xref href="System.Runtime.InteropServices.Marshal.GetLastPInvokeError" data-throw-if-not-resolved="false"></xref>`.

If `<xref href="FireBlade.WinInteropUtils.Window.SetWindowLongPtr(System.Int32%2cSystem.IntPtr)" data-throw-if-not-resolved="false"></xref>` has not been called previously, `<xref href="FireBlade.WinInteropUtils.Window.GetWindowLongPtr(System.Int32)" data-throw-if-not-resolved="false"></xref>` returns zero for values in the extra window or class memory.

Remarks

Reserve extra window memory by specifying a nonzero value in the `cbWndExtra` member of the [WNDCLASSEX](#) structure used with the [RegisterClassEx](#) function.

MoveToForeground()

Brings the thread that created the window into the foreground and activates the window. Keyboard input is directed to the window, and various visual cues are changed for the user. The system assigns a slightly higher priority to the thread that created the foreground window than it does to other threads.

```
public bool MoveToForeground()
```

Returns

[bool](#)

[true](#) if the window was brought to the foreground; otherwise, [false](#).

Remarks

The system restricts which processes can set the foreground window. A process can set the foreground window by calling [MoveToForeground\(\)](#) only if:

- All of the following conditions are true:
 - The calling process belongs to a desktop application, not a UWP app or a Windows Store app designed for Windows 8 or 8.1.
 - The foreground process has not disabled calls to [MoveToForeground\(\)](#) by a previous call to the [LockSetForegroundWindow](#) function.
 - No menus are active.
- Additionally, at least one of the following conditions is true:
 - The foreground lock time-out has expired.
 - The calling process is the foreground process.
 - The calling process was started by the foreground process.
 - There is currently no foreground window, and thus no foreground process.
 - The calling process received the last input event.
 - Either the foreground process or the calling process is being debugged.

It is possible for a process to be denied the right to set the foreground window even if it meets these conditions.

An application cannot force a window to the foreground while the user is working with another window. Instead, Windows flashes the taskbar button of the window to notify the user.

PostMessage(uint, nuint, nint)

Places (posts) a message in the message queue associated with the thread that created the specified window and returns without waiting for the thread to process the message.

```
public void PostMessage(uint uMsg, nuint wParam, nint lParam)
```

Parameters

uMsg [uint](#) 

The message to be posted.

For lists of the system-provided messages, see [System-Defined Messages](#) .

wParam [nuint](#) 

Additional message-specific information.

LPARAM [nint](#)

Additional message-specific information.

Examples

The following example shows how to post a private window message using the [PostMessage\(uint, uint, nint\)](#) function. Assume you defined a private window message called **WM_COMPLETE**:

```
private const int WM_USER = 0x0400;
private const int WM_COMPLETE = WM_USER + 0;
```

You can post a message to the message queue associated with the thread that created the specified window as shown below:

```
var time = DateTime.Now;
var window = Window.FromHandle(hwnd);

window.PostMessage(WM_COMPLETE, 0, (nint)time.ToFileTimeUtc());
```

Remarks

When a message is blocked by UIPI the last error is set to 5 (access denied).

Messages in a message queue are retrieved by calls to the **GetMessage** or **PeekMessage** function.

Applications that need to communicate using **HWND_BROADCAST** should use the [RegisterWindowMessage\(string\)](#) function to obtain a unique message for inter-application communication.

The system only does marshalling for system messages (those in the range 0 to (**WM_USER**-1)). To send other messages (those $\geq$ **WM_USER**) to another process, you must do custom marshalling.

If you send a message in the range below **WM_USER** to the asynchronous message functions ([PostMessage\(uint, uint, nint\)](#), **SendMessage**, and **SendMessageCallback**), its message parameters cannot include pointers. Otherwise, the operation will fail. The functions will return before the receiving thread has had a chance to process the message and the sender will free the memory before it is used.

Do not post the **WM_QUIT** message using [PostMessage\(uint, uint, nint\)](#); use the **PostQuitMessage** function.

An accessibility application can use [PostMessage\(uint, uint, uint\)](#) to post `WM_APPCOMMAND` messages to the shell to launch applications. This functionality is not guaranteed to work for other types of applications.

There is a limit of 10,000 posted messages per message queue. This limit should be sufficiently large. If your application exceeds the limit, it should be redesigned to avoid consuming so many system resources. To adjust this limit, modify the following registry key:

```
HKEY_LOCAL_MACHINE
SOFTWARE
Microsoft
Windows NT
CurrentVersion
Windows
USERPostMessageLimit
```

The minimum acceptable value is 4000.

Exceptions

[COMException](#)

The 10,000 message quota has been exceeded. There is not enough quota available to process this command.

[COMException](#)

An unknown error occurred.

SendMessage(uint, bool, bool)

Sends the specified message to the window. The [SendMessage\(uint, uint, uint\)](#) function calls the window procedure for the specified window and does not return until the window procedure has processed the message.

To send a message and return immediately, use the [SendMessageCallback](#) or [SendNotifyMessage](#) function. To post a message to a thread's message queue and return immediately, use the [PostMessage](#) or [PostThreadMessage](#) function.

```
public uint SendMessage(uint uMsg, bool wParam, bool lParam)
```

Parameters

uMsg [uint](#)

The message to be sent.

For lists of the system-provided messages, see

[System-Defined Messages](https://learn.microsoft.com/en-us/windows/desktop/winmsg/about-messages-and-message-queues).

wParam [bool](#)

Additional message-specific information.

lParam [bool](#)

Additional message-specific information.

Returns

[nint](#)

The return value specifies the result of the message processing; it depends on the message sent.

Remarks

Message sending is subject to UIPI. The thread of a process can send messages only to message queues of threads in processes of lesser or equal integrity level. When a message is blocked by UIPI the last error, retrieved with [GetLastError\(\)](#), is set to 5 (access denied).

Applications that need to communicate using HWND_BROADCAST should use the [RegisterWindowMessage](#) function to obtain a unique message for inter-application communication.

The system only does marshalling for system messages (those in the range 0 to (WM_USER-1)). To send other messages (those >= WM_USER) to another process, you must do custom marshalling.

If the specified window was created by the calling thread, the window procedure is called immediately as a subroutine. If the specified window was created by a different thread, the system switches to that thread and calls the appropriate window procedure. Messages sent between threads are processed only when the receiving thread executes message retrieval code. The sending thread is blocked until the receiving thread processes the message. However, the sending thread will process incoming nonqueued messages while waiting for its message to be processed. To prevent this, use [SendMessageTimeout](#) with SMTO_BLOCK set. For more information on non-queued messages, see [Messages and Message Queues](#).

An accessibility application can use [SendMessage\(uint, uint, uint\)](#) to send WM_APPCOMMAND messages to the shell to launch applications. This functionality is not guaranteed to work for other types of applications.

SendMessage(uint, bool, uint)

Sends the specified message to the window. The [SendMessage\(uint, uint, uint\)](#) function calls the window procedure for the specified window and does not return until the window procedure has processed the message.

To send a message and return immediately, use the [SendMessageCallback](#) or [SendNotifyMessage](#) function. To post a message to a thread's message queue and return immediately, use the [PostMessage](#) or [PostThreadMessage](#) function.

```
public uint SendMessage(uint uMsg, bool wParam, uint lParam)
```

Parameters

uMsg [uint](#)

The message to be sent.

For lists of the system-provided messages, see

<https://learn.microsoft.com/en-us/windows/desktop/winmsg/about-messages-and-message-queues>>System-Defined Messages.

wParam [bool](#)

Additional message-specific information.

lParam [uint](#)

Additional message-specific information.

Returns

[uint](#)

The return value specifies the result of the message processing; it depends on the message sent.

Remarks

Message sending is subject to UIPI. The thread of a process can send messages only to message queues of threads in processes of lesser or equal integrity level. When a message is blocked by UIPI the last error, retrieved with [GetLastError\(\)](#), is set to 5 (access denied).

Applications that need to communicate using HWND_BROADCAST should use the [RegisterWindowMessage](#) function to obtain a unique message for inter-application communication.

The system only does marshalling for system messages (those in the range 0 to (WM_USER-1)). To send other messages (those $\geq$ WM_USER) to another process, you must do custom marshalling.

If the specified window was created by the calling thread, the window procedure is called immediately as a subroutine. If the specified window was created by a different thread, the system switches to that thread and calls the appropriate window procedure. Messages sent between threads are processed only when the receiving thread executes message retrieval code. The sending thread is blocked until the receiving thread processes the message. However, the sending thread will process incoming nonqueued messages while waiting for its message to be processed. To prevent this, use [SendMessageTimeout](#) with SMTO_BLOCK set. For more information on non-queued messages, see [Messages and Message Queues](#).

An accessibility application can use [SendMessage\(uint, uint, uint\)](#) to send WM_APPCOMMAND messages to the shell to launch applications. This functionality is not guaranteed to work for other types of applications.

SendMessage(uint, uint, bool)

Sends the specified message to the window. The [SendMessage\(uint, uint, uint\)](#) function calls the window procedure for the specified window and does not return until the window procedure has processed the message.

To send a message and return immediately, use the [SendMessageCallback](#) or [SendNotifyMessage](#) function. To post a message to a thread's message queue and return immediately, use the [PostMessage](#) or [PostThreadMessage](#) function.

```
public uint SendMessage(uint uMsg, uint wParam, bool lParam)
```

Parameters

uMsg [uint](#)

The message to be sent.

For lists of the system-provided messages, see

System-Defined Messages.

wParam [nuint](#)

Additional message-specific information.

lParam [bool](#)

Additional message-specific information.

Returns

[nint](#)

The return value specifies the result of the message processing; it depends on the message sent.

Remarks

Message sending is subject to UIPI. The thread of a process can send messages only to message queues of threads in processes of lesser or equal integrity level. When a message is blocked by UIPI the last error, retrieved with [GetLastError\(\)](#), is set to 5 (access denied).

Applications that need to communicate using HWND_BROADCAST should use the [RegisterWindowMessage](#) function to obtain a unique message for inter-application communication.

The system only does marshalling for system messages (those in the range 0 to (WM_USER-1)). To send other messages (those >= WM_USER) to another process, you must do custom marshalling.

If the specified window was created by the calling thread, the window procedure is called immediately as a subroutine. If the specified window was created by a different thread, the system switches to that thread and calls the appropriate window procedure. Messages sent between threads are processed only when the receiving thread executes message retrieval code. The sending thread is blocked until the receiving thread processes the message. However, the sending thread will process incoming nonqueued messages while waiting for its message to be processed. To prevent this, use [SendMessageTimeout](#) with SMTO_BLOCK set. For more information on non-queued messages, see [Messages and Message Queues](#).

An accessibility application can use [SendMessage\(uint, nuint, nint\)](#) to send WM_APPCOMMAND messages to the shell to launch applications. This functionality is not guaranteed to work for other types of applications.

SendMessage(uint, nuint, nint)

Sends the specified message to the window. The [SendMessage\(uint, nuint, nint\)](#) function calls the window procedure for the specified window and does not return until the window procedure has processed the message.

To send a message and return immediately, use the [SendMessageCallback](#) or [SendNotifyMessage](#) function. To post a message to a thread's message queue and return immediately, use the [PostMessage](#) or [PostThreadMessage](#) function.

```
public nint SendMessage(uint uMsg, nuint wParam, nint lParam)
```

Parameters

uMsg [uint](#)

The message to be sent.

For lists of the system-provided messages, see

[System-Defined Messages](https://learn.microsoft.com/en-us/windows/desktop/winmsg/about-messages-and-message-queues).

wParam [nuint](#)

Additional message-specific information.

lParam [nint](#)

Additional message-specific information.

Returns

[nint](#)

The return value specifies the result of the message processing; it depends on the message sent.

Remarks

Message sending is subject to UIPI. The thread of a process can send messages only to message queues of threads in processes of lesser or equal integrity level. When a message is blocked by UIPI the last error, retrieved with [GetLastError\(\)](#), is set to 5 (access denied).

Applications that need to communicate using `HWND_BROADCAST` should use the [RegisterWindowMessage](#) function to obtain a unique message for inter-application communication.

The system only does marshalling for system messages (those in the range 0 to (`WM_USER`-1)). To send other messages (those $\geq$ `WM_USER`) to another process, you must do custom marshalling.

If the specified window was created by the calling thread, the window procedure is called immediately as a subroutine. If the specified window was created by a different thread, the system switches to that thread and calls the appropriate window procedure. Messages sent between threads are processed only when the receiving thread executes message retrieval code. The sending thread is blocked until the receiving thread processes the message. However, the sending thread will process incoming nonqueued messages while waiting for its message to be processed. To prevent this, use [SendMessageTimeout](#) with `SMTO_BLOCK` set. For more information on non-queued messages, see [Messages and Message Queues](#).

An accessibility application can use [SendMessage\(uint, uint, uint\)](#) to send `WM_APPCOMMAND` messages to the shell to launch applications. This functionality is not guaranteed to work for other types of applications.

SetActive()

Activates the window. The window must be attached to the calling thread's message queue.

```
public Window SetActive()
```

Returns

[Window](#)

The window that was previously active.

Remarks

The [SetActive\(\)](#) function activates a window, but not if the application is in the background. The window will be brought into the foreground (top of Z-Order) if its application is in the foreground when the system activates the window.

If the window was created by the calling thread, the active window status of the calling thread is set to the handle of the window. Otherwise, the active window status of the calling thread is set to [null](#).

Exceptions

SetPos(Point, Size, WindowSetPosOptions, Window?)

Changes the size, position, and Z order of a child, pop-up, or top-level window. These windows are ordered according to their appearance on the screen. The topmost window receives the highest rank and is the first window in the Z order.

```
public void SetPos(Point pos, Size size, WindowSetPosOptions options, Window? insertAfter  
= null)
```

Parameters

pos [Point](#)

The new position of the window, in client coordinates.

size [Size](#)

The new size of the window, in pixels.

options [WindowSetPosOptions](#)

The window sizing and positioning flags.

insertAfter [Window](#)

The window to precede the positioned window in the Z order. This parameter must be a window handle or one of the [ZOrderWindows](#).

Remarks

As part of the Vista re-architecture, all services were moved off the interactive desktop into Session 0. Window and window manager operations are only effective inside a session and cross-session attempts to manipulate the window will fail. For more information, see [The Windows Vista Developer Story: Application Compatibility Cookbook](#).

If you have changed certain window data using [SetWindowLongPtr\(int, nint\)](#), you must call [SetPos\(Point, Size, WindowSetPosOptions, Window?\)](#) for the changes to take effect. Use the following combination for **options**: [NoMove](#) | [NoSize](#) | [NoZOrder](#) | [FrameChanged](#).

A window can be made a topmost window either by setting the `insertAfter` parameter to [TopMost](#) and ensuring that the [NoZOrder](#) flag is not set, or by setting a window's position in the Z order so that it is above any existing topmost windows. When a non-topmost window is made topmost, its owned windows are also made topmost. Its owners, however, are not changed.

If neither the [NoActivate](#) nor [NoZOrder](#) flag is specified (that is, when the application requests that a window be simultaneously activated and its position in the Z order changed), the value specified in `insertAfter` is used only in the following circumstances:

- Neither the [TopMost](#) nor [NoTopMost](#) flag is specified in `insertAfter`.
- The window is not the active window.

An application cannot activate an inactive window without also bringing it to the top of the Z order. Applications can change an activated window's position in the Z order without restrictions, or it can activate a window and then move it to the top of the topmost or non-topmost windows.

If a topmost window is repositioned to the bottom ([Bottom](#)) of the Z order or after any non-topmost window, it is no longer topmost. When a topmost window is made non-topmost, its owners and its owned windows are also made non-topmost windows.

A non-topmost window can own a topmost window, but the reverse cannot occur. Any window (for example, a dialog box) owned by a topmost window is itself made a topmost window, to ensure that all owned windows stay above their owner.

If an application is not in the foreground, and should be in the foreground, it must call the [MoveToForeground\(\)](#) function.

To use [SetPos\(Point, Size, WindowSetPosOptions, Window?\)](#) to bring a window to the top, the process that owns the window must have `SetForegroundWindow` permission.

Exceptions

[Win32Exception](#) 

SetWindowLongPtr(int, nint)

Changes an attribute of the specified window. The function also sets a value at the specified offset in the extra window memory.

```
public nint SetWindowLongPtr(int nIndex, nint dwNewLong)
```

Parameters

nIndex [int](#)

The zero-based offset to the value to be set. Valid values are in the range zero through the number of bytes of extra window memory, minus the size of a LONG_PTR. To set any other value, specify one of the [WindowLongPtr](#).

dwNewLong [nint](#)

The replacement value.

Returns

[nint](#)

If the function succeeds, the return value is the previous value of the specified offset.

If the function fails, the return value is zero. To get extended error information, call [System.Runtime.InteropServices.Marshal.GetLastPInvokeError](#) data-throw-if-not-resolved="false".

If the previous value is zero and the function succeeds, the return value is zero, but the function does not clear the last error information.

To determine success or failure, clear the last error information by calling SetLastError with 0, then call [FireBlade.WinInteropUtils.Window.SetWindowLongPtr\(System.Int32%2cSystem.IntPtr\)"](#) data-throw-if-not-resolved="false". Function failure will be indicated by a return value of zero and a [System.Runtime.InteropServices.Marshal.GetLastPInvokeError](#) data-throw-if-not-resolved="false" result that is nonzero.

Remarks

Certain window data is cached, so changes you make using [SetWindowLongPtr\(int, nint\)](#) will not take effect until you call the [SetWindowPos](#) function.

If you use [SetWindowLongPtr\(int, nint\)](#) with the [WndProc](#) index to replace the window procedure, the window procedure must conform to the guidelines specified in the description of the [WndProc](#) callback function.

If you use [SetWindowLongPtr\(int, nint\)](#) with the [MessageResult](#) index to set the return value for a message processed by a dialog box procedure, the dialog box procedure should return [true](#) directly afterward. Otherwise, if you call any function that results in your dialog box procedure receiving a

window message, the nested window message could overwrite the return value you set by using [MessageResult](#).

Calling [SetWindowLongPtr\(int, nint\)](#) with the [WndProc](#) index creates a subclass of the window class used to create the window. An application can subclass a system class, but should not subclass a window class created by another process. The [SetWindowLongPtr\(int, nint\)](#) function creates the window subclass by changing the window procedure associated with a particular window class, causing the system to call the new window procedure instead of the previous one.

Do not call [SetWindowLongPtr\(int, nint\)](#) with the [HWNDParent](#) index to change the parent of a child window. Instead, use the [SetParent](#)  function. [HWNDParent](#) is used to change the owner of a top-level window, not the parent of a child window. A window can have either a parent or an owner, or neither, but never both simultaneously.

Calling [SetWindowLongPtr\(int, nint\)](#) to set the style on a progress bar will reset its position.

Show(WindowShowCommand)

Sets the window's show state.

```
public bool Show(WindowShowCommand cmd)
```

Parameters

cmd [WindowShowCommand](#)

Controls how the window is to be shown.

Returns

[bool](#) 

[true](#)  if the window was previously visible; otherwise, [false](#) .

ShowAsync(WindowShowCommand)

Sets the window's show state asynchronously.

```
public bool ShowAsync(WindowShowCommand cmd)
```

Parameters

cmd [WindowShowCommand](#)

Controls how the window is to be shown.

Returns

[bool](#)

[true](#) if the operation was successfully started; otherwise, [false](#).

Remarks

Note that this function is not "asynchronous" in the C# sense, but that it immediately returns, unlike [Show\(WindowShowCommand\)](#).

This function posts a show-window event to the message queue of the given window. An application can use this function to avoid becoming nonresponsive while waiting for a nonresponsive application to finish processing a show-window event.

ToString()

Returns the text of the window.

```
public override string ToString()
```

Returns

[string](#)

The text of the window.

Operators

explicit operator Window(nint)

Converts the window handle to a [Window](#).

```
public static explicit operator Window(nint hwnd)
```

Parameters

hwnd [nint](#)

The window handle to convert.

Returns

[Window](#)

implicit operator nint(Window)

Converts the [Window](#) to a handle.

```
public static implicit operator nint(Window wnd)
```

Parameters

wnd [Window](#)

The [Window](#) to convert.

Returns

[nint](#)

Class WindowClass

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Represents a window class.

```
public class WindowClass
```

Inheritance

[object](#)  ← WindowClass

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Fields

DlgWindowExtra

The value that should be assigned to [ExtraWindowSize](#) if the application uses [WindowClass](#) to register a dialog box created by using the **CLASS** directive in the resource file.

```
public const int DlgWindowExtra = 30
```

Field Value

[int](#) 

Properties

Atom

A class atom that uniquely identifies the class registered.

```
public ushort Atom { get; }
```

Property Value

[ushort](#)

Remarks

This value will be 0 before the [Register\(\)](#) function is called.

BackgroundBrush

The class background brush. If this member is [null](#), an application must paint its own background whenever it is requested to paint in its client area.

```
public SystemColorId? BackgroundBrush { get; set; }
```

Property Value

[SystemColorId?](#)

ClassIcon

The class icon. This member must be an icon resource. If this member is [null](#), the system provides a default icon.

```
public Icon? ClassIcon { get; set; }
```

Property Value

[Icon](#)

ClassName

A string representing the class name. The maximum length for [ClassName](#) is 256. If [ClassName](#) is greater than the maximum length, the [Register\(\)](#) function will fail.

```
public string ClassName { get; set; }
```

Property Value

[string](#)

ClassStyle

The class style(s). This member can be any combination of the [ClassStyle](#) enumeration.

```
public ClassStyle ClassStyle { get; set; }
```

Property Value

[ClassStyle](#)

ExtraSize

The number of extra bytes to allocate following the window-class structure. The system initializes the bytes to zero.

```
public int ExtraSize { get; set; }
```

Property Value

[int](#)

ExtraWindowSize

The number of extra bytes to allocate following the window instance. The system initializes the bytes to zero. If an application uses [WindowClass](#) to register a dialog box created by using the **CLASS** directive in the resource file, it must set this member to [DlgWindowExtra](#).

```
public int ExtraWindowSize { get; set; }
```

Property Value

[int](#)

Instance

Override the instance handle (HINSTANCE) where the window procedure is defined in.

```
public nint? Instance { get; set; }
```

Property Value

[nint](#)?

MenuName

The resource name of the class menu, as the name appears in the resource file. If you use an integer to identify the menu, use the [MakeIntResource\(int\)](#) macro. If this member is [null](#), windows belonging to this class have no default menu.

```
public string? MenuName { get; set; }
```

Property Value

[string](#)

SmallClassIcon

The small class icon. This member must be an icon resource. If this member is [null](#), the system provides a default icon.

```
public Icon? SmallClassIcon { get; set; }
```

Property Value

[Icon](#)

WindowProcedure

The window procedure. The system calls this function whenever a message is received.

```
public WindowClass.WndProc WindowProcedure { get; set; }
```

Property Value

[WindowClass.WndProc](#)

Methods

DefaultWndProc(Window, uint, uint, object)

Calls the default window procedure to provide default processing for any window messages that an application does not process. This function ensures that every message is processed. DefWindowProc is called with the same parameters received by the window procedure.

```
public static uint DefaultWndProc(Window wnd, uint uMsg, uint wParam, object lParam)
```

Parameters

wnd [Window](#)

The window that received the message.

uMsg [uint](#)

The message.

wParam [uint](#)

Additional message information. The content of this parameter depends on the value of the **uMsg** parameter.

lParam [object](#)

Additional message information. The content of this parameter depends on the value of the **uMsg** parameter.

Returns

[uint](#)

The return value is the result of the message processing and depends on the message.

FromExisting(string)

Gets a [WindowClass](#) instance for an existing window class.

```
public static WindowClass FromExisting(string className)
```

Parameters

className [string](#) 

The name of the class to get.

Returns

[WindowClass](#)

The [WindowClass](#) corresponding to the specified **className**.

FromExisting(string, nint)

Gets a [WindowClass](#) instance for an existing window class.

```
public static WindowClass FromExisting(string className, nint hInstance)
```

Parameters

className [string](#) 

The name of the class to get.

hInstance [nint](#) 

The handle to the module where the class is registered.

Returns

[WindowClass](#)

The [WindowClass](#) corresponding to the specified **className**.

Remarks

Note that the [Atom](#) property of the created [WindowClass](#) may be 0. This is not an error. Windows assigns an atom only when a class is registered through [Register\(\)](#). If the class name already exists — such as a built-in window class or one registered internally by Windows (e.g. through [InitCommonControls](#)) — the system does not return an atom, and Windows doesn't provide an API to query it afterward.

Because Windows does not expose class atoms for pre-existing classes, the library cannot provide an atom value for them and [Atom](#) is set to 0.

Register()

Registers the window class for subsequent use in calls to the [Window\(ExWindowStyles, string, string, uint, Point, Size, Window?, nint?, object?\)](#) function.

```
public void Register()
```

Delegate WindowClass.WndProc

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

A callback function, which you define in your application, that processes messages sent to a window. The **WndProc** name is a placeholder for the name of the function that you define in your application.

```
public delegate nint WindowClass.WndProc(Window window, uint uMsg, nuint wParam, object lParam)
```

Parameters

window [Window](#)

The window that received the message.

uMsg [uint](#)

The message. For lists of the system-provided messages, see [System-defined messages](#).

wParam [nuint](#)

Additional message information. The contents of the **wParam** parameter depends on the value of the **uMsg** parameter.

lParam [object](#)

Additional message information. The contents of the **lParam** parameter depends on the value of the **uMsg** parameter. This value is almost always a [nint](#); see exceptions in the Remarks.

Returns

[nint](#)

The result of the message processing, depending on the message sent.

Remarks

The **lParam** is almost always a [nint](#), but it can be different depending on the message received if **WinInteropUtils** automatically wraps it. The following messages are exceptions to this rule:

Message	LParam type
WM_CREATE	CreateStruct

Your window procedure should forward any unprocessed messages to the [DefaultWndProc\(Window, uint, uint, object\)](#) function.

If your application runs on a 32-bit version of Windows operating system, uncaught exceptions from the callback will be passed onto higher-level exception handlers of your application when available. The system then calls the unhandled exception filter to handle the exception prior to terminating the process. If the PCA is enabled, it will offer to fix the problem the next time you run the application.

However, if your application runs on a 64-bit version of Windows operating system or WOW64, you should be aware that a 64-bit operating system handles uncaught exceptions differently based on its 64-bit processor architecture, exception architecture, and calling convention. The following table summarizes all possible ways that a 64-bit Windows operating system or WOW64 handles uncaught exceptions:

1	The system suppresses any uncaught exceptions.
2	The system first terminates the process, and then the Program Compatibility Assistant (PCA) offers to fix it the next time you run the application. You can disable the PCA mitigation by adding a Compatibility section to the application manifest.
3	The system calls the exception filters but suppresses any uncaught exceptions when it leaves the callback scope, without invoking the associated handlers.

The following table shows how a 64-bit version of the Windows operating system, and WOW64, handles uncaught exceptions. Notice that behavior type 2 applies only to the 64-bit version of the Windows 7 operating system and later:

Windows XP - WOW64	3
Windows XP - x64	1
Windows Vista - WOW64	3
Windows Vista - x64	1
Windows Vista SP1 - WOW64	1
Windows Vista SP1 - x64	1

Windows 7+ - WOW64	1
Windows 7+ - x64	2

> [!NOTE] > On Windows 7 with SP1 (32-bit, 64-bit, or WOW64), the system calls the unhandled exception filter to handle the exception > prior to terminating the process. If the Program Compatibility Assistant (PCA) is enabled, then it will offer to fix the problem > the next time you run the application.

If you need to handle exceptions in your application, you can use structured exception handling to do so. For more information on how to use structured exception handling, see [Structured exception handling](#) .

Class WindowLongPtr

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Defines default index values for [SetWindowLongPtr\(int, nint\)](#) and [GetWindowLongPtr\(int\)](#).

```
public static class WindowLongPtr
```

Inheritance

[object](#)  ← WindowLongPtr

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Fields

AppHInstance

Gets or sets a new application instance handle. (GWLP_HINSTANCE)

```
public const int AppHInstance = -6
```

Field Value

[int](#) 

ExStyle

Gets or sets a new extended window style. (GWL_EXSTYLE)

```
public const int ExStyle = -20
```

Field Value

[int](#)

HWNDParent

Gets or sets a new owner for a top-level window. (GWLP_HWNDPARENT)

```
public const int HWNDParent = -8
```

Field Value

[int](#)

ID

Gets or sets a new identifier of the child window. The window cannot be a top-level window. (GWLP_ID)

```
public const int ID = -12
```

Field Value

[int](#)

MessageResult

Gets or sets the return value of a message processed in the dialog box procedure. (DWLP_MSGRESULT)

```
public const int MessageResult = 0
```

Field Value

[int](#)

Style

Gets or sets a new [window style](#). (GWL_STYLE)

```
public const int Style = -16
```

Field Value

[int](#)

UserData

Gets or sets the user data associated with the window. This data is intended for use by the application that created the window. Its value is initially zero. (GWLP_USERDATA)

```
public const int UserData = -21
```

Field Value

[int](#)

WndProc

Gets or sets a new address for the window procedure. (GWLP_WNDPROC)

```
public const int WndProc = -4
```

Field Value

[int](#)

Enum WindowSetPosOptions

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Window sizing and positioning flags for [SetPos\(Point, Size, WindowSetPosOptions, Window?\)](#).

[Flags]

```
public enum WindowSetPosOptions
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

AsyncWindowPos = 16384

If the calling thread and the thread that owns the window are attached to different input queues, the system posts the request to the thread that owns the window. This prevents the calling thread from blocking its execution while other threads process the request.

DeferErase = 8192

Prevents generation of the **WM_SYNCPAINT** message.

DrawFrame = 32

Draws a frame (defined in the window's class description) around the window.

FrameChanged = 32

Applies new frame styles set using the [SetWindowLongPtr\(int, nint\)](#) function. Sends a **WM_NCCALCSIZE** message to the window, even if the window's size is not being changed. If this flag is not specified, **WM_NCCALCSIZE** is sent only when the window's size is being changed.

HideWindow = 128

Hides the window.

NoActivate = 16

Does not activate the window. If this flag is not set, the window is activated and moved to the top of either the topmost or non-topmost group (depending on the setting of the **insertAfter** parameter).

NoCopyBits = 256

Discards the entire contents of the client area. If this flag is not specified, the valid contents of the client area are saved and copied back into the client area after the window is sized or repositioned.

NoMove = 2

Retains the current position (ignores the **pos** parameter).

NoOwnerZOrder = 512

Does not change the owner window's position in the Z order.

NoRedraw = 8

Does not redraw changes. If this flag is set, no repainting of any kind occurs. This applies to the client area, the nonclient area (including the title bar and scroll bars), and any part of the parent window uncovered as a result of the window being moved. When this flag is set, the application must explicitly invalidate or redraw any parts of the window and parent window that need redrawing.

NoReposition = 512

Same as the [NoOwnerZOrder](#) flag.

NoSendChanging = 1024

Prevents the window from receiving the **WM_WINDOWPOSCHANGING** message.

NoSize = 1

Retains the current size (ignores the **size** parameter).

NoZOrder = 4

Retains the current Z order (ignores the **insertAfter** parameter).

ShowWindow = 64

Displays the window.

Enum WindowShowCommand

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Represents the show command for [Show\(WindowShowCommand\)](#) and [ShowAsync\(WindowShowCommand\)](#).

```
public enum WindowShowCommand
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

ForceMinimize = 11

Minimizes a window, even if the thread that owns the window is not responding. This flag should only be used when minimizing windows from a different thread.

Hide = 0

Hides the window and activates another window.

Maximize = 3

Activates the window and displays it as a maximized window.

Minimize = 6

Minimizes the specified window and activates the next top-level window in the Z order.

Normal = 1

Activates and displays a window. If the window is minimized, maximized, or arranged, the system restores it to its original size and position. An application should specify this flag when displaying the window for the first time.

Restore = 9

Activates and displays the window. If the window is minimized, maximized, or arranged, the system restores it to its original size and position. An application should specify this flag when restoring a minimized window.

`Show = 5`

Activates the window and displays it in its current size and position.

`ShowDefault = 10`

Sets the show state based on the command value specified in the `STARTUPINFO` structure passed to the `CreateProcess` function by the program that started the application.

`ShowMaximized = 3`

Activates the window and displays it as a maximized window.

`ShowMinimized = 2`

Activates the window and displays it as a minimized window.

`ShowMinimizedNoActivate = 7`

Displays the window as a minimized window. This value is similar to [ShowMinimized](#), except the window is not activated.

`ShowNA = 8`

Displays the window in its current size and position. This value is similar to [Show](#), except that the window is not activated.

`ShowNoActivate = 4`

Displays a window in its most recent size and position. This value is similar to [ShowNormal](#), except that the window is not activated.

`ShowNormal = 1`

Activates and displays a window. If the window is minimized, maximized, or arranged, the system restores it to its original size and position. An application should specify this flag when displaying the window for the first time.

Enum WindowState

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Defines states a window can be in.

```
public enum WindowState
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Hidden = 3

The window is hidden.

Iconic = 2

The window is iconic (minimized).

Invisible = 3

The window is hidden.

Maximized = 1

The window is zoomed (maximized).

Minimized = 2

The window is iconic (minimized).

Normal = 0

The window is in the normal state.

Visible = 0

The window is in the normal state.

Zoomed = 1

The window is zoomed (maximized).

Class WindowStyles

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Represents the various standard styles that can be applied to a window (after the window has been created, these styles cannot be modified, except as documented).

```
public static class WindowStyles
```

Inheritance

[object](#)  ← WindowStyles

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Fields

Border

The window has a thin-line border.

```
public const uint Border = 8388608
```

Field Value

[uint](#) 

Caption

The window has a title bar.

```
public const uint Caption = 12582912
```

Field Value

[uint](#)

Child

The window is a child window. A window with this style cannot have a menu bar. This style cannot be used with the [Popup](#) style.

```
public const uint Child = 1073741824
```

Field Value

[uint](#)

ChildWindow

The window is a child window. A window with this style cannot have a menu bar. This style cannot be used with the [Popup](#) style.

```
public const uint ChildWindow = 1073741824
```

Field Value

[uint](#)

ClipChildren

Excludes the area occupied by child windows when drawing occurs within the parent window. This style is used when creating the parent window.

```
public const uint ClipChildren = 33554432
```

Field Value

[uint](#)

ClipSiblings

Clips child windows relative to each other; that is, when a particular child window receives a `WM_PAINT` message, the [ClipSiblings](#) style clips all other overlapping child windows out of the region of the child window to be updated. If [ClipSiblings](#) is not specified and child windows overlap, it is possible, when drawing within the client area of a child window, to draw within the client area of a neighboring child window.

```
public const uint ClipSiblings = 67108864
```

Field Value

[uint](#)

Disabled

The window is initially disabled. A disabled window cannot receive input from the user. To change this after a window has been created, use the [Enabled](#) property.

```
public const uint Disabled = 134217728
```

Field Value

[uint](#)

DlgFrame

The window has a border of a style typically used with dialog boxes. A window with this style cannot have a title bar.

```
public const uint DlgFrame = 4194304
```

Field Value

[uint](#)

Group

The window is the first control of a group of controls. The group consists of this first control and all controls defined after it, up to the next control with the [Group](#) style. The first control in each group usually has the [TabStop](#) style so that the user can move from group to group. The user can subsequently change the keyboard focus from one control in the group to the next control in the group by using the direction keys.

```
public const uint Group = 131072
```

Field Value

[uint](#)

Remarks

You can turn this style on and off to change dialog box navigation. To change this style after a window has been created, use the [SetWindowLongPtr\(int, nint\)](#) function.

HScrollBar

The window has a horizontal scroll bar.

```
public const uint HScrollBar = 1048576
```

Field Value

[uint](#)

Iconic

The window is initially minimized. Same as the [Minimize](#) style.

```
public const uint Iconic = 536870912
```

Field Value

[uint](#)

Maximize

The window is initially maximized.

```
public const uint Maximize = 16777216
```

Field Value

[uint](#)

MaximizeBox

The window has a maximize button. Cannot be combined with the [ContextHelp](#) style. The [SystemMenu](#) style must also be specified.

```
public const uint MaximizeBox = 65536
```

Field Value

[uint](#)

Minimize

The window is initially minimized. Same as the [Iconic](#) style.

```
public const uint Minimize = 536870912
```

Field Value

[uint](#)

MinimizeBox

The window has a minimize button. Cannot be combined with the [ContextHelp](#) style. The [SystemMenu](#) style must also be specified.

```
public const uint MinimizeBox = 131072
```

Field Value

[uint](#)↗

Overlapped

The window is an overlapped window. An overlapped window has a title bar and a border. Same as the [Tiled](#) style.

```
public const uint Overlapped = 0
```

Field Value

[uint](#)↗

OverlappedWindow

The window is an overlapped window. Same as the [TiledWindow](#) style.

```
public const uint OverlappedWindow = 13565952
```

Field Value

[uint](#)↗

Popup

The window is a pop-up window. This style cannot be used with the [Child](#) style.

```
public const uint Popup = 2147483648
```

Field Value

[uint](#)

PopupWindow

The window is a pop-up window. The [Caption](#) and [PopupWindow](#) styles must be combined to make the window menu visible.

```
public const uint PopupWindow = 2156396544
```

Field Value

[uint](#)

SizeBox

The window has a sizing border. Same as the [ThickFrame](#) style.

```
public const uint SizeBox = 262144
```

Field Value

[uint](#)

SystemMenu

The window has a window menu on its title bar. The [Caption](#) style must also be specified.

```
public const uint SystemMenu = 524288
```

Field Value

[uint](#)

TabStop

The window is a control that can receive the keyboard focus when the user presses the **TAB** key. Pressing the **TAB** key changes the keyboard focus to the next control with the [TabStop](#) style.

```
public const uint TabStop = 65536
```

Field Value

[uint](#)

ThickFrame

The window has a sizing border. Same as the [SizeBox](#) style.

```
public const uint ThickFrame = 262144
```

Field Value

[uint](#)

Tiled

The window is a tiled window. A tiled window has a title bar and a border. On Windows versions since Windows 2.0, same as the [Overlapped](#) style.

```
public const uint Tiled = 0
```

Field Value

[uint](#)

TiledWindow

The window is a tiled window. A tiled window has a title bar and a border. On Windows versions since Windows 2.0, same as the [OverlappedWindow](#) style.


```
public const uint TiledWindow = 13565952
```

Field Value

[uint](#)

VScrollBar

The window has a vertical scroll bar.

```
public const uint VScrollBar = 2097152
```

Field Value

[uint](#)

Visible

The window is initially visible.

```
public const uint Visible = 268435456
```

Field Value

[uint](#)

Remarks

This style can be turned on and off by using the [Visible](#) property or the [SetPos\(Point, Size, WindowSetPos Options, Window?\)](#) function.

Class WindowsFile

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Stores data about a Windows file.

```
[Obsolete("Use the new WinFile APIs instead. This class will be removed in the next version of WinInteropUtils.")]  
public class WindowsFile : IDisposable
```

Inheritance

[object](#)  ← WindowsFile

Implements

[IDisposable](#) 

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#) 

Constructors

WindowsFile(SHFILEINFO, string)

Stores data about a Windows file.

```
public WindowsFile(Shell32.SHFILEINFO info, string path)
```

Parameters

info [Shell32.SHFILEINFO](#)

The [Shell32.SHFILEINFO](#) to create the [WindowsFile](#) from.

path [string](#) 

The file path of the file.

Properties

Attributes

Gets the file or directory attributes.

```
public WindowsFileAttributes Attributes { get; }
```

Property Value

[WindowsFileAttributes](#)

Bytes

Gets the bytes of the file.

```
public byte[] Bytes { get; }
```

Property Value

[byte](#)  []

Remarks

Unlike the rest of the [WindowsFile](#) class, this value is not cached.

Content

Gets the text content of the file.

```
public string Content { get; }
```

Property Value

[string](#) 

Remarks

Unlike the rest of the [WindowsFile](#) class, this value is not cached.

DisplayName

Gets the display name of the file.

```
public string DisplayName { get; }
```

Property Value

[string](#)

Disposed

Set to [true](#) when the [WindowsFile](#) instance is disposed.

```
public bool Disposed { get; }
```

Property Value

[bool](#)

FilePath

Gets the file path this [WindowsFile](#) represents.

```
public string FilePath { get; }
```

Property Value

[string](#)

Icon

Gets the icon of the file.

```
public Icon Icon { get; }
```

Property Value

[Icon](#)

IconIndex

Gets the icon index of the file's icon in Windows's internal image list.

```
public int IconIndex { get; }
```

Property Value

[int](#)

TypeName

Gets the display name of the file type of the file that this [WindowsFile](#) stores, or a localized standard Windows value (such as "File folder") for folders.

```
public string TypeName { get; }
```

Property Value

[string](#)

Methods

Dispose()

Frees any resources associated with this [WindowsFile](#).

```
public void Dispose()
```

ToString()

Converts this [WindowsFile](#) to a string.

```
public override string ToString()
```

Returns

[string](#)

This [WindowsFile](#), converted to a string.

Enum WindowsFileAttributes

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Specifies attributes for Windows files.

[Flags]

```
public enum WindowsFileAttributes
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Archive = 32

A file or directory that is an archive file or directory. Applications typically use this attribute to mark files for backup or removal. (FILE_ATTRIBUTE_ARCHIVE)

ArchiveDir = Directory | Archive

An archive directory. Applications typically use this attribute to mark directories for backup or removal. This is a composite value.

Compressed = 2048

A file or directory that is compressed. For a file, all of the data in the file is compressed. For a directory, compression is the default for newly created files and subdirectories.

[Obsolete("This value is reserved for system use. Do not use.")] Device = 64

This value is reserved for system use. Do not use. (FILE_ATTRIBUTE_DEVICE)

Directory = 16

The handle that identifies a directory. (FILE_ATTRIBUTE_DIRECTORY)

Encrypted = 16384

A file or directory that is encrypted. For a file, all data streams in the file are encrypted. For a directory, encryption is the default for newly created files and subdirectories. (FILE_ATTRIBUTE_ENCRYPTED)

[Obsolete("This value is reserved for system use. Do not use.")] ExAttrib = 262144

A file or directory with extended attributes. This value is for internal use only. Do not use.
(FILE_ATTRIBUTE_EA)

Hidden = 2

The file or directory is hidden. It is not included in an ordinary directory listing.
(FILE_ATTRIBUTE_HIDDEN)

IntegrityStream = 32768

The directory or user data stream is configured with integrity (only supported on ReFS volumes). It is not included in an ordinary directory listing. The integrity setting persists with the file if it's renamed. If a file is copied the destination file will have integrity set if either the source file or destination directory have integrity set. **Windows Server 2008 R2, Windows 7, Windows Server 2008, Windows Vista, Windows Server 2003 and Windows XP:** This flag is not supported until *Windows Server 2012*.
(FILE_ATTRIBUTE_INTEGRITY_STREAM)

NoContentIndex = 8192

The file or directory is not to be indexed by the content indexing service.
(FILE_ATTRIBUTE_NOT_CONTENT_INDEXED)

NoScrubData = 131072

The user data stream not to be read by the background data integrity scanner (AKA scrubber). When set on a directory it only provides inheritance. This flag is only supported on Storage Spaces and ReFS volumes. It is not included in an ordinary directory listing. Windows Server 2008 R2, Windows 7, Windows Server 2008, Windows Vista, Windows Server 2003 and Windows XP: This flag is not supported until Windows 8 and Windows Server 2012. (FILE_ATTRIBUTE_NO_SCRUB_DATA)

None = 128

A file that does not have other attributes set. This attribute is valid only when used alone.
(FILE_ATTRIBUTE_NORMAL)

Offline = 4096

The data of a file is not available immediately. This attribute indicates that the file data is physically moved to offline storage. This attribute is used by Remote Storage, which is the hierarchical storage management software. Applications should not arbitrarily change this attribute.
(FILE_ATTRIBUTE_OFFLINE)

Pinned = 524288

This attribute indicates user intent that the file or directory should be kept fully present locally even when not being actively accessed. This attribute is for use with hierarchical storage management software. (FILE_ATTRIBUTE_PINNED)

ReadOnly = 1

A file that is read-only. Applications can read the file, but cannot write to it or delete it. This attribute is not honored on directories. For more information, see [You cannot view or change the Read-only or the System attributes of folders in Windows Server 2003, in Windows XP, in Windows Vista or in Windows 7](#). (FILE_ATTRIBUTE_READONLY)

RecallOnDataAccess = 4194304

When this attribute is set, it means that the file or directory is not fully present locally. For a file that means that not all of its data is on local storage (e.g. it may be sparse with some data still in remote storage). For a directory it means that some of the directory contents are being virtualized from another location. Reading the file / enumerating the directory will be more expensive than normal, e.g. it will cause at least some of the file/directory content to be fetched from a remote store. Only kernel-mode callers can set this bit. File system mini filters below the 180000 – 189999 altitude range (FSFilter HSM Load Order Group) must not issue targeted cached reads or writes to files that have this attribute set. This could lead to cache pollution and potential file corruption. For more information, see [Handling placeholders](#). (FILE_ATTRIBUTE_RECALL_ON_DATA_ACCESS)

RecallOnOpen = 262144

This attribute only appears in directory enumeration classes (FILE_DIRECTORY_INFORMATION, FILE_BOTH_DIR_INFORMATION, etc.). When this attribute is set, it means that the file or directory has no physical representation on the local system; the item is virtual. Opening the item will be more expensive than normal, e.g. it will cause at least some of it to be fetched from a remote store. (FILE_ATTRIBUTE_RECALL_ON_OPEN)

ReparsePoint = 1024

A file or directory that has an associated reparse point, or a file that is a symbolic link.

Sparse = 512

A file that is a sparse file. (FILE_ATTRIBUTE_SPARSE_FILE)

System = 4

A file or directory that the operating system uses as part of, or uses exclusively. (FILE_ATTRIBUTE_SYSTEM)

SystemFile = Hidden | System

A system file. A system file has the [Hidden](#) and [System](#) attributes. This is a composite value.

Temporary = 256

A file that is being used for temporary storage. File systems avoid writing data back to mass storage if sufficient cache memory is available, because typically, an application deletes a temporary file after the handle is closed. In that scenario, the system can entirely avoid writing the data. Otherwise, the data is written after the handle is closed. (FILE_ATTRIBUTE_TEMPORARY)

Unpinned = 1048576

This attribute indicates that the file or directory should not be kept fully present locally except when being actively accessed. This attribute is for use with hierarchical storage management software. (FILE_ATTRIBUTE_UNPINNED)

[Obsolete("This value is reserved for system use. Do not use.")] Virtual = 65536

This value is reserved for system use. Do not use. (FILE_ATTRIBUTE_VIRTUAL)

See Also

[File Attribute Constants](#)

Class ZOrderWindows

Namespace: [FireBlade.WinInteropUtils](#)

Assembly: WinInteropUtils.dll

Represents windows used for Z position ordering.

```
public static class ZOrderWindows
```

Inheritance

[object](#)  ← ZOrderWindows

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Remarks

Note that you cannot perform operations on these windows, as they are not valid window handles.

Fields

Bottom

Places the window at the bottom of the Z order. If the window identifies a topmost window, the window loses its topmost status and is placed at the bottom of all other windows.

```
public static readonly Window Bottom
```

Field Value

[Window](#)

NoTopMost

Places the window above all non-topmost windows (that is, behind all topmost windows). This flag has no effect if the window is already a non-topmost window.

```
public static readonly Window NoTopMost
```

Field Value

[Window](#)

Top

Places the window at the top of the Z order.

```
public static readonly Window Top
```

Field Value

[Window](#)

TopMost

Places the window above all non-topmost windows. The window maintains its topmost position even when it is deactivated.

```
public static readonly Window TopMost
```

Field Value

[Window](#)

Namespace FireBlade.WinInteropUtils. ComponentObjectModel

Classes

[COM](#)

Provides basic COM (Component-Object Model) methods from `01e32.dll` required to use COM.

Enums

[COM.COMInitOptions](#)

Specifies flags for [Initialize\(COMInitOptions\)](#).

[COM.CreateInstanceContext](#)

Values that are used in activation calls to indicate the execution contexts in which an object is to be run. Used in [CreateInstance<TCoInterface>\(Guid, object?, CreateInstanceContext\)](#).

Class COM

Namespace: [FireBlade.WinInteropUtils.ComponentObjectModel](#)

Assembly: WinInteropUtils.dll

Provides basic COM (Component-Object Model) methods from `ole32.dll` required to use COM.

```
public static class COM
```

Inheritance

[object](#)  ← COM

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Methods

CreateInstance<TCoInterface>(Guid, object?, CreateInstanceContext)

Creates and default-initializes a single object of the class associated with a specified CLSID.

Call [CreateInstance<TCoInterface>\(Guid, object?, CreateInstanceContext\)](#) when you want to create only one object on the local system. To create a single object on a remote system, call the `CoCreateInstanceEx` function. To create multiple objects based on a single CLSID, call the `CoGetClassObject` function.

```
public static TCoInterface? CreateInstance<TCoInterface>(Guid rclsid, object? pUnkOuter,  
COM.CreateInstanceContext dwClsContext)
```

Parameters

`rclsid` [Guid](#) 

The CLSID associated with the data and code that will be used to create the object.

`pUnkOuter` [object](#) 

If [null](#), indicates that the object is not being created as part of an aggregate. If non-[null](#), pointer to the aggregate object's `IUnknown` interface (the controlling `IUnknown`).

`dwClsContext` [COM.CreateInstanceContext](#)

Context in which the code that manages the newly created object will run. A bitwise combination of [COM.CreateInstanceContext](#) values.

Returns

`TCoInterface`

The requested interface, or [null](#).

Type Parameters

`TCoInterface`

The COM interface to create an instance of.

Remarks

The [CreateInstance<TCoInterface>\(Guid, object?, CreateInstanceContext\)](#) function provides a convenient shortcut by connecting to the class object associated with the specified CLSID, creating a default-initialized instance, and releasing the class object.

It is convenient to use [CreateInstance<TCoInterface>\(Guid, object?, CreateInstanceContext\)](#) when you need to create only a single instance of an object on the local machine. If you are creating an instance on remote computer, call `CoCreateInstanceEx`.

In the [COM.CreateInstanceContext](#) enumeration, you can specify the type of server used to manage the object. The constants can be [InprocServer](#), [InprocHandler](#), [LocalServer](#), [RemoteServer](#) or any combination of these values.

UWP applications

Although there are no restrictions on which CLSIDs a UWP application can pass to [CreateInstance<TCoInterface>\(Guid, object?, CreateInstanceContext\)](#), many objects will fail with [UnauthorizedAccessException](#) for security reasons, especially if they do not run in-process. Additionally, even if you can successfully create an object, it might fail at a later time due to UWP security constraints, app-model differences, etc. In particular, background tasks should limit the objects they communicate with to avoid hangs or other complications due to connected stand-by.

Exceptions

[InvalidOperationException](#)

TCoInterface must be an interface.

[ArgumentException](#)

`pUnkOuter` must be a COM object or [null](#).

[COMException](#)

This exception can get thrown in 2 cases:

- A specified class is not registered in the registration database. Also can indicate that the type of server you requested in the [COM.CreateInstanceContext](#) enumeration is not registered or the values for the server types in the registry are corrupt.
- This class cannot be created as part of an aggregate.

[InvalidCastException](#)

The specified class does not implement the requested interface, or the controlling `IUnknown` does not expose the requested interface.

[UnauthorizedAccessException](#)

The request may be denied because the COM object cannot be created in UWP applications.

CreateInstance<TCoInterface>(Guid, object?, CreateInstanceContext, Guid, out TCoInterface?)

Creates and default-initializes a single object of the class associated with a specified CLSID.

Call `CoCreateInstance` when you want to create only one object on the local system. To create a single object on a remote system, call the `CoCreateInstanceEx` function. To create multiple objects based on a single CLSID, call the `CoGetClassObject` function.

```
[Obsolete("Use the new overload instead. The new overload automatically detects the riid to prevent runtime errors, and uses .NET-style exceptions instead of the old HRESULT system.")]  
public static HRESULT CreateInstance<TCoInterface>(Guid rclsid, object? pUnkOuter,  
COM.CreateInstanceContext dwClsContext, Guid riid, out TCoInterface? ppv)
```

Parameters

`rclsid` [Guid](#)

The CLSID associated with the data and code that will be used to create the object.

pUnkOuter [object](#)

If [null](#), indicates that the object is not being created as part of an aggregate. If non-[null](#), pointer to the aggregate object's **IUnknown** interface (the controlling **IUnknown**).

dwClsContext [COM.CreateInstanceContext](#)

Context in which the code that manages the newly created object will run. A bitwise combination of [COM.CreateInstanceContext](#) values.

riid [Guid](#)

A reference to the identifier of the interface to be used to communicate with the object.

ppv TColInterface

Upon successful return, **ppv** contains the requested interface. Upon failure, **ppv** contains [null](#).

Returns

[HRESULT](#)

A [HRESULT](#). It can be one of the following values:

FireBlade.Win InteropUtils. HRESULT.S_OK	An instance of the specified object class was successfully created.
FireBlade.Win InteropUtils. HRESULT.REGDB_E_ CLASSNOTREG	A specified class is not registered in the registration database. Also can indicate that the type of server you requested in the COM.CreateInstanceContext enumeration is not registered or the values for the server types in the registry are corrupt.
FireBlade.Win InteropUtils. HRESULT.CLASS_E_ NOAGGREGATION	This class cannot be created as part of an aggregate.
FireBlade.Win InteropUtils. HRESULT.E_ NOINTERFACE	The specified class does not implement the requested interface, or the controlling IUnknown does not expose the requested interface.

FireBlade.Win
InteropUtils.
HResult.E_
POINTER

The `ppv` parameter is [null](#).

Type Parameters

TCoInterface

Remarks

The `CreateInstance` function provides a convenient shortcut by connecting to the class object associated with the specified CLSID, creating a default-initialized instance, and releasing the class object.

It is convenient to use `CreateInstance` when you need to create only a single instance of an object on the local machine. If you are creating an instance on remote computer, call `CoCreateInstanceEx`.

In the [COM.CreateInstanceContext](#) enumeration, you can specify the type of server used to manage the object. The constants can be [InprocServer](#), [InprocHandler](#), [LocalServer](#), [RemoteServer](#) or any combination of these values.

UWP applications

Although there are no restrictions on which CLSIDs a UWP application can pass to `CreateInstance`, many objects will fail with `FireBlade.WinInteropUtils.HResult.E_ACCESSDENIED` for security reasons, especially if they do not run in-process. Additionally, even if you can successfully create an object, it might fail at a later time due to UWP security constraints, app-model differences, etc. In particular, background tasks should limit the objects they communicate with to avoid hangs or other complications due to connected stand-by.

Exceptions

[InvalidOperationException](#)

TCoInterface must be an interface.

Initialize()

Initializes the COM library on the current thread and identifies the concurrency model as single-thread apartment (STA).

```
public static HRESULT Initialize()
```

Returns

[HRESULT](#)

This function can return the standard return values `FireBlade.WinInteropUtils.HRESULT.E_INVALIDARG`, `FireBlade.WinInteropUtils.HRESULT.E_OUTOFMEMORY`, and `FireBlade.WinInteropUtils.HRESULT.E_UNEXPECTED`, as well as the following values:

FireBlade.WinInteropUtils.HRESULT.S_OK	The COM library was initialized successfully on this thread.
FireBlade.WinInteropUtils.HRESULT.S_FALSE	The COM library is already initialized on this thread.
FireBlade.WinInteropUtils.HRESULT.RPC_E_CHANGED_MODE	A previous call to Initialize(COMInitOptions) specified the concurrency model for this thread as multithread apartment (MTA). This could also indicate that a change from neutral-threaded apartment to single-threaded apartment has occurred.

Remarks

New applications should call [Initialize\(COMInitOptions\)](#) instead of [Initialize\(\)](#).

If you want to use the Windows Runtime, you must call [RoInitialize](#) or [Windows::Foundation::Initialize](#) instead.

Initialize(COMInitOptions)

Initializes the COM library for use by the calling thread, sets the thread's concurrency model, and creates a new apartment for the thread if one is required.

```
public static HRESULT Initialize(COM.COMInitOptions options)
```

Parameters

options [COM.COMInitOptions](#)

The concurrency model and initialization options for the thread. Any combination of values from [COM.COMInitOptions](#) can be used, except that the [ApartmentThreaded](#) and [MultiThreaded](#) flags cannot both be set. The default is [MultiThreaded](#).

Returns

[HResult](#)

This function can return either `FireBlade.WinInteropUtils.HResult.S_OK` or `FireBlade.WinInteropUtils.HResult.S_FALSE`.

Remarks

[Initialize\(COMInitOptions\)](#) must be called at least once, and is usually called only once, for each thread that uses the COM library. Multiple calls to [Initialize\(COMInitOptions\)](#) by the same thread are allowed as long as they pass the same concurrency flag, but subsequent valid calls return `FireBlade.WinInteropUtils.HResult.S_FALSE`. If the concurrency flag does not match, then the call fails and returns `FireBlade.WinInteropUtils.HResult.RPC_E_CHANGED_MODE`. (For the purpose of this rule, a call to `Colnitalize` is equivalent to calling [Initialize\(COMInitOptions\)](#) with the [ApartmentThreaded](#) flag.) To uninitialized the COM library gracefully on a thread, each successful call to `Colnitalize` or [Initialize\(COMInitOptions\)](#), including any call that returns `FireBlade.WinInteropUtils.HResult.S_FALSE`, must be balanced by a corresponding call to [Uninitialize\(\)](#). Once COM has been uninitialized on a thread, you can reinitialize it in any mode, subject to the constraints above.

You need to initialize the COM library on a thread before you call any of the library functions except `CoGetMalloc`, to get a pointer to the standard allocator, and the memory allocation functions. Otherwise, the COM function will return `FireBlade.WinInteropUtils.HResult.CO_E_NOTINITIALIZED`.

Objects created in a single-threaded apartment (STA) receive method calls only from their apartment's thread, so calls are serialized and arrive only at message-queue boundaries (when the `PeekMessage` or `SendMessage` function is called).

Objects created on a COM thread in a multithread apartment (MTA) must be able to receive method calls from other threads at any time. You would typically implement some form of concurrency control in a multithreaded object's code using synchronization primitives such as critical sections, semaphores, or mutexes to help protect the object's data.

When an object that is configured to run in the neutral threaded apartment (NTA) is called by a thread that is in either an STA or the MTA, that thread transfers to the NTA. If this thread subsequently calls [Initialize\(COMInitOptions\)](#), the call fails and returns `FireBlade.WinInteropUtils.HResult.RPC_E_CHANGED_MODE`.

Because OLE technologies are not thread-safe, the OleInitialize function calls [Initialize\(COMInitOptions\)](#) with the [ApartmentThreaded](#) flag. As a result, an apartment that is initialized for multithreaded object concurrency cannot use the features enabled by OleInitialize.

Because there is no way to control the order in which in-process servers are loaded or unloaded, do not call CoInitialize, [Initialize\(COMInitOptions\)](#), or [Uninitialize\(\)](#) from the `DllMain` function.

QueryInterface<TCoInterface, TOut>(TCoInterface)

Queries a COM interface for another interface.

```
public static TOut QueryInterface<TCoInterface, TOut>(TCoInterface i) where TCoInterface :  
    notnull where TOut : notnull
```

Parameters

i TCoInterface

The interface to query.

Returns

TOut

The COM interface requested.

Type Parameters

TCoInterface

The COM interface to query.

TOut

The COM interface requested.

Exceptions

[ArgumentException](#) 

TCoInterface and TOut must be COM interfaces.

A different exception returned by the object's `QueryInterface` implementation.

Uninitialize()

Closes the COM library on the current thread, unloads all DLLs loaded by the thread, frees any other resources that the thread maintains, and forces all RPC connections on the thread to close.

```
public static extern void Uninitialize()
```

Remarks

A thread must call [Uninitialize\(\)](#) once for each successful call it has made to the [Initialize\(COMInitOptions\)](#) function, including any call that returns `FireBlade.WinInteropUtils.HResult.S_FALSE`. Only the [Uninitialize\(\)](#) call corresponding to the [Initialize\(COMInitOptions\)](#) call that initialized the library can close it.

Calls to `OleInitialize` must be balanced by calls to `OleUninitialize`. The `OleUninitialize` function calls [Uninitialize\(\)](#) internally, so applications that call `OleUninitialize` do not also need to call [Uninitialize\(\)](#).

[Uninitialize\(\)](#) should be called on application shutdown, as the last call made to the COM library after the application hides its main windows and falls through its main message loop. If there are open conversations remaining, [Uninitialize\(\)](#) starts a modal message loop and dispatches any pending messages from the containers or server for this COM application. By dispatching the messages, [Uninitialize\(\)](#) ensures that the application does not quit before receiving all of its pending messages. Non-COM messages are discarded.

Because there is no way to control the order in which in-process servers are loaded or unloaded, do not call [Initialize\(\)](#), [Initialize\(COMInitOptions\)](#), or [Uninitialize\(\)](#) from the `DllMain` function.

Enum COM.COMInitOptions

Namespace: [FireBlade.WinInteropUtils.ComponentObjectModel](#)

Assembly: WinInteropUtils.dll

Specifies flags for [Initialize\(COMInitOptions\)](#).

[Flags]

```
public enum COM.COMInitOptions
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

ApartmentThreaded = 2

Initializes the thread for STA (Single-Threaded Apartment).

DisableOle1DDE = 4

Disables DDE for OLE1 support.

MultiThreaded = 0

Initializes the thread for MTA (Multi-Threaded Apartment). This is the default.

SpeedOverMemory = 8

Increase memory usage in an attempt to increase performance.

Remarks

When a thread is initialized through a call to [Initialize\(COMInitOptions\)](#), you choose whether to initialize it as apartment-threaded or multithreaded by designating one of the members of [COM.COMInitOptions](#) as its parameter. This designates how incoming calls to any object created by that thread are handled, that is, the object's concurrency.

Apartment-threading, while allowing for multiple threads of execution, serializes all incoming calls by requiring that calls to methods of objects created by this thread always run on the same thread, i.e. the apartment/thread that created them. In addition, calls can arrive only at message-queue boundaries. Because of this serialization, it is not typically necessary to write concurrency control into the code for

the object, other than to avoid calls to [PeekMessage](#) and [SendMessage\(uint, uint, uint\)](#) during processing that must not be interrupted by other method invocations or calls to other objects in the same apartment/thread.

Multi-threading (also called free-threading) allows calls to methods of objects created by this thread to be run on any thread. There is no serialization of calls, i.e. many calls may occur to the same method or to the same object or simultaneously. Multi-threaded object concurrency offers the highest performance and takes the best advantage of multiprocessor hardware for cross-thread, cross-process, and cross-machine calling, since calls to objects are not serialized in any way. This means, however, that the code for objects must enforce its own concurrency model, typically through the use of synchronization primitives, such as critical sections, semaphores, or mutexes. In addition, because the object doesn't control the lifetime of the threads that are accessing it, no thread-specific state may be stored in the object (in [Thread Local Storage](#)).

Note: The multi-threaded apartment is intended for use by non-GUI threads. Threads in multi-threaded apartments should not perform UI actions. This is because UI threads require a message pump, and COM does not pump messages for threads in a multi-threaded apartment.

Enum COM.CreateInstanceContext

Namespace: [FireBlade.WinInteropUtils.ComponentObjectModel](#)

Assembly: WinInteropUtils.dll

Values that are used in activation calls to indicate the execution contexts in which an object is to be run.

Used in [CreateInstance<TCoInterface>\(Guid, object?, CreateInstanceContext\)](#).

[Flags]

```
public enum COM.CreateInstanceContext
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Activate32BitServer = 262144

Activate or connect to a 32-bit version of the server; fail if one is not registered.

Activate64BitServer = 524288

Activate or connect to a 64 bit version of the server; fail if one is not registered.

ActivateActivateAsActivatorAsUI = 8388608

Specify this flag for Interactive User activation behavior for As-Activator servers. A strongly named Medium IL Windows Store app can use this flag to launch an "As Activator" [COM](#) server without a strong name. Also, you can use this flag to bind to a running instance of the [COM](#) server that's launched by a desktop application.

The client must be Medium IL, it must be strongly named, which means that it has a SysAppID in the client token, it can't be in session 0, and it must have the same user as the session ID's user in the client token.

If the server is out-of-process and "As Activator", it launches the server with the token of the client token's session user. This token won't be strongly named.

If the server is out-of-process and RunAs "Interactive User", this flag has no effect.

If the server is out-of-process and is any other RunAs type, the activation fails.

This flag has no effect for in-process servers.

Off-machine activations fail when they use this flag.

ActivateArm32Server = 33554432

CLSCTX_ACTIVATE_ARM32_SERVER

ActivateX86Server = 262144

Activate or connect to a 32-bit version of the server; fail if one is not registered.

[Obsolete("This flag is reserved.", DiagnosticId = "WIU128")] AppContainer = 4194304

Indicates activation is for an app container.

This flag is reserved for internal use and is not intended to be used directly from your code.

DisableActivateAsActivator = 32768

Disables activate-as-activator (AAA) activations for this activation only. This flag overrides the setting of the EOAC_DISABLE_AAA flag from the EOLE_AUTHENTICATION_CAPABILITIES enumeration. This flag cannot be set at the same time as [EnableActivateAsActivator](#). Any activation where a server process would be launched under the caller's identity is known as an activate-as-activator (AAA) activation. Disabling AAA activations allows an application that runs under a privileged account (such as LocalSystem) to help prevent its identity from being used to launch untrusted components. Library applications that use activation calls should always set this flag during those calls. This helps prevent the library application from being used in an escalation-of-privilege security attack. This is the only way to disable AAA activations in a library application because the EOAC_DISABLE_AAA flag from the EOLE_AUTHENTICATION_CAPABILITIES enumeration is applied only to the server process and not to the library application.

EnableActivateAsActivator = 65536

Enables activate-as-activator (AAA) activations for this activation only. This flag overrides the setting of the EOAC_DISABLE_AAA flag from the EOLE_AUTHENTICATION_CAPABILITIES enumeration. This flag cannot be set at the same time as [DisableActivateAsActivator](#). Any activation where a server process would be launched under the caller's identity is known as an activate-as-activator (AAA) activation. Enabling this flag allows an application to transfer its identity to an activated component.

EnableCloaking = 1048576

When this flag is specified, [COM](#) uses the impersonation token of the thread, if one is present, for the activation request made by the thread. When this flag is not specified or if the thread does not have

an impersonation token, [COM](#) uses the process token of the thread's process for the activation request made by the thread.

EnableCodeDownload = 8192

Enables the downloading of code from the directory service or the Internet. This flag cannot be set at the same time as [NoCodeDownload](#).

FromDefaultContext = 131072

Begin this activation from the default context of the current apartment.

[Obsolete("This value is obsolete.")] Inproc16 = 8

Obsolete.

InprocHandler = 2

The code that manages objects of this class is an in-process handler. This is a DLL that runs in the client process and implements client-side structures of this class when instances of the class are accessed remotely.

[Obsolete("This value is obsolete.")] InprocHandler16 = 32

Obsolete.

InprocServer = 1

The code that creates and manages objects of this class is a DLL that runs in the same process as the caller of the function specifying the class context.

LocalServer = 4

The EXE code that creates and manages objects of this class runs on same machine but is loaded in a separate process space.

NoCodeDownload = 1024

Disables the downloading of code from the directory service or the Internet. This flag cannot be set at the same time as [EnableCodeDownload](#).

NoCustomMarshal = 4096

Specify if you want the activation to fail if it uses custom marshalling.

NoFailureLog = 16384

The [NoFailureLog](#) can be used to override the logging of failures in `CoCreateInstanceEx`.

If the `ActivationFailureLoggingLevel` is created, the following values can determine the status of event logging:

- 0 = Discretionary logging. Log by default, but clients can override by specifying [NoFailureLog](#) in `CoCreateInstanceEx`.
- 1 = Always log all failures no matter what the client specified.
- 2 = Never log any failures no matter what client specified. If the registry entry is missing, the default is 0. If you need to control customer applications, it is recommended that you set this value to 0 and write the client code to override failures. It is strongly recommended that you do not set the value to 2. If event logging is disabled, it is more difficult to diagnose problems.

`[Obsolete("This value is reserved.", DiagnosticId = "WIU128")] ProxyStubDll = -2147483648`

Used for loading Proxy/Stub DLLs. (CLSCTX_PS_DLL)

This flag is reserved for internal use and is not intended to be used directly from your code.

`RemoteServer = 16`

A remote context. The `LocalServer32` or `LocalService` code that creates and manages objects of this class is run on a different computer.

`[Obsolete("This value is reserved.", DiagnosticId = "WIU128")] Reserved1 = 64`

Reserved.

`[Obsolete("This value is reserved.", DiagnosticId = "WIU128")] Reserved2 = 128`

Reserved.

`[Obsolete("This value is reserved.", DiagnosticId = "WIU128")] Reserved3 = 256`

Reserved.

`[Obsolete("This value is reserved.", DiagnosticId = "WIU128")] Reserved4 = 512`

Reserved.

`[Obsolete("This value is reserved.", DiagnosticId = "WIU128")] Reserved5 = 2048`

Reserved.

`[Obsolete("This value is reserved.", DiagnosticId = "WIU128")] Reserved6 = 16777216`

This value is reserved. (CLSCTX_RESERVED6)

Remarks

Values from the [COM.CreateInstanceContext](#) enumeration are used in activation calls ([CreateInstance<TCoInterface>\(Guid, object?, CreateInstanceContext\)](#), [CoCreateInstanceEx](#), [CoGetClassObject](#), and so on) to indicate the preferred execution contexts (in-process, local, or remote) in which an object is to be run. They are also used in calls to [CoRegisterClassObject](#) to indicate the set of execution contexts in which a class object is to be made available for requests to construct instances.

To indicate that more than one context is acceptable, you can combine multiple values with Boolean ORs. The contexts are tried in the order in which they are listed.

Given a set of [COM.CreateInstanceContext](#) flags, the execution context to be used depends on the availability of registered class codes and other parameters according to the following algorithm.

1. If the call specifies one of the following, [RemoteServer](#) is implied and is added to the list of flags:
 - An explicit [COSERVERINFO](#) structure indicating a machine different from the current computer.
 - No explicit [COSERVERINFO](#) structure specified but the specified class is registered with either the [RemoteServerName](#) or [ActivateAtStorage](#) registry value.

The second case allows applications written prior to the release of distributed [COM](#) to be the configuration of classes for remote activation to be used by client applications available prior to DCOM and the [RemoteServer](#) flag. The cases in which there would be no explicit [COSERVERINFO](#) structure are when the value is specified as [null](#) or when it is not one of the function parameters (as in calls to [CreateInstance<TCoInterface>\(Guid, object?, CreateInstanceContext\)](#) and [CoGetClassObject](#)).

2. If the explicit [COSERVERINFO](#) parameter indicates the current computer, [RemoteServer](#) is removed if present.

The rest of the processing proceeds by looking at the value(s) in the following sequence:

1. If the flags include [RemoteServer](#) and no [COSERVERINFO](#) parameter is specified and if the activation request indicates a persistent state from which to initialize the object (with [CoGetInstanceFromFile](#), [CoGetInstanceFromIStorage](#), or, for a file moniker, in a call to [IMoniker::BindToObject](#)) and the class has an [ActivateAtStorage](#) subkey or no class registry information whatsoever, the request to activate and initialize is forwarded to the computer where the persistent state resides. (Refer to the remote activation functions listed in the See Also section for details.)
2. If the flags include [InprocServer](#), the class code in the DLL found under the class's [InprocServer32](#) key is used if this key exists. The class code will run within the same process as the caller.
3. If the flags include [InprocHandler](#), the class code in the DLL found under the class's [InprocHandler32](#) key is used if this key exists. The class code will run within the same process as the caller.
4. If the flags include [LocalServer](#), the class code in the service found under the class's [LocalService](#) key is used if this key exists. If no service is specified but an EXE is specified under that same key, the

class code associated with that EXE is used. The class code (in either case) will be run in a separate service process on the same computer as the caller.

5. If the flag is set to [RemoteServer](#) and an additional COSERVERINFO parameter to the function specifies a particular remote computer, a request to activate is forwarded to this remote computer with flags modified to set to [LocalServer](#). The class code will run in its own process on this specific computer, which must be different from that of the caller.
6. Finally, if the flags include [RemoteServer](#) and no COSERVERINFO parameter is specified and if a computer name is given under the class's [RemoteServerName](#) named-value, the request to activate is forwarded to this remote computer with the flags modified to be set to [LocalServer](#). The class code will run in its own process on this specific computer, which must be different from that of the caller.

[Activate32BitServer](#) and [Activate64BitServer](#)

The 64-bit versions of Windows introduce two new flags: [Activate32BitServer](#) and [Activate64BitServer](#). On a 64-bit computer, a 32-bit and 64-bit version of the same [COM](#) server may coexist. When a client requests an activation of an out-of-process server, these [COM.CreateInstanceContext](#) flags allow the client to specify a 32-bit or a 64-bit version of the server.

Usually, a client will not care whether it uses a 32-bit or a 64-bit version of the server. However, if the server itself loads an additional in-process server, then it and the in-process server must both be either 32-bit or 64-bit. For example, suppose that the client wants to use a server "A", which in turn loads an in-process server "B". If only a 32-bit version of server "B" is available, then the client must specify the 32-bit version of server "A". If only a 64-bit version of server "B" is available, then the client must specify the 64-bit version of server "A".

A server can specify its own architecture preference via the [PreferredServerBitness](#) registry key, but the client's preference, specified via a [Activate32BitServer](#) or [Activate64BitServer](#) flag, will override the server's preference. If the client does not specify a preference, then the server's preference will be used.

If neither the client nor the server specifies a preference, then:

- If the computer that hosts the server is running Windows Server 2003 with Service Pack 1 (SP1) or a later system, then [COM](#) will try to match the server architecture to the client architecture. In other words, for a 32-bit client, [COM](#) will activate a 32-bit server if available; otherwise it will activate a 64-bit version of the server. For a 64-bit client, [COM](#) will activate a 64-bit server if available; otherwise it will activate a 32-bit server.
- If the computer that hosts the server is running Windows XP or Windows Server 2003 without SP1 or later installed, then [COM](#) will prefer a 64-bit version of the server if available; otherwise it will activate a 32-bit version of the server.

If a [COM.CreateInstanceContext](#) enumeration has both the [Activate32BitServer](#) and [Activate64BitServer](#) flags set, then it is invalid and the activation will return `FireBlade.WinInteropUtils.HResult.E_INVALIDARG`.

The flags [Activate32BitServer](#) and [Activate64BitServer](#) flow across computer boundaries. If the computer that hosts the server is running the 64-bit Windows, then it will honor these flags; otherwise it will ignore them.

Namespace FireBlade.WinInteropUtils. ComponentObjectModel.Interfaces

Interfaces

[IFileDialogCustomize](#)

Exposes methods that allow an application to add controls to a common item dialog.

[IFileDialogOpenDialog](#)

Extends the `IFileDialog` interface by adding methods specific to the open dialog.

[IFileDialogSaveDialog](#)

Extends the `IFileDialog` interface by adding methods specific to the save dialog, which include those that provide support for the collection of metadata to be persisted with the file.

[IShellItem](#)

Exposes methods that retrieve information about a Shell item. [IShellItem](#) and [IShellItem2](#) are the preferred representations of items in any new code.

Enums

[SICHINTF](#)

Used to determine how to compare two Shell items. [Compare\(nint, SICHINTF, out int\)](#) uses this enumerated type.

[SIGDN](#)

Requests the form of an item's display name to retrieve through [GetDisplayName\(SIGDN, out nint\)](#).

Interface IFileDialogCustomize

Namespace: [FireBlade.WinInteropUtils.ComponentObjectModel.Interfaces](#)

Assembly: WinInteropUtils.dll

Exposes methods that allow an application to add controls to a common item dialog.

```
[InterfaceType(ComInterfaceType.InterfaceIsIUnknown)]  
[Guid("973510db-7d7f-452b-8975-74a85828d354")]  
public interface IFileDialogCustomize
```

Remarks

Controls are added to the dialog before the dialog is shown. Their layout is implied by the order in which they are added. Once the dialog is shown, controls cannot be added or removed, but the existing controls can be hidden or disabled at any time. Their labels can also be changed at any time.

Container controls are controls that can have items added to them. Container controls include combo boxes, menus, the drop-down list attached to the Open button, and any option button groups. The order that items appear in a container is the order in which they were added. There is no facility for reordering them. IDs are scoped to the parent control. Container controls, with the exception of menus, have a selected item.

Items with a container control cannot be changed after they have been created, except for their enabled and visible states. However, they can be added and removed at any time. For example, if you needed to change the text of a menu, you would have to remove the current menu and add another with the correct text.

Methods

AddCheckButton(uint, string, bool)

Adds a check button (check box) to the dialog.

```
HResult AddCheckButton(uint dwIDCtl, string pszLabel, bool bChecked)
```

Parameters

dwIDCtl [uint](#)

The ID of the check button to add.

pszLabel [string](#)

A [null](#)-terminated Unicode string that contains the button text.

bChecked [bool](#)

A [bool](#) indicating the current state of the check button. [true](#) if checked; [false](#) otherwise.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The default state for this control is enabled and visible.

AddComboBox(uint)

Adds a combo box to the dialog.

```
HResult AddComboBox(uint dwIDCtl)
```

Parameters

dwIDCtl [uint](#)

The ID of the combo box to add.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The default state for this control is enabled and visible.

AddControlItem(uint, uint, string)

Adds an item to a container control in the dialog.

```
HResult AddControlItem(uint dwIDCtl, uint dwIDItem, string pszLabel)
```

Parameters

dwIDCtl [uint](#)

The ID of the container control to which the item is to be added.

dwIDItem [uint](#)

The ID of the item.

pszLabel [string](#)

A [null](#)-terminated Unicode string that contains the item's text, which can be either a label or, in the case of a drop-down list, the item itself.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The default state for this item is enabled and visible. Items in control groups cannot be changed after they have been created, with the exception of their enabled and visible states.

Container controls include option button groups, combo boxes, drop-down lists on the Open or Save button, and menus.

AddEditBox(uint, string)

Adds an edit box control to the dialog.

```
HResult AddEditBox(uint dwIDCtl, string pszText)
```

Parameters

dwIDCtl [uint](#)

The ID of the edit box to add.

pszText [string](#)

A [null](#)-terminated Unicode string that provides the default text displayed in the edit box.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The default state for this control is enabled and visible.

To add a label next to the edit box, place it in a visual group with [StartVisualGroup\(uint, string\)](#).

AddMenu(uint, string)

Adds a menu to the dialog.

```
HResult AddMenu(uint dwIDCtl, string pszLabel)
```

Parameters

dwIDCtl [uint](#)

The ID of the menu to add.

pszLabel [string](#)

A [null](#)-terminated Unicode string that contains the menu name.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The default state for this control is enabled and visible.

To add items to this control, use [AddControlItem\(uint, uint, string\)](#).

AddPushButton(uint, string)

Adds a button to the dialog.

```
HResult AddPushButton(uint dwIDCtl, string pszLabel)
```

Parameters

dwIDCtl [uint](#) 

The ID of the button to add.

pszLabel [string](#) 

A [null](#) -terminated Unicode string that contains the button text.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The default state for this control is enabled and visible.

AddRadioButtonList(uint)

Adds an option button (also known as radio button) group to the dialog.

```
HResult AddRadioButtonList(uint dwIDCtl)
```

Parameters

dwIDCtl [uint](#)

The ID of the option button group to add.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The default state for this control is enabled and visible.

AddSeparator(uint)

Adds a separator to the dialog, allowing a visual separation of controls.

```
HResult AddSeparator(uint dwIDCtl)
```

Parameters

dwIDCtl [uint](#)

The control ID of the separator.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The default state for this control is enabled and visible.

AddText(uint, string)

Adds text content to the dialog.

```
HResult AddText(uint dwIDCtl, string pszText)
```

Parameters

dwIDCtl [uint](#) 

The ID of the text to add.

pszText [string](#) 

A [null](#) -terminated Unicode string that contains the text.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The default state for this control is enabled and visible.

EnableOpenDropDown(uint)

Enables a drop-down list on the Open or Save button in the dialog.

```
HResult EnableOpenDropDown(uint dwIDCtl)
```

Parameters

dwIDCtl [uint](#) 

The ID of the drop-down list.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The Open or Save button label takes on the text of the first item in the drop-down.

Use [AddControlItem\(uint, uint, string\)](#) to add items to the drop-down.

EndVisualGroup()

Stops the addition of elements to a visual group in the dialog.

```
HResult EndVisualGroup()
```

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

GetCheckButtonState(uint, out bool)

Gets the current state of a check button (check box) in the dialog.

```
HResult GetCheckButtonState(uint dwIDCtl, out bool pbChecked)
```

Parameters

dwIDCtl [uint](#) 

The ID of the check box.

pbChecked [bool](#)

The address of a [bool](#) value that indicates whether the box is checked. [true](#) means checked; [false](#), unchecked.

Returns

[HResult](#)

If this method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code.

GetControlItemState(uint, uint, out uint)

Gets the current state of an item in a container control found in the dialog.

```
HResult GetControlItemState(uint dwIDCtl, uint dwIDItem, out uint pdwState)
```

Parameters

dwIDCtl [uint](#)

The ID of the container control.

dwIDItem [uint](#)

The ID of the item.

pdwState [uint](#)

A pointer to a variable that receives one or more values from the CDCONTROLSTATE enumeration that indicate the current state of the control.

Returns

[HResult](#)

If this method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code.

Remarks

The default state of a control item is enabled and visible. Items in control groups cannot be changed after they have been created, with the exception of their enabled and visible states.

Container controls include option button groups, combo boxes, drop-down lists on the Open or Save button, and menus.

GetControlState(uint, out uint)

Gets the current visibility and enabled states of a given control.

```
HResult GetControlState(uint dwIDCtl, out uint pdwState)
```

Parameters

dwIDCtl [uint](#)

The ID of the control in question.

pdwState [uint](#)

A pointer to a variable that receives one or more values from the CDCONTROLSTATE enumeration that indicate the current state of the control.

Returns

[HResult](#)

If this method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code.

GetEditText(uint, out nint)

Gets the current text in an edit box control.

```
HResult GetEditText(uint dwIDCtl, out nint ppszText)
```

Parameters

dwIDCtl [uint](#)

The ID of the edit box.

`ppszText` [uint](#)

The address of a pointer to a [null](#)-terminated Unicode string that receives the text.

Returns

[HRESULT](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HRESULT](#) error code.

Remarks

It is the responsibility of the calling application to free the buffer referenced by `ppszText` when it is no longer needed. Use `CoTaskMemFree` to free the buffer.

GetSelectedItem(uint, out uint)

Gets a particular item from specified container controls in the dialog.

```
HResult GetSelectedItem(uint dwIDCtl, out uint pdwIDItem)
```

Parameters

`dwIDCtl` [uint](#)

The ID of the container control.

`pdwIDItem` [uint](#)

The ID of the item that the user selected in the control.

Returns

[HRESULT](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HRESULT](#) error code.

Remarks

To determine the user's final choice, this method can be called on option button groups, combo boxes, and drop-down lists on the Open or Save button after the dialog has closed. This method cannot be called on menus.

For option button groups and combo boxes, this method can also be called while the dialog is showing, to determine the current choice.

MakeProminent(uint)

Places a control in the dialog so that it stands out compared to other added controls.

```
HResult MakeProminent(uint dwIDCtl)
```

Parameters

dwIDCtl [uint](#) 

The ID of the control.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

This method causes the control to be placed near the Open or Save button instead of being grouped with the rest of the custom controls.

Only check buttons (check boxes), push buttons, combo boxes, and menus—or a visual group that contains only a single item of one of those types—can be made prominent.

Only one control can be marked in this way. If a dialog has only one added control, that control is marked as prominent by default.

RemoveControlItem(uint, uint)

Removes an item from a container control in the dialog.

```
HResult RemoveControlItem(uint dwIDCtl, uint dwIDItem)
```

Parameters

dwIDCtl [uint](#)

The ID of the container control from which the item is to be removed.

dwIDItem [uint](#)

The ID of the item.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

Container controls include option button groups, combo boxes, drop-down lists on the Open or Save button, and menus.

SetCheckButtonState(uint, bool)

Sets the state of a check button (check box) in the dialog.

```
HResult SetCheckButtonState(uint dwIDCtl, bool bChecked)
```

Parameters

dwIDCtl [uint](#)

The ID of the check box.

bChecked [bool](#)

A [bool](#) value that indicates whether the box is checked. [true](#) means checked; [false](#), unchecked.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

SetControlItemState(uint, uint, uint)

Sets the current state of an item in a container control found in the dialog.

```
HResult SetControlItemState(uint dwIDCtl, uint dwIDItem, uint dwState)
```

Parameters

`dwIDCtl` [uint](#)

The ID of the container control.

`dwIDItem` [uint](#)

The ID of the item.

`dwState` [uint](#)

One or more values from the `CDCONTROLSTATE` enumeration that indicate the new state of the control.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The default state of a control item is enabled and visible. Items in control groups cannot be changed after they have been created, with the exception of their enabled and visible states.

Container controls include option button groups, combo boxes, drop-down lists on the Open or Save button, and menus.

SetControlItemText(uint, uint, string)

Sets the text of a control item. For example, the text that accompanies a radio button or an item in a menu.

```
HResult SetControlItemText(uint dwIDCtl, uint dwIDItem, string pszLabel)
```

Parameters

dwIDCtl [uint](#)

The ID of the container control.

dwIDItem [uint](#)

The ID of the item.

pszLabel [string](#)

A [null](#)-terminated Unicode string that contains the text.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The default state of a control item is enabled and visible. Items in control groups cannot be changed after they have been created, with the exception of their enabled and visible states.

Container controls include option button groups, combo boxes, drop-down lists on the Open or Save button, and menus.

SetControlLabel(uint, string)

Sets the text associated with a control, such as button text or an edit box label.

```
HResult SetControlLabel(uint dwIDCtl, string pszLabel)
```

Parameters

dwIDCtl [uint](#)

The ID of the control whose text is to be changed.

pszLabel [string](#)

A null-terminated Unicode string that contains the text.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

Control labels can be changed at any time, including when the dialog is visible.

SetControlState(uint, uint)

Sets the current visibility and enabled states of a given control.

```
HResult SetControlState(uint dwIDCtl, uint dwState)
```

Parameters

dwIDCtl [uint](#)

The ID of the control in question.

dwState [uint](#)

One or more values from the `CDCONTROLSTATE` enumeration that indicate the current state of the control.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

When the dialog is shown, controls cannot be added or removed, but the existing controls can be hidden or disabled at any time.

SetEditBoxText(uint, string)

Sets the text in an edit box control found in the dialog.

```
HResult SetEditBoxText(uint dwIDCtl, string pszText)
```

Parameters

dwIDCtl [uint](#) 

The ID of the edit box.

pszText [string](#) 

A [null](#) -terminated Unicode string that contains the text.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

SetSelectedItem(uint, uint)

Sets the selected state of a particular item in an option button group or a combo box found in the dialog.

```
HResult SetSelectedItem(uint dwIDCtl, uint dwIDItem)
```

Parameters

dwIDCtl [uint](#)

The ID of the container control.

dwIDItem [uint](#)

The ID of the item to display as selected in the control.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

StartVisualGroup(uint, string)

Declares a visual group in the dialog. Subsequent calls to any "add" method add those elements to this group.

```
HResult StartVisualGroup(uint dwIDCtl, string pszLabel)
```

Parameters

dwIDCtl [uint](#)

The ID of the visual group.

pszLabel [string](#)

A [null](#)-terminated Unicode string that contains text that appears next to the visual group.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

Controls will continue to be added to this visual group until you call [EndVisualGroup\(\)](#).

A visual group can be hidden and disabled like any other control, except that doing so affects all of the controls within it. Individual members of the visual group can also be hidden and disabled singly.

Interface IFileOpenDialog

Namespace: [FireBlade.WinInteropUtils.ComponentObjectModel.Interfaces](#)

Assembly: WinInteropUtils.dll

Extends the [IFileDialog](#) interface by adding methods specific to the open dialog.

```
[InterfaceType(ComInterfaceType.InterfaceIsIUnknown)]  
[Guid("d57c7288-d4ad-4768-be02-9d969532d960")]  
public interface IFileOpenDialog
```

Methods

Advise(nint, out uint)

Assigns an event handler that listens for events coming from the dialog.

```
HResult Advise(nint pfde, out uint pdwCookie)
```

Parameters

pfde [nint](#)

A pointer to an [FileDialogEvents](#) implementation that will receive events from the dialog.

pdwCookie [uint](#)

A pointer to a **DWORD** that receives a value identifying this event handler. When the client is finished with the dialog, that client must call the [Unadvise\(uint\)](#) method with this value.

Returns

[HResult](#)

If this method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code.

Close(HResult)

Closes the dialog.

```
HResult Close(HResult hr)
```

Parameters

hr [HResult](#)

The code that will be returned by [Show\(nint\)](#) to indicate that the dialog was closed before a selection was made.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

An application can call this method from a callback method or function while the dialog is open. The dialog will close and the [Show\(nint\)](#) method will return with the [HResult](#) specified in **hr**.

If this method is called, there is no result available for the [GetResult\(out nint\)](#) or [GetResults\(out nint\)](#) methods, and they will fail if called.

GetCurrentSelection(out nint)

Gets the user's current selection in the dialog.

```
HResult GetCurrentSelection(out nint ppsi)
```

Parameters

ppsi [nint](#)

The address of a pointer to the interface that represents the item currently selected in the dialog. This item can be a file or folder selected in the view window, or something that the user has entered into

the dialog's edit box. The latter case may require a parsing operation (cancelable by the user) that blocks the current thread.

Returns

[HRESULT](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HRESULT](#) error code.

GetFileName(out nint)

Retrieves the text currently entered in the dialog's **File name** edit box.

```
HResult GetFileName(out nint pszName)
```

Parameters

`pszName` [nint](#)

The address of a pointer to a string that, when this method returns successfully, receives the text.

Returns

[HRESULT](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HRESULT](#) error code.

Remarks

The text in the **File name** edit box does not necessarily reflect the item the user chose. To get the item the user chose, use [GetResult\(out nint\)](#).

The calling application is responsible for releasing the retrieved buffer by using the `CoTaskMemFree` function.

GetFileTypeIndex(out uint)

Gets the currently selected file type.

```
HResult GetFileTypeIndex(out uint piFileType)
```

Parameters

piFileType [uint](#)

A pointer to a UINT value that receives the index of the selected file type in the file type array passed to [SetFileTypes\(uint, nint\)](#) in its cFileTypes parameter.

Returns

[HResult](#)

If this method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code.

Remarks

[GetFileTypeIndex\(out uint\)](#) can be called either while the dialog is open or after it has closed.

Note: The index returned by this function is a one-based index rather than zero-based.

GetFolder(out nint)

Gets either the folder currently selected in the dialog, or, if the dialog is not currently displayed, the folder that is to be selected when the dialog is opened.

```
HResult GetFolder(out nint ppsi)
```

Parameters

ppsi [nint](#)

The address of a pointer to the [IShellItem](#) interface that represents the folder.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The calling application is responsible for releasing the retrieved [IShellItem](#) when it is no longer needed.

GetOptions(out uint)

Gets the current flags that are set to control dialog behavior.

```
HResult GetOptions(out uint pfos)
```

Parameters

`pfos` [uint](#)

When this method returns successfully, points to a value made up of one or more of the `FILEOPENDIALOGOPTIONS` values.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

GetResult(out nint)

Gets the choice that the user made in the dialog.

```
HResult GetResult(out nint ppsi)
```

Parameters

`ppsi` [nint](#)

The address of a pointer to an [IShellItem](#) that represents the user's choice.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

[GetResult\(out nint\)](#) can be called after the dialog has closed or during the handling of an `OnFileOk` event. Calling this method at any other time will fail. If multiple items were chosen, this method will fail. In the case of multiple items, call [GetResults\(out nint\)](#).

[Show\(nint\)](#) must return a success code for a result to be available to [GetResult\(out nint\)](#).

GetResults(out nint)

Gets the user's choices in a dialog that allows multiple selection.

```
HResult GetResults(out nint ppenum)
```

Parameters

`ppenum` [nint](#) 

The address of a pointer to an `IShellItemArray` through which the items selected in the dialog can be accessed.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

This method can be used whether the selection consists of a single item or multiple items.

[GetResults\(out nint\)](#) can be called after the dialog has closed or during the handling of an `FileDialogEvents.OnFileOk` event. Calling this method at any other time will fail.

[Show\(nint\)](#) must return a success code for a result to be available to [GetResults\(out nint\)](#).

GetSelectedItems(out nint)

Gets the currently selected items in the dialog. These items may be items selected in the view, or text selected in the file name edit box.

```
HResult GetSelectedItems(out nint ppsai)
```

Parameters

ppsai [nint](#)

The address of a pointer to an IShellItemArray through which the selected items can be accessed.

Returns

[HResult](#)

If this method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code.

Remarks

This method can be used for single item or multiple item selections. If the user has entered new text in the file name field, this can be a time-consuming operation. When the application calls this method, the application parses the text in the filename field. For example, if this is a network share, the operation could take some time. However, this operation will not block the UI, since the user should be able to stop the operation, which will result in [GetSelectedItems\(out nint\)](#) returning a failure code).

SetDefaultFolder(nint)

Sets the folder used as a default if there is not a recently used folder value available.

```
HResult SetDefaultFolder(nint psi)
```

Parameters

psi [nint](#)

A pointer to the [IShellItem](#) interface that represents the folder.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

SetFileName(string)

Sets the file name that appears in the **File name** edit box when that dialog box is opened.

```
HResult SetFileName(string pszName)
```

Parameters

`pszName` [string](#) 

A pointer to the name of the file.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

SetFileTypeIndex(uint)

Sets the file type that appears as selected in the dialog.

```
HResult SetFileTypeIndex(uint iFileType)
```

Parameters

`iFileType` [uint](#) 

The index of the file type in the file type array passed to [SetFileTypes\(uint, nint\)](#) in its `cFileTypes` parameter. Note that this is a one-based index, not zero-based.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

SetFileTypes(uint, nint)

Sets the file types that the dialog can open or save.

```
HResult SetFileTypes(uint cFileTypes, nint rgFilterSpec)
```

Parameters

`cFileTypes` [uint](#)

The number of elements in the array specified by `rgFilterSpec`.

`rgFilterSpec` [nint](#)

A pointer to an array of `COMDLG_FILTERSPEC` structures, each representing a file type.

Returns

[HResult](#)

If the method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code, including the following:

<code>FireBlade.WinInteropUtils.HResult.E_UNEXPECTED</code>	SetFileTypes(uint, nint) has already been called.
<code>FireBlade.WinInteropUtils.HResult.E_UNEXPECTED</code>	The <code>FOS_PICKFOLDERS</code> flag was set in the SetOptions(uint) method.
<code>FireBlade.WinInteropUtils.HResult.E_INVALIDARG</code>	The <code>rgFilterSpec</code> parameter is null .

Examples

The following code example demonstrates the use of the array of COMDLG_FILTERSPEC structures in the context of this method. The example array consists of three COMDLG_FILTERSPEC structures. The first declares two patterns for the dialog filter, the second declares a single pattern, and the last shows files of all types. The variables szJPG, szBMP, and szAll are assumed to be previously declared strings that provide a friendly name for each filter. **Warning: This is C++ code, as the COMDLG_FILTERSPEC struct hasn't yet been ported to WinInteropUtils.**

```
COMDLG_FILTERSPEC rgSpec[] = { { szJPG, L".jpg;.jpeg" }, { szBMP, L".bmp" }, { szAll, L".*" }, };
```

Remarks

When using the Open dialog, the file types declared there are used to filter the view. When using the Save dialog, these values determine which file name extension is appended to the file name.

This method must be called before the dialog is shown and can only be called once for each dialog instance. File types cannot be modified once the Common Item dialog box is displayed.

SetFilter(nint)

Sets the filter. Deprecated since Windows 7.

```
HResult SetFilter(nint pFilter)
```

Parameters

pFilter [nint](#)

A pointer to the [IShellItemFilter](#) that is to be set.

Returns

[HResult](#)

If this method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code.

Remarks

This method can be used if the application needs to perform special filtering to remove some items from the dialog box's view. IncludeItem will be called for each item that would normally be included in the view. GetEnumFlagsForItem is not used. To filter by file type, [SetFileTypes\(uint, nint\)](#) should be used,

because in folders with a large number of items it may offer better performance than applying an `IShellItemFilter`.

This method is obsolete since Windows 7. Do not use it in new projects.

SetFolder(nint)

Sets a folder that is always selected when the dialog is opened, regardless of previous user action.

```
HResult SetFolder(nint psi)
```

Parameters

psi [nint](#) 

A pointer to the [IShellItem](#) interface that represents the folder.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

This folder overrides any "most recently used" folder. If this method is called while the dialog is displayed, it causes the dialog to navigate to the specified folder.

In general, we do not recommended the use of this method. If you call [SetFolder\(nint\)](#) before you display the dialog box, the most recent location that the user saved to or opened from is not shown. Unless there is a very specific reason for this behavior, it is not a good or expected user experience and should therefore be avoided. In almost all instances, [SetDefaultFolder\(nint\)](#) is the better method.

As of Windows 7, if the path of the folder specified through **psi** is the default path of a known folder, the known folder's current path is used in the dialog. That path might not be the same as the path specified in **psi**; for instance, if the known folder has been redirected. If the known folder is a library (virtual folders Documents, Music, Pictures, and Videos), the library's path is used in the dialog. If the specified library is hidden (as they are by default as of Windows 8.1), the library's default save location is used in the dialog, such as the Microsoft OneDrive Documents folder for the Documents library. Because of these

mappings, the folder location used in the dialog might not be exactly as you specified when you called this method.

SetOptions(uint)

Sets flags to control the behavior of the dialog.

```
HResult SetOptions(uint fos)
```

Parameters

fos [uint](#)

One or more of the FILEOPENDIALOGOPTIONS values.

Returns

[HResult](#)

If this method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code.

Remarks

Generally, this method should take the value that was retrieved by [GetOptions\(out uint\)](#) and modify it to include or exclude options by setting the appropriate flags.

SetTitle(string)

Sets the title of the dialog.

```
HResult SetTitle(string pszTitle)
```

Parameters

pszTitle [string](#)

A pointer to a string that contains the title text.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Show(nint)

Launches the modal window.

```
HResult Show(nint hwndOwner)
```

Parameters

`hwndOwner` [nint](#)

The handle of the owner window. This value can be [null](#).

Returns

[HResult](#)

If the method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code, including `Macros.HResultFromWin32(ERROR_CANCELLED)`, indicating the user closed the window by cancelling the operation.

Unadvise(uint)

Removes an event handler that was attached through the [Advise\(nint, out uint\)](#) method.

```
HResult Unadvise(uint dwCookie)
```

Parameters

`dwCookie` [uint](#)

The `DWORD` value that represents the event handler. This value is obtained through the `pdwCookie` parameter of the [Advise\(nint, out uint\)](#) method.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Interface IFileSaveDialog

Namespace: [FireBlade.WinInteropUtils.ComponentObjectModel.Interfaces](#)

Assembly: WinInteropUtils.dll

Extends the [IFileDialog](#) interface by adding methods specific to the save dialog, which include those that provide support for the collection of metadata to be persisted with the file.

```
[InterfaceType(ComInterfaceType.InterfaceIsIUnknown)]  
[Guid("84bccd23-5fde-4cdb-aea4-af64b83d78ab")]  
public interface IFileSaveDialog
```

Methods

Advise(nint, out uint)

Assigns an event handler that listens for events coming from the dialog.

```
HResult Advise(nint pfde, out uint pdwCookie)
```

Parameters

pfde [nint](#)

A pointer to an [IFileDialogEvents](#) implementation that will receive events from the dialog.

pdwCookie [uint](#)

A pointer to a **DWORD** that receives a value identifying this event handler. When the client is finished with the dialog, that client must call the [Unadvise\(uint\)](#) method with this value.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

ApplyProperties(nint, nint, nint, nint)

Applies a set of properties to an item using the Shell's copy engine.

```
HResult ApplyProperties(nint psi, nint pStore, nint hwnd, nint pSink)
```

Parameters

psi [nint](#)

Pointer to the [IShellItem](#) that represents the file being saved. This is usually the item retrieved by [GetResult\(out nint\)](#).

pStore [nint](#)

Pointer to the IPropertyStore that represents the property values to be applied to the file. This can be the property store returned by [GetProperties\(out nint\)](#).

hwnd [nint](#)

The handle of the application window.

pSink [nint](#)

Pointer to an optional IFileOperationProgressSink that the calling application can use if they want to be notified of the progress of the property stamping. This value may be [null](#).

Returns

[HResult](#)

If this method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code.

Remarks

This method should be used when the application has turned on property collection ([SetCollectedProperties\(nint, bool\)](#)), but does not persist the properties themselves into the saved file.

<i>Note: The file represented by the item specified in `<code class="paramref">psi</code>` must exist in physical storage before making the call to `<xref href="FireBlade.WinInteropUtils.ComponentObjectModel.Interfaces.IFileSaveDialog.ApplyPropert`

`ies(System.IntPtr%2cSystem.IntPtr%2cSystem.IntPtr%2cSystem.IntPtr)" data-throw-if-not-resolved="false"></xref>, so it must have been previously saved at some point.</i>`

Close(HResult)

Closes the dialog.

```
HResult Close(HResult hr)
```

Parameters

hr [HResult](#)

The code that will be returned by [Show\(nint\)](#) to indicate that the dialog was closed before a selection was made.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

An application can call this method from a callback method or function while the dialog is open. The dialog will close and the [Show\(nint\)](#) method will return with the [HResult](#) specified in **hr**.

If this method is called, there is no result available for the [GetResult\(out nint\)](#) or [GetResults\(out nint\)](#) methods, and they will fail if called.

GetCurrentSelection(out nint)

Gets the user's current selection in the dialog.

```
HResult GetCurrentSelection(out nint ppsi)
```

Parameters

`ppsi` [nint](#)

The address of a pointer to the interface that represents the item currently selected in the dialog. This item can be a file or folder selected in the view window, or something that the user has entered into the dialog's edit box. The latter case may require a parsing operation (cancelable by the user) that blocks the current thread.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

GetFileName(out nint)

Retrieves the text currently entered in the dialog's **File name** edit box.

```
HResult GetFileName(out nint pszName)
```

Parameters

`pszName` [nint](#)

The address of a pointer to a string that, when this method returns successfully, receives the text.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The text in the **File name** edit box does not necessarily reflect the item the user chose. To get the item the user chose, use [GetResult\(out nint\)](#).

The calling application is responsible for releasing the retrieved buffer by using the `CoTaskMemFree` function.

GetFileTypeIndex(out uint)

Gets the currently selected file type.

```
HResult GetFileTypeIndex(out uint piFileType)
```

Parameters

piFileType [uint](#)

A pointer to a UINT value that receives the index of the selected file type in the file type array passed to [SetFileTypes\(uint, nint\)](#) in its cFileTypes parameter.

Returns

[HResult](#)

If this method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code.

Remarks

[GetFileTypeIndex\(out uint\)](#) can be called either while the dialog is open or after it has closed.

Note: The index returned by this function is a one-based index rather than zero-based.

GetFolder(out nint)

Gets either the folder currently selected in the dialog, or, if the dialog is not currently displayed, the folder that is to be selected when the dialog is opened.

```
HResult GetFolder(out nint ppsi)
```

Parameters

ppsi [nint](#)

The address of a pointer to the [IShellItem](#) interface that represents the folder.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The calling application is responsible for releasing the retrieved [IShellItem](#) when it is no longer needed.

GetOptions(out uint)

Gets the current flags that are set to control dialog behavior.

```
HResult GetOptions(out uint pfos)
```

Parameters

`pfos` [uint](#)

When this method returns successfully, points to a value made up of one or more of the `FILEOPENDIALOGOPTIONS` values.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

GetProperties(out nint)

Retrieves the set of property values for a saved item or an item in the process of being saved.

```
HResult GetProperties(out nint ppStore)
```

Parameters

`ppStore` [nint](#)

Address of a pointer to an `IPropertyStore` that receives the property values.

Returns

[HRESULT](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HRESULT](#) error code.

Remarks

This method can be called while the dialog is showing to retrieve the current set of values in the metadata collection pane. It can also be called after the dialog has closed, to retrieve the final set of values.

The call to this method will fail unless property collection has been turned on with a call to [SetCollectedProperties\(nint, bool\)](#).

GetResult(out nint)

Gets the choice that the user made in the dialog.

```
HResult GetResult(out nint ppsi)
```

Parameters

`ppsi` [nint](#)

The address of a pointer to an [IShellItem](#) that represents the user's choice.

Returns

[HRESULT](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HRESULT](#) error code.

Remarks

[GetResult\(out nint\)](#) can be called after the dialog has closed or during the handling of an `OnFileOk` event. Calling this method at any other time will fail. If multiple items were chosen, this method will fail. In the

case of multiple items, call [GetResults\(out nint\)](#).

[Show\(nint\)](#) must return a success code for a result to be available to [GetResult\(out nint\)](#).

SetCollectedProperties(nint, bool)

Specifies which properties will be collected in the save dialog.

```
HResult SetCollectedProperties(nint pStore, bool fAppendDefault)
```

Parameters

pStore [nint](#) 

Pointer to the interface that represents the list of properties to collect. This parameter can be [null](#) .

fAppendDefault [bool](#) 

[true](#)  to show default properties for the currently selected filetype in addition to the properties specified by **pStore**. [false](#)  to show only properties specified by **pStore**.

Returns

[HResult](#)

If this method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code.

Remarks

The calling application can use the PSGetPropertyDescriptionListFromString function to construct an IPropertyDescriptionList from a string such as "prop:Comments;Subject;".

For more information about property schemas, see [Property Schemas](#) .

[SetCollectedProperties\(nint, bool\)](#) can be called at any time before the dialog is displayed or while it is visible. If different properties are to be collected depending on the chosen filetype, then [SetCollectedProperties\(nint, bool\)](#) can be called in response to OnTypeChange.

Note: By default, no properties are collected in the save dialog.

SetDefaultFolder(nint)

Sets the folder used as a default if there is not a recently used folder value available.

```
HResult SetDefaultFolder(nint psi)
```

Parameters

psi [nint](#)

A pointer to the [IShellItem](#) interface that represents the folder.

Returns

[HResult](#)

If this method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code.

SetFileName(string)

Sets the file name that appears in the **File name** edit box when that dialog box is opened.

```
HResult SetFileName(string pszName)
```

Parameters

pszName [string](#)

A pointer to the name of the file.

Returns

[HResult](#)

If this method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code.

SetFileTypeIndex(uint)

Sets the file type that appears as selected in the dialog.

```
HResult SetFileTypeIndex(uint iFileType)
```

Parameters

iFileType [uint](#)

The index of the file type in the file type array passed to [SetFileTypes\(uint, nint\)](#) in its cFileTypes parameter. Note that this is a one-based index, not zero-based.

Returns

[HResult](#)

If this method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code.

SetFileTypes(uint, nint)

Sets the file types that the dialog can open or save.

```
HResult SetFileTypes(uint cFileTypes, nint rgFilterSpec)
```

Parameters

cFileTypes [uint](#)

The number of elements in the array specified by **rgFilterSpec**.

rgFilterSpec [nint](#)

A pointer to an array of COMDLG_FILTERSPEC structures, each representing a file type.

Returns

[HResult](#)

If the method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code, including the following:

FireBlade.WinInteropUtils.HResult.E_UNEXPECTED	SetFileTypes(uint, nint) has already been called.
FireBlade.WinInteropUtils.HResult.E_UNEXPECTED	The FOS_PICKFOLDERS flag was set in the SetOptions(uint) method.
FireBlade.WinInteropUtils.HResult.E_INVALIDARG	The <code>rgFilterSpec</code> parameter is null .

Examples

The following code example demonstrates the use of the array of COMDLG_FILTERSPEC structures in the context of this method. The example array consists of three COMDLG_FILTERSPEC structures. The first declares two patterns for the dialog filter, the second declares a single pattern, and the last shows files of all types. The variables `szJPG`, `szBMP`, and `szAll` are assumed to be previously declared strings that provide a friendly name for each filter. **Warning: This is C++ code, as the COMDLG_FILTERSPEC struct hasn't yet been ported to WinInteropUtils.** `COMDLG_FILTERSPEC rgSpec[] = { { szJPG, L".jpg;jpeg" }, { szBMP, L".bmp" }, { szAll, L"*.*" }, {`

Remarks

When using the Open dialog, the file types declared there are used to filter the view. When using the Save dialog, these values determine which file name extension is appended to the file name.

This method must be called before the dialog is shown and can only be called once for each dialog instance. File types cannot be modified once the Common Item dialog box is displayed.

SetFilter(nint)

Sets the filter. Deprecated since Windows 7.

```
HResult SetFilter(nint pFilter)
```

Parameters

`pFilter` [nint](#)

A pointer to the [IShellItemFilter](#) that is to be set.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

This method can be used if the application needs to perform special filtering to remove some items from the dialog box's view. `IncludedItem` will be called for each item that would normally be included in the view. `GetEnumFlagsForItem` is not used. To filter by file type, [SetFileTypes\(uint, nint\)](#) should be used, because in folders with a large number of items it may offer better performance than applying an `IShellItemFilter`.

This method is obsolete since Windows 7. Do not use it in new projects.

SetFolder(nint)

Sets a folder that is always selected when the dialog is opened, regardless of previous user action.

```
HResult SetFolder(nint psi)
```

Parameters

psi [nint](#)

A pointer to the [IShellItem](#) interface that represents the folder.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

This folder overrides any "most recently used" folder. If this method is called while the dialog is displayed, it causes the dialog to navigate to the specified folder.

In general, we do not recommended the use of this method. If you call [SetFolder\(nint\)](#) before you display the dialog box, the most recent location that the user saved to or opened from is not shown. Unless there is a very specific reason for this behavior, it is not a good or expected user experience and should therefore be avoided. In almost all instances, [SetDefaultFolder\(nint\)](#) is the better method.

As of Windows 7, if the path of the folder specified through `psi` is the default path of a known folder, the known folder's current path is used in the dialog. That path might not be the same as the path specified in `psi`; for instance, if the known folder has been redirected. If the known folder is a library (virtual folders Documents, Music, Pictures, and Videos), the library's path is used in the dialog. If the specified library is hidden (as they are by default as of Windows 8.1), the library's default save location is used in the dialog, such as the Microsoft OneDrive Documents folder for the Documents library. Because of these mappings, the folder location used in the dialog might not be exactly as you specified when you called this method.

SetOptions(uint)

Sets flags to control the behavior of the dialog.

```
HResult SetOptions(uint fos)
```

Parameters

`fos` [uint](#)

One or more of the FILEOPENDIALOGOPTIONS values.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

Generally, this method should take the value that was retrieved by [GetOptions\(out uint\)](#) and modify it to include or exclude options by setting the appropriate flags.

SetProperties(nint)

Provides a property store that defines the default values to be used for the item being saved.

```
HResult SetPropertyStore(nint pStore)
```

Parameters

pStore [nint](#)

Pointer to the IPropertyStore interface that represents the property store that contains the associated metadata.

Returns

[HResult](#)

If this method succeeds, it returns FireBlade.WinInteropUtils.HResult.S_OK. Otherwise, it returns an [HResult](#) error code.

Remarks

This method can be called at any time before the dialog is opened or while the dialog is showing. If an item has inherent properties, this method should be called with those properties before showing the dialog.

When using Save As, the application should provide the properties of the item being saved to the Save dialog. Those properties should be retrieved from the original item by calling [GetPropertyStore](#) with the [GPS_HANDLERPROPERTIESONLY](#) flag.

To retrieve the properties of the saved item (which may have been modified by the user) after the dialog closes, call [GetProperties\(out nint\)](#).

To turn on property collection and indicate which properties should be displayed in the Save dialog, use [SetCollectedProperties\(nint, bool\)](#).

SetSaveAsItem(nint)

Sets an item to be used as the initial entry in a Save As dialog.

```
HResult SetSaveAsItem(nint psi)
```

Parameters

psi [nint](#)

Pointer to an [IShellItem](#) that represents the item.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Remarks

The name of the item is displayed in the file name edit box, and the containing folder is opened in the view. This would generally be used when the application is saving an item that already exists. For new items, use [SetFileName\(string\)](#).

SetTitle(string)

Sets the title of the dialog.

```
HResult SetTitle(string pszTitle)
```

Parameters

pszTitle [string](#)

A pointer to a string that contains the title text.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Show(nint)

Launches the modal window.

```
HResult Show(nint hwndOwner)
```

Parameters

hwndOwner [nint](#)

The handle of the owner window. This value can be [null](#).

Returns

[HResult](#)

If the method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code, including `Macros.HResultFromWin32(ERROR_CANCELLED)`, indicating the user closed the window by cancelling the operation.

Unadvise(uint)

Removes an event handler that was attached through the [Advise\(nint, out uint\)](#) method.

```
HResult Unadvise(uint dwCookie)
```

Parameters

dwCookie [uint](#)

The DWORD value that represents the event handler. This value is obtained through the **pdwCookie** parameter of the [Advise\(nint, out uint\)](#) method.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Interface IShellItem

Namespace: [FireBlade.WinInteropUtils.ComponentObjectModel.Interfaces](#)

Assembly: WinInteropUtils.dll

Exposes methods that retrieve information about a Shell item. [IShellItem](#) and [IShellItem2](#) are the preferred representations of items in any new code.

```
[InterfaceType(ComInterfaceType.InterfaceIsIUnknown)]  
[Guid("43826d1e-e718-42ee-bc55-a1e261c37bfe")]  
public interface IShellItem
```

Remarks

Third parties do not implement this interface; only use the implementation provided with the system.

Methods

BindToHandler(nint, ref Guid, ref Guid, out nint)

Binds to a handler for an item as specified by the handler ID value (BHID).

```
HResult BindToHandler(nint pbc, ref Guid bhid, ref Guid riid, out nint ppv)
```

Parameters

pbc [nint](#)

A pointer to an IBindCtx interface on a bind context object. Used to pass optional parameters to the handler. The contents of the bind context are handler-specific. For example, when binding to BHID_Stream, the STGM flags in the bind context indicate the mode of access desired (read or read/write).

bhid [Guid](#)

Reference to a GUID that specifies which handler will be created. The GUIDs are defined in [shlguid.h](#).

riid [Guid](#)

IID of the object type to retrieve.

ppv [nint](#)

When this method returns, contains a pointer of type `riid` that is returned by the handler specified by `bhid`.

Returns

[HResult](#)

If this method succeeds, it returns `FireBlade.WinInteropUtils.HResult.S_OK`. Otherwise, it returns an [HResult](#) error code.

Compare(nint, SICHINTF, out int)

Compares two [IShellItem](#) objects.

```
HResult Compare(nint psi, SICHINTF hint, out int piOrder)
```

Parameters

psi [nint](#)

A pointer to an [IShellItem](#) object to compare with the existing [IShellItem](#) object.

hint [SICHINTF](#)

One of the [SICHINTF](#) values that determines how to perform the comparison. See [SICHINTF](#) for the list of possible values for this parameter.

piOrder [int](#)

This parameter receives the result of the comparison. If the two items are the same this parameter equals zero; if they are different the parameter is nonzero.

Returns

[HResult](#)

`FireBlade.WinInteropUtils.HResult.S_OK` if the items are the same, `FireBlade.WinInteropUtils.HResult.S_FALSE` if they are different, or an error value otherwise.

GetAttributes(uint, out uint)

Gets a requested set of attributes of the [IShellItem](#) object.

```
HResult GetAttributes(uint sfgaoMask, out uint psfgaoAttribs)
```

Parameters

sfgaoMask [uint](#)

Specifies the attributes to retrieve. One or more of the SFGAO values. Use a bitwise OR operator to determine the attributes to retrieve.

psfgaoAttribs [uint](#)

A pointer to a value that, when this method returns successfully, contains the requested attributes. One or more of the SFGAO values. Only those attributes specified by **sfgaoMask** are returned; other attribute values are undefined.

Returns

[HResult](#)

FireBlade.WinInteropUtils.HResult.S_OK if the attributes returned exactly match those requested in **sfgaoMask**, FireBlade.WinInteropUtils.HResult.S_FALSE if the attributes do not exactly match, or a standard [COM](#) error value otherwise.

GetDisplayName(SIGDN, out nint)

Gets the display name of the [IShellItem](#) object.

```
HResult GetDisplayName(SIGDN sigdnName, out nint ppszName)
```

Parameters

sigdnName [SIGDN](#)

One of the [SIGDN](#) values that indicates how the name should look.

ppszName [nint](#)

A value that, when this function returns successfully, receives the address of a pointer to the retrieved display name.

Returns

[HResult](#)

FireBlade.WinInteropUtils.HResult.S_OK if successful; otherwise, an [HResult](#) error code.

GetParent(out nint)

Gets the parent of an [IShellItem](#) object.

```
HResult GetParent(out nint ppsi)
```

Parameters

ppsi [nint](#)

The address of a pointer to the parent of an [IShellItem](#) interface.

Returns

[HResult](#)

Returns FireBlade.WinInteropUtils.HResult.S_OK if successful, or an error value otherwise.

Enum SICHINTF

Namespace: [FireBlade.WinInteropUtils.ComponentObjectModel.Interfaces](#)

Assembly: WinInteropUtils.dll

Used to determine how to compare two Shell items. [Compare\(nint, SICHINTF, out int\)](#) uses this enumerated type.

[Flags]

```
public enum SICHINTF : uint
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

SICHINT_ALLFIELDS = 2147483648

Exact comparison of two instances of a Shell item.

SICHINT_CANONICAL = 268435456

This relates to the iOrder parameter of the [Compare\(nint, SICHINTF, out int\)](#) interface and indicates that the comparison is based on a canonical name.

SICHINT_DISPLAY = 0

This relates to the iOrder parameter of the [Compare\(nint, SICHINTF, out int\)](#) method and indicates that the comparison is based on the display in a folder view.

SICHINT_TEST_FILESYSPATH_IF_NOT_EQUAL = 536870912

Windows 7 and later. If the Shell items are not the same, test the file system paths.

Enum SIGDN

Namespace: [FireBlade.WinInteropUtils.ComponentObjectModel.Interfaces](#)

Assembly: WinInteropUtils.dll

Requests the form of an item's display name to retrieve through [GetDisplayName\(SIGDN, out nint\)](#).

```
[Flags]  
public enum SIGDN : uint
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

SIGDN_DESKTOPABSOLUTEEDITING = 2147794944

Returns the editing name relative to the desktop. In UI this name is suitable for display to the user.

SIGDN_DESKTOPABSOLUTEPARSING = 2147647488

Returns the parsing name relative to the desktop. This name is not suitable for use in UI.

SIGDN_FILESYSPATH = 2147844096

Returns the item's file system path, if it has one. Only items that report SFGAO_FILESYSTEM have a file system path. When an item does not have a file system path, a call to [GetDisplayName\(SIGDN, out nint\)](#) on that item will fail. In UI this name is suitable for display to the user in some cases, but note that it might not be specified for all items.

SIGDN_NORMALDISPLAY = 0

Returns the display name relative to the parent folder. In UI this name is generally ideal for display to the user.

SIGDN_PARENTRELATIVE = 2148007937

Returns the path relative to the parent folder.

SIGDN_PARENTRELATIVEEDITING = 2147684353

Returns the editing name relative to the parent folder. In UI this name is suitable for display to the user.

SIGDN_PARENTRELATIVEFORADDRESSBAR = 2147991553

Returns the path relative to the parent folder in a friendly format as displayed in an address bar. This name is suitable for display to the user.

SIGDN_PARENTRELATIVEFORUI = 2148089857

Returns the path relative to the parent folder for UI. Introduced in Windows 8.

SIGDN_PARENTRELATIVEPARSING = 2147581953

Returns the parsing name relative to the parent folder. This name is not suitable for use in UI.

SIGDN_URL = 2147909632

Returns the item's URL, if it has one. Some items do not have a URL, and in those cases a call to [Get DisplayName\(SIGDN, out nint\)](#) will fail. This name is suitable for display to the user in some cases, but note that it might not be specified for all items.

Namespace FireBlade.WinInteropUtils.Dialogs

Classes

[ColorPickerDialog](#)

Represents a dialog that allows the user to pick a color.

[DialogWindow<TShowReturn, TSelf>](#)

Represents a dialog window that can be shown modally or modelessly. This class is abstract.

[HelpEventArgs](#)

Provides event arguments for help events.

[WinMessageBox](#)

Provides methods for showing a Win32 message box.

Enums

[ColorPickerDialogFullOpenMode](#)

Defines how a [ColorPickerDialog](#) handles opening the full custom color picker.

[HelpContext](#)

The type of context for which help is requested.

[WinMessageBoxButtons](#)

Defines the buttons on a message box.

[WinMessageBoxIcon](#)

Specifies the icon in a message box.

[WinMessageBoxModality](#)

Specifies the modality of a [WinMessageBox](#).

[WinMessageBoxResult](#)

Represents the result of the message box.

Delegates

[ColorDialogHookProc](#)

Defines the hook procedure for a [ColorPickerDialog](#).

Delegate ColorDialogHookProc

Namespace: [FireBlade.WinInteropUtils.Dialogs](#)

Assembly: WinInteropUtils.dll

Defines the hook procedure for a [ColorPickerDialog](#).

```
public delegate uint ColorDialogHookProc(IntPtr hDlg, uint uMsg, IntPtr wParam, IntPtr lParam)
```

Parameters

hDlg [IntPtr](#)

A handle to the Color dialog box for which the message is intended.

uMsg [uint](#)

The identifier of the message being received.

wParam [IntPtr](#)

Additional information about the message. The exact meaning depends on the value of **uMsg**.

lParam [IntPtr](#)

Additional information about the message. The exact meaning depends on the value of **uMsg**.

Returns

[uint](#)

If the hook procedure returns zero, the default dialog box procedure processes the message.

If the hook procedure returns a nonzero value, the default dialog box procedure ignores the message.

Remarks

If the **uMsg** parameter indicates the WM_INITDIALOG message, then **lParam** is a garbage collector handle pointing to the instance of the [ColorPickerDialog](#) class. To get the instance, call the [FromIntPtr\(IntPtr\)](#) function and pass in the given **lParam**.

Class ColorPickerDialog

Namespace: [FireBlade.WinInteropUtils.Dialogs](#)

Assembly: WinInteropUtils.dll

Represents a dialog that allows the user to pick a color.

```
public class ColorPickerDialog : DialogWindow<Color?, ColorPickerDialog>
```

Inheritance

[object](#)  ← [DialogWindow](#)<[Color](#) , [ColorPickerDialog](#)> ← ColorPickerDialog

Inherited Members

[DialogWindow](#)<[Color](#)?, [ColorPickerDialog](#)>.WM_INITDIALOG ,
[DialogWindow](#)<[Color](#)?, [ColorPickerDialog](#)>.PackFlags(Dictionary<int, bool>),
[DialogWindow](#)<[Color](#)?, [ColorPickerDialog](#)>.Show() ,
[DialogWindow](#)<[Color](#)?, [ColorPickerDialog](#)>.Show(Window) ,
[DialogWindow](#)<[Color](#)?, [ColorPickerDialog](#)>.Show(ColorPickerDialog) ,
[DialogWindow](#)<[Color](#)?, [ColorPickerDialog](#)>.Show(ColorPickerDialog, Window) , [object.Equals\(object\)](#)  ,
[object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Properties

AnyColor

If [true](#) , causes the dialog box to display all available colors in the set of basic colors.

```
public bool AnyColor { get; set; }
```

Property Value

[bool](#) 

CustomColors

Represents the collection of custom colors.

```
public Color[] CustomColors { get; set; }
```

Property Value

[Color](#)[↗][]

Remarks

When the [Show\(\)](#) or [Show\(Window\)](#) functions return, this property will contain the custom colors set by the user. You can store this array to persist the custom colors between sessions. The array has a capacity of **16**.

DefaultColor

Specifies the default color chosen in the dialog.

```
public Color? DefaultColor { get; set; }
```

Property Value

[Color](#)[↗]?

FullOpenMode

Specifies how the dialog handles opening the full custom color picker.

```
public ColorPickerDialogFullOpenMode FullOpenMode { get; set; }
```

Property Value

[ColorPickerDialogFullOpenMode](#)

HookProcedure

Specifies the dialog hook procedure when [UseHookProc](#) is equal to [true](#)[↗].

```
public ColorDialogHookProc? HookProcedure { get; set; }
```

Property Value

[ColorDialogHookProc](#)

ShowHelp

Indicates whether to show the **Help** button.

```
public bool ShowHelp { get; set; }
```

Property Value

[bool](#)

Remarks

To implement help, set the [ShowHelp](#) property to [true](#), and call [RegisterWindowMessage\(string\)](#), passing in the string `commdlg_help`. Then, store the value returned by the function and check for it in your window procedure.

SolidColorsOnly

If [true](#), causes the dialog box to display only solid colors in the set of basic colors.

```
public bool SolidColorsOnly { get; set; }
```

Property Value

[bool](#)

UseHookProc

Gets or sets a value indicating whether to use the hook procedure defined in [HookProcedure](#).

```
public bool UseHookProc { get; set; }
```

Property Value

[bool](#)

Methods

Show()

Shows the color dialog modelessly.

```
public override Color? Show()
```

Returns

[Color](#)?

The chosen [Color](#), or [null](#) if the user cancelled the dialog.

Show(ColorPickerDialog)

Shows the [ColorPickerDialog](#).

```
public static Color? Show(ColorPickerDialog dlg)
```

Parameters

dlg [ColorPickerDialog](#)

The [ColorPickerDialog](#) to show.

Returns

[Color](#)?

The result of the dialog.

Show(ColorPickerDialog, Window)

Shows the [ColorPickerDialog](#) modally.

```
public static Color? Show(ColorPickerDialog dlg, Window wnd)
```

Parameters

dlg [ColorPickerDialog](#)

The [ColorPickerDialog](#) to show.

wnd [Window](#)

The owner window.

Returns

[Color](#)?

The result of the dialog.

Show(Window)

Shows the color dialog modally.

```
public override Color? Show(Window wnd)
```

Parameters

wnd [Window](#)

Returns

[Color](#)?

The chosen [Color](#), or [null](#) if the user cancelled the dialog.

Show(Window, Color)

Shows the color dialog modally.

```
public static Color? Show(Window wnd, Color defaultColor)
```

Parameters

wnd [Window](#)

The owner window.

defaultColor [Color](#)

The default color.

Returns

[Color](#)?

The chosen color.

Remarks

For more customization, create an instance of the [ColorPickerDialog](#) class.

Show(Window, Color, Color[])

Shows the color dialog modally.

```
public static Color? Show(Window wnd, Color defaultColor, Color[] customColors)
```

Parameters

wnd [Window](#)

The owner window.

defaultColor [Color](#)

The default color.

customColors [Color](#)[]

The array of custom colors in the dialog.

Returns

[Color](#)[?]

The chosen color.

Remarks

For more customization, create an instance of the [ColorPickerDialog](#) class.

Show(Color)

Shows the color dialog modelessly.

```
public static Color? Show(Color defaultColor)
```

Parameters

defaultColor [Color](#)[?]

The default color.

Returns

[Color](#)[?]

The chosen color.

Remarks

For more customization, create an instance of the [ColorPickerDialog](#) class.

Show(Color, Color[])

Shows the color dialog modelessly.

```
public static Color? Show(Color defaultColor, Color[] customColors)
```

Parameters

defaultColor [Color](#)[↗]

The default color.

customColors [Color](#)[↗][]

The array of custom colors in the dialog.

Returns

[Color](#)[↗]?

The chosen color.

Remarks

For more customization, create an instance of the [ColorPickerDialog](#) class.

Enum ColorPickerDialogFullOpenMode

Namespace: [FireBlade.WinInteropUtils.Dialogs](#)

Assembly: WinInteropUtils.dll

Defines how a [ColorPickerDialog](#) handles opening the full custom color picker.

```
public enum ColorPickerDialogFullOpenMode
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Disabled = 4

The **Define Custom Colors** button is disabled - the user can only pick basic colors.

Enabled = 0

The **Define Custom Colors** button is enabled, but the full color picker is not opened.

Opened = 2

The full custom color picker is opened.

Class DialogWindow<TShowReturn, TSelf>

Namespace: [FireBlade.WinInteropUtils.Dialogs](#)

Assembly: WinInteropUtils.dll

Represents a dialog window that can be shown modally or modelessly. This class is abstract.

```
public abstract class DialogWindow<TShowReturn, TSelf> where TSelf :  
    DialogWindow<TShowReturn, TSelf>
```

Type Parameters

TShowReturn

The return type of the [Show\(\)](#) and [Show\(Window\)](#) methods.

TSelf

The class that is inheriting [DialogWindow<TShowReturn, TSelf>](#).

Inheritance

[object](#)  ← [DialogWindow<TShowReturn, TSelf>](#)

Derived

[ColorPickerDialog](#), [WinMessageBox](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Fields

WM_INITDIALOG

The message sent to a dialog's hook procedure when it initializes.

```
protected const int WM_INITDIALOG = 272
```

Field Value

[int](#)

Methods

PackFlags(Dictionary<int, bool>)

Packs the specified flags for use in a dialog's struct's `dwFlags` or `Flags` member.

```
protected int PackFlags(Dictionary<int, bool> flags)
```

Parameters

flags [Dictionary](#) <[int](#), [bool](#)>

The dictionary of flags to pack. The key indicates the value of the flag, and the value indicates whether to pack the flag.

Returns

[int](#)

The packed flags, or 0 if no flags were marked as packable.

Show()

When overridden in a derived class, shows the dialog window.

```
public abstract TShowReturn Show()
```

Returns

TShowReturn

The result of the dialog.

Show(Window)

When overridden in a derived class, shows the dialog window modally.

```
public abstract TShowReturn Show(Window wnd)
```

Parameters

wnd [Window](#)

The owner window.

Returns

TShowReturn

The result of the dialog.

Show(TSelf)

Shows the **TSelf**.

```
public static TShowReturn Show(TSelf dlg)
```

Parameters

dlg TSelf

The **TSelf** to show.

Returns

TShowReturn

The result of the dialog.

Show(TSelf, Window)

Shows the **TSelf** modally.

```
public static TShowReturn Show(TSelf dlg, Window wnd)
```

Parameters

dlg TSelf

The TSelf to show.

wnd [Window](#)

The owner window.

Returns

TShowReturn

The result of the dialog.

Enum HelpContext

Namespace: [FireBlade.WinInteropUtils.Dialogs](#)

Assembly: WinInteropUtils.dll

The type of context for which help is requested.

```
public enum HelpContext
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

MenuItem = 2

Help requested for a menu item.

Window = 1

Help requested for a control or window.

Class HelpEventArgs

Namespace: [FireBlade.WinInteropUtils.Dialogs](#)

Assembly: WinInteropUtils.dll

Provides event arguments for help events.

```
public class HelpEventArgs : EventArgs
```

Inheritance

[object](#)  ← [EventArgs](#)  ← HelpEventArgs

Inherited Members

[EventArgs.Empty](#) , [object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Properties

ContextId

The help context identifier of the window or control.

```
public nint ContextId { get; }
```

Property Value

[nint](#) 

ContextType

The type of context for which help is requested.

```
public HelpContext ContextType { get; }
```

Property Value

ControlHandle

The handle of the child window, control, or menu item.

```
public nint ControlHandle { get; }
```

Property Value

[nint](#)

ControlId

The identifier of the window or control if [ContextType](#) is [Window](#), or identifier of the menu item if [ContextType](#) is [MenuItem](#).

```
public int ControlId { get; }
```

Property Value

[int](#)

MousePos

The [Point](#) that contains the screen coordinates of the mouse cursor. This is useful for providing help based on the position of the mouse cursor.

```
public Point MousePos { get; }
```

Property Value

[Point](#)

Class WinMessageBox

Namespace: [FireBlade.WinInteropUtils.Dialogs](#)

Assembly: WinInteropUtils.dll

Provides methods for showing a Win32 message box.

```
public class WinMessageBox : DialogWindow<WinMessageBoxResult, WinMessageBox>
```

Inheritance

[object](#)  ← [DialogWindow](#) <[WinMessageBoxResult](#), [WinMessageBox](#)> ← WinMessageBox

Inherited Members

[DialogWindow](#) <[WinMessageBoxResult](#), [WinMessageBox](#)>.WM_INITDIALOG ,
[DialogWindow](#) <[WinMessageBoxResult](#), [WinMessageBox](#)>.PackFlags(Dictionary<int, bool>),
[DialogWindow](#) <[WinMessageBoxResult](#), [WinMessageBox](#)>.Show(),
[DialogWindow](#) <[WinMessageBoxResult](#), [WinMessageBox](#)>.Show(Window),
[DialogWindow](#) <[WinMessageBoxResult](#), [WinMessageBox](#)>.Show(WinMessageBox),
[DialogWindow](#) <[WinMessageBoxResult](#), [WinMessageBox](#)>.Show(WinMessageBox, Window),
[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Properties

Buttons

Specifies the buttons on the message box.

```
public WinMessageBoxButtons Buttons { get; set; }
```

Property Value

[WinMessageBoxButtons](#)

Caption

Specifies the caption of the message box. If set to [null](#) , defaults to Error.

```
public string? Caption { get; set; }
```

Property Value

[string](#)[↗]

ContextHelpId

Specifies the help context. When [OnHelp](#) fires, this specifies the value that will be in the [ContextId](#) property of the [HelpEventArgs](#).

```
public uint? ContextHelpId { get; set; }
```

Property Value

[uint](#)[↗]?

Culture

Specifies the culture this message box uses. If [null](#)[↗], uses the default system culture.

```
public CultureInfo? Culture { get; set; }
```

Property Value

[CultureInfo](#)[↗]

Remarks

NOTE

Each localized release of Windows typically contains resources only for a limited set of languages. Thus, for example, the U.S. version offers **English**, the French version offers **French**, the German version offers **German**, and the Japanese version offers **Japanese**. Each version offers **Neutral**. This limits the set of values that can be used with the [Culture](#) parameter. Before specifying a language identifier, you should enumerate the locales that are installed on a system.

DefaultButton

Specifies the default button on the message box. This value must be in a range of 1-4.

```
public int DefaultButton { get; set; }
```

Property Value

[int](#)

Exceptions

[ArgumentOutOfRangeException](#)

The value is out of the range. The value must be in a range of 1-4.

DefaultDesktopOnly

Same as desktop of the interactive window station. For more information, see [Window Stations](#).

```
public bool DefaultDesktopOnly { get; set; }
```

Property Value

[bool](#)

Remarks

If the current input desktop is not the default desktop, [Show\(\)](#) and [Show\(Window\)](#) does not return until the user switches to the default desktop.

Icon

Gets or sets the icon displayed on the message box.

```
public WinMessageBoxIcon Icon { get; set; }
```

Property Value

Modality

Specifies the modality of the message box. The default is [ApplicationModal](#).

```
public WinMessageBoxModality Modality { get; set; }
```

Property Value

[WinMessageBoxModality](#).

RightAlign

If [true](#), the message box is right aligned. Otherwise, it is left-aligned. The default is [false](#).

```
public bool RightAlign { get; set; }
```

Property Value

[bool](#)

RightToLeft

If [true](#), displays message and caption text using right-to-left reading order on Hebrew and Arabic systems. (called RtlReading in Win32)

```
public bool RightToLeft { get; set; }
```

Property Value

[bool](#)

ServiceNotification

The caller is a service notifying the user of an event. The function displays a message box on the current active desktop, even if there is no user logged on to the computer.

```
public bool ServiceNotification { get; set; }
```

Property Value

[bool](#)[↗]

Remarks

NOTE

Terminal Services: If the calling thread has an impersonation token, the function directs the message box to the session specified in the impersonation token.

If this flag is set, the [Show\(\)](#) function should be called instead of [Show\(Window\)](#). This is so that the message box can appear on a desktop other than the desktop corresponding to the window corresponding to the specified `hWnd` in [Show\(Window\)](#).

For information on security considerations in regard to using this flag, see [Interactive Services](#)[↗]. In particular, be aware that this flag can produce interactive content on a locked desktop and should therefore be used for only a very limited set of scenarios, such as resource exhaustion.

SetForeground

If [true](#)[↗], the message box becomes the foreground window. Internally, the system calls the `SetForegroundWindow` function for the message box.

```
public bool SetForeground { get; set; }
```

Property Value

[bool](#)[↗]

ShowHelp

Adds a Help button to the message box. When the user clicks the Help button or presses F1, the message box fires the [OnHelp](#) event.

```
public bool ShowHelp { get; set; }
```

Property Value

[bool](#)

Text

Specifies the text of the message box.

```
public string Text { get; set; }
```

Property Value

[string](#)

TopMost

If [true](#), the message box is created as a top-most window (stays on top).

```
public bool TopMost { get; set; }
```

Property Value

[bool](#)

Methods

Show()

Shows the message box modelessly.


```
public override WinMessageBoxResult Show()
```

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the strings in the [Text](#) and [Caption](#) members should not be taken from a resource file, because an attempt to load the resource may fail.

Exceptions

[OutOfMemoryException](#)

There was not enough memory to create the message box.

Show(Window)

Shows the message box.

```
public override WinMessageBoxResult Show(Window wnd)
```

Parameters

wnd [Window](#)

The parent window to attach the message box to.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the strings in the [Text](#) and [Caption](#) members should not be taken from a resource file, because an attempt to load the resource may fail.

Exceptions

[OutOfMemoryException](#) 

There was not enough memory to create the message box.

Show(Window, string)

Shows a message box with the specified text and the default caption (Error) as a child of the window specified by `wnd`.

```
public static WinMessageBoxResult Show(Window wnd, string text)
```

Parameters

`wnd` [Window](#)

The parent window to attach the message box to.

`text` [string](#) 

The text on the message box.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the string in the `text` parameter should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the `FireBlade.WinInteropUtils.Dialogs.WinMessageBox.WinMessageBox()` constructor and show it with the [Show\(Window\)](#) method.

Exceptions

[OutOfMemoryException](#) 

There was not enough memory to create the message box.

Show(Window, string, WinMessageBoxButtons)

Shows a message box with the specified text and buttons as a child of the window specified by `wnd`.

```
public static WinMessageBoxResult Show(Window wnd, string text,
WinMessageBoxButtons buttons)
```

Parameters

`wnd` [Window](#)

The parent window to attach the message box to.

`text` [string](#) 

The text on the message box.

`buttons` [WinMessageBoxButtons](#)

The buttons on the message box. One of the [WinMessageBoxButtons](#) values.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the string in the `text` parameter should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the `FireBlade.WinInteropUtils.Dialogs.WinMessageBox.WinMessageBox()` constructor and show it with the [Show\(Window\)](#) method.

Exceptions

[OutOfMemoryException](#)

There was not enough memory to create the message box.

Show(Window, string, WinMessageBoxIcon)

Shows a message box with the specified text and icon as a child of the window specified by `wnd`.

```
public static WinMessageBoxResult Show(Window wnd, string text, WinMessageBoxIcon icon)
```

Parameters

`wnd` [Window](#)

The parent window to attach the message box to.

`text` [string](#) 

The text on the message box.

`icon` [WinMessageBoxIcon](#)

The icon displayed on the message box. One of the [WinMessageBoxIcon](#) values.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the string in the `text` parameter should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the `FireBlade.WinInteropUtils.Dialogs.WinMessageBox.WinMessageBox()` constructor and show it with the [Show\(Window\)](#) method.

Exceptions

[OutOfMemoryException](#)

There was not enough memory to create the message box.

Show(Window, string, WinMessageBoxIcon, WinMessageBoxButtons)

Shows a message box with the specified text, icon, and buttons as a child of the window specified by `wnd`.

```
public static WinMessageBoxResult Show(Window wnd, string text, WinMessageBoxIcon icon, WinMessageBoxButtons buttons)
```

Parameters

`wnd` [Window](#)

The parent window to attach the message box to.

text [string](#)

The text on the message box.

icon [WinMessageBoxIcon](#)

The icon displayed on the message box. One of the [WinMessageBoxIcon](#) values.

buttons [WinMessageBoxButtons](#)

The buttons on the message box. One of the [WinMessageBoxButtons](#) values.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the string in the **text** parameter should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the FireBlade.WinInteropUtils.Dialogs.WinMessageBox.WinMessageBox() constructor and show it with the [Show\(Window\)](#) method.

Exceptions

[OutOfMemoryException](#)

There was not enough memory to create the message box.

Show(Window, string, string?)

Shows a message box with the specified text and caption as a child of the window specified by **wnd**.

```
public static WinMessageBoxResult Show(Window wnd, string text, string? caption)
```

Parameters

wnd [Window](#)

The parent window to attach the message box to.

text [string](#) 

The text on the message box.

caption [string](#) 

The caption (title) of the message box.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the strings in the **text** and **caption** parameters should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the FireBlade.Win.InteropUtils.Dialogs.WinMessageBox.WinMessageBox() constructor and show it with the [Show\(Window\)](#) method.

Exceptions

[OutOfMemoryException](#) 

There was not enough memory to create the message box.

Show(Window, string, string?, WinMessageBoxButtons)

Shows a message box with the specified text, caption, and buttons as a child of the window specified by `wnd`.

```
public static WinMessageBoxResult Show(Window wnd, string text, string? caption, WinMessageBoxButtons buttons)
```

Parameters

`wnd` [Window](#)

The parent window to attach the message box to.

`text` [string](#) 

The text on the message box.

`caption` [string](#) 

The caption (title) of the message box.

`buttons` [WinMessageBoxButtons](#)

The buttons on the message box. One of the [WinMessageBoxButtons](#) values.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the strings in the `text` and `caption` parameters should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the `FireBlade.WinInteropUtils.Dialogs.WinMessageBox.WinMessageBox()` constructor and show it with the [Show\(Window\)](#).

method.

Exceptions

[OutOfMemoryException](#) 

There was not enough memory to create the message box.

Show(Window, string, string?, WinMessageBoxIcon)

Shows a message box with the specified text, caption, and icon as a child of the window specified by *wnd*.

```
public static WinMessageBoxResult Show(Window wnd, string text, string? caption,
WinMessageBoxIcon icon)
```

Parameters

wnd [Window](#)

The parent window to attach the message box to.

text [string](#) 

The text on the message box.

caption [string](#) 

The caption (title) of the message box.

icon [WinMessageBoxIcon](#)

The icon displayed on the message box. One of the [WinMessageBoxIcon](#) values.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the strings in the `text` and `caption` parameters should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the `FireBlade.WinInteropUtils.Dialogs.WinMessageBox.WinMessageBox()` constructor and show it with the [Show\(Window\)](#) method.

Exceptions

[OutOfMemoryException](#)

There was not enough memory to create the message box.

Show(Window, string, string?, WinMessageBoxIcon, WinMessageBoxButtons)

Shows a message box with the specified text, caption, icon, and buttons as a child of the window specified by `wnd`.

```
public static WinMessageBoxResult Show(Window wnd, string text, string? caption,
WinMessageBoxIcon icon, WinMessageBoxButtons buttons)
```

Parameters

`wnd` [Window](#)

The parent window to attach the message box to.

`text` [string](#) 

The text on the message box.

`caption` [string](#) 

The caption (title) of the message box.

`icon` [WinMessageBoxIcon](#)

The icon displayed on the message box. One of the [WinMessageBoxIcon](#) values.

buttons [WinMessageBoxButtons](#)

The buttons on the message box. One of the [WinMessageBoxButtons](#) values.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the strings in the **text** and **caption** parameters should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the FireBlade.Win.InteropUtils.Dialogs.WinMessageBox.WinMessageBox() constructor and show it with the [Show\(Window\)](#) method.

Exceptions

[OutOfMemoryException](#) 

There was not enough memory to create the message box.

Show(string)

Shows a modeless message box with the specified text and the default caption (Error).

```
public static WinMessageBoxResult Show(string text)
```

Parameters

text [string](#) 

The text on the message box.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the string in the `text` parameter should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the `FireBlade.WinInteropUtils.Dialogs.WinMessageBox.WinMessageBox()` constructor and show it with the [Show\(\)](#) method.

Exceptions

[OutOfMemoryException](#)

There was not enough memory to create the message box.

Show(string, WinMessageBoxButtons)

Shows a modeless message box with the specified text and buttons.

```
public static WinMessageBoxResult Show(string text, WinMessageBoxButtons buttons)
```

Parameters

`text` [string](#) 

The text on the message box.

`buttons` [WinMessageBoxButtons](#)

The buttons on the message box. One of the [WinMessageBoxButtons](#) values.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the string in the `text` parameter should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the `FireBlade.WinInteropUtils.Dialogs.WinMessageBox.WinMessageBox()` constructor and show it with the [Show\(\)](#) method.

Exceptions

[OutOfMemoryException](#)

There was not enough memory to create the message box.

Show(string, WinMessageBoxIcon)

Shows a modeless message box with the specified text and icon.

```
public static WinMessageBoxResult Show(string text, WinMessageBoxIcon icon)
```

Parameters

text [string](#) 

The text on the message box.

icon [WinMessageBoxIcon](#)

The icon displayed on the message box. One of the [WinMessageBoxIcon](#) values.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the string in the `text` parameter should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the `FireBlade.WinInteropUtils.Dialogs.WinMessageBox.WinMessageBox()` constructor and show it with the [Show\(\)](#) method.

Exceptions

[OutOfMemoryException](#) 

There was not enough memory to create the message box.

Show(string, WinMessageBoxIcon, WinMessageBoxButtons)

Shows a modeless message box with the specified text, icon, and buttons.

```
public static WinMessageBoxResult Show(string text, WinMessageBoxIcon icon,
WinMessageBoxButtons buttons)
```

Parameters

`text` [string](#) 

The text on the message box.

`icon` [WinMessageBoxIcon](#)

The icon displayed on the message box. One of the [WinMessageBoxIcon](#) values.

`buttons` [WinMessageBoxButtons](#)

The buttons on the message box. One of the [WinMessageBoxButtons](#) values.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the string in the `text` parameter should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the `FireBlade.WinInteropUtils.Dialogs.WinMessageBox.WinMessageBox()` constructor and show it with the [Show\(\)](#) method.

Exceptions

[OutOfMemoryException](#)

There was not enough memory to create the message box.

Show(string, string?)

Shows a modeless message box with the specified text and caption.

```
public static WinMessageBoxResult Show(string text, string? caption)
```

Parameters

`text` [string](#) 

The text on the message box.

`caption` [string](#) 

The caption (title) of the message box.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the strings in the `text` and `caption` parameters should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the `FireBlade.WinInteropUtils.Dialogs.WinMessageBox.WinMessageBox()` constructor and show it with the [Show\(\)](#) method.

Exceptions

[OutOfMemoryException](#)

There was not enough memory to create the message box.

Show(string, string?, WinMessageBoxButtons)

Shows a modeless message box with the specified text, caption, and buttons.

```
public static WinMessageBoxResult Show(string text, string? caption,
WinMessageBoxButtons buttons)
```

Parameters

`text` [string](#) 

The text on the message box.

`caption` [string](#) 

The caption (title) of the message box.

`buttons` [WinMessageBoxButtons](#)

The buttons on the message box. One of the [WinMessageBoxButtons](#) values.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the strings in the **text** and **caption** parameters should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the `FireBlade.WinInteropUtils.Dialogs.WinMessageBox.WinMessageBox()` constructor and show it with the [Show\(\)](#) method.

Exceptions

[OutOfMemoryException](#)

There was not enough memory to create the message box.

Show(string, string?, WinMessageBoxIcon)

Shows a modeless message box with the specified text, caption, and icon.

```
public static WinMessageBoxResult Show(string text, string? caption, WinMessageBoxIcon icon)
```

Parameters

text [string](#) 

The text on the message box.

caption [string](#) 

The caption (title) of the message box.

icon [WinMessageBoxIcon](#)

The icon displayed on the message box. One of the [WinMessageBoxIcon](#) values.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the strings in the **text** and **caption** parameters should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the FireBlade.WinInteropUtils.Dialogs.WinMessageBox.WinMessageBox() constructor and show it with the [Show\(\)](#) method.

Exceptions

[OutOfMemoryException](#)

There was not enough memory to create the message box.

Show(string, string?, WinMessageBoxIcon, WinMessageBoxButtons)

Shows a modeless message box with the specified text, caption, icon, and buttons.

```
public static WinMessageBoxResult Show(string text, string? caption, WinMessageBoxIcon icon, WinMessageBoxButtons buttons)
```

Parameters

text [string](#) 

The text on the message box.

caption [string](#)

The caption (title) of the message box.

icon [WinMessageBoxIcon](#)

The icon displayed on the message box. One of the [WinMessageBoxIcon](#) values.

buttons [WinMessageBoxButtons](#)

The buttons on the message box. One of the [WinMessageBoxButtons](#) values.

Returns

[WinMessageBoxResult](#)

The button that was clicked.

If a message box has a Cancel button, the function returns the [Cancel](#) value if either the ESC key is pressed or the Cancel button is selected. If the message box has no Cancel button, pressing ESC has no effect.

Remarks

When you use a system-modal message box to indicate that the system is low on memory, the strings in the **text** and **caption** parameters should not be taken from a resource file, because an attempt to load the resource may fail.

For extended customization, create an instance of the [WinMessageBox](#) class with the FireBlade.Win.InteropUtils.Dialogs.WinMessageBox.WinMessageBox() constructor and show it with the [Show\(\)](#) method.

Exceptions

[OutOfMemoryException](#)

There was not enough memory to create the message box.

Events

OnHelp

Fires when help is requested from the message box.

```
public event EventHandler<HelpEventArgs>? OnHelp
```

Event Type

[EventHandler](#)  <[HelpEventArgs](#)>

Enum WinMessageBoxButtons

Namespace: [FireBlade.WinInteropUtils.Dialogs](#)

Assembly: WinInteropUtils.dll

Defines the buttons on a message box.

```
public enum WinMessageBoxButtons : long
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

AbortRetryIgnore = 2

The message box contains three push buttons: Abort, Retry, and Ignore.

CancelRetryContinue = 6

The message box contains three push buttons: Cancel, Try Again, Continue. Use this instead of [Abort RetryIgnore](#).

Ok = 0

The message box contains one push button: OK. This is the default.

OkCancel = 1

The message box contains two push buttons: OK and Cancel.

RetryCancel = 5

The message box contains two push buttons: Retry and Cancel.

YesNo = 4

The message box contains two push buttons: Yes and No.

YesNoCancel = 3

The message box contains three push buttons: Yes, No, and Cancel.

Enum WinMessageBoxIcon

Namespace: [FireBlade.WinInteropUtils.Dialogs](#)

Assembly: WinInteropUtils.dll

Specifies the icon in a message box.

```
public enum WinMessageBoxIcon : long
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Asterisk = 64

An icon consisting of a lowercase letter *i* in a circle appears in the message box.

Error = 16

A stop-sign icon appears in the message box.

Exclamation = 48

An exclamation-point icon appears in the message box.

Hand = 16

A stop-sign icon appears in the message box.

Info = 64

An icon consisting of a lowercase letter *i* in a circle appears in the message box.

Information = 64

An icon consisting of a lowercase letter *i* in a circle appears in the message box.

None = 0

No icon is displayed in the message box.

[Obsolete("The question-mark message icon is not recommended; see documentation for details.")] Question = 32

A question-mark icon appears in the message box.

The question-mark message icon is no longer recommended because it does not clearly represent a specific type of message and because the phrasing of a message as a question could apply to any message type. In addition, users can confuse the message symbol question mark with Help information. Therefore, do not use this question mark message symbol in your message boxes. The system continues to support its inclusion only for backward compatibility.

Stop = 16

A stop-sign icon appears in the message box.

Warning = 48

An exclamation-point icon appears in the message box.

Enum WinMessageBoxModality

Namespace: [FireBlade.WinInteropUtils.Dialogs](#)

Assembly: WinInteropUtils.dll

Specifies the modality of a [WinMessageBox](#).

```
public enum WinMessageBoxModality : long
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

ApplicationModal = 0

The user must respond to the message box before continuing work in the window identified by the **hWnd** parameter. However, the user can move to the windows of other threads and work in those windows.

Depending on the hierarchy of windows in the application, the user may be able to move to other windows within the thread. All child windows of the parent of the message box are automatically disabled, but pop-up windows are not.

SystemModal = 4096

Same as [ApplicationModal](#) except that the message box is a top-most window (stays on top). Use system-modal message boxes to notify the user of serious, potentially damaging errors that require immediate attention (for example, running out of memory). This flag has no effect on the user's ability to interact with windows other than those associated with **hWnd**.

TaskModal = 8192

Same as [ApplicationModal](#) except that all the top-level windows belonging to the current thread are disabled if the **hWnd** parameter is [null](#). Use this flag when the calling application or library does not have a window handle available but still needs to prevent input to other windows in the calling thread without suspending other threads.

Enum WinMessageBoxResult

Namespace: [FireBlade.WinInteropUtils.Dialogs](#)

Assembly: WinInteropUtils.dll

Represents the result of the message box.

```
public enum WinMessageBoxResult
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Abort = 3

The Abort button was selected.

Cancel = 2

The Cancel button was selected.

Continue = 11

The Continue button was selected.

Ignore = 5

The Ignore button was selected.

No = 7

The No button was selected.

Ok = 1

The OK button was selected.

Retry = 4

The Retry button was selected.

TryAgain = 10

The Try Again button was selected.

Yes = 6

The Yes button was selected.

Namespace FireBlade.WinInteropUtils.File

System

Classes

[WinFile](#)

Represents a Windows file.

[WinFileIcon](#)

Represents the icon of a [WinFile](#).

[WinFileInfo](#)

Represents information about a [WinFile](#).

Enums

[ExecutableType](#)

Represents the type of an EXE file.

[WinFileAccess](#)

Represents the file access mode of a [WinFile](#).

[WinFileAttributes](#)

Specifies attributes for Windows files.

[WinFileInfoOptions](#)

Represents the options for [WinFileInfo\(string, WinFileInfoOptions, WinFileAttributes?\)](#).

[WinFileOpenMode](#)

Specifies how to open a [WinFile](#).

[WinFileOptions](#)

Represents options for a [WinFile](#).

[WinFileShareMode](#)

Represents the sharing mode of a [WinFile](#).

Enum ExecutableType

Namespace: [FireBlade.WinInteropUtils.FileSystem](#)

Assembly: WinInteropUtils.dll

Represents the type of an EXE file.

```
public enum ExecutableType
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Console = 4

Console application.

MSDos = 3

MS-DOS application (.COM).

NE = 2

16-bit New Executable file.

None = 0

Invalid type.

This value is used if the file is not an executable file or another error occurs.

PE = 1

32- or 64-bit Windows Portable Executable file.

Class WinFile

Namespace: [FireBlade.WinInteropUtils.FileSystem](#)

Assembly: WinInteropUtils.dll

Represents a Windows file.

```
public sealed class WinFile : IDisposable, IHandle
```

Inheritance

[object](#)  ← WinFile

Implements

[IDisposable](#) , [IHandle](#)

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

WinFile(string, WinFileAccess, WinFileShareMode, WinFileOpenMode, WinFileAttributes, WinFileOptions, WinFile?)

Initializes a new instance of the [WinFile](#) class with the specified path.

```
public WinFile(string path, WinFileAccess access = WinFileAccess.Read, WinFileShareMode  
shareMode = WinFileShareMode.None, WinFileOpenMode mode = WinFileOpenMode.OpenExisting,  
WinFileAttributes attribs = WinFileAttributes.None, WinFileOptions options =  
WinFileOptions.None, WinFile? templateFile = null)
```

Parameters

path [string](#) 

The path to the file.

access [WinFileAccess](#)

The requested access to the file. A bitwise combination of [WinFileAccess](#) values.

`shareMode` [WinFileShareMode](#)

The requested sharing mode of the file.

`mode` [WinFileOpenMode](#)

The action to take on a file that exists or does not exist.

`attribs` [WinFileAttributes](#)

The file attributes and flags.

`options` [WinFileOptions](#)

Additional options for opening the file.

`templateFile` [WinFile](#)

A valid template file with the [Read](#) access right. The template file supplies file attributes and extended attributes for the file that is being created. If opening an existing file, this parameter is ignored.

Properties

Bytes

Gets or sets the bytes of the file.

```
public byte[] Bytes { get; set; }
```

Property Value

[byte](#)[?][]

Remarks

Do not repeatedly get this property; instead, cache it, as reading files can be expensive.

Content

Gets the string content of the file.

```
public string Content { get; set; }
```

Property Value

[string](#)[↗]

Remarks

This property assumes the file is encoded with UTF-8. If the file is a different encoding, you may have to use [GetString\(byte\[\]\)](#)[↗] or [GetBytes\(string\)](#)[↗] instead.

FileInfo

Gets the information about this file.

```
public WinFileInfo FileInfo { get; }
```

Property Value

[WinFileInfo](#)

Handle

Gets the handle (**HFILE**) of this file.

```
public nint Handle { get; }
```

Property Value

[nint](#)[↗]

Path

Gets the path to this file.

```
public string Path { get; }
```

Property Value

[string](#)

Size

Gets the current size of the file, in bytes.

```
public long Size { get; }
```

Property Value

[long](#)

Methods

Dispose()

Releases this [WinFile](#).

```
public void Dispose()
```


Enum WinFileAccess

Namespace: [FireBlade.WinInteropUtils.FileSystem](#)

Assembly: WinInteropUtils.dll

Represents the file access mode of a [WinFile](#).

[Flags]

```
public enum WinFileAccess : uint
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

None = 0

Default access (no access).

When using this access mode, you can query certain information and metadata about the file, even if [Read](#) access would be denied.

Read = 2147483648

Read access.

Write = 1073741824

Write access.

Enum WinFileAttributes

Namespace: [FireBlade.WinInteropUtils.FileSystem](#)

Assembly: WinInteropUtils.dll

Specifies attributes for Windows files.

[Flags]

```
public enum WinFileAttributes
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Archive = 32

A file or directory that is an archive file or directory. Applications typically use this attribute to mark files for backup or removal.

ArchiveDir = Directory | Archive

An archive directory. Applications typically use this attribute to mark directories for backup or removal. This is a composite value.

Compressed = 2048

A file or directory that is compressed. For a file, all of the data in the file is compressed. For a directory, compression is the default for newly created files and subdirectories.

[Obsolete("This value is reserved for system use. Do not use.", DiagnosticId = "WIU128")] Device = 64

This value is reserved for system use. Do not use.

Directory = 16

The information queried identifies a directory.

Encrypted = 16384

A file or directory that is encrypted. For a file, all data streams in the file are encrypted. For a directory, encryption is the default for newly created files and subdirectories.

[Obsolete("This value is reserved for system use. Do not use.", DiagnosticId = "WIU128")] ExAttrib = 262144

A file or directory with extended attributes. This value is for internal use only. Do not use.

Hidden = 2

The file or directory is hidden. It is not included in an ordinary directory listing.

IntegrityStream = 32768

The directory or user data stream is configured with integrity (only supported on ReFS volumes). It is not included in an ordinary directory listing. The integrity setting persists with the file if it's renamed. If a file is copied the destination file will have integrity set if either the source file or destination directory have integrity set. **Windows Server 2008 R2, Windows 7, Windows Server 2008, Windows Vista, Windows Server 2003 and Windows XP:** This flag is not supported until *Windows Server 2012*.

NoContentIndex = 8192

The file or directory is not to be indexed by the content indexing service.

NoScrubData = 131072

The user data stream not to be read by the background data integrity scanner (AKA scrubber). When set on a directory it only provides inheritance. This flag is only supported on Storage Spaces and ReFS volumes. It is not included in an ordinary directory listing. Windows Server 2008 R2, Windows 7, Windows Server 2008, Windows Vista, Windows Server 2003 and Windows XP: This flag is not supported until Windows 8 and Windows Server 2012.

None = 128

A file that does not have other attributes set. This attribute is valid only when used alone.

Offline = 4096

The data of a file is not available immediately. This attribute indicates that the file data is physically moved to offline storage. This attribute is used by Remote Storage, which is the hierarchical storage management software. Applications should not arbitrarily change this attribute.

Pinned = 524288

This attribute indicates user intent that the file or directory should be kept fully present locally even when not being actively accessed. This attribute is for use with hierarchical storage management software.

ReadOnly = 1

A file that is read-only. Applications can read the file, but cannot write to it or delete it. This attribute is not honored on directories. You cannot view or change the Read-only or the System attributes of folders in Windows Server 2003, in Windows XP, in Windows Vista or in Windows 7.

RecallOnDataAccess = 4194304

When this attribute is set, it means that the file or directory is not fully present locally. For a file that means that not all of its data is on local storage (e.g. it may be sparse with some data still in remote storage). For a directory it means that some of the directory contents are being virtualized from another location. Reading the file/enumerating the directory will be more expensive than normal, e.g. it will cause at least some of the file/directory content to be fetched from a remote store. Only kernel-mode callers can set this bit. File system mini filters below the 180000–189999 altitude range (FSFilter HSM Load Order Group) must not issue targeted cached reads or writes to files that have this attribute set. This could lead to cache pollution and potential file corruption. For more information, see [Handling placeholders](#).

RecallOnOpen = 262144

This attribute only appears in directory enumeration classes. When this attribute is set, it means that the file or directory has no physical representation on the local system; the item is virtual. Opening the item will be more expensive than normal, e.g. it will cause at least some of it to be fetched from a remote store.

ReparsePoint = 1024

A file or directory that has an associated reparse point, or a file that is a symbolic link.

Sparse = 512

A file that is a sparse file.

System = 4

A file or directory that the operating system uses as part of, or uses exclusively.

SystemFile = Hidden | System

A system file. A system file has the [Hidden](#) and [System](#) attributes. This is a composite value.

Temporary = 256

A file that is being used for temporary storage. File systems avoid writing data back to mass storage if sufficient cache memory is available, because typically, an application deletes a temporary file after the

handle is closed. In that scenario, the system can entirely avoid writing the data. Otherwise, the data is written after the handle is closed.

`Unpinned = 1048576`

This attribute indicates that the file or directory should not be kept fully present locally except when being actively accessed. This attribute is for use with hierarchical storage management software.

`[Obsolete("This value is reserved for system use. Do not use.", DiagnosticId = "WIU128")] Virtual = 65536`

This value is reserved for system use. Do not use.

Class WinFileIcon

Namespace: [FireBlade.WinInteropUtils.FileSystem](#)

Assembly: WinInteropUtils.dll

Represents the icon of a [WinFile](#).

```
public sealed class WinFileIcon : IDisposable
```

Inheritance

[object](#)  ← WinFileIcon

Implements

[IDisposable](#) 

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

WinFileIcon(Icon, int)

Initializes a new instance of the [WinFileIcon](#) class with the specified icon.

```
public WinFileIcon(Icon icon, int sysIndex)
```

Parameters

icon [Icon](#) 

The icon.

sysIndex [int](#) 

The index of the icon within the system image list.

Properties

Icon

Gets the icon.

```
public Icon Icon { get; }
```

Property Value

[Icon](#)

SysIconIndex

Gets the index of the icon in the system image list.

```
public int SysIconIndex { get; }
```

Property Value

[int](#)

Methods

Dispose()

Releases the resources associated with the icon.

```
public void Dispose()
```

Class WinFileInfo

Namespace: [FireBlade.WinInteropUtils.FileSystem](#)

Assembly: WinInteropUtils.dll

Represents information about a [WinFile](#).

```
public sealed class WinFileInfo : IDisposable
```

Inheritance

[object](#)  ← WinFileInfo

Implements

[IDisposable](#) 

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

WinFileInfo(string, WinFileInfoOptions, WinFileAttributes?)

Retrieves the info of the file specified by **path**.

```
public WinFileInfo(string path, WinFileInfoOptions options = WinFileInfoOptions.None,  
WinFileAttributes? attribs = null)
```

Parameters

path [string](#) 

The path of the file to retrieve info about.

options [WinFileInfoOptions](#)

Specifies additional options for the file info to retrieve.

attribs [WinFileAttributes?](#)

File attributes. See **Remarks** for more info.

Remarks

If the `attrs` parameter is set, it indicates that the function should not attempt to access the file specified by `path`. Rather, it should act as if the file specified by `path` exists with the file attributes passed in `attrs`.

You should call this function from a background thread. Failure to do so could cause the UI to stop responding.

Properties

Attributes

Gets the attributes of the file.

```
public WinFileAttributes Attributes { get; }
```

Property Value

[WinFileAttributes](#)

Examples

To check if the file is a directory:

```
if (file.Attributes.HasFlag(WinFileAttributes.Directory))
{
    // TODO: Do stuff
}

file.Dispose();
```

DisplayName

Gets the name of the file as it appears in File Explorer.

```
public string DisplayName { get; }
```

Property Value

[string](#)

Remarks

Note that the display name can be affected by settings such as whether extensions are shown.

ExecutableType

Gets the type of the executable file.

```
public ExecutableType ExecutableType { get; }
```

Property Value

[ExecutableType](#)

A value from the [ExecutableType](#) enumeration if the file is an executable file; otherwise, [None](#).

IconLocation

Gets the location of the icon.

```
public string IconLocation { get; }
```

Property Value

[string](#)

IsDisposed

Gets or sets a value that indicates whether the [WinFileInfo](#) was disposed.

```
public bool IsDisposed { get; }
```

Property Value

[bool](#)

LargeIcon

Gets the large icon of the file.

```
public WinFileIcon LargeIcon { get; }
```

Property Value

[WinFileIcon](#)

ShellIcon

Gets the Shell-sized icon of the file.

```
public WinFileIcon ShellIcon { get; }
```

Property Value

[WinFileIcon](#)

SmallIcon

Gets the small icon of the file.

```
public WinFileIcon SmallIcon { get; }
```

Property Value

[WinFileIcon](#)

TypeName

Gets the string that describes the file's type.

```
public string TypeName { get; }
```

Property Value

[string](#) 

Remarks

For example, if the file is a .TXT file, the type name may be "Text file".

Methods

Dispose()

Releases all resources associated with this [WinFileInfo](#).

```
public void Dispose()
```

Enum WinFileInfoOptions

Namespace: [FireBlade.WinInteropUtils.FileSystem](#)

Assembly: WinInteropUtils.dll

Represents the options for [WinFileInfo\(string, WinFileInfoOptions, WinFileAttributes?\)](#).

[Flags]

```
public enum WinFileInfoOptions
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

IconOverlays = 32

Apply the appropriate overlays to the file's icon.

LinkOverlay = 32768

Add the link overlay to the file's icon.

None = 0

Fetch the file info normally.

OpenIcon = 2

Retrieve the file's open icon.

A container object displays an open icon to indicate that the container is open.

Selected = 65536

Blend the file's icon with the system highlight color.

Enum WinFileOpenMode

Namespace: [FireBlade.WinInteropUtils.FileSystem](#)

Assembly: WinInteropUtils.dll

Specifies how to open a [WinFile](#).

```
public enum WinFileOpenMode
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

CreateAlways = 2

Creates a new file, always. If the specified file exists and is writable, the file is truncated.

CreateNew = 1

Creates a new file, only if it does not already exist. If the specified file exists, [WinFile](#) throws an [IOException](#)[↗].

OpenAlways = 4

Opens a file, always. If the specified file does not exist and is a valid path to a writable location, the file is created.

OpenExisting = 3

Opens a file or device, only if it exists. If the specified file does not exist, [WinFile](#) throws a [FileNotFoundException](#)[↗].

TruncateExisting = 5

Opens a file and truncates it so that its size is zero bytes, only if it exists. If the specified file does not exist, [WinFile](#) throws a [FileNotFoundException](#)[↗].

The calling process must open the file with [Write](#) bit set as part of the `shareMode` parameter.

Enum WinFileOptions

Namespace: [FireBlade.WinInteropUtils.FileSystem](#)

Assembly: WinInteropUtils.dll

Represents options for a [WinFile](#).

[Flags]

```
public enum WinFileOptions
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

BackupSemantics = 33554432

The file is being opened or created for a backup or restore operation. The system ensures that the calling process overrides file security checks when the process has Backup and Restore privileges.

DeleteOnClose = 67108864

The file is to be deleted immediately after all of its handles are closed, which includes the specified handle and any other open or duplicated handles.

If there are existing open handles to a file, the call fails unless they were all opened with the [Delete](#) share mode. Subsequent open requests for the file fail, unless the [Delete](#) share mode is specified.

NoBuffering = 536870912

The file or device is being opened with no system caching for data reads and writes. This option does not affect hard disk caching or memory mapped files.

There are strict requirements for successfully working with [WinFiles](#) using the [NoBuffering](#) flag, for details see [File Buffering](#).[↗](#)

NoRecall = 1048576

The file data is requested, but it should continue to be located in remote storage. It should not be transported back to local storage. This option is for use by remote storage systems.

None = 0

Open or create the file with default options.

PosixSemantics = 16777216

Access will occur according to POSIX rules. This includes allowing multiple files with names, differing only in case, for file systems that support that naming.

Use care when using this option, because files created with this flag may not be accessible by applications that are written for MS-DOS or 16-bit Windows.

RandomAccess = 268435456

Access is intended to be random. The system can use this as a hint to optimize file caching.

This option has no effect if the file system does not support cached I/O and the [NoBuffering](#) option. For more information, see the [Filesystem Caching](#) page.

ReparsePoint = 2097152

Normal reparse point processing will not occur; [WinFile](#) will attempt to open the reparse point. When a file is opened, a file handle is returned, whether or not the filter that controls the reparse point is operational.

This option cannot be used with [CreateAlways](#). If the file is not a reparse point, then this flag is ignored. For more information, see the [Reparse Points](#) page.

SequentialScan = 134217728

Access is intended to be sequential from beginning to end. The system can use this as a hint to optimize file caching. This option should not be used if read-behind (that is, reverse scans) will be used.

This flag has no effect if the file system does not support cached I/O and the [NoBuffering](#) option. For more information, see the [Filesystem Caching](#) page.

SessionAware = 8388608

The file or device is being opened with session awareness. If this flag is not specified, then per-session devices (such as a device using RemoteFX USB Redirection) cannot be opened by processes running in session 0.

This flag has no effect for callers not in session 0. This flag is supported only on server editions of Windows.

WriteThrough = -2147483648

Write operations will not go through any intermediate cache, they will go directly to disk.

For more information, see the [Filesystem Caching](#) page.

Enum WinFileShareMode

Namespace: [FireBlade.WinInteropUtils.FileSystem](#)

Assembly: WinInteropUtils.dll

Represents the sharing mode of a [WinFile](#).

[Flags]

```
public enum WinFileShareMode
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Delete = 4

Enables subsequent open operations on a file or device to request delete access. Otherwise, no process can open the file or device if it requests delete access.

If this flag is not specified, but the file or device has been opened for delete access, the function fails.

> [!NOTE] > Delete access allows both delete and rename operations.

None = 0

Prevents subsequent open operations on a file or device if they request delete, read, or write access.

Read = 1

Enables subsequent open operations on a file or device to request read access. Otherwise, no process can open the file or device if it requests read access.

If this flag is not specified, but the file or device has been opened for read access, the function fails.

ReadWrite = Read | Write

Enables subsequent open operations on a file or device to request read or write access. Otherwise, no process can open the file or device if it requests read or write access. This is a composite value.

Write = 2

Enables subsequent open operations on a file or device to request write access. Otherwise, no process can open the file or device if it requests write access.

If this flag is not specified, but the file or device has been opened for write access or has a file mapping with write access, the function fails.

Namespace FireBlade.WinInteropUtils.Memory

Classes

[ComSmartPointer<T>](#)

Represents a smart COM pointer that automatically releases its associated COM interface on dispose.

[MemoryString](#)

Represents a string allocated in memory.

Enums

[MemoryStringEncoding](#)

Represents the encoding of a [MemoryString](#).

Class ComSmartPointer<T>

Namespace: [FireBlade.WinInteropUtils.Memory](#)

Assembly: WinInteropUtils.dll

Represents a smart COM pointer that automatically releases its associated COM interface on dispose.

```
public sealed class ComSmartPointer<T> : IDisposable where T : notnull
```

Type Parameters

T

The COM interface to wrap.

Inheritance

[object](#)  ← ComSmartPointer<T>

Implements

[IDisposable](#) 

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Remarks

This is a managed version of the `CComPtr` class from the ATL (Active Template Library).

Constructors

ComSmartPointer(Guid, object?, CreateInstanceContext)

Creates a new instance of [ComSmartPointer<T>](#) by creating a new instance of an interface.

```
public ComSmartPointer(Guid rclsid, object? pUnkOuter, COM.CreateInstanceContext  
dwClsContext)
```

Parameters

rclsid [Guid](#)

The CLSID associated with the data and code that will be used to create the object.

pUnkOuter [object](#)

If [null](#), indicates that the object is not being created as part of an aggregate. If non-[null](#), pointer to the aggregate object's **IUnknown** interface (the controlling **IUnknown**).

dwClsContext [COM.CreateInstanceContext](#)

Context in which the code that manages the newly created object will run. A bitwise combination of [COM.CreateInstanceContext](#) values.

Exceptions

[ArgumentException](#)

This exception can get thrown in 2 cases:

- **T** must be an interface.
- **pUnkOuter** must be a COM object or [null](#).

[COMException](#)

This exception can get thrown in 3 cases:

- Failed to create the COM interface.
- A specified class is not registered in the registration database. Also can indicate that the type of server you requested in the [COM.CreateInstanceContext](#) enumeration is not registered or the values for the server types in the registry are corrupt.
- This class cannot be created as part of an aggregate.

[InvalidCastException](#)

The specified class does not implement the requested interface, or the controlling **IUnknown** does not expose the requested interface.

[UnauthorizedAccessException](#)

The request may be denied because the COM object cannot be created in UWP applications.

ComSmartPointer(T)

Creates a new instance of the [ComSmartPointer<T>](#) class from an existing instance of an interface.

```
public ComSmartPointer(T it)
```

Parameters

it T

The interface to wrap.

Exceptions

[ArgumentException](#) 

T must be an interface.

Properties

Interface

The COM interface.

```
public T Interface { get; }
```

Property Value

T

IsDisposed

Gets a value indicating whether the COM interface was disposed.

```
public bool IsDisposed { get; }
```

Property Value

[bool](#) 

Methods

Dispose()

Releases the COM interface.

```
public void Dispose()
```

Operators

implicit operator T(ComSmartPointer<T>)

Converts this [ComSmartPointer<T>](#) to its COM interface.

```
public static implicit operator T(ComSmartPointer<T> ptr)
```

Parameters

ptr [ComSmartPointer](#)<T>

The [ComSmartPointer<T>](#) to extract the interface from.

Returns

T

Class MemoryString

Namespace: [FireBlade.WinInteropUtils.Memory](#)

Assembly: WinInteropUtils.dll

Represents a string allocated in memory.

```
public sealed class MemoryString : IDisposable
```

Inheritance

[object](#)  ← MemoryString

Implements

[IDisposable](#) 

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.ReferenceEquals\(object, object\)](#) 

Constructors

MemoryString(string, MemoryStringEncoding)

Allocates a new memory pointer to a string.

```
public MemoryString(string str, MemoryStringEncoding encode = MemoryStringEncoding.Unicode)
```

Parameters

str [string](#) 

The string to allocate a pointer to.

encode [MemoryStringEncoding](#)

The encoding to encode the string with.

Exceptions

[ArgumentNullException](#)

The string was [null](#).

[ArgumentException](#)

The encoding specified is invalid.

Properties

ByteCount

Gets the length, in bytes, of the allocated string.

```
public int ByteCount { get; }
```

Property Value

[int](#)

Encoding

Gets the encoding of the string.

```
public MemoryStringEncoding Encoding { get; }
```

Property Value

[MemoryStringEncoding](#)

Length

Gets the length, in Unicode [chars](#), of the allocated string.

```
public int Length { get; }
```

Property Value

[int](#)

OriginalString

Gets the original string.

```
public string OriginalString { get; }
```

Property Value

[string](#)

Pointer

Gets the in-memory pointer to the string.

```
public nint Pointer { get; }
```

Property Value

[nint](#)

Exceptions

[ObjectDisposedException](#)

The string was disposed.

Methods

Dispose()

Releases the memory string.

```
public void Dispose()
```

~MemoryString()

Releases the memory string.

```
protected ~MemoryString()
```

ToString()

Returns the original string.

```
public override string ToString()
```

Returns

[string](#)[↗]

The original allocated string.

Operators

implicit operator nint(MemoryString)

Converts this [MemoryString](#) to its pointer.

```
public static implicit operator nint(MemoryString str)
```

Parameters

str [MemoryString](#)

The [MemoryString](#) whose pointer is to be retrieved.

Returns

[nint](#)[↗]

Enum MemoryStringEncoding

Namespace: [FireBlade.WinInteropUtils.Memory](#)

Assembly: WinInteropUtils.dll

Represents the encoding of a [MemoryString](#).

```
public enum MemoryStringEncoding
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Ansi = 1

The string should be allocated as an ANSI string.

Unicode = 0

The string should be allocated as a Unicode (UTF-16) string.

Namespace FireBlade.WinInteropUtils.WinForms

Classes

[Divider](#)

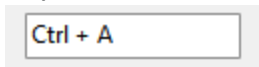
Represents a control that can act as a horizontal or vertical separator.

[FlagsEnumEditor](#)

Provides a type editor for enumerations with the [FlagsAttribute](#).

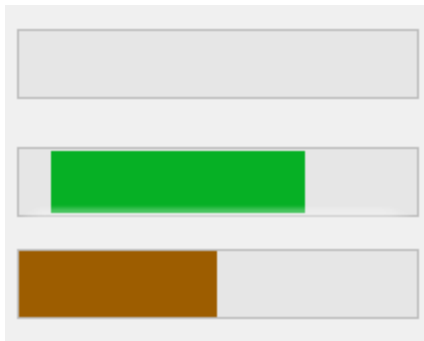
[HotKeyBox](#)

Represents a control that allows the user to enter a combination of keystrokes to be used as a hot key.



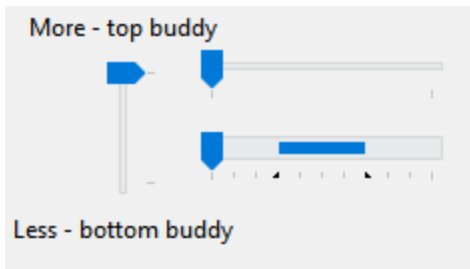
[ProgressBarEx](#)

Represents a Win32 progress bar control.



[Slider](#)

Represents a control that allows the user to slide a slider along a track, commonly referred to as a track bar.



[SliderActionList](#)

[SliderDesigner](#)

[SliderScrollEventArgs](#)

Provides event arguments for the [HorizontalScroll](#) and [VerticalScroll](#) events.

[SliderValueChangingEventArgs](#)

Defines event arguments for the [ValueChanging](#) event.

[WinInteropUtilsControlBase](#)

Represents the base class for WinInteropUtils.WinForms controls.

Enums

[HotKeyBoxModifiers](#)

Specifies the modifier keys in a [HotKeyBox](#).

[HotKeyBoxRules](#)

Specifies the invalid modifiers for a [HotKeyBox](#).

[HyperLinkLinkState](#)

Describes the state(s) of a FireBlade.WinInteropUtils.WinForms.Link.

[ProgressBarExStyle](#)

Specifies the style of a [ProgressBarEx](#) control.

[ProgressBarExType](#)

Specifies the [Type](#) of a [ProgressBarEx](#).

[SliderTickMode](#)

Specifies the type of ticks to use on a [Slider](#).

[SliderToolTipSide](#)

Specifies the side where the tooltip is located on a [Slider](#).

[SliderValueChangeReason](#)

Specifies the reason that a [Slider](#)'s value may be changing.

Class Divider

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Represents a control that can act as a horizontal or vertical separator.

```
public class Divider : Control, IDropTarget, ISynchronizeInvoke, IWin32Window,
    IBindableComponent, IComponent, IDisposable
```

Inheritance

[object](#) ← [MarshalByRefObject](#) ← [Component](#) ← [Control](#) ← Divider

Implements

[IDropTarget](#), [ISynchronizeInvoke](#), [IWin32Window](#), [IBindableComponent](#), [IComponent](#), [IDisposable](#)

Inherited Members

[Control.GetAccessibilityObjectById\(int\)](#), [Control.SetAutoSizeMode\(AutoSizeMode\)](#), [Control.GetAutoSizeMode\(\)](#), [Control.GetPreferredSize\(Size\)](#), [Control.AccessibilityNotifyClients\(AccessibleEvents, int\)](#), [Control.AccessibilityNotifyClients\(AccessibleEvents, int, int\)](#), [Control.BeginInvoke\(Delegate\)](#), [Control.BeginInvoke\(Action\)](#), [Control.BeginInvoke\(Delegate, params object\[\]\)](#), [Control.BringToFront\(\)](#), [Control.Contains\(Control\)](#), [Control.CreateAccessibilityInstance\(\)](#), [Control.CreateControlsInstance\(\)](#), [Control.CreateGraphics\(\)](#), [Control.CreateHandle\(\)](#), [Control.CreateControl\(\)](#), [Control.DefWndProc\(ref Message\)](#), [Control.DestroyHandle\(\)](#), [Control.Dispose\(bool\)](#), [Control.DoDragDrop\(object, DragDropEffects\)](#), [Control.DoDragDrop\(object, DragDropEffects, Bitmap, Point, bool\)](#), [Control.DrawToBitmap\(Bitmap, Rectangle\)](#), [Control.EndInvoke\(IAsyncResult\)](#), [Control.FindForm\(\)](#), [Control.GetTopLevel\(\)](#), [Control.RaiseKeyEvent\(object, KeyEventArgs\)](#), [Control.RaiseMouseEvent\(object, MouseEventArgs\)](#), [Control.Focus\(\)](#), [Control.FromChildHandle\(nint\)](#), [Control.FromHandle\(nint\)](#), [Control.GetChildAtPoint\(Point, GetChildAtPointSkip\)](#), [Control.GetChildAtPoint\(Point\)](#), [Control.GetContainerControl\(\)](#), [Control.GetScaledBounds\(Rectangle, SizeF, BoundsSpecified\)](#), [Control.GetNextControl\(Control, bool\)](#), [Control.GetStyle\(ControlStyles\)](#), [Control.Hide\(\)](#), [Control.InitLayout\(\)](#), [Control.Invalidate\(Region\)](#), [Control.Invalidate\(Region, bool\)](#), [Control.Invalidate\(\)](#), [Control.Invalidate\(bool\)](#), [Control.Invalidate\(Rectangle\)](#), [Control.Invalidate\(Rectangle, bool\)](#), [Control.Invoke\(Action\)](#), [Control.Invoke\(Delegate\)](#), [Control.Invoke\(Delegate, params object\[\]\)](#), [Control.Invoke<T>\(Func<T>\)](#), [Control.InvokePaint\(Control, PaintEventArgs\)](#),

[Control.InvokePaintBackground\(Control, PaintEventArgs\)](#), [Control.IsKeyLocked\(Keys\)](#), [Control.IsInputChar\(char\)](#), [Control.IsInputKey\(Keys\)](#), [Control.IsMnemonic\(char, string\)](#), [Control.LogicalToDeviceUnits\(int\)](#), [Control.LogicalToDeviceUnits\(Size\)](#), [Control.ScaleBitmapLogicalToDevice\(ref Bitmap\)](#), [Control.NotifyInvalidated\(Rectangle\)](#), [Control.InvokeOnClick\(Control, EventArgs\)](#), [Control.OnAutoSizeChanged\(EventArgs\)](#), [Control.OnBackColorChanged\(EventArgs\)](#), [Control.OnBackgroundImageChanged\(EventArgs\)](#), [Control.OnBackgroundImageLayoutChanged\(EventArgs\)](#), [Control.OnBindingContextChanged\(EventArgs\)](#), [Control.OnCausesValidationChanged\(EventArgs\)](#), [Control.OnContextMenuStripChanged\(EventArgs\)](#), [Control.OnCursorChanged\(EventArgs\)](#), [Control.OnDataContextChanged\(EventArgs\)](#), [Control.OnDockChanged\(EventArgs\)](#), [Control.OnEnabledChanged\(EventArgs\)](#), [Control.OnFontChanged\(EventArgs\)](#), [Control.OnForeColorChanged\(EventArgs\)](#), [Control.OnRightToLeftChanged\(EventArgs\)](#), [Control.OnNotifyMessage\(Message\)](#), [Control.OnParentBackColorChanged\(EventArgs\)](#), [Control.OnParentBackgroundImageChanged\(EventArgs\)](#), [Control.OnParentBindingContextChanged\(EventArgs\)](#), [Control.OnParentCursorChanged\(EventArgs\)](#), [Control.OnParentDataContextChanged\(EventArgs\)](#), [Control.OnParentEnabledChanged\(EventArgs\)](#), [Control.OnParentFontChanged\(EventArgs\)](#), [Control.OnParentForeColorChanged\(EventArgs\)](#), [Control.OnParentRightToLeftChanged\(EventArgs\)](#), [Control.OnParentVisibleChanged\(EventArgs\)](#), [Control.OnPrint\(PaintEventArgs\)](#), [Control.OnTabIndexChanged\(EventArgs\)](#), [Control.OnTabStopChanged\(EventArgs\)](#), [Control.OnTextChanged\(EventArgs\)](#), [Control.OnVisibleChanged\(EventArgs\)](#), [Control.OnParentChanged\(EventArgs\)](#), [Control.OnClick\(EventArgs\)](#), [Control.OnClientSizeChanged\(EventArgs\)](#), [Control.OnControlAdded\(ControlEventArgs\)](#), [Control.OnControlRemoved\(ControlEventArgs\)](#), [Control.OnCreateControl\(\)](#), [Control.OnHandleCreated\(EventArgs\)](#), [Control.OnLocationChanged\(EventArgs\)](#), [Control.OnHandleDestroyed\(EventArgs\)](#), [Control.OnDoubleClick\(EventArgs\)](#), [Control.OnDragEnter\(DragEventArgs\)](#), [Control.OnDragOver\(DragEventArgs\)](#), [Control.OnDragLeave\(EventArgs\)](#), [Control.OnDragDrop\(DragEventArgs\)](#), [Control.OnGiveFeedback\(GiveFeedbackEventArgs\)](#), [Control.OnEnter\(EventArgs\)](#), [Control.InvokeGotFocus\(Control, EventArgs\)](#), [Control.OnGotFocus\(EventArgs\)](#), [Control.OnHelpRequested\(HelpEventArgs\)](#), [Control.OnInvalidated\(InvalidateEventArgs\)](#), [Control.OnKeyDown\(KeyEventArgs\)](#), [Control.OnKeyPress\(KeyPressEventArgs\)](#), [Control.OnKeyUp\(KeyEventArgs\)](#), [Control.OnLayout\(LayoutEventArgs\)](#), [Control.OnLeave\(EventArgs\)](#), [Control.InvokeLostFocus\(Control, EventArgs\)](#), [Control.OnLostFocus\(EventArgs\)](#), [Control.OnMarginChanged\(EventArgs\)](#), [Control.OnMouseDoubleClick\(MouseEventArgs\)](#), [Control.OnMouseClick\(MouseEventArgs\)](#), [Control.OnMouseCaptureChanged\(EventArgs\)](#), [Control.OnMouseDown\(MouseEventArgs\)](#), [Control.OnMouseEnter\(EventArgs\)](#), [Control.OnMouseLeave\(EventArgs\)](#), [Control.OnDpiChangedBeforeParent\(EventArgs\)](#), [Control.OnDpiChangedAfterParent\(EventArgs\)](#), [Control.OnMouseHover\(EventArgs\)](#), [Control.OnMouseMove\(MouseEventArgs\)](#), [Control.OnMouseUp\(MouseEventArgs\)](#),

[Control.OnMouseWheel\(MouseEventArgs\)](#), [Control.OnMove\(EventArgs\)](#),
[Control.OnPaint\(PaintEventArgs\)](#), [Control.OnPaddingChanged\(EventArgs\)](#),
[Control.OnPaintBackground\(PaintEventArgs\)](#),
[Control.OnQueryContinueDrag\(QueryContinueDragEventArgs\)](#),
[Control.OnRegionChanged\(EventArgs\)](#), [Control.OnResize\(EventArgs\)](#),
[Control.OnPreviewKeyDown\(PreviewKeyDownEventArgs\)](#), [Control.OnSizeChanged\(EventArgs\)](#),
[Control.OnChangeUICues\(UICuesEventArgs\)](#), [Control.OnStyleChanged\(EventArgs\)](#),
[Control.OnSystemColorsChanged\(EventArgs\)](#), [Control.OnValidating\(CancelEventArgs\)](#),
[Control.OnValidated\(EventArgs\)](#), [Control.RescaleConstantsForDpi\(int, int\)](#),
[Control.PerformLayout\(\)](#), [Control.PerformLayout\(Control, string\)](#), [Control.PointToClient\(Point\)](#),
[Control.PointToScreen\(Point\)](#), [Control.PreProcessMessage\(ref Message\)](#),
[Control.PreProcessControlMessage\(ref Message\)](#), [Control.ProcessCmdKey\(ref Message, Keys\)](#),
[Control.ProcessDialogChar\(char\)](#), [Control.ProcessDialogKey\(Keys\)](#),
[Control.ProcessKeyEventArgs\(ref Message\)](#), [Control.ProcessKeyMessage\(ref Message\)](#),
[Control.ProcessKeyPreview\(ref Message\)](#), [Control.ProcessMnemonic\(char\)](#),
[Control.RaiseDragEvent\(object, DragEventArgs\)](#), [Control.RaisePaintEvent\(object, PaintEventArgs\)](#),
[Control.RecreateHandle\(\)](#), [Control.RectangleToClient\(Rectangle\)](#),
[Control.RectangleToScreen\(Rectangle\)](#), [Control.ReflectMessage\(nint, ref Message\)](#),
[Control.Refresh\(\)](#), [Control.ResetMouseEventArgs\(\)](#), [Control.ResetText\(\)](#), [Control.ResumeLayout\(\)](#),
[Control.ResumeLayout\(bool\)](#), [Control.Scale\(SizeF\)](#), [Control.ScaleControl\(SizeF, BoundsSpecified\)](#),
[Control.Select\(\)](#), [Control.Select\(bool, bool\)](#),
[Control.SelectNextControl\(Control, bool, bool, bool, bool\)](#), [Control.SendToBack\(\)](#),
[Control.SetBounds\(int, int, int, int\)](#), [Control.SetBounds\(int, int, int, int, BoundsSpecified\)](#),
[Control.SetBoundsCore\(int, int, int, int, BoundsSpecified\)](#), [Control.SetClientSizeCore\(int, int\)](#),
[Control.SizeFromClientSize\(Size\)](#), [Control.SetStyle\(ControlStyles, bool\)](#), [Control.SetTopLevel\(bool\)](#),
[Control.SetVisibleCore\(bool\)](#), [Control.RtlTranslateAlignment\(HorizontalAlignment\)](#),
[Control.RtlTranslateAlignment\(LeftRightAlignment\)](#),
[Control.RtlTranslateAlignment\(ContentAlignment\)](#),
[Control.RtlTranslateHorizontal\(HorizontalAlignment\)](#),
[Control.RtlTranslateLeftRight\(LeftRightAlignment\)](#), [Control.RtlTranslateContent\(ContentAlignment\)](#),
[Control.Show\(\)](#), [Control.SuspendLayout\(\)](#), [Control.Update\(\)](#), [Control.UpdateBounds\(\)](#),
[Control.UpdateBounds\(int, int, int, int\)](#), [Control.UpdateBounds\(int, int, int, int, int, int\)](#),
[Control.UpdateZOrder\(\)](#), [Control.UpdateStyles\(\)](#), [Control.WndProc\(ref Message\)](#),
[Control.OnImeModeChanged\(EventArgs\)](#), [Control.AccessibilityObject](#),
[Control.AccessibleDefaultActionDescription](#), [Control.AccessibleDescription](#),
[Control.AccessibleName](#), [Control.AccessibleRole](#), [Control.AllowDrop](#), [Control.Anchor](#),
[Control.AutoScrollOffset](#), [Control.LayoutEngine](#), [Control.DataContext](#), [Control.BackColor](#),
[Control.BackgroundImage](#), [Control.BackgroundImageLayout](#), [Control.BindingContext](#),
[Control.Bottom](#), [Control.Bounds](#), [Control.CanFocus](#), [Control.CanRaiseEvents](#),
[Control.CanSelect](#), [Control.Capture](#), [Control.CausesValidation](#),

[Control.CheckForIllegalCrossThreadCalls](#), [Control.ClientRectangle](#), [Control.ClientSize](#), [Control.CompanyName](#), [Control.ContainsFocus](#), [Control.ContextMenuStrip](#), [Control.Controls](#), [Control.Created](#), [Control.Cursor](#), [Control.DataBindings](#), [Control.DefaultBackColor](#), [Control.DefaultCursor](#), [Control.DefaultFont](#), [Control.DefaultForeColor](#), [Control.DefaultMargin](#), [Control.DefaultMaximumSize](#), [Control.DefaultMinimumSize](#), [Control.DefaultPadding](#), [Control.DeviceDpi](#), [Control.DisplayRectangle](#), [Control.IsDisposed](#), [Control.Disposing](#), [Control.Dock](#), [Control.DoubleBuffered](#), [Control.Enabled](#), [Control.Focused](#), [Control.Font](#), [Control.FontHeight](#), [Control.ForeColor](#), [Control.Handle](#), [Control.HasChildren](#), [Control.Height](#), [Control.IsHandleCreated](#), [Control.InvokeRequired](#), [Control.IsAccessible](#), [Control.IsAncestorSiteInDesignMode](#), [Control.IsMirrored](#), [Control.Left](#), [Control.Location](#), [Control.Margin](#), [Control.MaximumSize](#), [Control.MinimumSize](#), [Control.ModifierKeys](#), [Control.MouseButtons](#), [Control.MousePosition](#), [Control.Name](#), [Control.Parent](#), [Control.ProductName](#), [Control.ProductVersion](#), [Control.RecreatingHandle](#), [Control.Region](#), [Control.RenderRightToLeft](#), [Control.ResizeRedraw](#), [Control.Right](#), [Control.RightToLeft](#), [Control.ScaleChildren](#), [Control.Site](#), [Control.Size](#), [Control.TabIndex](#), [Control.TabStop](#), [Control.Tag](#), [Control.Text](#), [Control.Top](#), [Control.TopLevelControl](#), [Control.ShowKeyboardCues](#), [Control.ShowFocusCues](#), [Control.UseWaitCursor](#), [Control.Visible](#), [Control.Width](#), [Control.PreferredSize](#), [Control.Padding](#), [Control.CanEnableIme](#), [Control.DefaultImeMode](#), [Control.ImeMode](#), [Control.ImeModeBase](#), [Control.PropagatingImeMode](#), [Control.BackColorChanged](#), [Control.BackgroundImageChanged](#), [Control.BackgroundImageLayoutChanged](#), [Control.BindingContextChanged](#), [Control.CausesValidationChanged](#), [Control.ClientSizeChanged](#), [Control.ContextMenuStripChanged](#), [Control.CursorChanged](#), [Control.DockChanged](#), [Control.EnabledChanged](#), [Control.FontChanged](#), [Control.ForeColorChanged](#), [Control.LocationChanged](#), [Control.MarginChanged](#), [Control.RegionChanged](#), [Control.RightToLeftChanged](#), [Control.SizeChanged](#), [Control.TabIndexChanged](#), [Control.TabStopChanged](#), [Control.TextChanged](#), [Control.VisibleChanged](#), [Control.Click](#), [Control.ControlAdded](#), [Control.ControlRemoved](#), [Control.DataContextChanged](#), [Control.DragDrop](#), [Control.DragEnter](#), [Control.DragOver](#), [Control.DragLeave](#), [Control.GiveFeedback](#), [Control.HandleCreated](#), [Control.HandleDestroyed](#), [Control.HelpRequested](#), [Control.Invalidated](#), [Control.PaddingChanged](#), [Control.Paint](#), [Control.QueryContinueDrag](#), [Control.QueryAccessibilityHelp](#), [Control.DoubleClick](#), [Control.Enter](#), [Control.GotFocus](#), [Control.KeyDown](#), [Control.KeyPress](#), [Control.KeyUp](#), [Control.Layout](#), [Control.Leave](#), [Control.LostFocus](#), [Control.MouseClick](#), [Control.MouseDoubleClick](#), [Control.MouseCaptureChanged](#), [Control.MouseDown](#), [Control.MouseEnter](#), [Control.MouseLeave](#), [Control.DpiChangedBeforeParent](#), [Control.DpiChangedAfterParent](#), [Control.MouseHover](#), [Control.MouseMove](#), [Control.MouseUp](#), [Control.MouseWheel](#), [Control.Move](#), [Control.PreviewKeyDown](#), [Control.Resize](#), [Control.ChangeUICues](#), [Control.StyleChanged](#), [Control.SystemColorsChanged](#), [Control.Validating](#), [Control.Validated](#), [Control.ParentChanged](#), [Control.ImeModeChanged](#),

[Component.Dispose\(\)](#), [Component.GetService\(Type\)](#), [Component.ToString\(\)](#),
[Component.Container](#), [Component.DesignMode](#), [Component.Events](#), [Component.Disposed](#),
[MarshalByRefObject.GetLifetimeService\(\)](#), [MarshalByRefObject.InitializeLifetimeService\(\)](#),
[MarshalByRefObject.MemberwiseClone\(bool\)](#), [object.Equals\(object\)](#), [object.Equals\(object, object\)](#),
[object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#),
[object.ReferenceEquals\(object, object\)](#)

Constructors

Divider()

```
public Divider()
```

Properties

CreateParams

Gets the required creation parameters when the control handle is created.

```
protected override CreateParams CreateParams { get; }
```

Property Value

[CreateParams](#)

A [CreateParams](#) that contains the required creation parameters when the handle to the control is created.

DefaultSize

Gets the default size of the control.

```
protected override Size DefaultSize { get; }
```

Property Value

[Size](#)[↗]

The default [Size](#)[↗] of the control.

Orientation

Gets or sets the orientation of the [Divider](#).

```
public Orientation Orientation { get; set; }
```

Property Value

[Orientation](#)[↗]

Class FlagsEnumEditor

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Provides a type editor for enumerations with the [FlagsAttribute](#).

```
public class FlagsEnumEditor : UITypeEditor
```

Inheritance

[object](#) ← [UITypeEditor](#) ← FlagsEnumEditor

Inherited Members

[UITypeEditor.EditValue\(IServiceProvider, object\)](#), [UITypeEditor.GetEditStyle\(\)](#),
[UITypeEditor.GetPaintValueSupported\(\)](#),
[UITypeEditor.GetPaintValueSupported\(ITypeDescriptorContext\)](#),
[UITypeEditor.PaintValue\(object, Graphics, Rectangle\)](#), [UITypeEditor.PaintValue\(PaintValueEventArgs\)](#),
[UITypeEditor.IsDropDownResizable](#), [object.Equals\(object\)](#), [object.Equals\(object, object\)](#),
[object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#),
[object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Methods

EditValue(ITypeDescriptorContext?, IServiceProvider?, object?)

Edits the specified object's value using the editor style indicated by the [GetEditStyle\(\)](#) method.

```
public override object? EditValue(ITypeDescriptorContext? context, IServiceProvider?  
provider, object? value)
```

Parameters

context [ITypeDescriptorContext](#)

An [ITypeDescriptorContext](#) that can be used to gain additional context information.

provider [IServiceProvider](#)

An [IServiceProvider](#) that this editor can use to obtain services.

value [object](#)

The object to edit.

Returns

[object](#)

The new value of the object. If the value of the object has not changed, this should return the same object it was passed.

GetEditStyle(ITypeDescriptorContext?)

Gets the editor style used by the [EditValue\(IServiceProvider, object\)](#) method.

```
public override UITypeEditorEditStyle GetEditStyle(ITypeDescriptorContext? context)
```

Parameters

context [ITypeDescriptorContext](#)

An [ITypeDescriptorContext](#) that can be used to gain additional context information.

Returns

[UITypeEditorEditStyle](#)

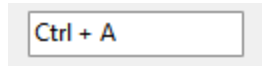
A [UITypeEditorEditStyle](#) value that indicates the style of editor used by the [EditValue\(IServiceProvider, object\)](#) method. If the [UITypeEditor](#) does not support this method, then [GetEditStyle\(\)](#) will return [None](#).

Class HotKeyBox

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Represents a control that allows the user to enter a combination of keystrokes to be used as a hot key.



```
public class HotKeyBox : WinInteropUtilsControlBase, IDropTarget, ISynchronizeInvoke,
    IWin32Window, IBindableComponent, IComponent, IDisposable
```

Inheritance

[object](#) ← [MarshalByRefObject](#) ← [Component](#) ← [Control](#) ← [WinInteropUtilsControlBase](#) ←
HotKeyBox

Implements

[IDropTarget](#), [ISynchronizeInvoke](#), [IWin32Window](#), [IBindableComponent](#), [IComponent](#), [IDisposable](#)

Inherited Members

[WinInteropUtilsControlBase.Window](#), [Control.GetAccessibilityObjectById\(int\)](#), [Control.SetAutoSizeMode\(AutoSizeMode\)](#), [Control.GetAutoSizeMode\(\)](#), [Control.GetPreferredSize\(Size\)](#), [Control.AccessibilityNotifyClients\(AccessibleEvents, int\)](#), [Control.AccessibilityNotifyClients\(AccessibleEvents, int, int\)](#), [Control.BeginInvoke\(Delegate\)](#), [Control.BeginInvoke\(Action\)](#), [Control.BeginInvoke\(Delegate, params object\[\]\)](#), [Control.BringToFront\(\)](#), [Control.Contains\(Control\)](#), [Control.CreateAccessibilityInstance\(\)](#), [Control.CreateControlsInstance\(\)](#), [Control.CreateGraphics\(\)](#), [Control.CreateHandle\(\)](#), [Control.CreateControl\(\)](#), [Control.DefWndProc\(ref Message\)](#), [Control.DestroyHandle\(\)](#), [Control.Dispose\(bool\)](#), [Control.DoDragDrop\(object, DragDropEffects\)](#), [Control.DoDragDrop\(object, DragDropEffects, Bitmap, Point, bool\)](#), [Control.DrawToBitmap\(Bitmap, Rectangle\)](#), [Control.EndInvoke\(IAsyncResult\)](#), [Control.FindForm\(\)](#), [Control.GetTopLevel\(\)](#), [Control.RaiseKeyEvent\(object, KeyEventArgs\)](#), [Control.RaiseMouseEvent\(object, MouseEventArgs\)](#), [Control.Focus\(\)](#), [Control.FromChildHandle\(nint\)](#), [Control.FromHandle\(nint\)](#), [Control.GetChildAtPoint\(Point, GetChildAtPointSkip\)](#), [Control.GetChildAtPoint\(Point\)](#), [Control.GetContainerControl\(\)](#), [Control.GetScaledBounds\(Rectangle,.SizeF, BoundsSpecified\)](#), [Control.GetNextControl\(Control, bool\)](#), [Control.GetStyle\(ControlStyles\)](#), [Control.Hide\(\)](#), [Control.InitLayout\(\)](#), [Control.Invalidate\(Region\)](#), [Control.Invalidate\(Region, bool\)](#), [Control.Invalidate\(\)](#), [Control.Invalidate\(bool\)](#), [Control.Invalidate\(Rectangle\)](#),

[Control.Invalidate\(Rectangle, bool\)](#), [Control.Invoke\(Action\)](#), [Control.Invoke\(Delegate\)](#),
[Control.Invoke\(Delegate, params object\[\]\)](#), [Control.Invoke<T>\(Func<T>\)](#),
[Control.InvokePaint\(Control, PaintEventArgs\)](#),
[Control.InvokePaintBackground\(Control, PaintEventArgs\)](#), [Control.IsKeyLocked\(Keys\)](#),
[Control.IsInputChar\(char\)](#), [Control.IsInputKey\(Keys\)](#), [Control.IsMnemonic\(char, string\)](#),
[Control.LogicalToDeviceUnits\(int\)](#), [Control.LogicalToDeviceUnits\(Size\)](#),
[Control.ScaleBitmapLogicalToDevice\(ref Bitmap\)](#), [Control.NotifyInvalidate\(Rectangle\)](#),
[Control.InvokeOnClick\(Control, EventArgs\)](#), [Control.OnAutoSizeChanged\(EventArgs\)](#),
[Control.OnBackColorChanged\(EventArgs\)](#), [Control.OnBackgroundImageChanged\(EventArgs\)](#),
[Control.OnBackgroundImageLayoutChanged\(EventArgs\)](#),
[Control.OnBindingContextChanged\(EventArgs\)](#), [Control.OnCausesValidationChanged\(EventArgs\)](#),
[Control.OnContextMenuStripChanged\(EventArgs\)](#), [Control.OnCursorChanged\(EventArgs\)](#),
[Control.OnDataContextChanged\(EventArgs\)](#), [Control.OnDockChanged\(EventArgs\)](#),
[Control.OnEnabledChanged\(EventArgs\)](#), [Control.OnFontChanged\(EventArgs\)](#),
[Control.OnForeColorChanged\(EventArgs\)](#), [Control.OnRightToLeftChanged\(EventArgs\)](#),
[Control.OnNotifyMessage\(Message\)](#), [Control.OnParentBackColorChanged\(EventArgs\)](#),
[Control.OnParentBackgroundImageChanged\(EventArgs\)](#),
[Control.OnParentBindingContextChanged\(EventArgs\)](#), [Control.OnParentCursorChanged\(EventArgs\)](#),
[Control.OnParentDataContextChanged\(EventArgs\)](#), [Control.OnParentEnabledChanged\(EventArgs\)](#),
[Control.OnParentFontChanged\(EventArgs\)](#), [Control.OnParentForeColorChanged\(EventArgs\)](#),
[Control.OnParentRightToLeftChanged\(EventArgs\)](#), [Control.OnParentVisibleChanged\(EventArgs\)](#),
[Control.OnPrint\(PaintEventArgs\)](#), [Control.OnTabIndexChanged\(EventArgs\)](#),
[Control.OnTabStopChanged\(EventArgs\)](#), [Control.OnTextChanged\(EventArgs\)](#),
[Control.OnVisibleChanged\(EventArgs\)](#), [Control.OnParentChanged\(EventArgs\)](#),
[Control.OnClick\(EventArgs\)](#), [Control.OnClientSizeChanged\(EventArgs\)](#),
[Control.OnControlAdded\(ControlEventArgs\)](#), [Control.OnControlRemoved\(ControlEventArgs\)](#),
[Control.OnCreateControl\(\)](#), [Control.OnLocationChanged\(EventArgs\)](#),
[Control.OnHandleDestroyed\(EventArgs\)](#), [Control.OnDoubleClick\(EventArgs\)](#),
[Control.OnDragEnter\(DragEventArgs\)](#), [Control.OnDragOver\(DragEventArgs\)](#),
[Control.OnDragLeave\(EventArgs\)](#), [Control.OnDragDrop\(DragEventArgs\)](#),
[Control.OnGiveFeedback\(GiveFeedbackEventArgs\)](#), [Control.OnEnter\(EventArgs\)](#),
[Control.InvokeGotFocus\(Control, EventArgs\)](#), [Control.OnGotFocus\(EventArgs\)](#),
[Control.OnHelpRequested\(HelpEventArgs\)](#), [Control.OnInvalidated\(InvalidateEventArgs\)](#),
[Control.OnKeyDown\(KeyEventArgs\)](#), [Control.OnKeyPress\(KeyPressEventArgs\)](#),
[Control.OnKeyUp\(KeyEventArgs\)](#), [Control.OnLayout\(LayoutEventArgs\)](#),
[Control.OnLeave\(EventArgs\)](#), [Control.InvokeLostFocus\(Control, EventArgs\)](#),
[Control.OnLostFocus\(EventArgs\)](#), [Control.OnMarginChanged\(EventArgs\)](#),
[Control.OnMouseDoubleClick\(MouseEventArgs\)](#), [Control.OnMouseClicked\(MouseEventArgs\)](#),
[Control.OnMouseCaptureChanged\(EventArgs\)](#), [Control.OnMouseDown\(MouseEventArgs\)](#),
[Control.OnMouseEnter\(EventArgs\)](#), [Control.OnMouseLeave\(EventArgs\)](#),

[Control.OnDpiChangedBeforeParent\(EventArgs\)](#), [Control.OnDpiChangedAfterParent\(EventArgs\)](#),
[Control.OnMouseHover\(EventArgs\)](#), [Control.OnMouseMove\(MouseEventArgs\)](#),
[Control.OnMouseUp\(MouseEventArgs\)](#), [Control.OnMouseWheel\(MouseEventArgs\)](#),
[Control.OnMove\(EventArgs\)](#), [Control.OnPaint\(PaintEventArgs\)](#),
[Control.OnPaddingChanged\(EventArgs\)](#), [Control.OnPaintBackground\(PaintEventArgs\)](#),
[Control.OnQueryContinueDrag\(QueryContinueDragEventArgs\)](#),
[Control.OnRegionChanged\(EventArgs\)](#), [Control.OnResize\(EventArgs\)](#),
[Control.OnPreviewKeyDown\(PreviewKeyDownEventArgs\)](#), [Control.OnSizeChanged\(EventArgs\)](#),
[Control.OnChangeUICues\(UICuesEventArgs\)](#), [Control.OnStyleChanged\(EventArgs\)](#),
[Control.OnSystemColorsChanged\(EventArgs\)](#), [Control.OnValidating\(CancelEventArgs\)](#),
[Control.OnValidated\(EventArgs\)](#), [Control.RescaleConstantsForDpi\(int, int\)](#),
[Control.PerformLayout\(\)](#), [Control.PerformLayout\(Control, string\)](#), [Control.PointToClient\(Point\)](#),
[Control.PointToScreen\(Point\)](#), [Control.PreProcessMessage\(ref Message\)](#),
[Control.PreProcessControlMessage\(ref Message\)](#), [Control.ProcessCmdKey\(ref Message, Keys\)](#),
[Control.ProcessDialogChar\(char\)](#), [Control.ProcessDialogKey\(Keys\)](#),
[Control.ProcessKeyEventArgs\(ref Message\)](#), [Control.ProcessKeyMessage\(ref Message\)](#),
[Control.ProcessKeyPreview\(ref Message\)](#), [Control.ProcessMnemonic\(char\)](#),
[Control.RaiseDragEvent\(object, DragEventArgs\)](#), [Control.RaisePaintEvent\(object, PaintEventArgs\)](#),
[Control.RecreateHandle\(\)](#), [Control.RectangleToClient\(Rectangle\)](#),
[Control.RectangleToScreen\(Rectangle\)](#), [Control.ReflectMessage\(nint, ref Message\)](#),
[Control.Refresh\(\)](#), [Control.ResetMouseEventArgs\(\)](#), [Control.ResetText\(\)](#), [Control.ResumeLayout\(\)](#),
[Control.ResumeLayout\(bool\)](#), [Control.Scale\(SizeF\)](#), [Control.ScaleControl\(SizeF, BoundsSpecified\)](#),
[Control.Select\(\)](#), [Control.Select\(bool, bool\)](#),
[Control.SelectNextControl\(Control, bool, bool, bool, bool\)](#), [Control.SendToBack\(\)](#),
[Control.SetBounds\(int, int, int, int\)](#), [Control.SetBounds\(int, int, int, int, BoundsSpecified\)](#),
[Control.SetBoundsCore\(int, int, int, int, BoundsSpecified\)](#), [Control.SetClientSizeCore\(int, int\)](#),
[Control.SizeFromClientSize\(Size\)](#), [Control.SetStyle\(ControlStyles, bool\)](#), [Control.SetTopLevel\(bool\)](#),
[Control.SetVisibleCore\(bool\)](#), [Control.RtlTranslateAlignment\(HorizontalAlignment\)](#),
[Control.RtlTranslateAlignment\(LeftRightAlignment\)](#),
[Control.RtlTranslateAlignment\(ContentAlignment\)](#),
[Control.RtlTranslateHorizontal\(HorizontalAlignment\)](#),
[Control.RtlTranslateLeftRight\(LeftRightAlignment\)](#), [Control.RtlTranslateContent\(ContentAlignment\)](#),
[Control.Show\(\)](#), [Control.SuspendLayout\(\)](#), [Control.Update\(\)](#), [Control.UpdateBounds\(\)](#),
[Control.UpdateBounds\(int, int, int, int\)](#), [Control.UpdateBounds\(int, int, int, int, int, int\)](#),
[Control.UpdateZOrder\(\)](#), [Control.UpdateStyles\(\)](#), [Control.OnImeModeChanged\(EventArgs\)](#),
[Control.AccessibilityObject](#), [Control.AccessibleDefaultActionDescription](#),
[Control.AccessibleDescription](#), [Control.AccessibleName](#), [Control.AccessibleRole](#),
[Control.AllowDrop](#), [Control.Anchor](#), [Control.AutoScrollOffset](#), [Control.LayoutEngine](#),
[Control.DataContext](#), [Control.BackColor](#), [Control.BackgroundImage](#),
[Control.BackgroundImageLayout](#), [Control.BindingContext](#), [Control.Bottom](#), [Control.Bounds](#),

[Control.CanFocus](#), [Control.CanRaiseEvents](#), [Control.CanSelect](#), [Control.Capture](#),
[Control.CausesValidation](#), [Control.CheckForIllegalCrossThreadCalls](#), [Control.ClientRectangle](#),
[Control.ClientSize](#), [Control.CompanyName](#), [Control.ContainsFocus](#), [Control.ContextMenuStrip](#),
[Control.Controls](#), [Control.Created](#), [Control.Cursor](#), [Control.DataBindings](#),
[Control.DefaultBackColor](#), [Control.DefaultCursor](#), [Control.DefaultFont](#),
[Control.DefaultForeColor](#), [Control.DefaultMargin](#), [Control.DefaultMaximumSize](#),
[Control.DefaultMinimumSize](#), [Control.DefaultPadding](#), [Control.DefaultSize](#), [Control.DeviceDpi](#),
[Control.DisplayRectangle](#), [Control.IsDisposed](#), [Control.Disposing](#), [Control.Dock](#),
[Control.DoubleBuffered](#), [Control.Enabled](#), [Control.Focused](#), [Control.Font](#),
[Control.FontHeight](#), [Control.ForeColor](#), [Control.Handle](#), [Control.HasChildren](#), [Control.Height](#),
[Control.IsHandleCreated](#), [Control.InvokeRequired](#), [Control.IsAccessible](#),
[Control.IsAncestorSiteInDesignMode](#), [Control.IsMirrored](#), [Control.Left](#), [Control.Location](#),
[Control.Margin](#), [Control.MaximumSize](#), [Control.MinimumSize](#), [Control.ModifierKeys](#),
[Control.MouseButtons](#), [Control.MousePosition](#), [Control.Name](#), [Control.Parent](#),
[Control.ProductName](#), [Control.ProductVersion](#), [Control.RecreatingHandle](#), [Control.Region](#),
[Control.RenderRightToLeft](#), [Control.ResizeRedraw](#), [Control.Right](#), [Control.RightToLeft](#),
[Control.ScaleChildren](#), [Control.Site](#), [Control.Size](#), [Control.TabIndex](#), [Control.TabStop](#),
[Control.Tag](#), [Control.Text](#), [Control.Top](#), [Control.TopLevelControl](#), [Control.ShowKeyboardCues](#),
[Control.ShowFocusCues](#), [Control.UseWaitCursor](#), [Control.Visible](#), [Control.Width](#),
[Control.PreferredSize](#), [Control.Padding](#), [Control.CanEnableIme](#), [Control.DefaultImeMode](#),
[Control.ImeMode](#), [Control.ImeModeBase](#), [Control.PropagatingImeMode](#),
[Control.BackColorChanged](#), [Control.BackgroundImageChanged](#),
[Control.BackgroundImageLayoutChanged](#), [Control.BindingContextChanged](#),
[Control.CausesValidationChanged](#), [Control.ClientSizeChanged](#),
[Control.ContextMenuStripChanged](#), [Control.CursorChanged](#), [Control.DockChanged](#),
[Control.EnabledChanged](#), [Control.FontChanged](#), [Control.ForeColorChanged](#),
[Control.LocationChanged](#), [Control.MarginChanged](#), [Control.RegionChanged](#),
[Control.RightToLeftChanged](#), [Control.SizeChanged](#), [Control.TabIndexChanged](#),
[Control.TabStopChanged](#), [Control.TextChanged](#), [Control.VisibleChanged](#), [Control.Click](#),
[Control.ControlAdded](#), [Control.ControlRemoved](#), [Control.DataContextChanged](#),
[Control.DragDrop](#), [Control.DragEnter](#), [Control.DragOver](#), [Control.DragLeave](#),
[Control.GiveFeedback](#), [Control.HandleCreated](#), [Control.HandleDestroyed](#),
[Control.HelpRequested](#), [Control.Invalidated](#), [Control.PaddingChanged](#), [Control.Paint](#),
[Control.QueryContinueDrag](#), [Control.QueryAccessibilityHelp](#), [Control.DoubleClick](#),
[Control.Enter](#), [Control.GotFocus](#), [Control.KeyDown](#), [Control.KeyPress](#), [Control.KeyUp](#),
[Control.Layout](#), [Control.Leave](#), [Control.LostFocus](#), [Control.MouseClick](#),
[Control.MouseDoubleClick](#), [Control.MouseCaptureChanged](#), [Control.MouseDown](#),
[Control.MouseEnter](#), [Control.MouseLeave](#), [Control.DpiChangedBeforeParent](#),
[Control.DpiChangedAfterParent](#), [Control.MouseHover](#), [Control.MouseMove](#), [Control.MouseUp](#),
[Control.MouseWheel](#), [Control.Move](#), [Control.PreviewKeyDown](#), [Control.Resize](#),

[Control.ChangeUICues](#), [Control.StyleChanged](#), [Control.SystemColorsChanged](#), [Control.Validating](#), [Control.Validated](#), [Control.ParentChanged](#), [Control.ImeModeChanged](#), [Component.Dispose\(\)](#), [Component.GetService\(Type\)](#), [Component.ToString\(\)](#), [Component.Container](#), [Component.DesignMode](#), [Component.Events](#), [Component.Disposed](#), [MarshalByRefObject.GetLifetimeService\(\)](#), [MarshalByRefObject.InitializeLifetimeService\(\)](#), [MarshalByRefObject.MemberwiseClone\(bool\)](#), [object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#)

Constructors

HotKeyBox()

```
public HotKeyBox()
```

Properties

CreateParams

Gets the required creation parameters when the control handle is created.

```
protected override CreateParams CreateParams { get; }
```

Property Value

[CreateParams](#)

A [CreateParams](#) that contains the required creation parameters when the handle to the control is created.

FallbackValue

Gets or sets the modifiers that the control should fall back to when the user inputs an invalid key combination.

```
public HotKeyBoxModifiers FallbackValue { get; set; }
```

Property Value

[HotKeyBoxModifiers](#)

Remarks

When a user enters an invalid key combination, as defined by rules specified in [Rules](#), the system uses the bitwise-OR operator to combine the keys entered by the user with the flags specified in [Fallback Value](#). The resulting key combination is converted into a string and then displayed in the hot key control.

IsExKey

Gets or sets a value indicating whether the control is using the extended key.

```
public bool IsExKey { get; set; }
```

Property Value

[bool](#)

Keys

Gets or sets the keys currently in the hot key box.

```
public Keys Keys { get; set; }
```

Property Value

[Keys](#)

Rules

Gets or sets the invalid modifiers for a hot key control.

```
public HotKeyBoxRules Rules { get; set; }
```

Property Value

Methods

OnHandleCreated(EventArgs)

Raises the [HandleCreated](#) event.

```
protected override void OnHandleCreated(EventArgs e)
```

Parameters

e [EventArgs](#)

An [EventArgs](#) that contains the event data.

WndProc(ref Message)

Processes Windows messages.

```
protected override void WndProc(ref Message m)
```

Parameters

m [Message](#)

The Windows [Message](#) to process.

Events

HotKeyChanged

Fires when the hot key in the control changes.

```
public event EventHandler? HotKeyChanged
```

Event Type

[EventHandler](#) 

Enum HotKeyBoxModifiers

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Specifies the modifier keys in a [HotKeyBox](#).

[Flags]

```
public enum HotKeyBoxModifiers
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Alt = 4

The Alt key.

Control = 2

The Ctrl key.

Ctrl = 2

The Ctrl key.

Extended = 8

The extended key.

None = 0

No modifiers.

Shift = 1

The Shift key.

Enum HotKeyBoxRules

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Specifies the invalid modifiers for a [HotKeyBox](#).

[Flags]

```
public enum HotKeyBoxRules
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Alt = 8

The Alt modifier is invalid.

Control = 4

The Ctrl modifier is invalid.

Ctrl = 4

The Ctrl modifier is invalid.

CtrlAlt = 64

The combination of the Ctrl and Alt modifiers are invalid.

None = 1

All modifiers are allowed.

Shift = 2

The Shift modifier is invalid.

ShiftAlt = 32

The combination of the Shift and Alt modifiers are invalid.

ShiftCtrl = 16

The combination of the Shift and Ctrl modifiers are invalid.

`ShiftCtrlAlt = 128`

The combination of the Shift, Ctrl, and Alt modifiers are invalid.

Enum HyperLinkLinkState

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Describes the state(s) of a FireBlade.WinInteropUtils.WinForms.Link.

[Flags]

```
public enum HyperLinkLinkState
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

DefaultColors = 16

Enable custom text colors to be used.

Enabled = 2

The link can respond to user input. This is the default unless the entire control is disabled. In this case, all links are disabled.

Focused = 1

The link has the keyboard focus. Pressing ENTER fires the FireBlade.WinInteropUtils.WinForms.HyperLink.LinkClicked event.

HotTrack = 8

Indicates that the [HyperLink](#) control will highlight in a different color ([Highlight](#)) when the mouse hovers over the control.

Visited = 4

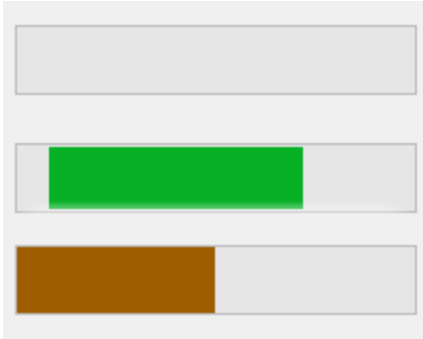
The link has been visited by the user. Changing the URL to one that has not been visited causes this flag to be cleared.

Class ProgressBarEx

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Represents a Win32 progress bar control.



```
public class ProgressBarEx : WinInteropUtilsControlBase, IDropTarget, ISynchronizeInvoke,
    IWin32Window, IBindableComponent, IComponent, IDisposable
```

Inheritance

[object](#) ← [MarshalByRefObject](#) ← [Component](#) ← [Control](#) ← [WinInteropUtilsControlBase](#) ← [ProgressBarEx](#)

Implements

[IDropTarget](#) , [ISynchronizeInvoke](#) , [IWin32Window](#) , [IBindableComponent](#) , [IComponent](#) , [IDisposable](#)

Inherited Members

[WinInteropUtilsControlBase.Window](#) , [Control.GetAccessibilityObjectById\(int\)](#) , [Control.SetAutoSizeMode\(AutoSizeMode\)](#) , [Control.GetAutoSizeMode\(\)](#) , [Control.GetPreferredSize\(Size\)](#) , [Control.AccessibilityNotifyClients\(AccessibleEvents, int\)](#) , [Control.AccessibilityNotifyClients\(AccessibleEvents, int, int\)](#) , [Control.BeginInvoke\(Delegate\)](#) , [Control.BeginInvoke\(Action\)](#) , [Control.BeginInvoke\(Delegate, params object\[\]\)](#) , [Control.BringToFront\(\)](#) , [Control.Contains\(Control\)](#) , [Control.CreateAccessibilityInstance\(\)](#) , [Control.CreateControlsInstance\(\)](#) , [Control.CreateGraphics\(\)](#) , [Control.CreateHandle\(\)](#) , [Control.CreateControl\(\)](#) , [Control.DefWndProc\(ref Message\)](#) , [Control.DestroyHandle\(\)](#) , [Control.Dispose\(bool\)](#) , [Control.DoDragDrop\(object, DragDropEffects\)](#) , [Control.DoDragDrop\(object, DragDropEffects, Bitmap, Point, bool\)](#) , [Control.DrawToBitmap\(Bitmap, Rectangle\)](#) , [Control.EndInvoke\(IAsyncResult\)](#) , [Control.FindForm\(\)](#) , [Control.GetTopLevel\(\)](#) , [Control.RaiseKeyEvent\(object, KeyEventArgs\)](#) , [Control.RaiseMouseEvent\(object, MouseEventArgs\)](#) , [Control.Focus\(\)](#)

[Control.FromChildHandle\(nint\)](#), [Control.FromHandle\(nint\)](#),
[Control.GetChildAtPoint\(Point, GetChildAtPointSkip\)](#), [Control.GetChildAtPoint\(Point\)](#),
[Control.GetContainerControl\(\)](#), [Control.GetScaledBounds\(Rectangle, SizeF, BoundsSpecified\)](#),
[Control.GetNextControl\(Control, bool\)](#), [Control.GetStyle\(ControlStyles\)](#), [Control.Hide\(\)](#),
[Control.InitLayout\(\)](#), [Control.Invalidate\(Region\)](#), [Control.Invalidate\(Region, bool\)](#),
[Control.Invalidate\(\)](#), [Control.Invalidate\(bool\)](#), [Control.Invalidate\(Rectangle\)](#),
[Control.Invalidate\(Rectangle, bool\)](#), [Control.Invoke\(Action\)](#), [Control.Invoke\(Delegate\)](#),
[Control.Invoke\(Delegate, params object\[\]\)](#), [Control.Invoke<T>\(Func<T>\)](#),
[Control.InvokePaint\(Control, PaintEventArgs\)](#),
[Control.InvokePaintBackground\(Control, PaintEventArgs\)](#), [Control.IsKeyLocked\(Keys\)](#),
[Control.IsInputChar\(char\)](#), [Control.IsInputKey\(Keys\)](#), [Control.IsMnemonic\(char, string\)](#),
[Control.LogicalToDeviceUnits\(int\)](#), [Control.LogicalToDeviceUnits\(Size\)](#),
[Control.ScaleBitmapLogicalToDevice\(ref Bitmap\)](#), [Control.NotifyInvalidate\(Rectangle\)](#),
[Control.OnClick\(Control, EventArgs\)](#), [Control.OnAutoSizeChanged\(EventArgs\)](#),
[Control.OnBackColorChanged\(EventArgs\)](#), [Control.OnBackgroundImageChanged\(EventArgs\)](#),
[Control.OnBackgroundImageLayoutChanged\(EventArgs\)](#),
[Control.OnBindingContextChanged\(EventArgs\)](#), [Control.OnCausesValidationChanged\(EventArgs\)](#),
[Control.OnContextMenuStripChanged\(EventArgs\)](#), [Control.OnCursorChanged\(EventArgs\)](#),
[Control.OnDataContextChanged\(EventArgs\)](#), [Control.OnDockChanged\(EventArgs\)](#),
[Control.OnEnabledChanged\(EventArgs\)](#), [Control.OnFontChanged\(EventArgs\)](#),
[Control.OnForeColorChanged\(EventArgs\)](#), [Control.OnRightToLeftChanged\(EventArgs\)](#),
[Control.OnNotifyMessage\(Message\)](#), [Control.OnParentBackColorChanged\(EventArgs\)](#),
[Control.OnParentBackgroundImageChanged\(EventArgs\)](#),
[Control.OnParentBindingContextChanged\(EventArgs\)](#), [Control.OnParentCursorChanged\(EventArgs\)](#),
[Control.OnParentDataContextChanged\(EventArgs\)](#), [Control.OnParentEnabledChanged\(EventArgs\)](#),
[Control.OnParentFontChanged\(EventArgs\)](#), [Control.OnParentForeColorChanged\(EventArgs\)](#),
[Control.OnParentRightToLeftChanged\(EventArgs\)](#), [Control.OnParentVisibleChanged\(EventArgs\)](#),
[Control.OnPrint\(PaintEventArgs\)](#), [Control.OnTabIndexChanged\(EventArgs\)](#),
[Control.OnTabStopChanged\(EventArgs\)](#), [Control.OnTextChanged\(EventArgs\)](#),
[Control.OnVisibleChanged\(EventArgs\)](#), [Control.OnParentChanged\(EventArgs\)](#),
[Control.OnClick\(EventArgs\)](#), [Control.OnClientSizeChanged\(EventArgs\)](#),
[Control.OnControlAdded\(ControlEventArgs\)](#), [Control.OnControlRemoved\(ControlEventArgs\)](#),
[Control.OnCreateControl\(\)](#), [Control.OnLocationChanged\(EventArgs\)](#),
[Control.OnHandleDestroyed\(EventArgs\)](#), [Control.OnDoubleClick\(EventArgs\)](#),
[Control.OnDragEnter\(DragEventArgs\)](#), [Control.OnDragOver\(DragEventArgs\)](#),
[Control.OnDragLeave\(EventArgs\)](#), [Control.OnDragDrop\(DragEventArgs\)](#),
[Control.OnGiveFeedback\(GiveFeedbackEventArgs\)](#), [Control.OnEnter\(EventArgs\)](#),
[Control.InvokeGotFocus\(Control, EventArgs\)](#), [Control.OnGotFocus\(EventArgs\)](#),
[Control.OnHelpRequested\(HelpEventArgs\)](#), [Control.OnInvalidated\(InvalidateEventArgs\)](#),
[Control.OnKeyDown\(KeyEventArgs\)](#), [Control.OnKeyPress\(KeyPressEventArgs\)](#),

[Control.OnKeyUp\(KeyEventArgs\)](#), [Control.OnLayout\(LayoutEventArgs\)](#),
[Control.OnLeave\(EventArgs\)](#), [Control.InvokeLostFocus\(Control, EventArgs\)](#),
[Control.OnLostFocus\(EventArgs\)](#), [Control.OnMarginChanged\(EventArgs\)](#),
[Control.OnMouseDoubleClick\(MouseEventArgs\)](#), [Control.OnMouseClicked\(MouseEventArgs\)](#),
[Control.OnMouseCaptureChanged\(EventArgs\)](#), [Control.OnMouseDown\(MouseEventArgs\)](#),
[Control.OnMouseEnter\(EventArgs\)](#), [Control.OnMouseLeave\(EventArgs\)](#),
[Control.OnDpiChangedBeforeParent\(EventArgs\)](#), [Control.OnDpiChangedAfterParent\(EventArgs\)](#),
[Control.OnMouseHover\(EventArgs\)](#), [Control.OnMouseMove\(MouseEventArgs\)](#),
[Control.OnMouseUp\(MouseEventArgs\)](#), [Control.OnMouseWheel\(MouseEventArgs\)](#),
[Control.OnMove\(EventArgs\)](#), [Control.OnPaint\(PaintEventArgs\)](#),
[Control.OnPaddingChanged\(EventArgs\)](#), [Control.OnPaintBackground\(PaintEventArgs\)](#),
[Control.OnQueryContinueDrag\(QueryContinueDragEventArgs\)](#),
[Control.OnRegionChanged\(EventArgs\)](#), [Control.OnResize\(EventArgs\)](#),
[Control.OnPreviewKeyDown\(PreviewKeyDownEventArgs\)](#), [Control.OnSizeChanged\(EventArgs\)](#),
[Control.OnChangeUICues\(UICuesEventArgs\)](#), [Control.OnStyleChanged\(EventArgs\)](#),
[Control.OnSystemColorsChanged\(EventArgs\)](#), [Control.OnValidating\(CancelEventArgs\)](#),
[Control.OnValidated\(EventArgs\)](#), [Control.RescaleConstantsForDpi\(int, int\)](#),
[Control.PerformLayout\(\)](#), [Control.PerformLayout\(Control, string\)](#), [Control.PointToClient\(Point\)](#),
[Control.PointToScreen\(Point\)](#), [Control.PreProcessMessage\(ref Message\)](#),
[Control.PreProcessControlMessage\(ref Message\)](#), [Control.ProcessCmdKey\(ref Message, Keys\)](#),
[Control.ProcessDialogChar\(char\)](#), [Control.ProcessDialogKey\(Keys\)](#),
[Control.ProcessKeyEventArgs\(ref Message\)](#), [Control.ProcessKeyMessage\(ref Message\)](#),
[Control.ProcessKeyPreview\(ref Message\)](#), [Control.ProcessMnemonic\(char\)](#),
[Control.RaiseDragEvent\(object, DragEventArgs\)](#), [Control.RaisePaintEvent\(object, PaintEventArgs\)](#),
[Control.RecreateHandle\(\)](#), [Control.RectangleToClient\(Rectangle\)](#),
[Control.RectangleToScreen\(Rectangle\)](#), [Control.ReflectMessage\(nint, ref Message\)](#),
[Control.Refresh\(\)](#), [Control.ResetMouseEventArgs\(\)](#), [Control.ResetText\(\)](#), [Control.ResumeLayout\(\)](#),
[Control.ResumeLayout\(bool\)](#), [Control.Scale\(SizeF\)](#), [Control.ScaleControl\(SizeF, BoundsSpecified\)](#),
[Control.Select\(\)](#), [Control.Select\(bool, bool\)](#),
[Control.SelectNextControl\(Control, bool, bool, bool, bool\)](#), [Control.SendToBack\(\)](#),
[Control.SetBounds\(int, int, int, int\)](#), [Control.SetBounds\(int, int, int, int, BoundsSpecified\)](#),
[Control.SetBoundsCore\(int, int, int, int, BoundsSpecified\)](#), [Control.SetClientSizeCore\(int, int\)](#),
[Control.SizeFromClientSize\(Size\)](#), [Control.SetStyle\(ControlStyles, bool\)](#), [Control.SetTopLevel\(bool\)](#),
[Control.SetVisibleCore\(bool\)](#), [Control.RtlTranslateAlignment\(HorizontalAlignment\)](#),
[Control.RtlTranslateAlignment\(LeftRightAlignment\)](#),
[Control.RtlTranslateAlignment\(ContentAlignment\)](#),
[Control.RtlTranslateHorizontal\(HorizontalAlignment\)](#),
[Control.RtlTranslateLeftRight\(LeftRightAlignment\)](#), [Control.RtlTranslateContent\(ContentAlignment\)](#),
[Control.Show\(\)](#), [Control.SuspendLayout\(\)](#), [Control.Update\(\)](#), [Control.UpdateBounds\(\)](#),
[Control.UpdateBounds\(int, int, int, int\)](#), [Control.UpdateBounds\(int, int, int, int, int, int\)](#),

[Control.UpdateZOrder\(\)](#), [Control.UpdateStyles\(\)](#), [Control.WndProc\(ref Message\)](#),
[Control.OnImeModeChanged\(EventArgs\)](#), [Control.AccessibilityObject](#),
[Control.AccessibleDefaultActionDescription](#), [Control.AccessibleDescription](#),
[Control.AccessibleName](#), [Control.AccessibleRole](#), [Control.AllowDrop](#), [Control.Anchor](#),
[Control.AutoScrollOffset](#), [Control.LayoutEngine](#), [Control.DataContext](#),
[Control.BackgroundImage](#), [Control.BackgroundImageLayout](#), [Control.BindingContext](#),
[Control.Bottom](#), [Control.Bounds](#), [Control.CanFocus](#), [Control.CanRaiseEvents](#),
[Control.CanSelect](#), [Control.Capture](#), [Control.CausesValidation](#),
[Control.CheckForIllegalCrossThreadCalls](#), [Control.ClientRectangle](#), [Control.ClientSize](#),
[Control.CompanyName](#), [Control.ContainsFocus](#), [Control.ContextMenuStrip](#), [Control.Controls](#),
[Control.Created](#), [Control.Cursor](#), [Control.DataBindings](#), [Control.DefaultBackColor](#),
[Control.DefaultCursor](#), [Control.DefaultFont](#), [Control.DefaultForeColor](#), [Control.DefaultMargin](#),
[Control.DefaultMaximumSize](#), [Control.DefaultMinimumSize](#), [Control.DefaultPadding](#),
[Control.DefaultSize](#), [Control.DeviceDpi](#), [Control.DisplayRectangle](#), [Control.IsDisposed](#),
[Control.Disposing](#), [Control.Dock](#), [Control.DoubleBuffered](#), [Control.Enabled](#), [Control.Focused](#),
[Control.Font](#), [Control.FontHeight](#), [Control.Handle](#), [Control.HasChildren](#), [Control.Height](#),
[Control.IsHandleCreated](#), [Control.InvokeRequired](#), [Control.IsAccessible](#),
[Control.IsAncestorSiteInDesignMode](#), [Control.IsMirrored](#), [Control.Left](#), [Control.Location](#),
[Control.Margin](#), [Control.MaximumSize](#), [Control.MinimumSize](#), [Control.ModifierKeys](#),
[Control.MouseButtons](#), [Control.MousePosition](#), [Control.Name](#), [Control.Parent](#),
[Control.ProductName](#), [Control.ProductVersion](#), [Control.RecreatingHandle](#), [Control.Region](#),
[Control.RenderRightToLeft](#), [Control.ResizeRedraw](#), [Control.Right](#), [Control.RightToLeft](#),
[Control.ScaleChildren](#), [Control.Site](#), [Control.Size](#), [Control.TabIndex](#), [Control.TabStop](#),
[Control.Tag](#), [Control.Text](#), [Control.Top](#), [Control.TopLevelControl](#), [Control.ShowKeyboardCues](#),
[Control.ShowFocusCues](#), [Control.UseWaitCursor](#), [Control.Visible](#), [Control.Width](#),
[Control.PreferredSize](#), [Control.Padding](#), [Control.CanEnableIme](#), [Control.DefaultImeMode](#),
[Control.ImeMode](#), [Control.ImeModeBase](#), [Control.PropagatingImeMode](#),
[Control.BackColorChanged](#), [Control.BackgroundImageChanged](#),
[Control.BackgroundImageLayoutChanged](#), [Control.BindingContextChanged](#),
[Control.CausesValidationChanged](#), [Control.ClientSizeChanged](#),
[Control.ContextMenuStripChanged](#), [Control.CursorChanged](#), [Control.DockChanged](#),
[Control.EnabledChanged](#), [Control.FontChanged](#), [Control.ForeColorChanged](#),
[Control.LocationChanged](#), [Control.MarginChanged](#), [Control.RegionChanged](#),
[Control.RightToLeftChanged](#), [Control.SizeChanged](#), [Control.TabIndexChanged](#),
[Control.TabStopChanged](#), [Control.TextChanged](#), [Control.VisibleChanged](#), [Control.Click](#),
[Control.ControlAdded](#), [Control.ControlRemoved](#), [Control.DataContextChanged](#),
[Control.DragDrop](#), [Control.DragEnter](#), [Control.DragOver](#), [Control.DragLeave](#),
[Control.GiveFeedback](#), [Control.HandleCreated](#), [Control.HandleDestroyed](#),
[Control.HelpRequested](#), [Control.Invalidated](#), [Control.PaddingChanged](#), [Control.Paint](#),
[Control.QueryContinueDrag](#), [Control.QueryAccessibilityHelp](#), [Control.DoubleClick](#),

[Control.Enter](#) , [Control.GotFocus](#) , [Control.KeyDown](#) , [Control.KeyPress](#) , [Control.KeyUp](#) , [Control.Layout](#) , [Control.Leave](#) , [Control.LostFocus](#) , [Control.MouseClick](#) , [Control.MouseDoubleClick](#) , [Control.MouseCaptureChanged](#) , [Control.MouseDown](#) , [Control.MouseEnter](#) , [Control.MouseLeave](#) , [Control.DpiChangedBeforeParent](#) , [Control.DpiChangedAfterParent](#) , [Control.MouseHover](#) , [Control.MouseMove](#) , [Control.MouseUp](#) , [Control.MouseWheel](#) , [Control.Move](#) , [Control.PreviewKeyDown](#) , [Control.Resize](#) , [Control.ChangeUICues](#) , [Control.StyleChanged](#) , [Control.SystemColorsChanged](#) , [Control.Validating](#) , [Control.Validated](#) , [Control.ParentChanged](#) , [Control.ImeModeChanged](#) , [Component.Dispose\(\)](#) , [Component.GetService\(Type\)](#) , [Component.ToString\(\)](#) , [Component.Container](#) , [Component.DesignMode](#) , [Component.Events](#) , [Component.Disposed](#) , [MarshalByRefObject.GetLifetimeService\(\)](#) , [MarshalByRefObject.InitializeLifetimeService\(\)](#) , [MarshalByRefObject.MemberwiseClone\(bool\)](#) , [object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#)

Constructors

ProgressBarEx()

Initializes a new instance of the [ProgressBarEx](#) class.

```
public ProgressBarEx()
```

Properties

BackColor

Gets or sets the background color of the progress bar.

```
public override Color BackColor { get; set; }
```

Property Value

[Color](#)

Remarks

Has no effect when visual styles are enabled.

CreateParams

Gets the required creation parameters when the control handle is created.

```
protected override CreateParams CreateParams { get; }
```

Property Value

[CreateParams](#)

A [CreateParams](#) that contains the required creation parameters when the handle to the control is created.

ForeColor

Gets or sets the bar color of the progress bar.

```
public override Color ForeColor { get; set; }
```

Property Value

[Color](#)

Remarks

Has no effect when visual styles are enabled.

MarqueeSpeed

Specifies the speed, in milliseconds, of the marquee animation when [Type](#) is set to [Marquee](#).

```
public int MarqueeSpeed { get; set; }
```

Property Value

[int](#)

Maximum

Specifies the maximum value the progress bar represents.

```
public int Maximum { get; set; }
```

Property Value

[int](#)

Minimum

Specifies the minimum value the progress bar represents.

```
public int Minimum { get; set; }
```

Property Value

[int](#)

Orientation

Specifies the orientation of the progress bar.

```
public Orientation Orientation { get; set; }
```

Property Value

[Orientation](#)

ReverseOnBackward

Determines the animation behavior that the progress bar should use when moving backward (from a higher value to a lower value). If this is set, then a "smooth" transition will occur, otherwise the control will "jump" to the lower value.

```
public bool ReverseOnBackward { get; set; }
```

Property Value

[bool](#)

Step

Gets or sets the amount that a call to [PerformStep\(\)](#) increases the progress bar's current position by.

```
public int Step { get; set; }
```

Property Value

[int](#)

Style

Gets or sets the style of the progress bar.

```
public ProgressBarExStyle Style { get; set; }
```

Property Value

[ProgressBarExStyle](#)

Type

Specifies the type of the progress bar.

```
public ProgressBarExType Type { get; set; }
```

Property Value

[ProgressBarExType](#)

UseDefaultBackColor

If [true](#), use the default progress bar background color.

```
public bool UseDefaultBackColor { get; set; }
```

Property Value

[bool](#)

Remarks

Has no effect when visual styles are enabled.

UseDefaultForeColor

If [true](#), use the default progress bar bar color.

```
public bool UseDefaultForeColor { get; set; }
```

Property Value

[bool](#)

Remarks

Has no effect when visual styles are enabled.

Value

Specifies the current value the progress bar displays.

```
public int Value { get; set; }
```

Property Value

[int](#)

Methods

OnHandleCreated(EventArgs)

Raises the [HandleCreated](#) event.

```
protected override void OnHandleCreated(EventArgs e)
```

Parameters

e [EventArgs](#)

An [EventArgs](#) that contains the event data.

PerformStep()

Advances the current position of the [ProgressBarEx](#) by the amount of the [Step](#) property, and redraws the control to reflect the new position.

```
public int PerformStep()
```

Returns

[int](#)

The previous position.

Remarks

When the position exceeds the maximum range value, this method resets the current position so that the progress indicator starts over again from the beginning.

Exceptions

[InvalidOperationException](#)

A step cannot be performed on a marquee progress bar as it does not use a value.

[InvalidOperationException](#)

Can't perform a step right now; the control handle has not been created yet.

StepBy(int)

Advances the current position of the progress bar by a specified increment and redraws the bar to reflect the new position.

```
public int StepBy(int step)
```

Parameters

step [int](#)[↗]

The amount to advance the position by.

Returns

[int](#)[↗]

The previous position.

Remarks

If the increment results in a value outside the range of the control, the position is set to the nearest boundary.

Exceptions

[InvalidOperationException](#)[↗]

A step cannot be performed on a marquee progress bar as it does not use a value.

[InvalidOperationException](#)[↗]

Can't perform a step right now; the control handle has not been created yet.

Enum ProgressBarExStyle

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Specifies the style of a [ProgressBarEx](#) control.

```
public enum ProgressBarExStyle
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Error = 2

Shows an error (red) progress bar.

Normal = 1

Shows a regular progress bar.

Paused = 3

Shows a paused (yellow) progress bar.

Enum ProgressBarExType

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Specifies the [Type](#) of a [ProgressBarEx](#).

```
public enum ProgressBarExType
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Blocks = 1

The progress bar displays progress status in blocks that fill up accross the bar. This is the default.

Marquee = 2

The progress indicator does not grow in size but instead moves repeatedly along the length of the bar, indicating activity without specifying what proportion of the progress is complete.

NOTE

Requires visual styles. Make sure that you include [EnableVisualStyles\(\)](#) inside your Program class before starting your application.

Smooth = 0

The progress bar displays progress status in a smooth scrolling bar instead of the default segmented bar.

NOTE

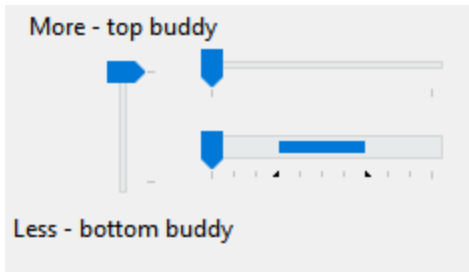
This style is supported only in the **Windows Classic** theme. All other themes override this style.

Class Slider

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Represents a control that allows the user to slide a slider along a track, commonly referred to as a track bar.



```
public class Slider : WinInteropUtilsControlBase, IDropTarget, ISynchronizeInvoke,
    IWin32Window, IBindableComponent, IComponent, IDisposable
```

Inheritance

[object](#) ← [MarshalByRefObject](#) ← [Component](#) ← [Control](#) ← [WinInteropUtilsControlBase](#) ← Slider

Implements

[IDropTarget](#), [ISynchronizeInvoke](#), [IWin32Window](#), [IBindableComponent](#), [IComponent](#), [IDisposable](#)

Inherited Members

[WinInteropUtilsControlBase.Window](#), [Control.GetAccessibilityObjectById\(int\)](#), [Control.SetAutoSizeMode\(AutoSizeMode\)](#), [Control.GetAutoSizeMode\(\)](#), [Control.GetPreferredSize\(Size\)](#), [Control.AccessibilityNotifyClients\(AccessibleEvents, int\)](#), [Control.AccessibilityNotifyClients\(AccessibleEvents, int, int\)](#), [Control.BeginInvoke\(Delegate\)](#), [Control.BeginInvoke\(Action\)](#), [Control.BeginInvoke\(Delegate, params object\[\]\)](#), [Control.BringToFront\(\)](#), [Control.Contains\(Control\)](#), [Control.CreateAccessibilityInstance\(\)](#), [Control.CreateControlsInstance\(\)](#), [Control.CreateGraphics\(\)](#), [Control.CreateHandle\(\)](#), [Control.CreateControl\(\)](#), [Control.DefWndProc\(ref Message\)](#), [Control.DestroyHandle\(\)](#), [Control.Dispose\(bool\)](#), [Control.DoDragDrop\(object, DragDropEffects\)](#), [Control.DoDragDrop\(object, DragDropEffects, Bitmap, Point, bool\)](#), [Control.DrawToBitmap\(Bitmap, Rectangle\)](#), [Control.EndInvoke\(IAsyncResult\)](#), [Control.FindForm\(\)](#), [Control.GetTopLevel\(\)](#), [Control.RaiseKeyEvent\(object, KeyEventArgs\)](#), [Control.RaiseMouseEvent\(object, MouseEventArgs\)](#), [Control.Focus\(\)](#), [Control.FromChildHandle\(nint\)](#), [Control.FromHandle\(nint\)](#),

[Control.GetChildAtPoint\(Point, GetChildAtPointSkip\)](#), [Control.GetChildAtPoint\(Point\)](#),
[Control.GetContainerControl\(\)](#), [Control.GetScaledBounds\(Rectangle, SizeF, BoundsSpecified\)](#),
[Control.GetNextControl\(Control, bool\)](#), [Control.GetStyle\(ControlStyles\)](#), [Control.Hide\(\)](#),
[Control.InitLayout\(\)](#), [Control.Invalidate\(Region\)](#), [Control.Invalidate\(Region, bool\)](#),
[Control.Invalidate\(\)](#), [Control.Invalidate\(bool\)](#), [Control.Invalidate\(Rectangle\)](#),
[Control.Invalidate\(Rectangle, bool\)](#), [Control.Invoke\(Action\)](#), [Control.Invoke\(Delegate\)](#),
[Control.Invoke\(Delegate, params object\[\]\)](#), [Control.Invoke<T>\(Func<T>\)](#),
[Control.InvokePaint\(Control, PaintEventArgs\)](#),
[Control.InvokePaintBackground\(Control, PaintEventArgs\)](#), [Control.IsKeyLocked\(Keys\)](#),
[Control.IsInputChar\(char\)](#), [Control.IsInputKey\(Keys\)](#), [Control.IsMnemonic\(char, string\)](#),
[Control.LogicalToDeviceUnits\(int\)](#), [Control.LogicalToDeviceUnits\(Size\)](#),
[Control.ScaleBitmapLogicalToDevice\(ref Bitmap\)](#), [Control.NotifyInvalidate\(Rectangle\)](#),
[Control.OnClick\(Control, EventArgs\)](#), [Control.OnAutoSizeChanged\(EventArgs\)](#),
[Control.OnBackgroundImageChanged\(EventArgs\)](#),
[Control.OnBackgroundImageLayoutChanged\(EventArgs\)](#),
[Control.OnBindingContextChanged\(EventArgs\)](#), [Control.OnCausesValidationChanged\(EventArgs\)](#),
[Control.OnContextMenuStripChanged\(EventArgs\)](#), [Control.OnCursorChanged\(EventArgs\)](#),
[Control.OnDataContextChanged\(EventArgs\)](#), [Control.OnDockChanged\(EventArgs\)](#),
[Control.OnEnabledChanged\(EventArgs\)](#), [Control.OnFontChanged\(EventArgs\)](#),
[Control.OnForeColorChanged\(EventArgs\)](#), [Control.OnRightToLeftChanged\(EventArgs\)](#),
[Control.OnNotifyMessage\(Message\)](#), [Control.OnParentBackColorChanged\(EventArgs\)](#),
[Control.OnParentBackgroundImageChanged\(EventArgs\)](#),
[Control.OnParentBindingContextChanged\(EventArgs\)](#), [Control.OnParentCursorChanged\(EventArgs\)](#),
[Control.OnParentDataContextChanged\(EventArgs\)](#), [Control.OnParentEnabledChanged\(EventArgs\)](#),
[Control.OnParentFontChanged\(EventArgs\)](#), [Control.OnParentForeColorChanged\(EventArgs\)](#),
[Control.OnParentRightToLeftChanged\(EventArgs\)](#), [Control.OnParentVisibleChanged\(EventArgs\)](#),
[Control.OnPrint\(PaintEventArgs\)](#), [Control.OnTabIndexChanged\(EventArgs\)](#),
[Control.OnTabStopChanged\(EventArgs\)](#), [Control.OnTextChanged\(EventArgs\)](#),
[Control.OnVisibleChanged\(EventArgs\)](#), [Control.OnParentChanged\(EventArgs\)](#),
[Control.OnClick\(EventArgs\)](#), [Control.OnClientSizeChanged\(EventArgs\)](#),
[Control.OnControlAdded\(ControlEventArgs\)](#), [Control.OnControlRemoved\(ControlEventArgs\)](#),
[Control.OnCreateControl\(\)](#), [Control.OnLocationChanged\(EventArgs\)](#),
[Control.OnHandleDestroyed\(EventArgs\)](#), [Control.OnDoubleClick\(EventArgs\)](#),
[Control.OnDragEnter\(DragEventArgs\)](#), [Control.OnDragOver\(DragEventArgs\)](#),
[Control.OnDragLeave\(EventArgs\)](#), [Control.OnDragDrop\(DragEventArgs\)](#),
[Control.OnGiveFeedback\(GiveFeedbackEventArgs\)](#), [Control.OnEnter\(EventArgs\)](#),
[Control.InvokeGotFocus\(Control, EventArgs\)](#), [Control.OnGotFocus\(EventArgs\)](#),
[Control.OnHelpRequested\(HelpEventArgs\)](#), [Control.OnInvalidated\(InvalidateEventArgs\)](#),
[Control.OnKeyDown\(KeyEventArgs\)](#), [Control.OnKeyPress\(KeyPressEventArgs\)](#),
[Control.OnKeyUp\(KeyEventArgs\)](#), [Control.OnLayout\(LayoutEventArgs\)](#),

[Control.OnLeave\(EventArgs\)](#), [Control.InvokeLostFocus\(Control, EventArgs\)](#),
[Control.OnLostFocus\(EventArgs\)](#), [Control.OnMarginChanged\(EventArgs\)](#),
[Control.OnMouseDoubleClick\(MouseEventArgs\)](#), [Control.OnMouseClick\(MouseEventArgs\)](#),
[Control.OnMouseCaptureChanged\(EventArgs\)](#), [Control.OnMouseDown\(MouseEventArgs\)](#),
[Control.OnMouseEnter\(EventArgs\)](#), [Control.OnMouseLeave\(EventArgs\)](#),
[Control.OnDpiChangedBeforeParent\(EventArgs\)](#), [Control.OnDpiChangedAfterParent\(EventArgs\)](#),
[Control.OnMouseHover\(EventArgs\)](#), [Control.OnMouseMove\(MouseEventArgs\)](#),
[Control.OnMouseUp\(MouseEventArgs\)](#), [Control.OnMouseWheel\(MouseEventArgs\)](#),
[Control.OnMove\(EventArgs\)](#), [Control.OnPaint\(PaintEventArgs\)](#),
[Control.OnPaddingChanged\(EventArgs\)](#), [Control.OnPaintBackground\(PaintEventArgs\)](#),
[Control.OnQueryContinueDrag\(QueryContinueDragEventArgs\)](#),
[Control.OnRegionChanged\(EventArgs\)](#), [Control.OnResize\(EventArgs\)](#),
[Control.OnPreviewKeyDown\(PreviewKeyDownEventArgs\)](#), [Control.OnSizeChanged\(EventArgs\)](#),
[Control.OnChangeUICues\(UICuesEventArgs\)](#), [Control.OnStyleChanged\(EventArgs\)](#),
[Control.OnSystemColorsChanged\(EventArgs\)](#), [Control.OnValidating\(CancelEventArgs\)](#),
[Control.OnValidated\(EventArgs\)](#), [Control.RescaleConstantsForDpi\(int, int\)](#),
[Control.PerformLayout\(\)](#), [Control.PerformLayout\(Control, string\)](#), [Control.PointToClient\(Point\)](#),
[Control.PointToScreen\(Point\)](#), [Control.PreProcessMessage\(ref Message\)](#),
[Control.PreProcessControlMessage\(ref Message\)](#), [Control.ProcessCmdKey\(ref Message, Keys\)](#),
[Control.ProcessDialogChar\(char\)](#), [Control.ProcessDialogKey\(Keys\)](#),
[Control.ProcessKeyEventArgs\(ref Message\)](#), [Control.ProcessKeyMessage\(ref Message\)](#),
[Control.ProcessKeyPreview\(ref Message\)](#), [Control.ProcessMnemonic\(char\)](#),
[Control.RaiseDragEvent\(object, DragEventArgs\)](#), [Control.RaisePaintEvent\(object, PaintEventArgs\)](#),
[Control.RecreateHandle\(\)](#), [Control.RectangleToClient\(Rectangle\)](#),
[Control.RectangleToScreen\(Rectangle\)](#), [Control.ReflectMessage\(nint, ref Message\)](#),
[Control.Refresh\(\)](#), [Control.ResetMouseEventArgs\(\)](#), [Control.ResetText\(\)](#), [Control.ResumeLayout\(\)](#),
[Control.ResumeLayout\(bool\)](#), [Control.Scale\(SizeF\)](#), [Control.ScaleControl\(SizeF, BoundsSpecified\)](#),
[Control.Select\(\)](#), [Control.Select\(bool, bool\)](#),
[Control.SelectNextControl\(Control, bool, bool, bool, bool\)](#), [Control.SendToBack\(\)](#),
[Control.SetBounds\(int, int, int, int\)](#), [Control.SetBounds\(int, int, int, int, BoundsSpecified\)](#),
[Control.SetBoundsCore\(int, int, int, int, BoundsSpecified\)](#), [Control.SetClientSizeCore\(int, int\)](#),
[Control.SizeFromClientSize\(Size\)](#), [Control.SetStyle\(ControlStyles, bool\)](#), [Control.SetTopLevel\(bool\)](#),
[Control.SetVisibleCore\(bool\)](#), [Control.RtlTranslateAlignment\(HorizontalAlignment\)](#),
[Control.RtlTranslateAlignment\(LeftRightAlignment\)](#),
[Control.RtlTranslateAlignment\(ContentAlignment\)](#),
[Control.RtlTranslateHorizontal\(HorizontalAlignment\)](#),
[Control.RtlTranslateLeftRight\(LeftRightAlignment\)](#), [Control.RtlTranslateContent\(ContentAlignment\)](#),
[Control.Show\(\)](#), [Control.SuspendLayout\(\)](#), [Control.Update\(\)](#), [Control.UpdateBounds\(\)](#),
[Control.UpdateBounds\(int, int, int, int\)](#), [Control.UpdateBounds\(int, int, int, int, int, int\)](#),
[Control.UpdateZOrder\(\)](#), [Control.UpdateStyles\(\)](#), [Control.OnImeModeChanged\(EventArgs\)](#),

[Control.AccessibilityObject](#), [Control.AccessibleDefaultActionDescription](#),
[Control.AccessibleDescription](#), [Control.AccessibleName](#), [Control.AccessibleRole](#),
[Control.AllowDrop](#), [Control.Anchor](#), [Control.AutoScrollOffset](#), [Control.LayoutEngine](#),
[Control.DataContext](#), [Control.BackColor](#), [Control.BackgroundImage](#),
[Control.BackgroundImageLayout](#), [Control.BindingContext](#), [Control.Bottom](#), [Control.Bounds](#),
[Control.CanFocus](#), [Control.CanRaiseEvents](#), [Control.CanSelect](#), [Control.Capture](#),
[Control.CausesValidation](#), [Control.CheckForIllegalCrossThreadCalls](#), [Control.ClientRectangle](#),
[Control.ClientSize](#), [Control.CompanyName](#), [Control.ContainsFocus](#), [Control.ContextMenuStrip](#),
[Control.Controls](#), [Control.Created](#), [Control.Cursor](#), [Control.DataBindings](#),
[Control.DefaultBackColor](#), [Control.DefaultCursor](#), [Control.DefaultFont](#),
[Control.DefaultForeColor](#), [Control.DefaultMargin](#), [Control.DefaultMaximumSize](#),
[Control.DefaultMinimumSize](#), [Control.DefaultPadding](#), [Control.DefaultSize](#), [Control.DeviceDpi](#),
[Control.DisplayRectangle](#), [Control.IsDisposed](#), [Control.Disposing](#), [Control.Dock](#),
[Control.DoubleBuffered](#), [Control.Enabled](#), [Control.Focused](#), [Control.Font](#),
[Control.FontHeight](#), [Control.ForeColor](#), [Control.Handle](#), [Control.HasChildren](#), [Control.Height](#),
[Control.IsHandleCreated](#), [Control.InvokeRequired](#), [Control.IsAccessible](#),
[Control.IsAncestorSiteInDesignMode](#), [Control.IsMirrored](#), [Control.Left](#), [Control.Location](#),
[Control.Margin](#), [Control.MaximumSize](#), [Control.MinimumSize](#), [Control.ModifierKeys](#),
[Control.MouseButtons](#), [Control.MousePosition](#), [Control.Name](#), [Control.Parent](#),
[Control.ProductName](#), [Control.ProductVersion](#), [Control.RecreatingHandle](#), [Control.Region](#),
[Control.RenderRightToLeft](#), [Control.ResizeRedraw](#), [Control.Right](#), [Control.RightToLeft](#),
[Control.ScaleChildren](#), [Control.Site](#), [Control.Size](#), [Control.TabIndex](#), [Control.TabStop](#),
[Control.Tag](#), [Control.Text](#), [Control.Top](#), [Control.TopLevelControl](#), [Control.ShowKeyboardCues](#),
[Control.ShowFocusCues](#), [Control.UseWaitCursor](#), [Control.Visible](#), [Control.Width](#),
[Control.PreferredSize](#), [Control.Padding](#), [Control.CanEnableIme](#), [Control.DefaultImeMode](#),
[Control.ImeMode](#), [Control.ImeModeBase](#), [Control.PropagatingImeMode](#),
[Control.BackColorChanged](#), [Control.BackgroundImageChanged](#),
[Control.BackgroundImageLayoutChanged](#), [Control.BindingContextChanged](#),
[Control.CausesValidationChanged](#), [Control.ClientSizeChanged](#),
[Control.ContextMenuStripChanged](#), [Control.CursorChanged](#), [Control.DockChanged](#),
[Control.EnabledChanged](#), [Control.FontChanged](#), [Control.ForeColorChanged](#),
[Control.LocationChanged](#), [Control.MarginChanged](#), [Control.RegionChanged](#),
[Control.RightToLeftChanged](#), [Control.SizeChanged](#), [Control.TabIndexChanged](#),
[Control.TabStopChanged](#), [Control.TextChanged](#), [Control.VisibleChanged](#), [Control.Click](#),
[Control.ControlAdded](#), [Control.ControlRemoved](#), [Control.DataContextChanged](#),
[Control.DragDrop](#), [Control.DragEnter](#), [Control.DragOver](#), [Control.DragLeave](#),
[Control.GiveFeedback](#), [Control.HandleCreated](#), [Control.HandleDestroyed](#),
[Control.HelpRequested](#), [Control.Invalidated](#), [Control.PaddingChanged](#), [Control.Paint](#),
[Control.QueryContinueDrag](#), [Control.QueryAccessibilityHelp](#), [Control.DoubleClick](#),
[Control.Enter](#), [Control.GotFocus](#), [Control.KeyDown](#), [Control.KeyPress](#), [Control.KeyUp](#),

[Control.Layout](#), [Control.Leave](#), [Control.LostFocus](#), [Control.MouseClick](#), [Control.MouseDoubleClick](#), [Control.MouseCaptureChanged](#), [Control.MouseDown](#), [Control.MouseEnter](#), [Control.MouseLeave](#), [Control.DpiChangedBeforeParent](#), [Control.DpiChangedAfterParent](#), [Control.MouseHover](#), [Control.MouseMove](#), [Control.MouseUp](#), [Control.MouseWheel](#), [Control.Move](#), [Control.PreviewKeyDown](#), [Control.Resize](#), [Control.ChangeUICues](#), [Control.StyleChanged](#), [Control.SystemColorsChanged](#), [Control.Validating](#), [Control.Validated](#), [Control.ParentChanged](#), [Control.ImeModeChanged](#), [Component.Dispose\(\)](#), [Component.GetService\(Type\)](#), [Component.ToString\(\)](#), [Component.Container](#), [Component.DesignMode](#), [Component.Events](#), [Component.Disposed](#), [MarshalByRefObject.GetLifetimeService\(\)](#), [MarshalByRefObject.InitializeLifetimeService\(\)](#), [MarshalByRefObject.MemberwiseClone\(bool\)](#), [object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#)

Constructors

Slider()

```
public Slider()
```

Properties

CreateParams

Gets the required creation parameters when the control handle is created.

```
protected override CreateParams CreateParams { get; }
```

Property Value

[CreateParams](#)

A [CreateParams](#) that contains the required creation parameters when the handle to the control is created.

FixedLength

Gets or sets a value indicating whether the length of the slider thumb can be overridden via the [ThumbLength](#) property.

```
public bool FixedLength { get; set; }
```

Property Value

[bool](#)

IsThumbVisible

Gets or sets whether the slider thumb is visible.

```
public bool IsThumbVisible { get; set; }
```

Property Value

[bool](#)

LargeChange

Gets or sets the amount of ticks that the slider will move when a large change occurs, such as moving the slider with the [PageUp](#)/[PageDown](#) keys.

```
public int LargeChange { get; set; }
```

Property Value

[int](#)

Remarks

Also commonly referred to as the "page size".

LeftBuddy

Gets or sets the buddy to the left of the slider if the slider is horizontal, or above the slider if the slider is vertical.

```
public Control? LeftBuddy { get; set; }
```

Property Value

[Control](#)[↗]

Exceptions

[ArgumentException](#)[↗]

The buddy can't be set to a Form.

[ArgumentException](#)[↗]

The buddy can't be set to the same Slider that the buddy is being set on.

Maximum

Gets or sets the maximum value the slider can represent.

```
public int Maximum { get; set; }
```

Property Value

[int](#)[↗]

Minimum

Gets or sets the minimum value the slider can represent.

```
public int Minimum { get; set; }
```

Property Value

[int](#)[↗]

Orientation

Gets or sets the orientation of the slider.

```
public Orientation Orientation { get; set; }
```

Property Value

[Orientation](#)

ReverseInput

Gets or sets whether input is reversed.

```
public bool ReverseInput { get; set; }
```

Property Value

[bool](#)

Remarks

By default, the control uses down equal to right and up equal to left. Enable to reverse the default, making down equal left and up equal right.

Reversed

Gets or sets whether the slider is reversed, where smaller numbers mean higher and larger numbers mean lower.

```
public bool Reversed { get; set; }
```

Property Value

[bool](#)

RightBuddy

Gets or sets the buddy to the right of the slider if the slider is horizontal, or below the slider if the slider is vertical.

```
public Control? RightBuddy { get; set; }
```

Property Value

[Control](#)

Exceptions

[ArgumentException](#)

The buddy can't be set to a Form.

[ArgumentException](#)

The buddy can't be set to the same Slider that the buddy is being set on.

SelectionRangeEnd

Gets or sets the end of the slider's selection range.

```
public int SelectionRangeEnd { get; set; }
```

Property Value

[int](#)

Remarks

The slider's selection range does not affect its functionality in any way. It is up to the application to implement the range. You might do this in one of the following ways:

- Use a selection range to enable the user to set maximum and minimum values for some parameter. For example, the user could move the slider to a position and then click a button labeled "Max". The application then sets the selection range to show the values chosen by the user.
- Limit the movement of the slider to a predetermined subrange within the control, by handling the [ValueChanged](#) event and disallowing any movement outside the selection range. You might do this,

for example, if the range of values available to the user can change because of other choices the user has made, or according to available resources.

SelectionRangeStart

Gets or sets the start of the slider's selection range.

```
public int SelectionRangeStart { get; set; }
```

Property Value

[int](#)

Remarks

The slider's selection range does not affect its functionality in any way. It is up to the application to implement the range. You might do this in one of the following ways:

- Use a selection range to enable the user to set maximum and minimum values for some parameter. For example, the user could move the slider to a position and then click a button labeled "Max". The application then sets the selection range to show the values chosen by the user.
- Limit the movement of the slider to a predetermined subrange within the control, by handling the [ValueChanging](#) event and disallowing any movement outside the selection range. You might do this, for example, if the range of values available to the user can change because of other choices the user has made, or according to available resources.

ShowSelectionRange

Gets or sets a value indicating whether the slider's selection range should be shown.

```
public bool ShowSelectionRange { get; set; }
```

Property Value

[bool](#)

Remarks

The slider's selection range does not affect its functionality in any way. It is up to the application to implement the range. You might do this in one of the following ways:

- Use a selection range to enable the user to set maximum and minimum values for some parameter. For example, the user could move the slider to a position and then click a button labeled "Max". The application then sets the selection range to show the values chosen by the user.
- Limit the movement of the slider to a predetermined subrange within the control, by handling the [ValueChanging](#) event and disallowing any movement outside the selection range. You might do this, for example, if the range of values available to the user can change because of other choices the user has made, or according to available resources.

ShowToolTip

Gets or sets a value indicating whether to show a tooltip showing the value of the slider.

```
public bool ShowToolTip { get; set; }
```

Property Value

[bool](#)

SmallChange

Gets or sets the amount of ticks that the slider will move when a small change occurs, such as moving the slider with the arrow keys.

```
public int SmallChange { get; set; }
```

Property Value

[int](#)

Remarks

Also commonly referred to as the "line size".

ThumbLength

Gets or sets the length of the thumb in pixels. Only applies when [FixedLength](#) is set to [true](#).

```
public int ThumbLength { get; set; }
```

Property Value

[int](#)

ThumbRect

Gets the size and position of the bounding rectangle for the slider.

```
public Rectangle ThumbRect { get; }
```

Property Value

[Rectangle](#)

TickFrequency

Gets or sets the frequency of ticks on the slider. Only applies when [TickMode](#) is [Custom](#).

```
public int TickFrequency { get; set; }
```

Property Value

[int](#)

TickMode

Specifies the types of ticks displayed on the slider.

```
public SliderTickMode TickMode { get; set; }
```

Property Value

TickStyle

Gets or sets the position of ticks in the slider.

```
public TickStyle TickStyle { get; set; }
```

Property Value

[TickStyle](#)↗

ToolTipSide

Gets or sets the side where the tooltip should be displayed.

```
public SliderToolTipSide ToolTipSide { get; set; }
```

Property Value

[SliderToolTipSide](#)

Value

Gets or sets the currently selected value of the slider.

```
public int Value { get; set; }
```

Property Value

[int](#)↗

Methods

AddTick(int)

Adds a tick to the [Slider](#) at the specified position.

```
public void AddTick(int pos)
```

Parameters

pos [int](#)↗

The position of the tick.

ClearSelectionRange()

Clears the slider's selection range and redraws the control.

```
public void ClearSelectionRange()
```

ClearSelectionRange(bool)

Clears the slider's selection range and optionally redraws the control.

```
public void ClearSelectionRange(bool redraw)
```

Parameters

redraw [bool](#)↗

Whether to redraw the control.

ClearTicks()

Removes the current tick marks from the slider. This message does not remove the first and last tick marks, which are created automatically by the slider.

```
public void ClearTicks()
```

GetTickCount()

Retrieves the number of tick marks in a slider.

```
public int? GetTickCount()
```

Returns

[int](#)?

If [TickMode](#) isn't set to [Custom](#), returns 2 for the beginning and ending ticks. If it's set to [None](#), returns zero. Otherwise, it takes the difference between the range minimum and maximum, divides by the tick frequency, and adds 2.

Remarks

The [GetTickCount\(\)](#) method counts all of the tick marks, including the first and last tick marks created by the slider.

GetTickPosition(int)

Retrieves the logical position of a tick mark in a slider. The logical position can be any of the integer values in the slider's range of minimum to maximum slider positions.

```
public int GetTickPosition(int idx)
```

Parameters

idx [int](#)

Zero-based index identifying a tick mark. Valid indexes are in the range from zero to two less than the tick count returned by the [GetTickCount\(\)](#) method.

Returns

[int](#)

The logical position of the specified tick mark, or -1 if **idx** does not specify a valid index, or the handle of the slider hasn't been created.

GetTickPositions()

Retrieves an array that contains the positions of the tick marks for a slider.

```
public uint[]? GetTickPositions()
```

Returns

[uint](#)  []

An [uint](#)  array. The elements of the array specify the logical positions of the slider's tick marks, not including the first and last tick marks created by the slider. The logical positions can be any of the integer values in the slider's range of minimum to maximum slider positions.

Remarks

The number of elements in the array is two less than the tick count returned by the [GetTickCount\(\)](#) method. Note that the values in the array may include duplicate positions and may not be in sequential order. The returned array is valid until you change the slider's tick marks.

OnBackColorChanged(EventArgs)

Raises the [BackColorChanged](#)  event.

```
protected override void OnBackColorChanged(EventArgs e)
```

Parameters

e [EventArgs](#) 

An [EventArgs](#)  that contains the event data.

OnHandleCreated(EventArgs)

Raises the [HandleCreated](#)  event.

```
protected override void OnHandleCreated(EventArgs e)
```


Parameters

e [EventArgs](#)

An [EventArgs](#) that contains the event data.

WndProc(ref Message)

Processes Windows messages.

```
protected override void WndProc(ref Message m)
```

Parameters

m [Message](#)

The Windows [Message](#) to process.

Events

HorizontalScroll

Fires when the slider scrolls horizontally.

```
public event EventHandler<SliderScrollEventArgs>? HorizontalScroll
```

Event Type

[EventHandler](#) <[SliderScrollEventArgs](#)>

ValueChanging

Fires before the slider's value changes and allows validation.

```
public event EventHandler<SliderValueChangingEventArgs>? ValueChanging
```

Event Type

[EventHandler](#) <[SliderValueChangingEventArgs](#)>

VerticalScroll

Fires when the slider scrolls vertically.

```
public event EventHandler<SliderScrollEventArgs> VerticalScroll
```

Event Type

[EventHandler](#) <[SliderScrollEventArgs](#)>

Class SliderActionList

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

```
public class SliderActionList : DesignerActionList
```

Inheritance

[object](#) ← [DesignerActionList](#) ← SliderActionList

Inherited Members

[DesignerActionList.GetService\(Type\)](#), [DesignerActionList.AutoShow](#),
[DesignerActionList.Component](#), [object.Equals\(object\)](#), [object.Equals\(object, object\)](#),
[object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#),
[object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Constructors

SliderActionList(IComponent?)

```
public SliderActionList(IComponent? comp)
```

Parameters

comp [IComponent](#)

Methods

ClearSelectionRange()

```
public void ClearSelectionRange()
```

GetSortedActionItems()

Returns the collection of [DesignerActionItem](#) objects contained in the list.

```
public override DesignerActionItemCollection GetSortedActionItems()
```

Returns

[DesignerActionItemCollection](#)

A [DesignerActionItem](#) array that contains the items in this list.

Class SliderDesigner

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

```
public class SliderDesigner : ControlDesigner, ITreeDesigner, IDesigner, IDisposable,
    IDesignerFilter, IComponentInitializer
```

Inheritance

[object](#) ← [ComponentDesigner](#) ← [ControlDesigner](#) ← SliderDesigner

Implements

[ITreeDesigner](#), [IDesigner](#), [IDisposable](#), [IDesignerFilter](#), [IComponentInitializer](#)

Inherited Members

[ControlDesigner.InvalidPoint](#), [ControlDesigner.accessibilityObj](#),
[ControlDesigner.NumberOfInternalControlDesigners\(\)](#),
[ControlDesigner.InternalControlDesigner\(int\)](#), [ControlDesigner.BaseWndProc\(ref Message\)](#),
[ControlDesigner.CanBeParentedTo\(IDesigner\)](#), [ControlDesigner.DefWndProc\(ref Message\)](#),
[ControlDesigner.DisplayError\(Exception\)](#), [ControlDesigner.Dispose\(bool\)](#),
[ControlDesigner.EnableDesignMode\(Control, string\)](#), [ControlDesigner.EnableDragDrop\(bool\)](#),
[ControlDesigner.GetControlGlyph\(GlyphSelectionType\)](#),
[ControlDesigner.GetGlyphs\(GlyphSelectionType\)](#), [ControlDesigner.GetHitTest\(Point\)](#),
[ControlDesigner.HookChildControls\(Control\)](#), [ControlDesigner.Initialize\(IComponent\)](#),
[ControlDesigner.InitializeExistingComponent\(IDictionary\)](#),
[ControlDesigner.InitializeNewComponent\(IDictionary\)](#), [ControlDesigner.OnSetComponentDefaults\(\)](#),
[ControlDesigner.OnContextMenu\(int, int\)](#), [ControlDesigner.OnCreateHandle\(\)](#),
[ControlDesigner.OnDragEnter\(DragEventArgs\)](#), [ControlDesigner.OnDragComplete\(DragEventArgs\)](#),
[ControlDesigner.OnDragDrop\(DragEventArgs\)](#), [ControlDesigner.OnDragLeave\(EventArgs\)](#),
[ControlDesigner.OnDragOver\(DragEventArgs\)](#),
[ControlDesigner.OnGiveFeedback\(GiveFeedbackEventArgs\)](#),
[ControlDesigner.OnMouseDownBegin\(int, int\)](#), [ControlDesigner.OnMouseDownEnd\(bool\)](#),
[ControlDesigner.OnMouseDownMove\(int, int\)](#), [ControlDesigner.OnMouseDownEnter\(\)](#),
[ControlDesigner.OnMouseDownHover\(\)](#), [ControlDesigner.OnMouseDownLeave\(\)](#),
[ControlDesigner.OnPaintAdornments\(PaintEventArgs\)](#), [ControlDesigner.OnSetCursor\(\)](#),
[ControlDesigner.PreFilterProperties\(IDictionary\)](#), [ControlDesigner.UnhookChildControls\(Control\)](#),
[ControlDesigner.WndProc\(ref Message\)](#), [ControlDesigner.BehaviorService](#),
[ControlDesigner.AssociatedComponents](#), [ControlDesigner.AccessibilityObject](#),
[ControlDesigner.Control](#), [ControlDesigner.EnableDragRect](#), [ControlDesigner.ParentComponent](#),

[ControlDesigner.ParticipatesWithSnapLines](#), [ControlDesigner.AutoResizeHandles](#),
[ControlDesigner.SnapLines](#), [ControlDesigner.InheritanceAttribute](#), [ComponentDesigner.Dispose\(\)](#),
[ComponentDesigner.DoDefaultAction\(\)](#),
[ComponentDesigner.InvokeGetInheritanceAttribute\(ComponentDesigner\)](#),
[ComponentDesigner.InitializeNonDefault\(\)](#), [ComponentDesigner.GetService\(Type\)](#),
[ComponentDesigner.PostFilterAttributes\(IDictionary\)](#),
[ComponentDesigner.PostFilterEvents\(IDictionary\)](#),
[ComponentDesigner.PostFilterProperties\(IDictionary\)](#),
[ComponentDesigner.PreFilterAttributes\(IDictionary\)](#),
[ComponentDesigner.PreFilterEvents\(IDictionary\)](#),
[ComponentDesigner.RaiseComponentChanged\(MemberDescriptor, object, object\)](#),
[ComponentDesigner.RaiseComponentChanging\(MemberDescriptor\)](#),
[ComponentDesigner.Inherited](#), [ComponentDesigner.ShadowProperties](#),
[ComponentDesigner.Component](#), [ComponentDesigner.Verbs](#), [object.Equals\(object\)](#),
[object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#),
[object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Properties

ActionLists

Gets the design-time action lists supported by the component associated with the designer.

```
public override DesignerActionListCollection ActionLists { get; }
```

Property Value

[DesignerActionListCollection](#)

The design-time action lists supported by the component associated with the designer.

SelectionRules

Gets the selection rules that indicate the movement capabilities of a component.

```
public override SelectionRules SelectionRules { get; }
```

Property Value

[SelectionRules](#)[↗]

A bitwise combination of [SelectionRules](#)[↗] values.

Class SliderScrollEventArgs

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Provides event arguments for the [HorizontalScroll](#) and [VerticalScroll](#) events.

```
public class SliderScrollEventArgs : EventArgs
```

Inheritance

[object](#)  ← [EventArgs](#)  ← SliderScrollEventArgs

Inherited Members

[EventArgs.Empty](#) , [object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Properties

Reason

Gets the reason for the event.

```
public SliderValueChangeReason Reason { get; }
```

Property Value

[SliderValueChangeReason](#)

Source

The [Slider](#) control that was the source of the event. You should rely on the `sender` parameter passed to your event handler function instead of using this property for a more reliable way of checking the event source.

```
public Slider? Source { get; }
```


Property Value

[Slider](#)

Value

Specifies the current position of the slider.

```
public int Value { get; }
```

Property Value

[int](#)

Remarks

Specifies the current position of the slider if the [Reason](#) is [Thumb](#) or [ThumbDrag](#). For all other [Slider ValueChangeReason](#) values, the value of this property is zero; check the [Value](#) property to determine the slider position.

Enum SliderTickMode

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Specifies the type of ticks to use on a [Slider](#).

```
public enum SliderTickMode
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Auto = 0

The control has a tick mark for each increment in its range of values.

Custom = 1

The control has a custom amount of ticks dictated by the [TickFrequency](#) property.

None = 16

No ticks.

Enum SliderToolTipSide

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Specifies the side where the tooltip is located on a [Slider](#).

```
public enum SliderToolTipSide
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Bottom = 2

The tooltip control will be positioned below the slider.

Left = 1

The tooltip control will be positioned to the left of the slider.

Right = 3

The tooltip control will be positioned to the right of the trackbar.

Top = 0

The tooltip control will be positioned above the slider.

Enum SliderValueChangeReason

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Specifies the reason that a [Slider](#)'s value may be changing.

```
public enum SliderValueChangeReason
```

Extension Methods

[EnumExtensions.GetValues\(Enum\)](#)

Fields

Bottom = 7

The user has set the slider to the minimum value, for example by pressing the **Home** key.

LargeDecrease = 3

The user has decreased the value of the slider by clicking the track.

LargeIncrease = 2

The user has incremented the slider by clicking the track.

SmallDecrease = 1

The user has decreased the slider using the arrow keys.

SmallIncrease = 0

The user has incremented the slider using the arrow keys.

Thumb = 4

The user has finished dragging the thumb and released the mouse button.

ThumbDrag = 5

The user is currently dragging the thumb and moving it around. Fires continuously while the thumb is being dragged.

ThumbEnd = 8

The user released the thumb after dragging.

Use this reason only for cleanup, as it doesn't give you the actual moved-to value. For validation, please use the [Thumb](#) reason instead.

Top = 6

The user has set the slider to the maximum value, for example by pressing the **End** key.

See Also

[Slider](#)

Class SliderValueChangingEventArgs

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Defines event arguments for the [ValueChanging](#) event.

```
public class SliderValueChangingEventArgs : CancelEventArgs
```

Inheritance

[object](#)  ← [EventArgs](#)  ← [CancelEventArgs](#)  ← SliderValueChangingEventArgs

Inherited Members

[CancelEventArgs.Cancel](#) , [EventArgs.Empty](#) , [object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Properties

NewValue

Gets the new position of the thumb (value of the slider).

```
public int NewValue { get; }
```

Property Value

[int](#) 

Reason

Gets the reason for the event.

```
public SliderValueChangeReason Reason { get; }
```

Property Value

Class WinInteropUtilsControlBase

Namespace: [FireBlade.WinInteropUtils.WinForms](#)

Assembly: WinInteropUtils.WinForms.dll

Represents the base class for WinInteropUtils.WinForms controls.

```
public abstract class WinInteropUtilsControlBase : Control, IDropTarget, ISynchronizeInvoke,
    IWin32Window, IBindableComponent, IComponent, IDisposable
```

Inheritance

[object](#) ← [MarshalByRefObject](#) ← [Component](#) ← [Control](#) ← WinInteropUtilsControlBase

Implements

[IDropTarget](#), [ISynchronizeInvoke](#), [IWin32Window](#), [IBindableComponent](#), [IComponent](#), [IDisposable](#)

Derived

[HotKeyBox](#), [ProgressBarEx](#), [Slider](#)

Inherited Members

[Control.GetAccessibilityObjectById\(int\)](#), [Control.SetAutoSizeMode\(AutoSizeMode\)](#), [Control.GetAutoSizeMode\(\)](#), [Control.GetPreferredSize\(Size\)](#), [Control.AccessibilityNotifyClients\(AccessibleEvents, int\)](#), [Control.AccessibilityNotifyClients\(AccessibleEvents, int, int\)](#), [Control.BeginInvoke\(Delegate\)](#), [Control.BeginInvoke\(Action\)](#), [Control.BeginInvoke\(Delegate, params object\[\]\)](#), [Control.BringToFront\(\)](#), [Control.Contains\(Control\)](#), [Control.CreateAccessibilityInstance\(\)](#), [Control.CreateControlsInstance\(\)](#), [Control.CreateGraphics\(\)](#), [Control.CreateHandle\(\)](#), [Control.CreateControl\(\)](#), [Control.DefWndProc\(ref Message\)](#), [Control.DestroyHandle\(\)](#), [Control.Dispose\(bool\)](#), [Control.DoDragDrop\(object, DragDropEffects\)](#), [Control.DoDragDrop\(object, DragDropEffects, Bitmap, Point, bool\)](#), [Control.DrawToBitmap\(Bitmap, Rectangle\)](#), [Control.EndInvoke\(IAsyncResult\)](#), [Control.FindForm\(\)](#), [Control.GetTopLevel\(\)](#), [Control.RaiseKeyEvent\(object, KeyEventArgs\)](#), [Control.RaiseMouseEvent\(object, MouseEventArgs\)](#), [Control.Focus\(\)](#), [Control.FromChildHandle\(nint\)](#), [Control.FromHandle\(nint\)](#), [Control.GetChildAtPoint\(Point, GetChildAtPointSkip\)](#), [Control.GetChildAtPoint\(Point\)](#), [Control.GetContainerControl\(\)](#), [Control.GetScaledBounds\(Rectangle, SizeF, BoundsSpecified\)](#), [Control.GetNextControl\(Control, bool\)](#), [Control.GetStyle\(ControlStyles\)](#), [Control.Hide\(\)](#), [Control.InitLayout\(\)](#), [Control.Invalidate\(Region\)](#), [Control.Invalidate\(Region, bool\)](#), [Control.Invalidate\(\)](#), [Control.Invalidate\(bool\)](#), [Control.Invalidate\(Rectangle\)](#),

[Control.Invalidate\(Rectangle, bool\)](#), [Control.Invoke\(Action\)](#), [Control.Invoke\(Delegate\)](#),
[Control.Invoke\(Delegate, params object\[\]\)](#), [Control.Invoke<T>\(Func<T>\)](#),
[Control.InvokePaint\(Control, PaintEventArgs\)](#),
[Control.InvokePaintBackground\(Control, PaintEventArgs\)](#), [Control.IsKeyLocked\(Keys\)](#),
[Control.IsInputChar\(char\)](#), [Control.IsInputKey\(Keys\)](#), [Control.IsMnemonic\(char, string\)](#),
[Control.LogicalToDeviceUnits\(int\)](#), [Control.LogicalToDeviceUnits\(Size\)](#),
[Control.ScaleBitmapLogicalToDevice\(ref Bitmap\)](#), [Control.NotifyInvalidate\(Rectangle\)](#),
[Control.OnClick\(Control, EventArgs\)](#), [Control.OnAutoSizeChanged\(EventArgs\)](#),
[Control.OnBackColorChanged\(EventArgs\)](#), [Control.OnBackgroundImageChanged\(EventArgs\)](#),
[Control.OnBackgroundImageLayoutChanged\(EventArgs\)](#),
[Control.OnBindingContextChanged\(EventArgs\)](#), [Control.OnCausesValidationChanged\(EventArgs\)](#),
[Control.OnContextMenuStripChanged\(EventArgs\)](#), [Control.OnCursorChanged\(EventArgs\)](#),
[Control.OnDataContextChanged\(EventArgs\)](#), [Control.OnDockChanged\(EventArgs\)](#),
[Control.OnEnabledChanged\(EventArgs\)](#), [Control.OnFontChanged\(EventArgs\)](#),
[Control.OnForeColorChanged\(EventArgs\)](#), [Control.OnRightToLeftChanged\(EventArgs\)](#),
[Control.OnNotifyMessage\(Message\)](#), [Control.OnParentBackColorChanged\(EventArgs\)](#),
[Control.OnParentBackgroundImageChanged\(EventArgs\)](#),
[Control.OnParentBindingContextChanged\(EventArgs\)](#), [Control.OnParentCursorChanged\(EventArgs\)](#),
[Control.OnParentDataContextChanged\(EventArgs\)](#), [Control.OnParentEnabledChanged\(EventArgs\)](#),
[Control.OnParentFontChanged\(EventArgs\)](#), [Control.OnParentForeColorChanged\(EventArgs\)](#),
[Control.OnParentRightToLeftChanged\(EventArgs\)](#), [Control.OnParentVisibleChanged\(EventArgs\)](#),
[Control.OnPrint\(PaintEventArgs\)](#), [Control.OnTabIndexChanged\(EventArgs\)](#),
[Control.OnTabStopChanged\(EventArgs\)](#), [Control.OnTextChanged\(EventArgs\)](#),
[Control.OnVisibleChanged\(EventArgs\)](#), [Control.OnParentChanged\(EventArgs\)](#),
[Control.OnClick\(EventArgs\)](#), [Control.OnClientSizeChanged\(EventArgs\)](#),
[Control.OnControlAdded\(ControlEventArgs\)](#), [Control.OnControlRemoved\(ControlEventArgs\)](#),
[Control.OnCreateControl\(\)](#), [Control.OnHandleCreated\(EventArgs\)](#),
[Control.OnLocationChanged\(EventArgs\)](#), [Control.OnHandleDestroyed\(EventArgs\)](#),
[Control.OnDoubleClick\(EventArgs\)](#), [Control.OnDragEnter\(DragEventArgs\)](#),
[Control.OnDragOver\(DragEventArgs\)](#), [Control.OnDragLeave\(EventArgs\)](#),
[Control.OnDragDrop\(DragEventArgs\)](#), [Control.OnGiveFeedback\(GiveFeedbackEventArgs\)](#),
[Control.OnEnter\(EventArgs\)](#), [Control.InvokeGotFocus\(Control, EventArgs\)](#),
[Control.OnGotFocus\(EventArgs\)](#), [Control.OnHelpRequested\(HelpEventArgs\)](#),
[Control.OnInvalidated\(InvalidateEventArgs\)](#), [Control.OnKeyDown\(KeyEventArgs\)](#),
[Control.OnKeyPress\(KeyPressEventArgs\)](#), [Control.OnKeyUp\(KeyEventArgs\)](#),
[Control.OnLayout\(LayoutEventArgs\)](#), [Control.OnLeave\(EventArgs\)](#),
[Control.InvokeLostFocus\(Control, EventArgs\)](#), [Control.OnLostFocus\(EventArgs\)](#),
[Control.OnMarginChanged\(EventArgs\)](#), [Control.OnMouseDoubleClick\(MouseEventArgs\)](#),
[Control.OnMouseClick\(MouseEventArgs\)](#), [Control.OnMouseCaptureChanged\(EventArgs\)](#),
[Control.OnMouseDown\(MouseEventArgs\)](#), [Control.OnMouseEnter\(EventArgs\)](#),

[Control.OnMouseLeave\(EventArgs\)](#), [Control.OnDpiChangedBeforeParent\(EventArgs\)](#),
[Control.OnDpiChangedAfterParent\(EventArgs\)](#), [Control.OnMouseHover\(EventArgs\)](#),
[Control.OnMouseMove\(MouseEventArgs\)](#), [Control.OnMouseUp\(MouseEventArgs\)](#),
[Control.OnMouseWheel\(MouseEventArgs\)](#), [Control.OnMove\(EventArgs\)](#),
[Control.OnPaint\(PaintEventArgs\)](#), [Control.OnPaddingChanged\(EventArgs\)](#),
[Control.OnPaintBackground\(PaintEventArgs\)](#),
[Control.OnQueryContinueDrag\(QueryContinueDragEventArgs\)](#),
[Control.OnRegionChanged\(EventArgs\)](#), [Control.OnResize\(EventArgs\)](#),
[Control.OnPreviewKeyDown\(PreviewKeyDownEventArgs\)](#), [Control.OnSizeChanged\(EventArgs\)](#),
[Control.OnChangeUICues\(UICuesEventArgs\)](#), [Control.OnStyleChanged\(EventArgs\)](#),
[Control.OnSystemColorsChanged\(EventArgs\)](#), [Control.OnValidating\(CancelEventArgs\)](#),
[Control.OnValidated\(EventArgs\)](#), [Control.RescaleConstantsForDpi\(int, int\)](#),
[Control.PerformLayout\(\)](#), [Control.PerformLayout\(Control, string\)](#), [Control.PointToClient\(Point\)](#),
[Control.PointToScreen\(Point\)](#), [Control.PreProcessMessage\(ref Message\)](#),
[Control.PreProcessControlMessage\(ref Message\)](#), [Control.ProcessCmdKey\(ref Message, Keys\)](#),
[Control.ProcessDialogChar\(char\)](#), [Control.ProcessDialogKey\(Keys\)](#),
[Control.ProcessKeyEventArgs\(ref Message\)](#), [Control.ProcessKeyMessage\(ref Message\)](#),
[Control.ProcessKeyPreview\(ref Message\)](#), [Control.ProcessMnemonic\(char\)](#),
[Control.RaiseDragEvent\(object, DragEventArgs\)](#), [Control.RaisePaintEvent\(object, PaintEventArgs\)](#),
[Control.RecreateHandle\(\)](#), [Control.RectangleToClient\(Rectangle\)](#),
[Control.RectangleToScreen\(Rectangle\)](#), [Control.ReflectMessage\(nint, ref Message\)](#),
[Control.Refresh\(\)](#), [Control.ResetMouseEventArgs\(\)](#), [Control.ResetText\(\)](#), [Control.ResumeLayout\(\)](#),
[Control.ResumeLayout\(bool\)](#), [Control.Scale\(SizeF\)](#), [Control.ScaleControl\(SizeF, BoundsSpecified\)](#),
[Control.Select\(\)](#), [Control.Select\(bool, bool\)](#),
[Control.SelectNextControl\(Control, bool, bool, bool, bool\)](#), [Control.SendToBack\(\)](#),
[Control.SetBounds\(int, int, int, int\)](#), [Control.SetBounds\(int, int, int, int, BoundsSpecified\)](#),
[Control.SetBoundsCore\(int, int, int, int, BoundsSpecified\)](#), [Control.SetClientSizeCore\(int, int\)](#),
[Control.SizeFromClientSize\(Size\)](#), [Control.SetStyle\(ControlStyles, bool\)](#), [Control.SetTopLevel\(bool\)](#),
[Control.SetVisibleCore\(bool\)](#), [Control.RtlTranslateAlignment\(HorizontalAlignment\)](#),
[Control.RtlTranslateAlignment\(LeftRightAlignment\)](#),
[Control.RtlTranslateAlignment\(ContentAlignment\)](#),
[Control.RtlTranslateHorizontal\(HorizontalAlignment\)](#),
[Control.RtlTranslateLeftRight\(LeftRightAlignment\)](#), [Control.RtlTranslateContent\(ContentAlignment\)](#),
[Control.Show\(\)](#), [Control.SuspendLayout\(\)](#), [Control.Update\(\)](#), [Control.UpdateBounds\(\)](#),
[Control.UpdateBounds\(int, int, int, int\)](#), [Control.UpdateBounds\(int, int, int, int, int, int\)](#),
[Control.UpdateZOrder\(\)](#), [Control.UpdateStyles\(\)](#), [Control.WndProc\(ref Message\)](#),
[Control.OnImeModeChanged\(EventArgs\)](#), [Control.AccessibilityObject](#),
[Control.AccessibleDefaultActionDescription](#), [Control.AccessibleDescription](#),
[Control.AccessibleName](#), [Control.AccessibleRole](#), [Control.AllowDrop](#), [Control.Anchor](#),
[Control.AutoScrollOffset](#), [Control.LayoutEngine](#), [Control.DataContext](#), [Control.BackColor](#),

[Control.BackgroundImage](#), [Control.BackgroundImageLayout](#), [Control.BindingContext](#), [Control.Bottom](#), [Control.Bounds](#), [Control.CanFocus](#), [Control.CanRaiseEvents](#), [Control.CanSelect](#), [Control.Capture](#), [Control.CausesValidation](#), [Control.CheckForIllegalCrossThreadCalls](#), [Control.ClientRectangle](#), [Control.ClientSize](#), [Control.CompanyName](#), [Control.ContainsFocus](#), [Control.ContextMenuStrip](#), [Control.Controls](#), [Control.Created](#), [Control.CreateParams](#), [Control.Cursor](#), [Control.DataBindings](#), [Control.DefaultBackColor](#), [Control.DefaultCursor](#), [Control.DefaultFont](#), [Control.DefaultForeColor](#), [Control.DefaultMargin](#), [Control.DefaultMaximumSize](#), [Control.DefaultMinimumSize](#), [Control.DefaultPadding](#), [Control.DefaultSize](#), [Control.DeviceDpi](#), [Control.DisplayRectangle](#), [Control.IsDisposed](#), [Control.Disposing](#), [Control.Dock](#), [Control.DoubleBuffered](#), [Control.Enabled](#), [Control.Focused](#), [Control.Font](#), [Control.FontHeight](#), [Control.ForeColor](#), [Control.Handle](#), [Control.HasChildren](#), [Control.Height](#), [Control.IsHandleCreated](#), [Control.InvokeRequired](#), [Control.IsAccessible](#), [Control.IsAncestorSiteInDesignMode](#), [Control.IsMirrored](#), [Control.Left](#), [Control.Location](#), [Control.Margin](#), [Control.MaximumSize](#), [Control.MinimumSize](#), [Control.ModifierKeys](#), [Control.MouseButtons](#), [Control.MousePosition](#), [Control.Name](#), [Control.Parent](#), [Control.ProductName](#), [Control.ProductVersion](#), [Control.RecreatingHandle](#), [Control.Region](#), [Control.RenderRightToLeft](#), [Control.ResizeRedraw](#), [Control.Right](#), [Control.RightToLeft](#), [Control.ScaleChildren](#), [Control.Site](#), [Control.Size](#), [Control.TabIndex](#), [Control.TabStop](#), [Control.Tag](#), [Control.Text](#), [Control.Top](#), [Control.TopLevelControl](#), [Control.ShowKeyboardCues](#), [Control.ShowFocusCues](#), [Control.UseWaitCursor](#), [Control.Visible](#), [Control.Width](#), [Control.PreferredSize](#), [Control.Padding](#), [Control.CanEnableIme](#), [Control.DefaultImeMode](#), [Control.ImeMode](#), [Control.ImeModeBase](#), [Control.PropagatingImeMode](#), [Control.BackColorChanged](#), [Control.BackgroundImageChanged](#), [Control.BackgroundImageLayoutChanged](#), [Control.BindingContextChanged](#), [Control.CausesValidationChanged](#), [Control.ClientSizeChanged](#), [Control.ContextMenuStripChanged](#), [Control.CursorChanged](#), [Control.DockChanged](#), [Control.EnabledChanged](#), [Control.FontChanged](#), [Control.ForeColorChanged](#), [Control.LocationChanged](#), [Control.MarginChanged](#), [Control.RegionChanged](#), [Control.RightToLeftChanged](#), [Control.SizeChanged](#), [Control.TabIndexChanged](#), [Control.TabStopChanged](#), [Control.TextChanged](#), [Control.VisibleChanged](#), [Control.Click](#), [Control.ControlAdded](#), [Control.ControlRemoved](#), [Control.DataContextChanged](#), [Control.DragDrop](#), [Control.DragEnter](#), [Control.DragOver](#), [Control.DragLeave](#), [Control.GiveFeedback](#), [Control.HandleCreated](#), [Control.HandleDestroyed](#), [Control.HelpRequested](#), [Control.Invalidated](#), [Control.PaddingChanged](#), [Control.Paint](#), [Control.QueryContinueDrag](#), [Control.QueryAccessibilityHelp](#), [Control.DoubleClick](#), [Control.Enter](#), [Control.GotFocus](#), [Control.KeyDown](#), [Control.KeyPress](#), [Control.KeyUp](#), [Control.Layout](#), [Control.Leave](#), [Control.LostFocus](#), [Control.MouseClick](#), [Control.MouseDoubleClick](#), [Control.MouseCaptureChanged](#), [Control.MouseDown](#), [Control.MouseEnter](#), [Control.MouseLeave](#), [Control.DpiChangedBeforeParent](#),

[Control.DpiChangedAfterParent](#), [Control.MouseHover](#), [Control.MouseMove](#), [Control.MouseUp](#), [Control.MouseWheel](#), [Control.Move](#), [Control.PreviewKeyDown](#), [Control.Resize](#), [Control.ChangeUICues](#), [Control.StyleChanged](#), [Control.SystemColorsChanged](#), [Control.Validating](#), [Control.Validated](#), [Control.ParentChanged](#), [Control.ImeModeChanged](#), [Component.Dispose\(\)](#), [Component.GetService\(Type\)](#), [Component.ToString\(\)](#), [Component.Container](#), [Component.DesignMode](#), [Component.Events](#), [Component.Disposed](#), [MarshalByRefObject.GetLifetimeService\(\)](#), [MarshalByRefObject.InitializeLifetimeService\(\)](#), [MarshalByRefObject.MemberwiseClone\(bool\)](#), [object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#)

Properties

Window

Gets the WinInteropUtils [Window](#) for the current control.

```
[Browsable(false)]  
public Window Window { get; }
```

Property Value

[Window](#)